Classical Algebraic and Analytic Number Theory

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Contents

Part I

Multiplicative Number Theory

Chapter 1

Arithmetic Functions

1.1 Multiplicative and Additive Functions

Any function

$$f : \mathbb{Z}_+ \to \mathbb{C},$$

is said to be **arithmetic**. That is, arithmetic functions are maps from the positive integers to the complex numbers. If an arithmetic function f satisfies

$$f(nm) = f(n)f(m) \quad \text{or} \quad f(nm) = f(n) + f(m),$$

whenever n and m are relatively prime, we say f is **multiplicative** or **additive** respectively. If either condition holds regardless of the relatively primality of n and m then we say f is **completely multiplicative** or **completely additive**. If f is additive or multiplicative then f is uniquely determined by its values on prime powers and if f is completely additive or completely multiplicative then it is uniquely determined by its values on primes. Moreover, we have

$$f(1) = 1 \quad \text{or} \quad f(1) = 0,$$

according to if f is multiplicative or additive. Most important arithmetic functions are either multiplicative or additive. Below is a list of the most important arithmetic functions (some of these functions are restrictions of common functions but we define them here as arithmetic functions for completeness):

(i) The **constant function**: The function $\mathbf{1}(n)$. This function is neither additive nor multiplicative.

(ii) The **indicator function**: The function $\varepsilon(n)$ defined by

$$\varepsilon(n) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n \geq 2. \end{cases}$$

This function is completely multiplicative.

(iii) The **identity function**: The function $\mathrm{id}(n)$. This function is completely multiplicative.

(iv) The **logarithm**: The function $\log(n)$. This function is completely additive.

(v) The **Möbius function**: The function $\mu(n)$ defined by

$$\mu(n) = \begin{cases} 1 & \text{if } n \text{ is square-free with an even number of prime factors,} \\ -1 & \text{if } n \text{ is square-free with an odd number of prime factors,} \\ 0 & \text{if } n \text{ is not square-free.} \end{cases}$$

This function is multiplicative.

(vi) The **characteristic function of square-free integers**: The square of the Möbius function $\mu^2(n)$. This function is multiplicative.

(vii) **Liouville's function**: The function $\lambda(n)$ defined by

$$\lambda(n) = \begin{cases} 1 & \text{if } n = 1, \\ (-1)^k & \text{if } n \text{ is composed of } k \text{ not necessarily distinct prime factors.} \end{cases}$$

This function is completely multiplicative.

(viii) **Euler's totient function**: The function $\varphi(n)$ defined by

$$\varphi(n) = \sum_{\substack{a \pmod n \\ (a,n)=1}} 1.$$

This function is multiplicative.

(ix) The **divisor function**: The function $\sigma_0(n)$ defined by

$$\sigma_0(n) = \sum_{d|n} 1.$$

This function is multiplicative.

(x) The **sum of divisors function**: The function $\sigma_1(n)$ defined by

$$\sigma_1(n) = \sum_{d|n} d.$$

This function is multiplicative.

(xi) The **generalized sum of divisors function**: The function $\sigma_s(n)$ defined by

$$\sigma_s(n) = \sum_{d|n} d^s,$$

for any $s \in \mathbb{C}$. This function is multiplicative.

(xii) The **number of distinct prime factors function**: The function $\omega(n)$ defined by

$$\omega(n) = \sum_{p|n} 1.$$

This function is additive.

(xiii) The **total number of prime divisors function**: The function $\Omega(n)$ defined by

$$\Omega(n) = \sum_{p^m|n} 1.$$

This function is completely additive.

(xiv) The **von Mangoldt function**: The function $\Lambda(n)$ defined by

$$\Lambda(n) = \begin{cases} 0 & \text{if } n \text{ is not a prime power,} \\ \log(p) & \text{if } n = p^m \text{ for some prime } p \text{ and positive integer } m. \end{cases}$$

This function is neither additive or multiplicative.

1.2 Dirichlet Convolution and Möbius Inversion

If f and g are two arithmetic functions then we can define a new arithmetic function $f * g$ called their **Dirichlet convolution** defined by

$$(f * g)(n) = \sum_{d|n} f(d)g\left(\frac{n}{d}\right).$$

As the sum is over all divisors, Dirichlet convolution is a symmetric operation. Dirichlet convolution preserves multiplicative functions.

Proposition 1.2.1. *If f and g are multiplicative arithmetic functions then so is their Dirichlet convolution $f * g$.*

Proof. Let n and m be relatively prime positive integers. Every divisor d of nm is of the form $d = d'd''$ with d' a divisor of n, d'' a divisor of m, and d' and d'' relatively prime. A short computation shows

$$\sum_{d|nm} f(d)g\left(\frac{nm}{d}\right) = \left(\sum_{d'|n} f(d')g\left(\frac{n}{d'}\right)\right)\left(\sum_{d''|m} f(d'')g\left(\frac{m}{d''}\right)\right),$$

which is equivalent to the claim. $\square$

This result makes the set of multiplicative functions into a commutative semigroup under Dirichlet convolution. It is actually a commutative monoid since the indicator function ε acts as an identity. This means

$$f * \varepsilon = f.$$

A certain case of interest for Dirichlet convolution is when the Möbius function is convolved with the constant function.

Proposition 1.2.2. *We have*

$$\sum_{d|n} \mu(d) = \begin{cases} 1 & \text{if } n = 1, \\ 0 & \text{if } n \geq 2. \end{cases}$$

In particular,

$$\mu * \mathbf{1} = \varepsilon.$$

Proof. The first statement is equivalent to the second, so it suffices to prove the first. The sum $\sum_{d|n} \mu(d)$ is multiplicative by Proposition 1.2.1 so we may assume $n = p^r$ is a nonnegative power of a prime. When $r = 0$, $d = 1$ and the sum is 1. When $r \geq 1$, d runs over $1, p, \ldots, p^r$. Every value is zero except $\mu(1) = 1$ and $\mu(p) = -1$. This proves that the sum is zero. □

With this result we can prove **Möbius inversion**:

Theorem (Möbius inversion). *If f and g are arithmetic functions, then*

$$g(n) = \sum_{d|n} f(d) \quad \text{if and only if} \quad f(n) = \sum_{d|n} g(d)\mu\left(\frac{n}{d}\right).$$

In particular,

$$g = f * \mathbf{1} \quad \text{if and only if} \quad f = g * \mu.$$

Proof. As the first statement is equivalent to the second, we will prove the second statement. Convolving $g = f * \mathbf{1}$ with μ gives $f = g * \mu$ in view of Proposition 1.2.2. This proves the forward implication. The reverse implication follows by convolving $f = g * \mu$ with $\mathbf{1}$ and arguing analogously. □

Chapter 2

Dirichlet Characters and Exponential Sums

We will make minor use of Poisson summation as developed in Appendix B.3.

2.1 Dirichlet Characters

Let m be a positive integer. A multiplicative homomorphism

$$\chi : \mathbb{Z} \to \mathbb{C},$$

is said to be a **Dirichlet character** modulo m if it is m-periodic and such that $\chi(a) = 0$ if and only if $(a, m) > 1$. We call m the **modulus** of χ. A Dirichlet character is necessarily a completely multiplicative arithmetic function when restricted to the positive integers.

We say a Dirichlet character χ is **principal** if it only takes values 0 or 1. There is always a unique principal Dirichlet character modulo m, denoted $\chi_{m,0}$, defined by

$$\chi_{m,0}(a) = \begin{cases} 1 & (a, m) = 1, \\ 0 & (a, m) > 1. \end{cases}$$

When the modulus is 1, the principal Dirichlet character is identically 1 and we call this the **trivial Dirichlet character**. This is the only Dirichlet character modulo 1.

By Euler's little theorem, $a^{\varphi(m)} \equiv 1 \pmod{m}$ whenever $(a, m) = 1$. This forces $\chi(a)^{\varphi(m)} = 1$ and so the nonzero values of any Dirichlet character modulo m are $\varphi(m)$-th roots of unity. This implies that there are only finitely many Dirichlet characters of any fixed modulus. Given two Dirichlet character χ and ψ modulo m, the functions

$$\chi\psi : \mathbb{Z} \to \mathbb{C} \quad \text{and} \quad \overline{\chi} : \mathbb{Z} \to \mathbb{C},$$

are also Dirichlet characters modulo m. This turns the set of such Dirichlet characters into an abelian group denote by X_m where the identity is the principal Dirichlet character modulo m and the inverse is given by the conjugate as the nonzero values of Dirichlet characters are roots of unity.

This is all strikingly similar to characters on $(\mathbb{Z}/m\mathbb{Z})^*$ and there is indeed a connection. By the multiplicativity and m-periodicity of χ, it induces a character of $(\mathbb{Z}/m\mathbb{Z})^*$. Conversely, if we are given a character on $(\mathbb{Z}/m\mathbb{Z})^*$ we can extend it to a Dirichlet character χ by defining it to be m-periodic with $\chi(a) = 0$ if $(a, m) > 1$. We call this extension a **zero extension**. This argument shows that Dirichlet characters modulo m are exactly the zero extensions of characters on $(\mathbb{Z}/m\mathbb{Z})^*$. As abelian groups are isomorphic to their character groups, we deduce that the group of Dirichlet characters modulo m is isomorphic to $(\mathbb{Z}/m\mathbb{Z})^*$. Therefore there are $\varphi(m)$ Dirichlet characters modulo m and we identify them with the characters on $(\mathbb{Z}/m\mathbb{Z})^*$. Just as for characters of abelian groups, we have orthogonality relations called the **Dirichlet orthogonality relations**.

Proposition (Dirichlet orthogonality relations). *Let χ and ψ be Dirichlet characters modulo m and let $a, b \in (\mathbb{Z}/m\mathbb{Z})^*$. Then*

$$\sum_{a \pmod m} (\chi\overline{\psi})(a) = \varphi(m)\delta_{\chi,\psi} \quad \textit{and} \quad \sum_{\chi \pmod m} \chi(a\bar{b}) = \varphi(m)\delta_{a,b}.$$

In particular,

$$\sum_{a \pmod m} \chi(a) = \varphi(m)\delta_{\chi,\chi_{m,0}} \quad \textit{and} \quad \sum_{\chi \pmod m} \chi(a) = \varphi(m)\delta_{a,1}.$$

Proof. If $\chi = \psi$ then the first sum is clearly $\phi(m)$. If not, let $b \in (\mathbb{Z}/m\mathbb{Z})^*$ be such that $(\chi\overline{\psi})(b) \neq 1$. A short computation shows

$$(\chi\overline{\psi})(b) \sum_{a \pmod m} (\chi\overline{\psi})(a) = \sum_{a \pmod m} (\chi\overline{\psi})(a),$$

in which case the sum vanishes. This proves the first identity. For the second, if $a = b$ then the second sum is clearly $\varphi(m)$. If $a \neq b$, we claim that there exists a Dirichlet character ψ modulo m with $\psi(a\bar{b}) \neq 1$. Set $g = a\bar{b}$. The cyclic subgroup $\langle g \rangle$ of $(\mathbb{Z}/m\mathbb{Z})^*$ has some order $d > 1$. Consider the homomorphism

$$\psi_d : \langle g \rangle \to \mathbb{C} \qquad g^k \mapsto e^{\frac{2\pi i k}{d}}.$$

By the structure theorem for finite abelian groups, $(\mathbb{Z}/m\mathbb{Z})^* \cong \langle g \rangle \times H$ for some subgroup H. Whence we define a Dirichlet character ψ modulo m by

$$\psi : (\mathbb{Z}/m\mathbb{Z})^* \to \mathbb{C} \qquad g^k h \mapsto e^{\frac{2\pi i k}{d}}.$$

This ψ is the desired Dirichlet character, A short computation shows

$$\psi(a\bar{b}) \sum_{\chi \pmod m} \chi(a\bar{b}) = \sum_{\chi \pmod m} \chi(a\bar{b}),$$

in which case the sum vanishes. This proves the second identity and the first statement in its entirety. The second statement follows from the first upon taking $\psi = \chi_{m,0}$ and $b = 1$ respectively. $\square$

It is possible for Dirichlet characters of a fixed modulus to arise from Dirichlet characters of a smaller modulus. Suppose χ and χ^* are Dirichlet characters modulo m and d respectively with $d \mid m$. We say χ is induced **induced** from χ^* if

$$\chi(a) = \begin{cases} \chi^*(a) & \text{if } (a, m) = 1, \\ 0 & \text{if } (a, m) > 1. \end{cases}$$

This means χ is a Dirichlet character whose values are given by those of χ^*. Necessarily χ is d-periodic and its nonzero values are $\varphi(d)$-th roots of unity. We say a Dirichlet character is **primitive** if it is not induced by any Dirichlet character other than itself and **imprimitive** otherwise. The principal Dirichlet characters are precisely those induced from the trivial Dirichlet character and the only primitive one is the trivial Dirichlet character itself. Moreover, a Dirichlet character is primitive if and only if its conjugate is. Our primary aim will be to show that every Dirichlet character is induced from a unique primitive Dirichlet character.

Theorem 2.1.1. *Suppose χ is a Dirichlet character modulo m. There exists a unique primitive Dirichlet character $\widetilde{\chi}$ such that χ is induced from $\widetilde{\chi}$.*

Proof. Let q be the positive integer given by

$$q = \min\{d \mid m : \chi(a) = \chi(b) \text{ for } a \equiv b \pmod{d} \text{ with } (ab, m) = 1\},$$

and let $\widetilde{\chi} : \mathbb{Z} \to \mathbb{C}$ be defined by

$$\widetilde{\chi}(a) = \begin{cases} \chi(a) & \text{if } (a, q) = 1, \\ 0 & \text{if } (a, q) > 1. \end{cases}$$

The definition of q implies $\widetilde{\chi}$ is well-defined and hence q-periodic. In fact, $\widetilde{\chi}$ is a Dirichlet character modulo q and minimality forces $\widetilde{\chi}$ to be primitive. By construction χ is indued from $\widetilde{\chi}$ and this proves existence. Now suppose $\widetilde{\chi}_1$ and $\widetilde{\chi}_2$ are two primitive Dirichlet characters modulo q_1 and q_2 respectively both of which induce χ. Then $\widetilde{\chi}_1(a) = \widetilde{\chi}_2(a)$ whenever $(a, m) = 1$. Setting $q = (q_1, q_2)$, we also have $\widetilde{\chi}_1(a) = \widetilde{\chi}_2(a)$ whenever $(a, q) = 1$. Hence $\widetilde{\chi}_1$ and $\widetilde{\chi}_2$ are both induced from the same Dirichlet character modulo q. Primitivity implies $q_1 = q_2$ and $\widetilde{\chi}_1 = \widetilde{\chi}_2$. This proves uniqueness. $\square$

In light of this result, we define the **conductor** q of a Dirichlet character χ modulo m to be the modulus of the unique primitive Dirichlet character $\widetilde{\chi}$ inducing χ. By the proof, the conductor is given by

$$q = \min\{d \mid m : \chi(a) = \chi(b) \text{ for } a \equiv b \pmod{d} \text{ with } (ab, m) = 1\}.$$

Also observe that χ is q-periodic, q is the minimal period of χ, and the nonzero values of χ are $\varphi(q)$-th roots of unity. Moreover,

$$\chi = \widetilde{\chi}\chi_{\frac{m}{q},0},$$

and χ is primitive if and only if its conductor and modulus are equal.

As not every Dirichlet character of a fixed modulus is primitive, it is natural to ask how many primitive Dirichlet characters there are for a given modulus. Let $N(m)$ be the number of primitive Dirichlet characters modulo m. Then $N(m)$ is easily determined via Möbius inversion.

Proposition 2.1.2. *For any positive integer m, we have*

$$\phi(m) = \sum_{d|m} N(d),$$

where $N(d)$ be the number of primitive Dirichlet characters modulo d. In particular, $N(m)$ is given by

$$N(m) = \sum_{d|m} \phi(d)\mu\left(\frac{m}{d}\right)$$

Proof. To prove the first formula, the right-hand side counts the number of Dirichlet characters modulo m since every such Dirichlet character is induced from a unique primitive Dirichlet character whose modulus divides m by Theorem 2.1.1. The left hand side also counts the number of Dirichlet characters modulo m as are $\phi(m)$ many. This proves the first formula. The second follows by Möbius inversion. $\square$

Primitive Dirichlet characters also behave well with respect to multiplication if the conductors are relatively prime as the following proposition shows:

Proposition 2.1.3. *Suppose χ_1 and χ_2 are Dirichlet characters modulo m_1 and m_2 respectively with $(m_1, m_2) = 1$. Set $\chi = \chi_1\chi_2$. Then χ is a primitive if and only if χ_1 and χ_2 both are.*

Proof. By construction, χ is a Dirichlet character modulo m_1m_2. Let q_1 and q_2 be the conductors of χ_1 and χ_2 respectively and let q be the conductor of χ. Then χ is q_1q_2-periodic and $q \mid q_1q_2$.

For the forward implication, suppose χ is primitive so that $q = m_1m_2$ whence $m_1m_2 \mid q_1q_2$. This forces $m_1 = q_1$ and $m_2 = q_2$ proving χ_1 and χ_2 are both primitive. For the reverse implication, suppose χ_1 and χ_2 are both primitive so that $q_1 = m_1$ and $q_2 = m_2$. Then $(q_1, q_1) = 1$ and the Chinese remainder theorem implies that χ is q_1-periodic on those integers that are congruent to 1 modulo q_2 and q_2-periodic on those integers that are congruent to 1 modulo q_1. This forces $q_1 \mid q$ and $q_2 \mid q$ which together imply $q = q_1q_2$ and thus χ is primitive. $\square$

We would now like to distinguish Dirichlet characters based on their nonzero values. We say χ is **real** if it is real-valued. This means the nonzero values of χ are 1 or -1 since they are the only real roots of unity. We say χ is **complex** if it is not real. More commonly, we distinguish Dirichlet characters by the roots of unity that their nonzero values take. We say χ is of **order** n if the nonzero values of χ are all n-th roots of unity. In the cases of small order we will often use the latin derived names **quadratic**, **cubic**, etc. To connect these two naming conventions observe that a nontrivial Dirichlet character is quadratic if and only if it is real and an other nontrivial Dirichlet character is necessarily complex. Also note that quadratic Dirichlet characters are their own conjugates.

We will also distinguish Dirichlet characters by their parity. By multiplicativity, we must have $\chi(-1) = \pm 1$. Accordingly, we say χ is **even** if $\chi(-1) = 1$ and **odd** if $\chi(-1) = -1$. Then even Dirichlet characters are even functions while odd Dirichlet characters are odd functions. Note that conjugate and induced Dirichlet characters necessarily have the same parity. The parity is also expressed via the formula

$$\frac{\chi(1) - \chi(-1)}{2} = \begin{cases} 0 & \text{if } \chi \text{ is even,} \\ 1 & \text{if } \chi \text{ is odd.} \end{cases}$$

2.2 Quadratic Dirichlet Characters

Quadratic Dirichlet characters deserve special attention as it is possible to classify all of them explicitly. This is due to the fact that they arise from Jacobi and Kronecker symbols. For a positive odd integer m, define

$$\chi_m(a) = \left(\frac{a}{m}\right).$$

By definition of the Jacobi symbol, χ_m becomes a quadratic Dirichlet character modulo m. Unfortunately, the quadratic Dirichlet characters constructed in this manner do not exhaust all possible examples. To accomplish this we need to use Kronecker symbols. An integer D is said to be a **fundamental discriminant** if it is of the form

$$D = \begin{cases} d & \text{if } D \equiv 1 \pmod 4, \\ 4d & \text{if } D \equiv 8, 12 \pmod{16}, \end{cases}$$

for some square-free integer d. Necessarily $d \equiv 1 \pmod 4$ or $d \equiv 2, 3 \pmod 4$ respectively and thus is nonzero. We define $\chi_D : \mathbb{Z} \to \mathbb{C}$ by

$$\chi_D(a) = \left(\frac{D}{a}\right).$$

It turns out that χ_D defines a primitive quadratic Dirichlet character modulo $|D|$, provided $D \neq 1$, and exhausts all such possibilities.

Theorem 2.2.1. *If D is a fundamental discriminant and $D \neq 1$ then χ_D is a primitive quadratic Dirichlet character of conductor $|D|$. Moreover, all primitive quadratic Dirichlet characters are of this form.*

Proof. We first show that χ_D is a primitive quadratic Dirichlet character modulo $|D|$. If $D \equiv 1 \pmod 4$, the sign in quadratic reciprocity is always 1. Then

$$\chi_D(a) = \left(\frac{a}{|D|}\right),$$

which is a Dirichlet character modulo $|D|$. If $D \equiv 12 \pmod{16}$ then $\frac{D}{4} \equiv 3 \pmod 4$ and the sign in quadratic reciprocity is $\left(\frac{-1}{a}\right)$ which is the primitive quadratic Dirichlet character modulo 4 as there are only two such Dirichlet characters and $\left(\frac{-1}{a}\right)$ is not principal. Whence

$$\chi_D(a) = \left(\frac{-1}{a}\right)\left(\frac{a}{\left|\frac{D}{4}\right|}\right),$$

which is a Dirichlet character modulo $|D|$. If $D \equiv 8 \pmod{16}$ first observe that $\left(\frac{D}{a}\right) = \left(\frac{8}{a}\right)\left(\frac{\frac{D}{8}}{a}\right)$ where $\left(\frac{8}{a}\right)$ is one of the two primitive quadratic Dirichlet character modulo 8 (the other is $\left(\frac{-8}{a}\right)$). As $\frac{D}{8} \equiv 1, 3 \pmod 4$, the sign in quadratic reciprocity is either 1 or $\left(\frac{-1}{a}\right)$ according to these two cases. Thus

$$\chi_D(a) = \left(\frac{8}{a}\right)\left(\frac{a}{\left|\frac{D}{8}\right|}\right) \quad \text{or} \quad \chi_D(a) = \left(\frac{-8}{a}\right)\left(\frac{a}{\left|\frac{D}{8}\right|}\right),$$

according to if $\frac{D}{8} \equiv 1, 3 \pmod 4$ respectively, and in either case is a Dirichlet character modulo $|D|$. We can compactly express all of these cases as follows:

$$\chi_D(a) = \begin{cases} \left(\frac{a}{|D|}\right) & \text{if } D \equiv 1 \pmod 4, \\ \left(\frac{-1}{a}\right)\left(\frac{a}{\left|\frac{D}{4}\right|}\right) & \text{if } \frac{D}{4} \equiv 3 \pmod 4, \\ \left(\frac{8}{a}\right)\left(\frac{a}{\left|\frac{D}{8}\right|}\right) & \text{if } \frac{D}{8} \equiv 1 \pmod 4, \\ \left(\frac{-8}{a}\right)\left(\frac{a}{\left|\frac{D}{8}\right|}\right) & \text{if } \frac{D}{8} \equiv 3 \pmod 4. \end{cases}$$

This proves χ_D is a Dirichlet characters modulo $|D|$. It is not hard to see that χ_D is primitive. Indeed, we have already mentioned that the characters $\left(\frac{-1}{a}\right)$, $\left(\frac{8}{a}\right)$, and $\left(\frac{-8}{a}\right)$ are all primitive. Since D, $\frac{D}{4}$, and $\frac{D}{8}$ are square-free according to their equivalences modulo 4 and $D \neq 1$, it suffices to show that χ_p is primitive for primes p with $p \neq 2$ by Proposition 2.1.3. This is immediate since p is prime and χ_p is not principal.

We now show that every primitive quadratic Dirichlet character is of the form χ_D for some fundamental discriminant D. By Proposition 2.1.3 again, it suffices to consider primitive quadratic Dirichlet character modulo a prime power p^m.

First suppose $p \neq 2$. Then $(\mathbb{Z}/p^m\mathbb{Z})^*$ is cyclic, generated by say g, and every $a \in (\mathbb{Z}/p^m\mathbb{Z})^*$ is of the form $a = g^\nu$ for some $\nu \in (\mathbb{Z}/\varphi(p^m)\mathbb{Z})$. It follows that every corresponding Dirichlet character χ is of the form

$$\chi(a) = e^{\frac{2\pi i k\nu}{\varphi(p^m)}},$$

for an integer k modulo $\varphi(p^m)$. Moreover, χ is primitive if and only if $k \not\equiv 0 \pmod p$ for otherwise χ is a Dirichlet character modulo p^{m-1}. Similarly, χ is quadratic if and only if $k \equiv \frac{\varphi(p^m)}{2} \pmod{\varphi(p^m)}$. Such a k exists and is unique because $p \neq 2$. We also see that if χ is quadratic then it is imprimitive unless $m = 1$ for then $\varphi(p) = p - 1$ which is not a multiple of p. To summarized, there is a unique quadratic Dirichlet character modulo p^m and it is primitive if and only if $m = 1$. Necessarily, this unique primitive quadratic Dirichlet character modulo p is given by χ_D for the fundamental discriminant $D = p$ if $p \equiv 1 \pmod 4$ and $D = -p$ if $p \equiv 3 \pmod 4$.

Now suppose $p = 2$ so that $p^m = 2^m$. If $m = 1$ then $\varphi(2) = 1$ and there are no primitive quadratic Dirichlet characters as the only Dirichlet character is principal. If $m = 2$ then $\varphi(4) = 2$ so that there are two Dirichlet characters. They are both quadratic but only one is primitive namely the aforementioned $\left(\frac{-1}{a}\right)$. This primitive quadratic Dirichlet character is given by χ_D for the fundamental discriminant $D = -4$. For $m \geq 3$ there is an isomorphism $(\mathbb{Z}/2^m\mathbb{Z})^* \cong C_2 \times C_{2^{m-2}}$ where C_2 and $C_{2^{m-2}}$ are the cyclic groups of order 2 and 2^{m-2} respectively. Therefore every $a \in (\mathbb{Z}/2^m\mathbb{Z})^*$ is of the form $a = (-1)^\mu 5^\nu$ for some $\mu \in \mathbb{Z}/2\mathbb{Z}$ and $\nu \in \mathbb{Z}/2^{m-2}\mathbb{Z}$. Then every corresponding Dirichlet character χ is of the form

$$\chi(a) = e^{\frac{2\pi i j\mu}{2}} e^{\frac{2\pi i k\nu}{2^{m-2}}},$$

for integers j modulo 2 and k modulo 2^{m-2}. Similarly to the case for $p \neq 2$, χ is primitive if and only if $k \not\equiv 0 \pmod{2^{m-2}}$. Moreover, χ is quadratic if and only if $k \equiv 0 \pmod{2^{m-3}}$. These congruences together imply that a primitive quadratic Dirichlet character exists if and only if $m = 3$. In this case there are four Dirichlet characters. They are all quadratic but only two are primitive, namely the aforementioned $\left(\frac{8}{a}\right)$ and $\left(\frac{-8}{a}\right)$. These two primitive quadratic Dirichlet characters are given by χ_D for the fundamental discriminants $D = 8$ and $D = -8$ respectively. $\square$

It follows from Theorem 2.2.1 that all quadratic Dirichlet characters are induced from some χ_D including $D = 1$ since this corresponds to the trivial Dirichlet character. In particular, so too are the quadratic Dirichlet characters given by Jacobi symbols.

2.3 Ramanujan and Gauss Sums

For integers b and m with m positive, the **Ramanujan sum** $r(b, m)$ is defined by

$$r(b, m) = \sum_{\substack{a \pmod m \\ (a,m)=1}} e^{\frac{2\pi i ab}{m}}.$$

The Ramanujan sum is a finite sum of m-th roots of unity. When $b \mid m$ the summands are all 1 and the Ramanujan sum has the simple evaluation

$$r(b, m) = \varphi(m).$$

For a general idex the Ramanujan sums can be computed explicitly by means of the Möbius function.

Proposition 2.3.1. *For integers b and m with m positive, we have*

$$r(b, m) = \sum_{d \mid (b,m)} d\mu\left(\frac{m}{d}\right).$$

Proof. The identity is obvious if $m = 1$ since the Ramanujan sum is 1. So assume $m > 1$. Every a modulo m is of the form $a = a'd$ for some divisor d of m and a' modulo $\frac{m}{d}$ with $\left(a', \frac{m}{d}\right) = 1$. So summing $r(b,d)$ over the divisors d of m gives

$$\sum_{d|m} r(b,d) = \sum_{a \pmod{m}} e^{\frac{2\pi iab}{m}},$$

If $m \mid b$ the latter sum is m while if $m \nmid b$ the sum vanishes as it is the sum of all m-th roots of unity. Thus

$$\sum_{d|m} r(b,d) = \begin{cases} m & \text{if } m \mid b, \\ 0 & \text{if } m \nmid b. \end{cases}$$

Now apply Möbius inversion. □

More general Ramanujan sums can be constructed by introducing a Dirichlet character. Let χ be a Dirichlet character modulo m. For any integer b, the **Ramanujan sum** $\tau(n,\chi)$ associated to χ is given by

$$\tau(b,\chi) = \sum_{a \pmod{m}} \chi(a) e^{\frac{2\pi iab}{m}}.$$

This generalizes the previous Ramanujan sum as

$$r(b,m) = \tau(b,\chi_{m,0}).$$

When $b \mid m$ the summands are all 1 and the Dirichlet orthogonality relations imply

$$\tau(b,\chi) = \varphi(m)\delta_{\chi,\chi_{m,0}}.$$

When $b = 1$ we simply write $\tau(\chi) = \tau(1,\chi)$ and call $\tau(\chi)$ the **Gauss sum** associated to χ. The following proposition develops the basic properties of these Ramanujan sums:

Proposition 2.3.2. *Let χ and ψ be nontrivial Dirichlet characters modulo m and n respectively and let b be an integer. Then the following properties hold:*

(i) $\overline{\tau(b,\overline{\chi})} = \chi(-1)\tau(b,\chi)$.

(ii) If $(b,m) = 1$ *then* $\tau(b,\chi) = \overline{\chi}(b)\tau(\chi)$.

(iii) If $(b,m) > 1$ *and χ is primitive then* $\tau(b,\chi) = 0$.

(iv) If $(m,n) = 1$ *then* $\tau(b,\chi\psi) = \chi(n)\psi(m)\tau(b,\chi)\tau(b,\psi)$.

(v) If $\widetilde{\chi}$ is the primitive Dirichlet character of conductor q inducing χ, then

$$\tau(\chi) = \mu\left(\frac{m}{q}\right)\widetilde{\chi}\left(\frac{m}{q}\right)\tau(\widetilde{\chi}).$$

Proof. We will prove the properties separately.

(i) This follows by direct computation.

(ii) This follows by direct computation.

(iii) Let c be an integer with $(c, m) = 1$ and satisfying $\chi(c) \neq 1$. Such a c exists because otherwise χ the principal Dirichlet character modulo m and thus imprimitive. A short computation shows

$$\chi(c)\tau(b, \chi) = \tau(b, \chi),$$

whence $\tau(b, \chi) = 0$.

(iv) Since $(m, n) = 1$, the Chinese remainder theorem implies that any a modulo mn is of the form $a = a'n + a''m$ with a' modulo m and a'' modulo n. Whence

$$(\chi\psi)(a) = \chi(a'n)\psi(a''m).$$

Using this fact, a short computation shows

$$\sum_{a \pmod{mn}} (\chi\psi)(a) e^{\frac{2\pi i ab}{mn}} = \chi(n)\psi(m) \cdot \left(\sum_{a' \pmod{m}} \chi(a') e^{\frac{2\pi i ab}{m}} \right) \left(\sum_{a'' \pmod{n}} \psi(a'') e^{\frac{2\pi i a'' b}{n}} \right),$$

which is equivalent to the claim.

(v) First consider the case when $\left(\frac{m}{q}, q\right) = 1$. In view of $\chi = \widetilde{\chi}\chi_{\frac{m}{q},0}$, we use (iv) to obtain

$$\tau(\chi) = \tau(\chi_{\frac{m}{q},0}) \widetilde{\chi}\left(\frac{m}{q}\right) \tau(\widetilde{\chi}).$$

As $\tau(\chi_{\frac{m}{q},0}) = r\left(1, \frac{m}{q}\right)$, we use Proposition 2.3.1 to compute $\tau(\chi_{\frac{m}{q},0}) = \mu\left(\frac{m}{q}\right)$. Whence

$$\tau(\chi) = \mu\left(\frac{m}{q}\right) \widetilde{\chi}\left(\frac{m}{q}\right) \tau(\widetilde{\chi}).$$

Now suppose $\left(\frac{m}{q}, q\right) > 1$. In this case the right-hand side is zero because $\widetilde{\chi}\left(\frac{m}{q}\right) = 0$. So we must show $\tau(\chi) = 0$. Now there exists a prime p with $p \mid \frac{m}{q}$ and $p \mid q$. For any a modulo m write $a = a'\frac{m}{p} + a''$ with a' modulo p and a'' modulo $\frac{m}{p}$. Moreover, as $p \mid \frac{m}{q}$ we know $q \mid \frac{m}{p}$. These two facts and a short computation together show

$$\tau(\chi) = \left(\sum_{a' \pmod{p}} e^{\frac{2\pi i a'}{p}} \right) \left(\sum_{a'' \pmod{\frac{m}{p}}} \widetilde{\chi}(a'') e^{\frac{2\pi i a''}{m}} \right).$$

The first sum vanishes since it is the sum of all p-th roots of unity. This proves $\tau(\chi) = 0$. ∎

These properties help to reduce the evaluation of Ramanujan and Gauss sums. However, even evaluating Gauss sums for arbitrary primitive Dirichlet characters is a very difficult problem much of which is still open. Yet it is not difficult to determine the modulus of the Gauss sum when χ is primitive.

Theorem 2.3.3. *Let χ be a primitive Dirichlet character of conductor q. Then*

$$|\tau(\chi)| = \sqrt{q}.$$

Proof. If χ is the trivial Dirichlet character the claim is obvious since the Gauss sum is 1. So assume χ is nontrivial whence $q > 1$. Consider instead $|\tau(\chi)|^2 = \tau(\chi)\overline{\tau(\chi)}$. Expanding the Gauss sum $\overline{\tau(\chi)}$ and invoking Proposition 2.3.2 (ii), a short computation shows

$$|\tau(\chi)|^2 = \sum_{a \pmod q} \tau(a,\chi)e^{-\frac{2\pi i a}{q}}.$$

Upon expanding the Ramanujan sum, another short computation gives

$$|\tau(\chi)|^2 = \sum_{a' \pmod q} \chi(a')\left(\sum_{a \pmod q} e^{\frac{2\pi i a(a'-1)}{q}}\right).$$

If $a' \equiv 1 \pmod q$ the inner sum is q and otherwise vanishes as it is the sum of all q-th roots of unity. It follows that the double sum is q whence $|\tau(\chi)|^2 = q$. This is equivalent to the claim. $\square$

As an almost immediate corollary, we deduce a useful expression for primitive Dirichlet characters as exponential sums.

Corollary 2.3.4. *Let χ be a primitive Dirichlet character of conductor q. Then for any integer b, we have*

$$\tau(b,\chi) = \overline{\chi}(b)\tau(\chi).$$

In particular,

$$\chi(b) = \frac{1}{\tau(\overline{\chi})}\sum_{a \pmod q} \overline{\chi}(a)e^{\frac{2\pi i ab}{q}}.$$

Proof. For the first statement, the identity is obvious if χ is the trivial character as the Ramanujan sum is 1. So assume χ is nontrivial. If $(b,q) = 1$ then this is exactly Proposition 2.3.2 (ii). If $(b,q) > 1$ then the identity follows from Proposition 2.3.2 (iii) and that $\overline{\chi}(b) = 0$. This proves the first statement in full. For the second statement, observe that $\tau(\chi) \neq 0$ by Theorem 2.3.3. The second identity follows upon replacing χ with $\overline{\chi}$, dividing by $\tau(\chi)$, and expanding the Ramanujan sum. $\square$

For a Dirichlet character χ modulo m, we define the **epsilon factor** ε_χ by

$$\varepsilon_\chi = \frac{\tau(\chi)}{\sqrt{m}}.$$

When χ is primitive, the epsilon factor lies on the unit circle by Theorem 2.3.3. The question of the evaluation of Gauss sums, and hence Ramanujan sums, boils down to determining what value the epsilon factor is. This is the real difficultly in evaluating Gauss sums. However, when χ is primitive there is a simple relationship between the epsilon factors ε_χ and $\varepsilon_{\overline{\chi}}$:

Proposition 2.3.5. *Let χ be a primitive Dirichlet character of conductor q. Then*

$$\varepsilon_\chi \varepsilon_{\overline{\chi}} = \chi(-1).$$

Proof. If χ is trivial this is obvious since both epsilon factors are 1. So assume χ is nontrivial. On the one hand, $\varepsilon_{\overline{\chi}}$ lies on the unit circle so that

$$\varepsilon_{\overline{\chi}}^{-1} = \overline{\frac{\tau(\chi)}{\sqrt{q}}}.$$

On the other hand, Proposition 2.3.2 (i) implies

$$\varepsilon_\chi = \chi(-1)\overline{\frac{\tau(\chi)}{\sqrt{q}}}.$$

Combining these identities gives the result. $\square$

2.4 Quadratic Gauss Sums

Our primary aim will be to evaluate the epsilon factor of the Gauss sum for quadratic Dirichlet characters defined by Jacobi symbols. To accomplish this we will study and auxiliary exponential sum. For integers b and m with m positive, the **quadratic Gauss sum** $g(b,m)$ is defined by

$$g(b,m) = \sum_{a \pmod m} e^{\frac{2\pi i a^2 b}{m}}.$$

When $b \mid m$ the summands are all 1 and the quadratic Gauss sum evaluates to

$$g(b,m) = m.$$

If $b = 1$ we write $g(m) = g(1,m)$. It turns out that for square-free m the Ramanujan sum attached to the quadratic Dirichlet character modulo m given by the Jacobi symbol is precisely the quadratic Gauss sum. This takes some work to prove. The first step is to reduce to the case when $(b,m) = 1$.

Proposition 2.4.1. *Let b and m be integers with m positive. Then*

$$g(b,m) = (b,m)g\left(\frac{b}{(b,m)}, \frac{m}{(b,m)}\right).$$

Proof. Any a modulo m is of the form $a = a'\frac{m}{(b,m)} + a''$ with a' modulo (b,m) and a'' modulo $\frac{m}{(b,m)}$. A short computation shows

$$\sum_{a \pmod m} e^{\frac{2\pi i a^2 b}{m}} = (b,m) \sum_{a'' \pmod{\frac{m}{(b,m)}}} e^{\frac{2\pi i (a'')^2 \frac{b}{(b,m)}}{\frac{m}{(b,m)}}}.$$

The remaining sum is exactly $g\left(\frac{b}{(b,m)}, \frac{m}{(b,m)}\right)$ and the desired identity follows. □

The second step is to deduce an equivalent formulation of the Ramanujan sum associated to quadratic Dirichlet characters given by Jacobi symbols. This will imply equivalence between the aforementioned exponential sums when the modulus is an odd prime.

Proposition 2.4.2. *Let b and m be integers with m positive, odd, and such that $(b,m) = 1$. Let χ_m be the quadratic Dirichlet character modulo m given by the Jacobi symbol. Then*

$$\tau(b, \chi_m) = \sum_{a \pmod m} \left(1 + \left(\frac{a}{m}\right)\right) e^{\frac{2\pi i a b}{m}}.$$

When $m = p$ is prime,

$$\tau(b, \chi_p) = g(b,p).$$

Proof. The first statement is obvious when $m = 1$ since the Ramanujan sum is 1. So assume $m > 1$. Now write the sum as

$$\sum_{a \pmod m} e^{\frac{2\pi i a b}{m}} + \sum_{a \pmod m} \left(\frac{a}{m}\right) e^{\frac{2\pi i a b}{m}}.$$

The first sum vanishes as it is the sum of all m-th roots of unity. This proves the first identity. Now let $m = p$ be an odd prime. Then

$$1 + \left(\frac{a}{p}\right) = \begin{cases} 2 & \text{if } a \text{ is a quadratic residue modulo } p, \\ 0 & \text{if } a \text{ is not quadratic residue modulo } p, \\ 1 & \text{if } p \mid a. \end{cases}$$

Moreover, when a is a quadratic residue modulo p there exists an a' modulo p with $a \equiv (a')^2 \pmod p$. As there are $\frac{p-1}{2}$ such residues, the first statement implies

$$\tau(b, \chi_p) = 1 + \sum_{\substack{a' \pmod p \\ a' \neq 0}} e^{\frac{2\pi i (a')^2 b}{p}}.$$

This is exactly $g(b,p)$ which proves the second statement. □

The third step is to generalize the second statement in Proposition 2.4.2 for square-free m. To accomplish this we will need to develop some properties of quadratic Gauss sums.

Proposition 2.4.3. *Let b, m, and n be integers with m and n positive and let p be an odd prime Then the following properties hold:*

(i) If $(b,p)=1$ then $g(b,p^r)=pg(b,p^{r-2})$ provided $r\geq 2$.

(ii) If $(m,n)=1$ and $(b,mn)=1$ then $g(b,mn)=g(bn,m)g(bm,n)$.

(iii) If m is odd and $(b,m)=1$ then $g(b,m)=\left(\frac{b}{m}\right)g(m)$ where $\left(\frac{b}{m}\right)$ is the Jacobi symbol.

Proof. We will prove the properties separately.

(i) Every a modulo p satisfies $(a,p)=1$ or is a multiple of p. Whence

$$g(b,p^r)=\sum_{\substack{a \pmod{p^r}\\(a,p)=1}} e^{\frac{2\pi i a^2 b}{p^r}}+\sum_{a \pmod{p^{r-1}}} e^{\frac{2\pi i a^2 b}{p^{r-2}}},$$

since every a modulo p satisfies $(a,p)=1$ or not. Every a modulo p^{r-1} is of the form $a=a'p^{r-2}+a''$ with a' modulo p and a'' modulo p^{r-2}. A short computation shows

$$\sum_{a \pmod{p^{r-1}}} e^{\frac{2\pi i a^2 b}{p^{r-2}}}=p\sum_{a'' \pmod{p}} e^{\frac{2\pi i (a'')^2 b}{p^{r-2}}}.$$

The remaining sum is exactly $g(b,p^{r-2})$. So it suffices to show that the first sum vanishes. This sum is exactly $r(b,p^r)$ and Proposition 2.3.1 implies $r(b,p^r)=\mu(p^r)$ which is zero as $r\geq 2$.

(ii) Since $(m,n)=1$, the Chinese remainder theorem implies that any a modulo mn is of the form $a=a'n+a''m$ with a' modulo m and a'' modulo n. Whence

$$\left(\sum_{a' \pmod{m}} e^{\frac{2\pi i (a')^2 bn}{m}}\right)\left(\sum_{a'' \pmod{n}} e^{\frac{2\pi i (a'')^2 bm}{n}}\right)=\sum_{a \pmod{mn}} e^{\frac{2\pi i a^2 b}{mn}}.$$

This is equivalent to the claim.

(iii) The claim is obvious if $m=1$ because the quadratic Gauss sum is 1. So assume $m>1$. By multiplicativity of the Jacobi symbol and (ii), it suffices to prove the claim when $m=p^r$ is an odd prime power. The case when $r=1$ follows from Proposition 2.4.2, Proposition 2.3.2 (ii), and that quadratic Dirichlet characters are their own conjugates. The case when $r\geq 2$ follows by induction using (i). □

At last we can prove our Ramanujan and quadratic Gauss sums agree.

Theorem 2.4.4. *Suppose m is a square-free positive odd integer and let χ_m be the quadratic Dirichlet character modulo m given by the Jacobi symbol. Then for any integer b with $(b,m)=1$, we have*

$$\tau(b,\chi_m)=g(b,m).$$

Proof. The claim is obvious if $m = 1$ because the Ramanujan and Gauss sums are both 1. So suppose $m > 1$. It suffices to assume $b = 1$ by Proposition 2.3.2 (ii), Proposition 2.4.3 (iii), and that quadratic Dirichlet characters are their own conjugates. Let $m = p_1 p_2 \cdots p_k$ be the prime decomposition of m. Repeated application of Proposition 2.3.2 (iv) shows

$$\tau(\chi) = \left(\prod_{i<j} \chi_{p_j}(p_i)\chi_{p_i}(p_j)\right)\left(\prod_i \tau(\chi_{p_i})\right).$$

By Proposition 2.4.2, we may write

$$\left(\prod_{i<j} \chi_{p_j}(p_i)\chi_{p_i}(p_j)\right)\left(\prod_i \tau(\chi_{p_i})\right) = \left(\prod_{i<j} \left(\frac{p_i}{p_j}\right)\left(\frac{p_j}{p_i}\right)\right)\left(\prod_i g(p_i)\right).$$

Repeated application of Proposition 2.4.3 (ii) gives

$$\left(\prod_{i<j} \left(\frac{p_i}{p_j}\right)\left(\frac{p_j}{p_i}\right)\right)\left(\prod_i g(p_i)\right) = g(m).$$

This completes the proof. □

The properties in Proposition 2.4.3 help to reduce the evaluation of quadratic Gauss sums. Thankfully, it is possible to completely evaluate quadratic Gauss sums when $b = 1$ and therefore even some Ramanujan sums. As with the Gauss sum, we first deduce a fact about the modulus.

Theorem 2.4.5. *Let m be a positive odd integer. Then*

$$|g(m)| = \sqrt{m}.$$

Proof. The claim is obvious if $m = 1$ because the quadratic Gauss sum is 1. So assume $m > 1$. By Proposition 2.4.3 (ii) and (iii), it suffices to prove the claim when $m = p^r$ is an odd prime power. The case when $r = 1$ follows from Theorems 2.3.3 and 2.4.4. The case when $r \geq 2$ is proved by induction and Proposition 2.4.3 (i). □

For any integer m, we define the **epsilon factor** ε_m by

$$\varepsilon_m = \frac{g(m)}{\sqrt{m}}.$$

Theorem 2.4.5 says that this value lies on the unit circle when m is odd. This evaluation of these epsilon factors was completely resolved by Gauss in 1808. Modern proofs use analytic techniques by expressing $g(m)$ in a form where Poisson summation can be applied.

Theorem 2.4.6. *Let $m \geq 1$. Then*

$$\varepsilon_m = \begin{cases} (1+i) & \textit{if } m \equiv 0 \pmod 4, \\ 1 & \textit{if } m \equiv 1 \pmod 4, \\ 0 & \textit{if } m \equiv 2 \pmod 4, \\ i & \textit{if } m \equiv 3 \pmod 4. \end{cases}$$

Proof. We will write the quadratic Gauss in a form to which Poisson summation can be applied. For then we can evaluate the quadratic Gauss sum by means of Fourier transforms. To this end, consider the function

$$f(x) = \begin{cases} e^{\frac{2\pi i x^2}{m}} & \text{if } x \in [0,m], \\ 0 & \text{if } x \notin [0,m], \end{cases}$$

which is piecewise smooth and compactly supported on $\mathbb{R}$. Therefore Poisson summation applies. On the one hand, this means

$$\sum_{n\in\mathbb{Z}} f(n) = g(m).$$

On the other hand, we have

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{t\in\mathbb{Z}} \sqrt{m} e^{-\frac{2\pi i t^2 m}{4}} \int_{\frac{t\sqrt{m}}{2}}^{\sqrt{m}+\frac{t\sqrt{m}}{2}} e^{2\pi i x^2}\,dx.$$

As $t \equiv 0, 1 \pmod 4$ according to if t is even or odd, the subsums according to this parity are

$$\sqrt{m}\int_{-\infty}^{\infty} e^{2\pi i x^2}\,dx \quad \text{and} \quad \sqrt{m}e^{-\frac{2\pi i m}{4}}\int_{-\infty}^{\infty} e^{2\pi i x^2}\,dx,$$

respectively. As the sum is ordered with respect to this parity, we have

$$\sum_{n\in\mathbb{Z}} f(n) = \sqrt{m}\left(1+e^{-\frac{2\pi i m}{4}}\right)\int_{-\infty}^{\infty} e^{2\pi i x^2}\,dx.$$

Equating our two expressions gives

$$g(m) = \sqrt{m}\left(1+e^{-\frac{2\pi i m}{4}}\right)\int_{-\infty}^{\infty} e^{2\pi i x^2}\,dx.$$

To compute the remaining integral we take $m = 1$. As the Gauss sum is 1 and $e^{-\frac{2\pi i}{4}} = -i$, we find that

$$\int_{-\infty}^{\infty} e^{2\pi i x^2}\,dx = \frac{1}{1-i}.$$

Therefore

$$\varepsilon_m = \frac{1+e^{-\frac{2\pi i m}{4}}}{1-i} = \begin{cases} (1+i) & \text{if } m \equiv 0 \pmod 4, \\ 1 & \text{if } m \equiv 1 \pmod 4, \\ 0 & \text{if } m \equiv 2 \pmod 4, \\ i & \text{if } m \equiv 3 \pmod 4, \end{cases}$$

as desired. $\square$

As an immediate corollary, we can evaluate the epsilon factor ε_{χ_p} for the quadratic Dirichlet character χ_p modulo p given by the Jacobi symbol when p is an odd prime.

Corollary 2.4.7. *Let p be an odd prime and χ_p be the quadratic Dirichlet character modulo p given by the Jacobi symbol. Then*

$$\varepsilon_{\chi_p} = \begin{cases} 1 & \text{if } p \equiv 1 \pmod 4, \\ i & \text{if } p \equiv 3 \pmod 4. \end{cases}$$

Proof. The claim follows immediately from Theorem 2.4.6 and Proposition 2.4.2. $\square$

Part II

Algebraic Number Theory

Chapter 3

Arithmetic of Number Fields

Throughout, all rings will be understood to be commutative and with unity.

3.1 Integrality

Let $\mathcal{O}/\mathcal{o}$ be an extension of integral domains. We say that $\beta \in \mathcal{O}$ is **integral** over $\mathcal{o}$ if β is the root of a monic polynomial $f(x) \in \mathcal{o}[x]$. In other words, β satisfies an equation of the form

$$\beta^n + \alpha_{n-1}\beta^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha_i \in \mathcal{o}$. We say that $\mathcal{O}$ is **integral** over $\mathcal{o}$ if every element of $\mathcal{O}$ is integral over $\mathcal{o}$. As we may expect, the set of integral elements form a ring.

Proposition 3.1.1. *Let $\mathcal{O}/\mathcal{o}$ be an extension of integral domains. Then $\beta_1, \ldots, \beta_n \in \mathcal{O}$ are all integral over $\mathcal{o}$ if and only if $\mathcal{o}[\beta_1, \ldots, \beta_n]$ is a finitely generated $\mathcal{o}$-module. In particular, the elements of $\mathcal{O}$ that are integral over $\mathcal{o}$ form a ring.*

Proof. We will prove the forward implication by induction. First suppose $\beta \in \mathcal{O}$ is integral over $\mathcal{o}$. Then there exists a monic polynomial $f(x) \in \mathcal{o}[x]$ of say degree n such that $f(\beta) = 0$. Let $g(x) \in \mathcal{o}[x]$ and write

$$g(x) = q(x)f(x) + r(x),$$

with $q(x), r(x) \in \mathcal{o}[x]$ where the degree of $r(x)$ is strictly less than n. Then $g(\beta) = r(\beta)$. Letting

$$r(x) = \alpha_{n-1}x^{n-1} + \cdots + \alpha_1 x + a_0,$$

with $\alpha_i \in \mathcal{o}$, it follows that

$$g(\beta) = \alpha_{n-1}\beta^{n-1} + \cdots + \alpha_1\beta + \alpha_0.$$

As $g(x)$ was arbitrary, we see that $1, \beta, \ldots, \beta^{n-1}$ generate $\mathcal{o}[\beta]$ as an $\mathcal{o}$-module. Now assume by induction that $\mathcal{o}[\beta_1, \ldots, \beta_{n-1}]$ is a finitely generated $\mathcal{o}$-module. Then $\mathcal{o}[\beta_1, \ldots, \beta_n]$ is a finitely generated $\mathcal{o}$-module. This proves the forward implication of the first statement. For the reverse implication, suppose $\mathcal{o}[\beta_1, \dots, \beta_n]$ is a finitely generated $\mathcal{o}$-module and let $\omega_1, \dots, \omega_r$ be generators. Then for any $\beta \in \mathcal{o}[\beta_1, \dots, \beta_n]$, we have

$$\beta\omega_i = \sum_j \alpha_{i,j}\omega_j,$$

with $\alpha_{i,j} \in \mathcal{o}$. These r equations are equivalent to the identity

$$\begin{pmatrix} \beta - \alpha_{1,1} & \alpha_{1,2} & \cdots & -\alpha_{1,r} \\ -\alpha_{2,1} & \beta - \alpha_{2,2} & & \\ \vdots & & \ddots & \\ -\alpha_{r,1} & & & \beta - \alpha_{r,r} \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \vdots \\ \omega_r \end{pmatrix} = \mathbf{0}.$$

The determinant of the matrix on the left-hand side must be zero. This shows that β is the root of the characteristic polynomial $\det(xI - (\alpha_{i,j}))$ which is a monic polynomial with coefficients in $\mathcal{o}$. Hence β is integral over $\mathcal{o}$. As β was arbitrary, the elements $\beta_1, \dots, \beta_n$ are all integral over $\mathcal{o}$ and that the sum and product of elements that are integral over $\mathcal{o}$ are also integral over $\mathcal{o}$. This proves the reverse implication and the second statement. □

Integrality is also transitive.

Proposition 3.1.2. *Let $\widetilde{\mathcal{O}}/\mathcal{O}/\mathcal{o}$ be a tower of integral domains. If $\widetilde{\mathcal{O}}$ is integral over $\mathcal{O}$ and $\mathcal{O}$ is integral over $\mathcal{o}$ then $\widetilde{\mathcal{O}}$ is integral over $\mathcal{o}$.*

Proof. Let $\gamma \in \widetilde{\mathcal{O}}$. Since $\widetilde{\mathcal{O}}$ is integral over $\mathcal{O}$, we have

$$\gamma^n + \beta_{n-1}\gamma^{n-1} + \cdots + \beta_0 = 0,$$

for some positive integer n and $\beta_i \in \mathcal{O}$. Set $R = \mathcal{o}[\beta_0, \dots, \beta_{n-1}]$. Then $R[\gamma]$ is a finitely generated R-module. As $\mathcal{O}$ is integral over $\mathcal{o}$, Proposition 3.1.1 implies that $R[\gamma]$ is also a finitely generated $\mathcal{o}$-module whence γ is integral over $\mathcal{o}$. As γ was arbitrary, this proves $\widetilde{\mathcal{O}}$ is integral over $\mathcal{o}$. □

In light of this result, we define the **integral closure** $\overline{\mathcal{o}}$ of $\mathcal{o}$ in $\mathcal{O}$ by

$$\overline{\mathcal{o}} = \{\beta \in \mathcal{O} : \beta \text{ is integral over } \mathcal{o}\}.$$

Then $\overline{\mathcal{o}}$ is a subring of $\mathcal{O}$ by Proposition 3.1.1. Clearly $\mathcal{o} \subseteq \overline{\mathcal{o}}$. Moreover, we say that $\mathcal{o}$ is **integrally closed** in $\mathcal{O}$ if $\mathcal{o} = \overline{\mathcal{o}}$. As $\mathcal{o} \subseteq \overline{\mathcal{o}} \subseteq \overline{\overline{\mathcal{o}}}$, Proposition 3.1.2 implies that $\overline{\mathcal{o}}$ is automatically integrally closed in $\mathcal{O}$. It will often be more fruitful to work with the integral closure rather than a generic ring.

There is a particular situation of integrally closed rings which is deserving of additional interest. Suppose $\mathcal{o}$ is an integral domain with field of fractions K. We call the integral closure $\overline{\mathcal{o}}$ of $\mathcal{o}$ in K the **normalization** of $\mathcal{o}$ and simply say that $\mathcal{o}$ is an **integrally closed domain** if $\mathcal{o}$ is equal to its normalization. It turns out that every unique factorization domain is an integrally closed domain.

Lemma 3.1.3. *Let $\mathcal{o}$ be a unique factorization domain with field of fractions K. Then $\mathcal{o}$ is an integrally closed domain. In particular, every principal ideal domain is an integrally closed domain.*

Proof. Let $\kappa \in K$ be such that

$$\kappa^n + \alpha_{n-1}\kappa^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha_i \in \mathcal{o}$. Write $\kappa = \frac{\alpha}{\beta}$ for $\alpha, \beta \in \mathcal{o}$ with β nonzero and $(\alpha, \beta) = 1$. Multiplying by β^n and isolating the leading term shows

$$\alpha^n = -(\alpha_{n-1}\beta\alpha^{n-1} + \cdots + \alpha_0\beta^n).$$

As β divides the right-hand side it divides the left-hand side as well. But then β is a unit in $\mathcal{o}$ since $(\alpha, \beta) = 1$. This means $\kappa \in \mathcal{o}$ whence $\mathcal{o}$ is an integrally closed domain. This proves the first statement. The second statement is immediate since every principal ideal domain is a unique factorization domain. □

We will often consider the more refined setting where $\mathcal{o}$ is an integrally closed domain with field of fractions K, L/K is a finite separable extension, and $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L. In this setting, L is the field of fractions of $\mathcal{O}$ and the elements of L which are integral over $\mathcal{o}$ have a simple description.

Proposition 3.1.4. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then every $\lambda \in L$ is of the form*

$$\lambda = \frac{\beta}{\alpha},$$

for some $\beta \in \mathcal{O}$ and nonzero $\alpha \in \mathcal{o}$. In particular, $\mathcal{O}$ is an integrally closed domain with field of fractions L. Moreover, $\lambda \in L$ is integral over $\mathcal{o}$ if and only if the minimal polynomial $m_\lambda(x)$ of λ over K has coefficients in $\mathcal{o}$.

Proof. As L/K is finite, it is necessarily algebraic so that any $\lambda \in L$ satisfies

$$\alpha\lambda^n + \alpha_{n-1}\lambda^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha, \alpha_i \in \mathcal{o}$ with α nonzero. We claim that $\alpha\lambda$ is integral over $\mathcal{o}$. Indeed, multiplying the previous identity by α^{n-1} yields

$$(\alpha\lambda)^n + \alpha'_{n-1}(\alpha\lambda)^{n-1} + \cdots + \alpha'_0 = 0,$$

where $\alpha'_i = \alpha_i\alpha^{n-1-i}$. Whence $\alpha\lambda$ is integral over $\mathcal{o}$ and thus $\alpha\lambda \in \mathcal{O}$. Then $\alpha\lambda = \beta$ for some $\beta \in \mathcal{O}$ which is equivalent to $\lambda = \frac{\beta}{\alpha}$. Therefore L is the field of fractions of $\mathcal{O}$. Now $\mathcal{O}$ is an integral domain as it is a subring of a field. Being the integral closure of $\mathcal{o}$ in L, $\mathcal{O}$ is an integrally closed domain with field of fractions L. It remains to prove the last statement. For the forward implication, suppose λ is integral over $\mathcal{o}$. Then λ is a root of a monic polynomial $f(x) \in \mathcal{o}[x]$. As $m_\lambda(x)$ divides $f(x)$ in $\mathcal{o}[x]$, all of the roots of $m_\lambda(x)$ are integral over $\mathcal{o}$ too. By Vieta's formulas, the coefficients of $m_\lambda(x)$ integral over $\mathcal{o}$ as well whence $m_\lambda(x) \in \mathcal{o}[x]$. For the reverse implication, if the minimal polynomial $m_\lambda(x)$ of λ over K has coefficients in $\mathcal{o}$ then λ is automatically integral over $\mathcal{o}$. □

The previous result can also be generalized to towers in light of the fact that integral closure is transitive.

Proposition 3.1.5. *Let $\mathcal{o}$ be an integrally closed domains with field of fractions K, let $M/L/K$ be a finite separable tower, and suppose $\mathcal{O}$ and $\widetilde{\mathcal{O}}$ are the integral closures of $\mathcal{o}$ in L and M respectively. Then $\mathcal{O}$ and $\widetilde{\mathcal{O}}$ are integrally closed domains. In particular, $\widetilde{\mathcal{O}}$ is the integral closure of $\mathcal{O}$ in L.*

Proof. The first statement follows from two applications of Proposition 3.1.4. To prove the second statement, we must show $\widetilde{\mathcal{O}} = \overline{\mathcal{O}}$. The forward inclusion is obvious as $\mathcal{o}$ is a subset of $\mathcal{O}$. For the reverse inclusion, Proposition 3.1.2 implies that $\overline{\mathcal{O}}$ is integral over $\mathcal{o}$. As $\widetilde{\mathcal{O}}$ is the integral closure of $\mathcal{o}$ in M, the reverse inclusion follows.□

Number Fields

Number fields are a particular instance of the aforementioned setting. A **number field** K is a field extension of $\mathbb{Q}$ of finite degree. That is, K is a finite dimensional $\mathbb{Q}$-vector space. In particular, $K/\mathbb{Q}$ is a finite separable extension since $\mathbb{Q}$ is perfect whence the primitive element theorem applies. Moreover, $K/\mathbb{Q}$ is Galois if and only if it is normal. We say that the **degree** of K is $[K : \mathbb{Q}]$ which is simply the degree of K as a $\mathbb{Q}$-vector space. In the cases of small degrees we will often use the latin derived names **quadratic**, **cubic**, etc. Any $\kappa \in K$ is called an **algebraic number**. We define the **ring of integers** $\mathcal{O}_K$ of K to be the integral closure of $\mathbb{Z}$ in K. In other words,

$$\mathcal{O}_K = \{\kappa \in K : \kappa \text{ is integral over } \mathbb{Z}\}.$$

Any $\alpha \in \mathcal{O}_K$ is called an **algebraic integer**. Note that α is an algebraic integer if and only if it is the root of a monic polynomial $f(x) \in \mathbb{Z}[x]$. It follows from Proposition 3.1.4 that every $\kappa \in K$ is of the form

$$\kappa = \frac{\alpha}{a},$$

for some algebraic integer α and nonzero integer $\mathcal{o}$. In particular, K is the field of fractions of $\mathcal{O}_K$ and $\mathcal{O}_K$ is an integrally closed domain. Moreover, κ is an algebraic integer if and only if the minimal polynomial $m_\kappa(x)$ of κ over $\mathbb{Q}$ has coefficients in $\mathbb{Z}$.

Remark 3.1.6. By Lemma 3.1.3, $\mathbb{Z}$ is an integrally closed domain and therefore the ring of integers of $\mathbb{Q}$ is exactly $\mathbb{Z}$.

3.2 Traces and Norms

Let $\mathcal{O}/\mathcal{o}$ be an extension of integral domains such that $\mathcal{O}$ is a free $\mathcal{o}$-module of rank n. Recall that by fixing a basis for $\mathcal{O}/\mathcal{o}$, there is an isomorphism

$$\mathrm{End}_{\mathcal{O}/\mathcal{o}} \cong \mathrm{Mat}_n(\mathcal{o})$$

This permits us to view $\mathcal{o}$-linear operators from $\mathcal{O}$ to itself as $n \times n$ matrices with coefficients in $\mathcal{o}$. To any $\beta \in \mathcal{O}$, we define the $\mathcal{o}$-linear operator T_β on $\mathcal{O}$ by

$$T_\beta : \mathcal{O} \to \mathcal{O} \qquad x \mapsto \beta x.$$

This is simply multiplication by β. The trace and determinant of T_β are deserving of special interest. For instance, let $f_\beta(x)$ denote the characteristic polynomial of T_β so that

$$f_\beta(x) = \det(xI - T_\beta) = x^n - \alpha_{n-1}x^{n-1} + \cdots + (-1)^n\alpha_0,$$

with $\alpha_i \in \mathcal{o}$. Then the trace and determinant have the alternative representations

$$\mathrm{trace}(T_\beta) = \alpha_{n-1} \quad \text{and} \quad \det(T_\beta) = \alpha_0,$$

We define the **trace** and **norm** of $\mathcal{O}/\mathcal{o}$, denoted $\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}$ and $\mathrm{N}_{\mathcal{O}/\mathcal{o}}$, by

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}} : \mathcal{O} \to \mathcal{o} \qquad \beta \mapsto \mathrm{trace}(T_\beta) \quad \text{and} \quad \mathrm{N}_{\mathcal{O}/\mathcal{o}} : \mathcal{O} \to \mathcal{o} \qquad \beta \mapsto \det(T_\beta).$$

Our primary aim will be to develop a sufficient understanding of the trace and norm. The trace is additive while the norm is multiplicative because they correspond to the trace and determinant of matrices. Whence

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta + \beta') = \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta) + \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta') \quad \text{and} \quad \mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta\beta') = \mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta)\,\mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta'),$$

for all $\beta, \beta' \in \mathcal{O}$ making the trace and norm into homomorphisms. In fact, we have the additional relations

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\alpha\beta) = \alpha\,\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta) \quad \text{and} \quad \mathrm{N}_{\mathcal{O}/\mathcal{o}}(\alpha\beta) = \alpha^n\,\mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta),$$

for all $\alpha \in \mathcal{o}$ by scalar multiplication of matrices. The trace and norm also behave well with respect to direct sums. If $\mathcal{O} = \mathcal{O}_1 \oplus \mathcal{O}_2$ and we have a decomposition $\beta = \beta_1 + \beta_2$ with $\beta_1 \in \mathcal{O}_1$ and $\beta_2 \in \mathcal{O}_2$ then

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta) = \mathrm{Tr}_{\mathcal{O}_1/\mathcal{o}}(\beta_1) + \mathrm{Tr}_{\mathcal{O}_2/\mathcal{o}}(\beta_2) \quad \text{and} \quad \mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta) = \mathrm{N}_{\mathcal{O}_1/\mathcal{o}}(\beta_1)\,\mathrm{N}_{\mathcal{O}_2/\mathcal{o}}(\beta_2),$$

because the corresponding matrix for β is block diagonal.

In the case of a degree n field extension L/K, we call $\mathrm{Tr}_{L/K}$ and $\mathrm{N}_{L/K}$ the **field trace** and **field norm** of L/K. For any $\lambda \in L$, the **field trace** and **field norm** of λ are $\mathrm{Tr}_{L/K}(\lambda)$ and $\mathrm{N}_{L/K}(\lambda)$. In this case, $\mathrm{N}_{L/K}(\lambda) = 0$ if and only if $\lambda = 0$ because otherwise T_λ has inverse $T_{\lambda^{-1}}$ and hence a nonzero determinant. Therefore we obtain homomorphisms

$$\mathrm{Tr}_{L/K} : L \to K \quad \text{and} \quad \mathrm{N}_{L/K} : L^* \to K^*.$$

When L/K is also separable, we can derive alternative descriptions of the field trace and field norm of L/K. This additional assumption is weak because we are mostly interested in field extensions of $\mathbb{Q}$ and $\mathbb{F}_p$ of finite degree which are always separable as these fields are perfect. In any case, to do this we need to work in the algebraic closure $\overline{K}$ of K. As L/K is a degree n separable extension, there are exactly n distinct K-embeddings $\sigma_1, \ldots, \sigma_n$ of L into $\overline{K}$. That is, there are n elements of $\mathrm{Hom}_K(L, \overline{K})$. These K-embeddings are constructed by letting θ be a primitive element for L/K and sending θ to one of its conjugate roots in the minimal polynomial $m_\theta(x)$ of θ over K. These K-embeddings are deeply connected to the field trace and field norm.

Proposition 3.2.1. *Let L/K be a degree n separable extension and let σ run over the elements of* $\mathrm{Hom}_K(L, \overline{K})$*. For any $\lambda \in L$, the characteristic polynomial $f_\lambda(x)$ of T_λ over K is a power of the minimal polynomial $m_\lambda(x)$ of λ over K and satisfies*

$$f_\lambda(x) = \prod_\sigma (x - \sigma(\lambda)).$$

We also have

$$\mathrm{Tr}_{L/K}(\lambda) = \sum_\sigma \sigma(\lambda) \quad \textit{and} \quad \mathrm{N}_{L/K}(\lambda) = \prod_\sigma \sigma(\lambda).$$

Moreover, if L/K is Galois and $\lambda_1, \ldots, \lambda_n$ are the conjugates of λ then

$$\mathrm{Tr}_{L/K}(\lambda) = \sum_i \lambda_i \quad \textit{and} \quad \mathrm{N}_{L/K}(\lambda) = \prod_i \lambda_i.$$

Proof. Let m and d be the degrees of $K(\lambda)/K$ and $L/K(\lambda)$ respectively. Write

$$m_\lambda(x) = x^m + \kappa_{m-1}x^{m-1} + \cdots + \kappa_0,$$

with $\kappa_i \in K$. We claim

$$f_\lambda(x) = m_\lambda(x)^d.$$

Indeed, recall that $1, \lambda, \ldots, \lambda^{m-1}$ is a basis of $K(\lambda)/K$. If $\alpha_1, \ldots, \alpha_d$ is a basis for $L/K(\lambda)$ then

$$\alpha_1, \alpha_1\lambda, \ldots, \alpha_1\lambda^{m-1}, \ldots, \alpha_d, \alpha_d\lambda, \ldots, \alpha_d\lambda^{m-1},$$

is a basis for L/K. The matrix of T_λ with respect to this basis is block diagonal with d blocks each of the form

$$\begin{pmatrix} & 1 & & \\ & & \ddots & \\ & & & 1 \\ -\kappa_0 & -\kappa_1 & \cdots & -\kappa_{m-1} \end{pmatrix}.$$

This matrix is the companion matrix to $m_\lambda(x)$ and hence the characteristic polynomial is $m_\lambda(x)$ as well. Our claim follows since the characteristic polynomial of a block diagonal matrix is the product of the characteristic polynomials of the blocks. Since λ is algebraic over K of degree m, $K(\lambda)$ is the splitting field of $m_\lambda(x)$ and there are m elements of $\mathrm{Hom}_K(K(\lambda), \overline{K})$. Then the elements of $\mathrm{Hom}_K(L, \overline{K})$ are partitioned into m many equivalence classes each of size d where two K-embeddings are equivalent if and only if they take the same value at λ. If τ runs over the elements of $\mathrm{Hom}_K(K(\lambda), \overline{K})$ then this a complete set of representatives. As

$$m_\lambda(x) = \prod_\tau (x - \tau(\lambda))$$

it follows that

$$f_\lambda(x) = \prod_\sigma (x - \sigma(\lambda)).$$

The formulas for the field trace and field norm follow from Vieta's formulas applied to this product for $f_\lambda(x)$. Now suppose L/K is Galois. Then $\mathrm{Hom}_K(L, \overline{K}) = \mathrm{Gal}(L/K)$. Therefore the conjugates of λ are exactly the images of λ under these K-embeddings and the last statement follows. $\square$

As an application of this result, we can show how the trace and norm act when $\mathcal{o}$ is an integrally closed domain with field of fractions K, L/K is a finite separable extension, and $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L.

Proposition 3.2.2. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. If $\beta \in \mathcal{O}$ then $\mathrm{Tr}_{L/K}(\beta) \in \mathcal{o}$ and $\mathrm{N}_{L/K}(\beta) \in \mathcal{o}$.*

Proof. By Proposition 3.1.4, the minimal polynomial $m_\beta(x)$ of β over K has coefficients in $\mathcal{o}$. From Proposition 3.2.1, the characteristic polynomial $f_\beta(x)$ is a power of $m_\beta(x)$ and hence $f_\beta(x)$ has coefficients in $\mathcal{o}$ too. Whence $\mathrm{Tr}_{L/K}(\beta) \in \mathcal{o}$ and $\mathrm{N}_{L/K}(\beta) \in \mathcal{o}$ as they are coefficients of $f_\beta(x)$ up to sign. $\square$

In this setting it is also easy to classify the units of $\mathcal{O}$ in terms of the units of $\mathcal{o}$.

Proposition 3.2.3. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then $\beta \in \mathcal{O}$ is a unit if and only if $\mathrm{N}_{L/K}(\beta) \in \mathcal{o}$ is a unit.*

Proof. For the forward implication, if $\beta \in \mathcal{O}$ is a unit then $\frac{1}{\beta} \in \mathcal{O}$. Whence

$$\mathrm{N}_{L/K}(\beta)\,\mathrm{N}_{L/K}\left(\frac{1}{\beta}\right) = 1.$$

By Proposition 3.2.2, both of the field norms on the left-hand side are in $\mathcal{o}$. This proves $\mathrm{N}_{L/K}(\beta) \in \mathcal{o}$ is a unit. For the reverse implication, we know by assumption that β is nonzero. Also, Proposition 3.1.4 implies that the minimal polynomial $m_\beta(x)$ of β over K has coefficients in $\mathcal{o}$. The constant term is a unit since it is $\mathrm{N}_{L/K}(\beta)$ up to sign. Letting the degree of $m_\beta(x)$ be m, we have shown that

$$m_\beta(x) = x^m + \alpha_{m-1}x^{m-1} + \cdots + \alpha,$$

with $\alpha, \alpha_i \in \mathcal{o}$ where α is a unit. Dividing this relation by $\alpha\beta^m$, we find that $\frac{1}{\beta}$ is a root of the monic polynomial

$$f(x) = x^m + \frac{\alpha_1}{\alpha}x^{m-1} + \cdots + \frac{1}{\alpha},$$

whose coefficients are in $\mathcal{o}$. Hence $\frac{1}{\beta} \in \mathcal{O}$ and thus β is a unit. $\square$

Having introduced traces and norms, we turn to discussing discriminants of free modules. Let $\mathcal{O}/\mathcal{o}$ be an extension of integral domains such that $\mathcal{O}$ is a free $\mathcal{o}$-module of rank n. If $\beta_1, \ldots, \beta_n$ is a basis for $\mathcal{O}/\mathcal{o}$, we define its **trace matrix** $\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)$ by

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n) = \begin{pmatrix} \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1\beta_1) & \cdots & \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1\beta_n) \\ \vdots & & \vdots \\ \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_n\beta_1) & \cdots & \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_n\beta_n) \end{pmatrix}.$$

The **discriminant** $d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)$ of $\beta_1, \ldots, \beta_n$ is defined to be the determinant of the trace matrix. That is,

$$d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n) = \det(\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)).$$

In particular, $d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n) \in \mathcal{o}$ since the entries of the trace matrix belong to $\mathcal{o}$. It is also independent of the choice of basis up to elements of $(\mathcal{o}^*)^2$. For if $\beta_1', \ldots, \beta_n'$ is another basis for $\mathcal{O}/\mathcal{o}$, we have

$$\beta_i' = \sum_j \alpha_{i,j}\beta_j,$$

with $\alpha_{i,j} \in \mathcal{o}$. Then $(\alpha_{i,j})$ is the base change matrix from $\beta_1, \ldots, \beta_n$ to $\beta_1', \ldots, \beta_n'$ and so has nonzero determinant. Thus $\det((\alpha_{i,j})) \in \mathcal{o}^*$. Moreover,

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1', \ldots, \beta_n') = (\alpha_{i,j})\,\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)(\alpha_{i,j})^t,$$

which, upon taking the determinant, shows

$$d_{\mathcal{O}/\mathcal{o}}(\beta_1', \ldots, \beta_n') = \det((\alpha_{i,j}))^2 d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n). \tag{3.1}$$

Accordingly, we define the **discriminant** $d_{\mathcal{O}/\mathcal{o}}$ of $\mathcal{O}/\mathcal{o}$ to be the coset in $\mathcal{o}/(\mathcal{o}^*)^2$ represented by any discriminant $d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)$. In other words,

$$d_{\mathcal{O}/\mathcal{o}} = d_{\mathcal{O}/\mathcal{o}}(\beta_1, \ldots, \beta_n)(\mathcal{o}^*)^2.$$

In particular, $d_{\mathcal{O}/\mathcal{o}} = 0$ is independent of the choice of representative. The discriminant is also multiplicative with respect to direct sums.

Proposition 3.2.4. *Let $\mathcal{O}/\mathcal{o}$ be an extension of integral domains such that $\mathcal{O}$ is a free $\mathcal{o}$-module of rank n. Suppose we have a direct sum decomposition*

$$\mathcal{O} = \mathcal{O}_1 \oplus \mathcal{O}_2,$$

for free $\mathcal{o}$-modules $\mathcal{O}_1$ and $\mathcal{O}_2$ of ranks n_1 and n_2 with bases $\beta_{1,1}, \ldots, \beta_{n_1,1}$ and $\beta_{1,2}, \ldots, \beta_{n_2,2}$ over $\mathcal{o}$ respectively. Then $\beta_{1,1}, \ldots, \beta_{n_1,1}, \beta_{1,2}, \ldots, \beta_{n_2,2}$ is a basis of $\mathcal{O}/\mathcal{o}$ and

$$d_{\mathcal{O}/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1}, \beta_{1,2}, \ldots, \beta_{n_2,2}) = d_{\mathcal{O}_1/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1})d_{\mathcal{O}_2/\mathcal{o}}(\beta_{1,2}, \ldots, \beta_{n_2,2}).$$

Proof. Clearly $\beta_{1,1}, \ldots, \beta_{n_1,1}, \beta_{1,2}, \ldots, \beta_{n_2,2}$ is a basis for $\mathcal{O}/\mathcal{o}$ and $\beta_{i,1}\beta_{j,2} = 0$. It follows that $d_{\mathcal{O}/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1}, \beta_{1,2}, \ldots, \beta_{n_2,2})$ is the determinant of the block diagonal matrix

$$\begin{pmatrix} \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1}) & \\ & \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_{1,2}, \ldots, \beta_{n_2,2}) \end{pmatrix}.$$

Recall that

$$\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_1) = \mathrm{Tr}_{\mathcal{O}_1/\mathcal{o}}(\beta_1) \quad \text{and} \quad \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta_2) = \mathrm{Tr}_{\mathcal{O}_2/\mathcal{o}}(\beta_2),$$

for any $\beta_1 \in \mathcal{O}_1$ and $\beta_2 \in \mathcal{O}_2$. Whence the block diagonal matrix above is

$$\begin{pmatrix} \mathrm{Tr}_{\mathcal{O}_1/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1}) & \\ & \mathrm{Tr}_{\mathcal{O}_2/\mathcal{o}}(\beta_{1,2}, \ldots, \beta_{n_2,2}) \end{pmatrix}.$$

The determinant of this matrix is $d_{\mathcal{O}_1/\mathcal{o}}(\beta_{1,1}, \ldots, \beta_{n_1,1})d_{\mathcal{O}_2/\mathcal{o}}(\beta_{1,2}, \ldots, \beta_{n_2,2})$ which completes the proof. $\square$

We now specialize to the setting of a degree n separable extension L/K. In this case, it turns out that the discriminant of a basis is nonzero. To see this will require some work. First, we define the **trace form** of L/K to be the map

$$\mathrm{Tr}_{L/K} : L \times L \to K \qquad (\lambda, \eta) \mapsto \mathrm{Tr}_{L/K}(\lambda\eta).$$

In other words, this is just the field trace of L/K considered as a pairing. Clearly this is a symmetric bilinear form. In fact, the trace form is also nondegenerate as Proposition 3.2.1 implies

$$\mathrm{Tr}_{L/K}(1) = n,$$

and we can pair any $\lambda \in L$ with its inverse. Using the trace form, we can show that the discriminant of any basis for L/K is nonzero.

Proposition 3.2.5. *Let L/K be a degree n separable extension and let $\lambda_1, \ldots, \lambda_n$ be a basis for L/K. Then $d_{L/K}(\lambda_1, \ldots, \lambda_n) \neq 0$.*

Proof. Suppose by contradiction that $d_{L/K}(\lambda_1, \ldots, \lambda_n) = 0$. Then the trace matrix $\mathrm{Tr}_{L/K}(\lambda_1, \ldots, \lambda_n)$ is not invertible. Therefore

$$\mathrm{Tr}_{L/K}(\lambda_1, \ldots, \lambda_n) \begin{pmatrix} \kappa_1 \\ \vdots \\ \kappa_n \end{pmatrix} = \mathbf{0},$$

for some $\kappa_i \in K$ not all of which are zero. Then the element

$$\lambda = \sum_j \kappa_j \lambda_j.$$

is nonzero. Moreover, the previous identity is equivalent to the fact that $\mathrm{Tr}_{L/K}(\lambda\lambda_i) = 0$. As $\lambda_1, \ldots, \lambda_n$ is a basis for L/K, it follows that the trace form is degenerate which is a contradiction. Hence $d_{L/K}(\lambda_1, \ldots, \lambda_n) \neq 0$. $\square$

In addition to the discriminant $d_{L/K}(\lambda_1, \ldots, \lambda_n)$ being nonzero, we can also write it in a more useful form. To do this, we define the **embedding matrix** $M(\lambda_1, \ldots, \lambda_n)$ of the basis $\lambda_1, \ldots, \lambda_n$ by

$$M(\lambda_1, \ldots, \lambda_n) = \begin{pmatrix} \sigma_1(\lambda_1) & \cdots & \sigma_1(\lambda_n) \\ \vdots & & \vdots \\ \sigma_n(\lambda_1) & \cdots & \sigma_n(\lambda_n) \end{pmatrix},$$

where $\sigma_1, \ldots, \sigma_n$ are the elements of $\mathrm{Hom}_K(L, \overline{K})$. The discriminant turns out to be the square of the determinant of the embedding matrix.

Proposition 3.2.6. *Let L/K be a degree n separable extension. Then for any basis $\lambda_1, \ldots, \lambda_n$ of L/K, we have*

$$d_{L/K}(\lambda_1, \ldots, \lambda_n) = \det(M(\lambda_1, \ldots, \lambda_n))^2.$$

Proof. Recall that the ij-entry of $M(\lambda_1, \ldots, \lambda_n)^t M(\lambda_1, \ldots, \lambda_n)$ is the dot product of the i-th and j-th columns of $M(\lambda_1, \ldots, \lambda_n)$. Then a short computation shows

$$\det(M(\lambda_1, \ldots, \lambda_n)^t M(\lambda_1, \ldots, \lambda_n)) = \det\left(\left(\sum_{\sigma \in \mathrm{Hom}_K(L,\overline{K})} \sigma(\lambda_i\lambda_j)\right)_{i,j}\right).$$

By Proposition 3.2.1, this identity is equivalent to the claim. □

In general, it is difficult to compute the discriminant of a basis for L/K. However, if the basis is of the form $1, \lambda, \ldots, \lambda^{n-1}$ (take $\lambda = \theta$ for a primitive element θ of L/K) then the discriminant of this basis can be easily computed. Indeed, the embedding matrix becomes

$$M(1, \lambda, \ldots, \lambda^{n-1}) = \begin{pmatrix} 1 & \sigma_1(\lambda) & \cdots & \sigma_1(\lambda)^{n-1} \\ \vdots & \vdots & & \vdots \\ 1 & \sigma_n(\lambda) & \cdots & \sigma_n(\lambda)^{n-1} \end{pmatrix},$$

which is a Vandermonde matrix. By Proposition 3.2.6, we have

$$d_{L/K}(1, \lambda, \ldots, \lambda^{n-1}) = \prod_{i \le j} (\sigma_i(\lambda) - \sigma_j(\lambda))^2, \tag{3.2}$$

which is the square of the Vandermonde determinant of $M(1, \lambda, \ldots, \lambda^{n-1})$.

Let us now consider the situation when $\mathcal{o}$ is an integrally closed domain with field of fractions K, L/K be a degree n separable extension, and $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L. If $\lambda_1, \ldots, \lambda_n$ is a basis for L/K then using Proposition 3.1.4 we may multiply by a nonzero element of $\mathcal{o}$, if necessary, to ensure that this basis belongs to $\mathcal{O}$. In many instances this will be useful. As a first application, the discriminant of such a basis relates inclusion of $\mathcal{O}$ relative to $\mathcal{o}$.

Lemma 3.2.7. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a degree n separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. If $\lambda_1, \dots, \lambda_n$ is a basis for L/K that is contained in $\mathcal{O}$ then*

$$d_{L/K}(\lambda_1, \dots, \lambda_n)\mathcal{O} \subseteq \mathcal{o}\lambda_1 + \cdots + \mathcal{o}\lambda_n.$$

In particular, $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module of rank at most n.

Proof. Let $\beta \in \mathcal{O}$ and write
$$\beta = \sum_j \kappa_j \lambda_j.$$
for some $\kappa_j \in K$. Linearity of the trace implies

$$\mathrm{Tr}_{L/K}(\lambda_i \beta) = \sum_j \kappa_j \mathrm{Tr}_{L/K}(\lambda_i \lambda_j),$$

These n equations are equivalent to the identity

$$\begin{pmatrix} \mathrm{Tr}_{L/K}(\beta\lambda_1) \\ \vdots \\ \mathrm{Tr}_{L/K}(\beta\lambda_n) \end{pmatrix} = \mathrm{Tr}_{L/K}(\lambda_1, \dots, \lambda_n) \begin{pmatrix} \kappa_1 \\ \vdots \\ \kappa_n \end{pmatrix}.$$

By Proposition 3.2.2, the column vector on the left-hand side and the trace matrix have entries in $\mathcal{o}$. Cramer's rule implies $d_{L/K}(\lambda_1, \dots, \lambda_n)\kappa_i \in \mathcal{o}$. As β was arbitrary, this means
$$d_{L/K}(\lambda_1, \dots, \lambda_n)\mathcal{O} \subseteq \mathcal{o}\lambda_1 + \cdots + \mathcal{o}\lambda_n,$$
proving the first statement. The second statement is immediate. □

In this situation, we say that $\beta_1, \dots, \beta_n$ is an **integral basis** for $\mathcal{O}/\mathcal{o}$ if

$$\mathcal{O} = \mathcal{o}\beta_1 + \cdots + \mathcal{o}\beta_n.$$

Equivalently, $\mathcal{O}$ is a free $\mathcal{o}$-module of rank n. An integral basis is necessarily a basis for L/K by Proposition 3.1.4. However, an integral basis need not always exist. For if $\lambda_1, \dots, \lambda_n$ is a basis of L/K, Proposition 3.1.4 implies that we can multiply by a nonzero element of $\mathcal{o}$ to ensure that this basis is contained in $\mathcal{O}$. However, $\lambda_1, \dots, \lambda_n$ need not also be a basis of $\mathcal{O}/\mathcal{o}$. Nevertheless, if $\mathcal{o}$ is a principal ideal domain then we can ensure the existence of an integral basis.

Theorem 3.2.8. *Let $\mathcal{o}$ be a principal ideal domain with field of fractions K, let L/K be a degree n separable extension, and let $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then $\mathcal{O}/\mathcal{o}$ admits an integral basis. Moreover, every finitely generated nonzero $\mathcal{O}$-submodule M of L is a free $\mathcal{o}$-module of rank n.*

Proof. Let $\lambda_1, \ldots, \lambda_n$ be a basis for L/K contained in $\mathcal{O}$. Then Lemma 3.2.7 implies

$$d_{L/K}(\lambda_1, \ldots, \lambda_n)\mathcal{O} \subseteq \mathcal{o}\lambda_1 + \cdots + \mathcal{o}\lambda_n.$$

It follows from the structure theorem for finitely generated modules over principal ideal domains that $\mathcal{O}$ is also a free $\mathcal{o}$-module of rank at most n. But any basis for $\mathcal{O}/\mathcal{o}$ must also be a basis for L/K by Proposition 3.1.4. Hence the rank is exactly n and $\mathcal{O}$ admits an integral basis over $\mathcal{o}$ which proves the first statement.

Now let M is a finitely generated nonzero $\mathcal{O}$-submodule of L with generators $\omega_1, \ldots, \omega_r$. By Proposition 3.1.4, there exists a nonzero $\alpha \in \mathcal{o}$ such that $\alpha\omega_i \in \mathcal{O}$. Whence $\alpha M \subseteq \mathcal{O}$. But then from our previous work, we have

$$\alpha d_{L/K}(\lambda_1, \ldots, \lambda_n)M \subseteq \mathcal{o}\lambda_1 + \cdots + \mathcal{o}\lambda_n.$$

By the structure theorem for finitely generated modules over principal ideal domains again, M is a free $\mathcal{o}$-module of rank at most n. To see that the rank is at least n, let $\lambda \in M$ be nonzero. Then $\lambda\lambda_1, \ldots, \lambda\lambda_n$ is a basis for L/K contained in M. Thus the rank of M is at least n, and in particular it must be n. This proves the second statement. □

Recall that if L_1/K and L_2/K are finite separable extensions then the composite L of L_1 and L_2 is such that L/K is also a finite separable extension. Integral bases behave well with respect to composite fields provided the fields are linearly disjoint.

Proposition 3.2.9. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L_1/K and L_2/K be degree n_1 and n_2 separable extensions, and $\mathcal{O}_1$ and $\mathcal{O}_2$ be the integral closures of $\mathcal{o}$ in L_1 and L_2 respectively. Suppose L_1 and L_2 are linearly disjoint over K in $\overline{K}$ and that $\mathcal{O}_1/\mathcal{o}$ and $\mathcal{O}_2/\mathcal{o}$ admit integral bases $\beta_{1,1}, \ldots, \beta_{n_1,1}$ and $\beta_{1,2}, \ldots, \beta_{n_2,2}$ with*

$$\alpha_1 d_{L_1/K}(\beta_{1,1}, \ldots, \beta_{n_1,1}) + \alpha_2 d_{L_2/K}(\beta_{1,2}, \ldots, \beta_{n_2,2}) = 1,$$

for some $\alpha_1, \alpha_2 \in \mathcal{o}$. Let L be the composite of L_1 and L_2 and let $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then $\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}$ is an integral basis for $\mathcal{O}/\mathcal{o}$ and

$$\mathcal{O} = \mathcal{O}_1\mathcal{O}_2.$$

In particular,

$$d_{L/K}(\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}) = d_{L_1/K}(\beta_{1,1}, \ldots, \beta_{n_1,1})^{n_2} d_{L_2/K}(\beta_{1,2}, \ldots, \beta_{n_2,2})^{n_1}.$$

Proof. We will start by proving $\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}$ is an integral basis for $\mathcal{O}/\mathcal{o}$. Since L_1 and L_2 are linearly disjoint over K in $\overline{K}$, it must be that $\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}$ is a basis for L/K. Therefore any $\beta \in \mathcal{O}$ is of the form

$$\beta = \sum_{i,j} \kappa_{i,j}\beta_{i,1}\beta_{j,2},$$

for some $\kappa_{i,j} \in K$. We need to show that $\kappa_{i,j} \in \mathcal{o}$. To this end, let

$$\alpha_{i,2} = \sum_j \kappa_{i,j}\beta_{j,2} \quad \text{and} \quad \alpha_{j,1} = \sum_i \kappa_{i,j}\beta_{i,1},$$

so that

$$\beta = \sum_i \alpha_{i,2}\beta_{i,1} \quad \text{and} \quad \beta = \sum_j \alpha_{j,1}\beta_{j,2}.$$

In particular, $\alpha_{i,2} \in L_2$ and $\alpha_{j,1} \in L_1$. By linear disjointness, we have

$$\mathrm{Hom}_K(L, \overline{K}) \cong \mathrm{Hom}_K(L_1, \overline{K}) \times \mathrm{Hom}_K(L_2, \overline{K}).$$

So letting $\sigma_{1,1}, \ldots, \sigma_{n_1,1}$ and $\sigma_{1,2}, \ldots, \sigma_{n_2,2}$ be the elements of $\mathrm{Hom}_K(L_1, \overline{K})$ and $\mathrm{Hom}_K(L_2, \overline{K})$ respectively, $\sigma_{1,1}\sigma_{1,2}, \ldots, \sigma_{n_1,1}\sigma_{n_2,2}$ are the elements of $\mathrm{Hom}_K(L, \overline{K})$. In particular, we may view $\sigma_{1,1}, \ldots, \sigma_{n_1,1}$ and $\sigma_{1,2}, \ldots, \sigma_{n_2,2}$ as elements of $\mathrm{Hom}_K(L, \overline{K})$ that act as the identity on L_2 and L_1 respectively. Then the n_1 and n_2 equations

$$\sum_k \sigma_{i,1}(\beta_{k,1})\alpha_{k,2} = \sigma_{i,1}(\beta) \quad \text{and} \quad \sum_\ell \sigma_{j,2}(\beta_{\ell,2})\alpha_{\ell,1} = \sigma_{j,2}(\beta),$$

are equivalent to the identities

$$M(\beta_{1,1}, \ldots, \beta_{n_1,1}) \begin{pmatrix} \alpha_{1,2} \\ \vdots \\ \alpha_{n_1,2} \end{pmatrix} = \begin{pmatrix} \sigma_{1,1}(\beta) \\ \vdots \\ \sigma_{n_1,1}(\beta) \end{pmatrix},$$

and

$$M(\beta_{1,2}, \ldots, \beta_{n_2,2}) \begin{pmatrix} \alpha_{1,1} \\ \vdots \\ \alpha_{n_2,1} \end{pmatrix} = \begin{pmatrix} \sigma_{1,2}(\beta) \\ \vdots \\ \sigma_{n_2,2}(\beta) \end{pmatrix},$$

respectively. The embedding matrices and column vectors on the right-hand sides all have entries in $\mathcal{O}$. Cramer's rule together with Proposition 3.2.6 imply that $d_{L_1/K}(\beta_{1,1}, \ldots, \beta_{n_1,1})\alpha_{i,2} \in \mathcal{O}$ and $d_{L_2/K}(\beta_{1,2}, \ldots, \beta_{n_2,2})\alpha_{j,1} \in \mathcal{O}$. As $\mathcal{O}_1 = L_1 \cap \mathcal{O}$ and $\mathcal{O}_2 = L_2 \cap \mathcal{O}$, we have $\alpha_{i,2} \in \mathcal{O}_2$ and $\alpha_{j,1} \in \mathcal{O}_1$. Expanding $\alpha_{i,2}$ and $\alpha_{j,1}$ show that $d_{L_1/K}(\beta_{1,1}, \ldots, \beta_{n_1,1})\kappa_{i,j} \in \mathcal{o}$ and $d_{L_2/K}(\beta_{1,2}, \ldots, \beta_{n_2,2})\kappa_{i,j} \in \mathcal{o}$. From the identity

$$k_{i,j} = k_{i,j}(\alpha_1 d_{L_1/K}(\beta_{1,1}, \ldots, \beta_{n_1,1}) + \alpha_2 d_{L_2/K}(\beta_{1,2}, \ldots, \beta_{n_2,2})),$$

we conclude $k_{i,j} \in \mathcal{o}$. This proves $\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}$ is an integral basis for $\mathcal{O}/\mathcal{o}$. The fact that

$$\mathcal{O} = \mathcal{O}_1\mathcal{O}_2,$$

follows at once. It remains to prove the discriminant formula which will be achieved by decomposing the embedding matrix into blocks. Recall that $\sigma_{1,1}\sigma_{1,2}, \ldots, \sigma_{n_1,1}\sigma_{n_2,2}$ are the elements of $\mathrm{Hom}_K(L, \overline{K})$, we have

$$M(\beta_{1,1}\beta_{1,2}, \ldots, \beta_{n_1,1}\beta_{n_2,2}) = \begin{pmatrix} \sigma_{1,1}(\beta_{1,1})\sigma_{1,2}(\beta_{1,2}) & \cdots & \sigma_{1,1}(\beta_{n_1,1})\sigma_{1,2}(\beta_{n_2,2}) \\ \vdots & & \vdots \\ \sigma_{n_1,1}(\beta_{1,1})\sigma_{n_2,2}(\beta_{1,2}) & \cdots & \sigma_{n_1,1}(\beta_{n_1,1})\sigma_{n_2,2}(\beta_{n_2,2}) \end{pmatrix}.$$

This left-hand side admits the block factorization

$$\begin{pmatrix} M(\beta_{1,1},\dots,\beta_{n_1,1}) & & \\ & \ddots & \\ & & M(\beta_{1,1},\dots,\beta_{n_1,1}) \end{pmatrix} \begin{pmatrix} I\sigma_{1,2}(\beta_{1,2}) & \cdots & I\sigma_{1,2}(\beta_{n_2,2}) \\ \vdots & & \vdots \\ I\sigma_{n_2,2}(\beta_{1,2}) & \cdots & I\sigma_{n_2,2}(\beta_{n_2,2}) \end{pmatrix},$$

where the second matrix is the Kronecker product $M(\beta_{1,2},\dots,\beta_{n_2,2}) \otimes I$. As this Kronecker product can be expressed as $(I \otimes M(\beta_{1,2},\dots,\beta_{n_2,2}))P$ for some permutation matrix P, it takes the form

$$\begin{pmatrix} M(\beta_{1,2},\dots,\beta_{n_2,2}) & & \\ & \ddots & \\ & & M(\beta_{1,2},\dots,\beta_{n_2,2}) \end{pmatrix} P,$$

with $\det(P) = \pm 1$. Putting these decompositions together and applying Proposition 3.2.6 shows

$$d_{L/K}(\beta_{1,1}\beta_{1,2},\dots,\beta_{n_1,1}\beta_{n_2,2}) = d_{L_1/K}(\beta_{1,1},\dots,\beta_{n_1,1})^{n_2} d_{L_2/K}(\beta_{1,2},\dots,\beta_{n_2,2})^{n_1}.$$

This proves the discriminant formula. □

It is generally a difficult problem to write down an integral basis explicitly. However, there is one instance in which this is possible. We say that $\mathcal{O}/\mathcal{o}$ is **monogenic** if $\mathcal{O} = \mathcal{o}[\beta]$ for some $\beta \in \mathcal{O}$. It follows immediately that $1, \beta, \dots, \beta^{n-1}$ is an integral basis for $\mathcal{O}/\mathcal{o}$. The discriminant of this basis is then given by Equation (3.2).

Regardless of if $\mathcal{O}/\mathcal{o}$ is monogenic or not, if an integral basis exists then the field trace $\mathrm{Tr}_{L/K}$ and field norm $\mathrm{N}_{L/K}$ agree with the trace $\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}$ and norm $\mathrm{N}_{\mathcal{O}/\mathcal{o}}$ respectively.

Proposition 3.2.10. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a degree n separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. If $\mathcal{O}$ admits an integral basis over $\mathcal{o}$ then*

$$\mathrm{Tr}_{L/K}(\beta) = \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta) \quad \text{and} \quad \mathrm{N}_{L/K}(\beta) = \mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta),$$

for all $\beta \in \mathcal{O}$.

Proof. Let $\beta_1,\dots,\beta_n$ be an integral basis for $\mathcal{O}/\mathcal{o}$. Then $\beta_1,\dots,\beta_n$ is also a basis for L/K. It follows that the multiplication by β map in L has the same matrix representation as it does in $\mathcal{O}$. Whence

$$\mathrm{Tr}_{L/K}(\beta) = \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\beta) \quad \text{and} \quad \mathrm{N}_{L/K}(\beta) = \mathrm{N}_{\mathcal{O}/\mathcal{o}}(\beta).$$ □

Number Fields

We now turn to the case of a number field K of degree n. We write $\mathrm{Tr}_K = \mathrm{Tr}_{K/\mathbb{Q}}$ and $\mathrm{N}_K = \mathrm{N}_{K/\mathbb{Q}}$ and call these the **field trace** and **field norm** of K. Moreover, for any $\kappa \in K$ we call $\mathrm{Tr}_K(\kappa)$ and $\mathrm{N}_K(\kappa)$ the **field trace** and **field norm** of κ. Observe from Propositions 3.2.2 and 3.2.3 that the field trace and field norm of algebraic integers are themselves integers and any $\kappa \in \mathcal{O}_K$ is a unit if and only if $\mathrm{N}_K(\kappa) = \pm 1$. We also call the nondegenerate symmetric bilinear form

$$\mathrm{Tr}_{K/\mathbb{Q}} : K \times K \to \mathbb{Q} \qquad (\kappa, \lambda) \mapsto \mathrm{Tr}_{K/\mathbb{Q}}(\kappa\lambda),$$

the **trace form** of K. Moreover, since $\mathbb{Z}$ is a principal ideal domain Theorem 3.2.8 implies that $\mathcal{O}_K/\mathbb{Z}$ admits an integral basis. Whence $\mathcal{O}_K$ is a free $\mathbb{Z}$-module of rank n. Accordingly, we say that $\alpha_1, \dots, \alpha_n$ is an **integral basis** for K if it is an integral basis for $\mathcal{O}_K/\mathbb{Z}$. Accordingly, we define the **discriminant** Δ_K of K to be the discriminant of $\mathcal{O}_K/\mathbb{Z}$. As $(\mathbb{Z}^*)^2 = \{1\}$, Δ_K is an integer and satisfies

$$\Delta_K = d_{L/K}(\alpha_1, \dots, \alpha_n),$$

for any integral basis $\alpha_1, \dots, \alpha_n$ for K. Moreover, Δ_K is nonzero since the trace form is nondegenerate and may very well be negative. In light of Proposition 3.2.6, we also have

$$|\det(M(\alpha_1, \dots, \alpha_n))| = \sqrt{|\Delta_K|}. \tag{3.3}$$

Lastly, we say K is **monogenic** if $\mathcal{O}_K$ is monogenic over $\mathbb{Z}$.

3.3 Dedekind Domains

Let $\mathcal{o}$ be an integral domain with field of fractions K. Any nonzero ideal $\mathfrak{a}$ of $\mathcal{o}$ is said to be an **integral ideal** of $\mathcal{o}$. We call any prime integral ideal $\mathfrak{p}$ of $\mathcal{o}$ a **prime** of $\mathcal{o}$. If $\mathfrak{p}$ is principal with $\mathfrak{p} = \alpha\mathcal{o}$ for some nonzero $\alpha \in K$ we will also refer to α as the prime instead of $\mathfrak{p}$. In any case, an integral ideal is just an $\mathcal{o}$-submodule of $\mathcal{o}$. More generally, we say $\mathfrak{f}$ is a **fractional ideal** of $\mathcal{o}$ if it is a nonzero $\mathcal{o}$-submodule of K such that there is a nonzero $\delta \in \mathcal{o}$ with $\delta\mathfrak{f} \in \mathcal{o}$. This simply means $\delta\mathfrak{f}$ is an integral ideal. Conversely, if $\mathfrak{a}$ is an integral ideal and $\delta \in \mathcal{o}$ is nonzero then

$$\mathfrak{f} = \frac{1}{\delta}\mathfrak{a},$$

is a fractional ideal. So every fractional ideal is of this form. Clearly fractional ideals are closed under multiplication. The fractional ideal $\mathfrak{f}$ is said to be **principal** if it is generated by a single element. That is, if $\mathfrak{f} = \kappa\mathcal{o}$ for some nonzero $\kappa \in K$. In particular, all integral ideals are fractional ideals by taking $\delta = 1$ and all principal integral ideals are principal fractional ideals. The **inverse** $\mathfrak{f}^{-1}$ of $\mathfrak{f}$ is defined by

$$\mathfrak{f}^{-1} = \{\kappa \in K : \kappa\mathfrak{f} \subseteq \mathcal{o}\}.$$

Note that $\delta \in \mathfrak{f}^{-1}$ whence $\delta\mathfrak{f}^{-1}$ is an integral ideal so that $\mathfrak{f}^{-1}$ is a fractional ideal. It is the largest fractional ideal such $\mathfrak{f}\mathfrak{f}^{-1} \subseteq \mathcal{O}$. We say that $\mathfrak{f}$ is **invertible** if

$$\mathfrak{f}\mathfrak{f}^{-1} = \mathcal{O}.$$

A priori, a fractional ideal need not be invertible nor should we expect it to be. In order to guarantee invertibility, we need to consider a more restrictive setting than mere integral domains. This will allow for integral ideals to factor uniquely into a product of primes and for the fractional ideals to form an abelian group under multiplication in which every fractional ideal is invertible and $\mathcal{O}$ is the identity. This will essentially permit us to treat integral ideas as we would integers.

Accordingly, $\mathcal{O}$ is said to be a **Dedekind domain** if the following properties are satisfied:

(i) $\mathcal{O}$ is an integrally closed domain.

(ii) $\mathcal{O}$ is noetherian.

(iii) Every prime of $\mathcal{O}$ is maximal.

Observe that $\mathcal{O}$ being noetherian forces every integral ideal $\mathfrak{a}$ to be a finitely generated $\mathcal{O}$-module. In fact, as every fractional ideal $\mathfrak{f}$ is of the form $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$, for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{a}$, $\mathfrak{f}$ is a finitely generated $\mathcal{O}$-submodule of K. Also, the condition that every prime of $\mathcal{O}$ is maximal is equivalent to $\mathcal{O}$ being one-dimensional as the longest chain of strict prime ideals is

$$0 \subsetneq \mathfrak{p} \subsetneq \mathcal{O},$$

for any prime $\mathfrak{p}$, and this is equivalent to $\mathfrak{p}$ being maximal as every maximal ideal is necessarily prime. In particular, a Dedekind domain is just a one-dimensional noetherian domain that is also an integrally closed domain. One-dimensional noetherian domains will also play a prominent role alongside Dedekind domains.

Remark 3.3.1. As $\mathbb{Z}$ is a principal ideal domain which is an integrally closed domain by Remark 3.1.6, $\mathbb{Z}$ is a Dedekind domain. The primes of $\mathbb{Z}$ are exactly the primes p.

In view of principal ideal domains being integrally closed domains by Lemma 3.1.3, they are Dedekind domains. In fact, Dedekind domains should be thought of as generalizations of principal ideal domains. As $\mathbb{Z}$ is a prototypical example of a principal ideal domain, a Dedekind domain $\mathcal{O}$ should be thought of as a generalization of $\mathbb{Z}$. In fact, we will show that being a principal ideal domain is equivalent to being a unique factorization domain for $\mathcal{O}$. So if $\mathcal{O}$ is not a principal ideal domain then unique factorization fails. However, it will be possible to remedy most of this as one of our primary aims is to prove that integral ideals factor uniquely into a product of primes. This clearly generalizes unique factorization in $\mathbb{Z}$ which will be enough. To this end, we first show inclusion in one direction for integral ideals. Interesting, this only requires that $\mathcal{O}$ be a noetherian domain.

Lemma 3.3.2. *Let $\mathcal{O}$ be a noetherian domain. For every integral ideal $\mathfrak{a}$ of $\mathcal{O}$, there exist primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$ of $\mathcal{O}$ such that*

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k \subseteq \mathfrak{a}.$$

Proof. Let $\mathcal{S}$ be the set of integral ideals which do not contain a product of primes. It suffices to show $\mathcal{S}$ is empty. Assuming otherwise and ordering $\mathcal{S}$ by inclusion, there exists a maximal element $\mathfrak{a} \in \mathcal{S}$ as $\mathcal{O}$ is noetherian. Moreover $\mathfrak{a}$ cannot be prime. Then there exist $\alpha_1, \alpha_2 \in \mathcal{O}$ with $\alpha_1\alpha_2 \in \mathfrak{a}$ and such that $\alpha_1, \alpha_2 \notin \mathfrak{a}$. Define integral ideals

$$\mathfrak{b}_1 = \mathfrak{a} + \alpha_1\mathcal{O} \quad \text{and} \quad \mathfrak{b}_2 = \mathfrak{a} + \alpha_2\mathcal{O}.$$

Note that $\mathfrak{b}_1$ and $\mathfrak{b}_2$ strictly contain $\mathfrak{a}$. Moreover, $\mathfrak{b}_1\mathfrak{b}_2 \subseteq \mathfrak{a}$ because

$$\mathfrak{b}_1\mathfrak{b}_2 = \mathfrak{a}^2 + \alpha_1\mathfrak{a} + \alpha_2\mathfrak{a} + \alpha_1\alpha_2\mathcal{O}.$$

Maximality of $\mathfrak{a}$ implies that there exist primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$ and $\mathfrak{q}_1, \ldots, \mathfrak{q}_\ell$ such that

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k \subseteq \mathfrak{b}_1 \quad \text{and} \quad \mathfrak{q}_1 \cdots \mathfrak{q}_\ell \subseteq \mathfrak{b}_2.$$

But then

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k\mathfrak{q}_1 \cdots \mathfrak{q}_\ell \subseteq \mathfrak{a},$$

which contradicts the fact that $\mathfrak{a} \in \mathcal{S}$. Hence $\mathcal{S}$ is empty. □

More work will be necessary obtain reverse inclusion and this requires the full strength of Dedekind domains. We being by showing that every prime $\mathfrak{p}$ is invertible. Note that unlike integral ideals, $1 \in \mathfrak{p}^{-1}$ so that $\mathfrak{p}^{-1}$ contains units.

Lemma 3.3.3. *Let $\mathcal{O}$ be a Dedekind domain and $\mathfrak{p}$ be a prime of $\mathcal{O}$. Then*

$$\mathcal{O} \subset \mathfrak{p}^{-1} \quad \textit{and} \quad \mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}.$$

Proof. We will first prove the inclusion. As $\mathcal{O} \subseteq \mathfrak{p}^{-1}$, it suffices to show $\mathfrak{p}^{-1} - \mathcal{O}$ is nonempty. To this end, let $\alpha \in \mathfrak{p}$ be nonzero. By Lemma 3.3.2 let k be the smallest positive integer such that there exist primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$ with

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k \subseteq \alpha\mathcal{O}.$$

As $\alpha\mathcal{O} \subseteq \mathfrak{p}$ and $\mathfrak{p}$ is prime, there must be some $\mathfrak{p}_i$ such that $\mathfrak{p}_i \subseteq \mathfrak{p}$. Without loss of generality, we may assume $\mathfrak{p}_1 \subseteq \mathfrak{p}$. As primes are maximal in $\mathcal{O}$, we conclude $\mathfrak{p}_1 = \mathfrak{p}$. Minimality of k forces

$$\mathfrak{p}_2 \cdots \mathfrak{p}_k \not\subseteq \alpha\mathcal{O}.$$

Hence there exists $\beta \in \mathfrak{p}_2 \cdots \mathfrak{p}_k$ with $\beta \notin \alpha\mathcal{O}$. We claim $\beta\alpha^{-1} \in \mathfrak{p}^{-1} - \mathcal{O}$. Since $\mathfrak{p}_1 = \mathfrak{p}$, what we have previously shown implies $\beta\mathfrak{p} \subseteq \alpha\mathcal{O}$ whence $\beta\alpha^{-1}\mathfrak{p} \in \mathcal{O}$ which means $\beta\alpha^{-1} \in \mathfrak{p}^{-1}$. But as $\beta \notin \alpha\mathcal{O}$, we also have $\beta\alpha^{-1} \notin \mathcal{O}$. Hence $\beta\alpha^{-1} \in \mathfrak{p}^{-1} - \mathcal{O}$ as desired.

Now let us prove the identity. Observe that

$$\mathfrak{p} \subseteq \mathfrak{p}\mathfrak{p}^{-1} \subseteq \mathcal{O}.$$

Since $\mathfrak{p}$ is maximal, it follows that $\mathfrak{p}\mathfrak{p}^{-1}$ is either $\mathfrak{p}$ or $\mathcal{O}$. So it suffices to show that the first case cannot hold. Assume by contradiction that $\mathfrak{p}\mathfrak{p}^{-1} = \mathfrak{p}$. Let $\omega_1, \dots, \omega_r$ be generators of $\mathfrak{p}$ as an $\mathcal{O}$-module and let $\alpha \in \mathfrak{p}^{-1} - \mathcal{O}$. Then $\alpha\omega_i \in \mathfrak{p}\mathfrak{p}^{-1}$ whence $\alpha\mathfrak{p} \subseteq \mathfrak{p}\mathfrak{p}^{-1}$ and $\alpha\mathfrak{p} \subseteq \mathfrak{p}$. But then

$$\alpha\omega_i = \sum_j \alpha_{i,j}\omega_j,$$

with $\alpha_{i,j} \in \mathcal{O}$. These r equations are equivalent to the identity

$$\begin{pmatrix} \alpha - \alpha_{1,1} & \alpha_{1,2} & \cdots & -\alpha_{1,r} \\ -\alpha_{2,1} & \alpha - \alpha_{2,2} & & \\ \vdots & & \ddots & \\ -\alpha_{r,1} & & & \alpha - \alpha_{r,r} \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \vdots \\ \omega_r \end{pmatrix} = \mathbf{0}.$$

Thus the determinant of the matrix on the left-hand side must be zero. But this means α is a root of the characteristic polynomial $\det(xI - (\alpha_{i,j}))$ which is a monic polynomial with coefficients $\mathcal{O}$. As $\mathcal{O}$ is an integrally closed domain, $\alpha \in \mathcal{O}$ which is a contradiction. Thus $\mathfrak{p}\mathfrak{p}^{-1} = \mathcal{O}$. $\square$

We can now show that every integral ideal factors uniquely into a product of primes.

Theorem 3.3.4. *Let $\mathcal{O}$ be a Dedekind domain. Then for every integral ideal $\mathfrak{a}$ of $\mathcal{O}$ there exist primes $\mathfrak{p}_1, \dots, \mathfrak{p}_k$ of $\mathcal{O}$ such that $\mathfrak{a}$ factors as*

$$\mathfrak{a} = \mathfrak{p}_1 \cdots \mathfrak{p}_k.$$

Moreover, this factorization is unique up to reordering of the factors.

Proof. We first prove existence and then uniqueness. For existence, let $\mathcal{S}$ be the set of integral ideals that are not a product of primes. We will show that $\mathcal{S}$ is empty. Assuming otherwise and ordering $\mathcal{S}$ by inclusion, there exists a maximal element $\mathfrak{a} \in \mathcal{S}$ as $\mathcal{O}$ is noetherian. Necessarily $\mathfrak{a}$ is not prime. Since primes are maximal in $\mathcal{O}$, there is some prime $\mathfrak{p}_1$ for which $\mathfrak{a} \subset \mathfrak{p}_1$. Then Lemma 3.3.3 implies

$$\mathfrak{a} \subset \mathfrak{a}\mathfrak{p}_1^{-1} \subset \mathcal{O}.$$

In particular, $\mathfrak{a}\mathfrak{p}_1^{-1}$ is an integral ideal. By maximality of $\mathfrak{a}$, $\mathfrak{a}\mathfrak{p}_1^{-1}$ factors into a product of primes. That is, there exist primes $\mathfrak{p}_2, \dots, \mathfrak{p}_k$ such that

$$\mathfrak{a}\mathfrak{p}_1^{-1} = \mathfrak{p}_2, \dots, \mathfrak{p}_k.$$

Hence

$$\mathfrak{a} = \mathfrak{p}_1, \dots, \mathfrak{p}_k,$$

which is a contradiction. Therefore $\mathcal{S}$ is empty thus proving the existence of such a factorization.

Now we prove uniqueness. Suppose $\mathfrak{a}$ admits factorizations

$$\mathfrak{a} = \mathfrak{p}_1, \ldots, \mathfrak{p}_k \quad \text{and} \quad \mathfrak{a} = \mathfrak{q}_1, \ldots, \mathfrak{q}_\ell,$$

for primes $\mathfrak{p}_i$ and $\mathfrak{q}_j$. Since $\mathfrak{p}_1$ is prime, there is some j for which $\mathfrak{q}_j \subseteq \mathfrak{p}_1$. Without loss of generality, we may assume $\mathfrak{q}_1 \subseteq \mathfrak{p}_1$ and since primes are maximal in $\mathcal{O}$ we have $\mathfrak{q}_1 = \mathfrak{p}_1$. Then

$$\mathfrak{p}_2, \ldots, \mathfrak{p}_k = \mathfrak{q}_2, \ldots, \mathfrak{q}_\ell.$$

Repeating this process, we see that the factorizations are the same. This proves uniqueness of the factorization. □

As a near immediate corollary, fractional ideal admits analogous factorizations which implies that they are invertible.

Corollary 3.3.5. *Let $\mathcal{O}$ be a Dedekind domain. Then for every fractional ideal $\mathfrak{f}$ of $\mathcal{O}$ there exist primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$ and $\mathfrak{q}_1, \ldots, \mathfrak{q}_\ell$ of $\mathcal{O}$ such that $\mathfrak{f}$ factors as*

$$\mathfrak{f} = \mathfrak{p}_1 \cdots \mathfrak{p}_k \mathfrak{q}_1^{-1}, \ldots, \mathfrak{q}_\ell^{-1}.$$

Moreover, this factorization is unique up to reordering of the factors.

Proof. Write $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{a}$. By Theorem 3.3.4, $\mathfrak{a}$ and $\delta\mathcal{O}$ admit unique factorizations

$$\mathfrak{a} = \mathfrak{p}_1 \cdots \mathfrak{p}_k \quad \text{and} \quad \delta\mathcal{O} = \mathfrak{q}_1, \ldots, \mathfrak{q}_\ell,$$

for some primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$ and $\mathfrak{q}_1, \ldots, \mathfrak{q}_\ell$ up to reordering of the factors. Hence

$$\mathfrak{f} = \mathfrak{p}_1 \cdots \mathfrak{p}_k \mathfrak{q}_1^{-1}, \ldots, \mathfrak{q}_\ell^{-1}.$$ □

So for any fractional ideal $\mathfrak{f}$ there exist distinct prime $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ such that $\mathfrak{f}$ admits a factorization

$$\mathfrak{f} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r},$$

for some nonzero integers e_i. This is called the **prime factorization** of $\mathfrak{f}$ with **prime factors** $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$. In particular, the prime factorization of an integral ideal $\mathfrak{a}$ is of the form

$$\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r},$$

for some positive integers e_i. Accordingly, for any two integral ideal $\mathfrak{a}$ and $\mathfrak{b}$ we say that $\mathfrak{b}$ **divides** $\mathfrak{a}$ and write $\mathfrak{b} \mid \mathfrak{a}$ if $\mathfrak{a} \subseteq \mathfrak{b}$. Sometimes this condition is expressed as *to divide is to contain*. By prime factorization, this is equivalent to the fact that every prime power factor of $\mathfrak{b}$ appears in the prime factorization of $\mathfrak{a}$. In the case of a prime $\mathfrak{p}$, we also say that $\mathfrak{p}^e$ **exactly divides** $\mathfrak{a}$ and write $\mathfrak{p}^e \mid\mid \mathfrak{a}$ if $\mathfrak{p}^e$ divides $\mathfrak{a}$ but no higher power of $\mathfrak{p}$ divides $\mathfrak{a}$. This is equivalent to the fact that $\mathfrak{p}^e$ is the power of $\mathfrak{p}$ appearing in the prime factorization of $\mathfrak{a}$. Moreover, if $\mathfrak{a} \mid \mathfrak{p}$ then $\mathfrak{a} = \mathfrak{p}$. The

greatest common divisor $(\mathfrak{a}, \mathfrak{b})$ of $\mathfrak{a}$ and $\mathfrak{b}$ is defined to be the integral ideal that all other common integral ideal divisors divide. Since to divide is to contain, $(\mathfrak{a}, \mathfrak{b})$ is the smallest ideal that contains both $\mathfrak{a}$ and $\mathfrak{b}$. This is $\mathfrak{a}+\mathfrak{b}$ and so $(\mathfrak{a}, \mathfrak{b}) = \mathfrak{a}+\mathfrak{b}$. The **least common multiple** $[\mathfrak{a}, \mathfrak{b}]$ of $\mathfrak{a}$ and $\mathfrak{b}$ is defined to be the integral ideal that divides all other common multiples. Since to divide is to contain, $[\mathfrak{a}, \mathfrak{b}]$ is the largest integral ideal that is contained in both $\mathfrak{a}$ and $\mathfrak{b}$. This is $\mathfrak{a} \cap \mathfrak{b}$ and so $[\mathfrak{a}, \mathfrak{b}] = \mathfrak{a} \cap \mathfrak{b}$. Lastly, we say that $\mathfrak{a}$ and $\mathfrak{b}$ are **relatively prime** if $(\mathfrak{a}, \mathfrak{b}) = \mathcal{O}$. In other words, $\mathfrak{a}$ and $\mathfrak{b}$ are comaximal which is to say

$$\mathfrak{a}+\mathfrak{b} = \mathcal{O}.$$

This is equivalent to the prime factorizations of $\mathfrak{a}$ and $\mathfrak{b}$ containing distinct primes. In particular, distinct primes and their powers are relatively prime. As these integral ideals are comaximal, we also have $\mathfrak{a} \cap \mathfrak{b} = \mathfrak{a}\mathfrak{b}$. In terms of the least common multiple, this means $[\mathfrak{a}, \mathfrak{b}] = \mathfrak{a}\mathfrak{b}$.

Just as it is common to suppress the fundamental theorem of arithmetic and just state the prime factorization of an integer, we suppress referencing Theorem 3.3.4 and simply state the prime factorization of a fractional ideal.

With the prime factorization in hand, we will discuss the group structure of the fractional ideals of $\mathcal{O}$. Let $I(\mathcal{O})$ denote the set of fractional ideals of $\mathcal{O}$. This is indeed an abelian group.

Theorem 3.3.6. *Let $\mathcal{O}$ be a Dedekind domain with field of fractions K. Then $I(\mathcal{O})$ is an abelian group with identity $\mathcal{O}$.*

Proof. Associativity and commutativity of $I(\mathcal{O})$ are obvious. The identity is $\mathcal{O}$ because every fractional ideal is a finitely generated $\mathcal{O}$-submodule of K. It remains to show that every fractional ideal $\mathfrak{f}$ is invertible. Write $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{a}$. Then we must show $\mathfrak{a}$ is invertible. In view of Lemma 3.3.3, if

$$\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r},$$

is the prime factorization of $\mathfrak{a}$ then let

$$\mathfrak{b} = \mathfrak{p}_1^{-e_1} \cdots \mathfrak{p}_r^{-e_r}.$$

We must show $\mathfrak{b} = \mathfrak{a}^{-1}$. For the forward inclusion, $\mathfrak{a}\mathfrak{b} = \mathcal{O}$ implies $\mathfrak{b} \subseteq \mathfrak{a}^{-1}$ by the definition of $\mathfrak{a}^{-1}$. For the reverse inclusion, suppose $\kappa \in \mathfrak{a}^{-1}$. Then $\kappa\mathfrak{a} \subseteq \mathcal{O}$ and multiplying by $\mathfrak{b}$ shows $\kappa\mathcal{O} \subseteq \mathfrak{b}$ whence $\kappa \in \mathfrak{b}$. $\square$

It follows from this result that every fractional ideal is invertible. We will use this fact so often that we suppress it for legibility.

Let $P(\mathcal{O})$ denote the subgroup of principal fractional ideals of $I(\mathcal{O})$. Since $I(\mathcal{O})$ is abelian by the previous result, $P(\mathcal{O})$ is normal. The **ideal class group** $\mathrm{Cl}(\mathcal{O})$ of $\mathcal{O}$ is defined to be the quotient

$$\mathrm{Cl}(\mathcal{O}) = I(\mathcal{O})/P(\mathcal{O}).$$

A class of $\mathrm{Cl}(\mathcal{O})$ is called an **ideal class** of $\mathcal{O}$. As every fractional ideal $\mathfrak{f}$ can be expressed as $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero δino and integral ideal $\mathfrak{a}$, we have $\delta\mathfrak{f} = \mathfrak{a}$ and it follows that every ideal class can be represented by an integral ideal. The **class number** $h_{\mathcal{O}}$ of $\mathcal{O}$ is defined by

$$h_{\mathcal{O}} = |\,\mathrm{Cl}(\mathcal{O})|.$$

The ideal class group encodes how much $\mathcal{O}$ fails to be a principal ideal domain and the class number is a measure of the degree of failure. Indeed, $\mathcal{O}$ is a principal ideal domain if and only if $h_{\mathcal{O}} = 1$.

Remark 3.3.7. The class number $h_{\mathcal{O}}$ need not be finite for a general Dedekind domain $\mathcal{O}$.

The **unit group** of $\mathcal{O}$ is defined to be $\mathcal{O}^*$. That is, the unit group is the group of units in $\mathcal{O}$. The ideal class and unit groups of $\mathcal{O}$ fit into a short exact sequence.

Proposition 3.3.8. *Let $\mathcal{O}$ be a Dedekind domain with field of fractions K. Then there is a short exact sequence*

$$1 \longrightarrow K^*/\mathcal{O}^* \longrightarrow I(\mathcal{O}) \longrightarrow \mathrm{Cl}(\mathcal{O}) \longrightarrow 1,$$

induced by natural inclusions and projections and where the second map is induced by $\kappa \mapsto \kappa\mathcal{O}$.

Proof. As the second map is injective and the third map is surjective, the sequence is exact at $K^*/\mathcal{O}^*$ and $\mathrm{Cl}(\mathcal{O})$. So it remains to prove exactness at $I(\mathcal{O})$. This follows because the principal fractional ideals represent the identity class of $\mathrm{Cl}(\mathcal{O})$ and these are generated by a complete set of representatives for $K^*/\mathcal{O}^*$. $\square$

Thinking of the third map in this exact sequence as passing from numbers in K^* to fractional ideals in $I(\mathcal{O})$, exactness means that unit group is measuring the contraction (how many numbers are annihilated) taking place during this process while the ideal class group is measuring the expansion (how many fractional ideals are created).

Remark 3.3.9. The class number $h_{\mathcal{O}}$ and unit group $\mathcal{O}^*$ are two of the most difficult pieces of algebraic data of $\mathcal{O}$ to compute.

We will now turn to proving some interesting results about Dedekind domains. As a first result, being a principal ideal domain is equivalent to being a unique factorization domain for Dedekind domains.

Proposition 3.3.10. *Let $\mathcal{O}$ be a Dedekind domain. Then $\mathcal{O}$ is a principal ideal domain if and only if it is a unique factorization domain.*

Proof. The forward implication is immediate as every principal ideal domain is a unique factorization domain. For the reverse implication, suppose $\mathcal{O}$ is a unique factorization domain. As every integral ideal factors into a product of primes by Theorem 3.3.4 and the product of principal integral ideals is principal, it suffices to show that primes are principal. Let $\mathfrak{p}$ be a prime. Then there exists nonzero $\alpha \in \mathfrak{p}$ and α is not a unit since $\mathfrak{p}$ a proper ideal. Since $\mathcal{O}$ is a unique factorization domain, let $\alpha = \varepsilon\pi_1^{e_1}\cdots\pi_r^{e_r}$ be the prime factorization of α. Since $\mathfrak{p}$ is prime, it follows that there is some $\pi_i \in \mathfrak{p}$. Without loss of generality, we may assume $\pi_1 \in \mathfrak{p}$. Then the integral ideal $\pi_1\mathcal{O}$ satisfies $\pi_1\mathcal{O} \subseteq \mathfrak{p}$. As π_1 is prime, $\pi_1\mathcal{O}$ is a prime and hence maximal since prime ideals are maximal in $\mathcal{O}$. Whence $\pi_1\mathcal{O} = \mathfrak{p}$ proving $\mathfrak{p}$ is principal. $\square$

The Chinese remainder theorem also applies to Dedekind domains which will allow us to deduce some more results. Suppose $\mathfrak{a}$ is an integral ideal with prime factorization

$$\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r}.$$

As powers of distinct primes are relatively prime, the integral ideals $\mathfrak{p}_1^{e_1}, \ldots, \mathfrak{p}_r^{e_r}$ are pairwise relatively prime. Whence they are pairwise comaximal so that the Chinese remainder theorem gives an isomorphism

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_i \mathcal{O}/\mathfrak{p}_i^{e_i},$$

induced by natural projection. In particular, for any choice of $\alpha_i \in \mathcal{O}$ there exists a $\alpha \in \mathcal{O}$ such that

$$\alpha \equiv \alpha_i \pmod{\mathfrak{p}_i^{e_i}}.$$

We will use this fact to prove a few useful lemmas about Dedekind domains. Our first lemma shows that given two integral ideals, we can multiply by a relatively prime integral ideal to produce a principal integral ideal.

Lemma 3.3.11. *Let $\mathcal{O}$ be a Dedekind domain and $\mathfrak{a}$ and $\mathfrak{b}$ be integral ideals of $\mathcal{O}$. Then there exists an integral ideal $\mathfrak{c}$ of $\mathcal{O}$ relatively prime to $\mathfrak{b}$ such that $\mathfrak{ac}$ is principal.*

Proof. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ be the prime factors of both $\mathfrak{a}$ and $\mathfrak{b}$ so that

$$\mathfrak{a} = \mathfrak{p}_1^{e_1} \cdots \mathfrak{p}_r^{e_r} \quad \text{and} \quad \mathfrak{b} = \mathfrak{p}_1^{f_1} \cdots \mathfrak{p}_r^{f_r},$$

for some nonnegative integers e_i and f_i. By prime factorization, there exists $\alpha_i \in \mathfrak{p}_i^{e_i} - \mathfrak{p}_i^{e_i+1}$. As $\mathfrak{p}_1^{e_1+1}, \ldots, \mathfrak{p}_r^{e_r+1}$ are pairwise relatively prime, the Chinese remainder theorem implies the existence of an $\alpha \in \mathcal{O}$ such that $\alpha \equiv \alpha_i \pmod{\mathfrak{p}_i^{e_i+1}}$. But then

$$\alpha \equiv 0 \pmod{\mathfrak{p}_i^{e_i}} \quad \text{and} \quad \alpha \equiv \alpha_i \pmod{\mathfrak{p}_i^{e_i+1}}.$$

Whence $\mathfrak{p}_i^{e_i} \mid\mid \alpha\mathcal{O}$. It follows that $(\alpha\mathcal{O}, \mathfrak{ab}) = \mathfrak{a}$. Then there exists an integral ideal $\mathfrak{c}$ such that $\alpha\mathcal{O} = \mathfrak{ac}$ and necessarily $(\mathfrak{ac}, \mathfrak{ab}) = \mathfrak{a}$. Thus $(\mathfrak{c}, \mathfrak{b}) = \mathcal{O}$ which is to say that $\mathfrak{c}$ must be relatively prime to $\mathfrak{b}$. □

Our second lemma shows that multiplying by fractional ideals does not affect quotients.

Lemma 3.3.12. *Let $\mathcal{O}$ be a Dedekind domain where $\mathfrak{f}$, $\mathfrak{g}$, and $\mathfrak{h}$ be fractional ideals of $\mathcal{O}$ with $\mathfrak{g} \subseteq \mathfrak{f}$. Then we have an isomorphism*

$$\mathfrak{f}/\mathfrak{g} \cong \mathfrak{fh}/\mathfrak{gh},$$

induced by multiplication and natural projection. In particular, there is an isomorphism

$$\mathfrak{a}^{-1}/\mathcal{O} \cong \mathcal{O}/\mathfrak{a},$$

induced by multiplication and natural projection for any integral ideal $\mathfrak{a}$ of $\mathcal{O}$.

Proof. Write $\mathfrak{f} = \frac{1}{\alpha}\mathfrak{a}$, $\mathfrak{g} = \frac{1}{\beta}\mathfrak{b}$, and $\mathfrak{h} = \frac{1}{\gamma}\mathfrak{c}$ for some nonzero $\alpha, \beta, \gamma \in \mathcal{O}$ and integral ideals $\mathfrak{a}$, $\mathfrak{b}$, and $\mathfrak{c}$ with $\mathfrak{b} \subseteq \mathfrak{a}$. In view of the isomorphisms

$$\mathfrak{f}/\mathfrak{g} \cong \beta\mathfrak{a}/\alpha\mathfrak{b} \quad \text{and} \quad \mathfrak{f}\mathfrak{h}/\mathfrak{g}\mathfrak{h} \cong \beta\mathfrak{a}\mathfrak{c}/\alpha\mathfrak{b}\mathfrak{c},$$

it suffices to show

$$\mathfrak{a}/\mathfrak{b} \cong \mathfrak{a}\mathfrak{c}/\mathfrak{b}\mathfrak{c}.$$

To this end, as $\mathfrak{a} \mid \mathfrak{b}$ we conclude $\mathfrak{b}\mathfrak{a}^{-1}$ is an integral ideal. Then Lemma 3.3.11 implies the existence of an integral ideal $\mathfrak{d}$ relatively prime to $\mathfrak{b}\mathfrak{a}^{-1}$ such that $\mathfrak{c}\mathfrak{d} = \delta\mathcal{O}$ for some nonzero $\delta \in \mathcal{O}$. Whence

$$\mathfrak{d} + \mathfrak{b}\mathfrak{a}^{-1} = \mathcal{O} \quad \text{and} \quad \mathfrak{d} \cap \mathfrak{b}\mathfrak{a}^{-1} = \mathfrak{d}\mathfrak{b}\mathfrak{a}^{-1}.$$

Writing $\mathfrak{d} = \delta\mathfrak{c}^{-1}$, short computations show that these identities imply

$$\delta\mathfrak{a} + \mathfrak{b}\mathfrak{c} = \mathfrak{a}\mathfrak{c} \quad \text{and} \quad \delta^{-1}\mathfrak{b}\mathfrak{c} \cap \mathfrak{a} = \mathfrak{b}.$$

Consider the homomorphism

$$\phi : \mathfrak{a} \to \mathfrak{a}\mathfrak{c}/\mathfrak{b}\mathfrak{c} \qquad \alpha \mapsto \delta\alpha + \mathfrak{b}\mathfrak{c},$$

induced by multiplication and natural projection. By the first isomorphism theorem, it suffices to show ϕ is surjective and $\ker\phi = \mathfrak{b}$. Surjectivity follows from the first identity above. As $\ker\phi = \delta^{-1}\mathfrak{b}\mathfrak{c} \cap \mathfrak{a}$, the second identity above implies $\ker\phi = \mathfrak{b}$. This proves the first statement. For the second statement, take $\mathfrak{f} = \mathfrak{a}^{-1}$, $\mathfrak{g} = \mathcal{O}$, and $\mathfrak{h} = \mathfrak{a}$. □

If $\mathcal{O}$ is a Dedekind domain and $\mathfrak{p}$ is a prime of $\mathcal{O}$, we call $\mathbb{F}_{\mathfrak{p}} = \mathcal{O}/\mathfrak{p}$ the **residue class field** of $\mathcal{O}$ by $\mathfrak{p}$. This is field as mfp is maximal. Our next lemma gives an isomorphism between the residue class field and quotients by powers of a prime.

Lemma 3.3.13. *Let $\mathcal{O}$ be a Dedekind domain. Then for any prime $\mathfrak{p}$ of $\mathcal{O}$ and non-negative integer n, we have an isomorphism*

$$\mathbb{F}_{\mathfrak{p}} \cong \mathfrak{p}^n/\mathfrak{p}^{n+1},$$

induced by multiplication and natural projection.

Proof. By uniqueness of prime factorizations of fractional ideals, there exists $\beta \in \mathfrak{p}^n - \mathfrak{p}^{n+1}$. Consider the homomorphism

$$\phi : \mathcal{O} \to \mathfrak{p}^n/\mathfrak{p}^{n+1} \qquad \alpha \mapsto \beta\alpha + \mathfrak{p}^{n+1},$$

induced by multiplication and natural projection. By the first isomorphism theorem, it suffices to show ϕ is surjective and $\ker\phi = \mathfrak{p}$. By our choice of β, we find that $\beta\mathfrak{p}^{-n}$ is an integral ideal relatively prime to $\mathfrak{p}$. Whence

$$\beta\mathfrak{p}^{-n} + \mathfrak{p} = \mathcal{O} \quad \text{and} \quad \beta\mathfrak{p}^{-n} \cap \mathfrak{p} = \beta\mathfrak{p}^{1-n}.$$

Short computations show that these identities imply

$$\beta\mathcal{O} + \mathfrak{p}^{n+1} = \mathfrak{p}^n \quad \text{and} \quad \beta^{-1}\mathfrak{p}^{n+1} \cap \mathcal{O} = \mathfrak{p}.$$

Surjectivity follows from the first identity. As $\ker\phi = \beta^{-1}\mathfrak{p}^{n+1} \cap \mathcal{O}$, the second identity implies $\ker\phi = \mathfrak{p}$. □

We now state a few additional interesting propositions about Dedekind domains. The first is that any Dedekind domain with only finitely many primes is a principal ideal domain.

Proposition 3.3.14. *Let $\mathcal{o}$ be a Dedekind domain. If there are only finitely many primes of $\mathcal{o}$ then $\mathcal{o}$ is a principal ideal domain.*

Proof. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ be the primes of $\mathcal{o}$. Then for any integral ideal $\mathfrak{a}$, Lemma 3.3.11 implies the existence of an integral ideal $\mathfrak{b}$ relatively prime to $\mathfrak{p}_1 \cdots \mathfrak{p}_r$ such that $\mathfrak{a}\mathfrak{b} = \alpha\mathcal{o}$ for some nonzero $\alpha \in \mathcal{o}$. As $\mathfrak{b}$ is relatively prime to all of the primes of $\mathcal{o}$ we must have $\mathfrak{b} = \mathcal{o}$. Then $\mathfrak{a} = \alpha\mathcal{o}$. As $\mathfrak{a}$ was arbitrary, $\mathcal{o}$ is a principal ideal domain. □

The second is that any Dedekind domain is not far off from being a principal ideal domain in terms of the number of generators needed for an integral ideal. In fact, any fractional ideal is generated by at most two elements.

Proposition 3.3.15. *Let $\mathcal{o}$ be a Dedekind domain. Then every fractional ideal $\mathfrak{f}$ of $\mathcal{o}$ is generated by at most two elements.*

Proof. Writing $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{o}$ and integral ideal $\mathfrak{a}$, it suffices to prove the claim $\mathfrak{a}$. Let $\alpha \in \mathfrak{a}$ be nonzero. By Lemma 3.3.11, there exists an integral ideal $\mathfrak{b}$ relatively prime to $\alpha\mathcal{o}$ such that $\mathfrak{a}\mathfrak{b} = \beta\mathcal{o}$ for some nonzero $\beta \in \mathcal{o}$. Whence

$$\alpha\mathcal{o} + \mathfrak{b} = \mathcal{o}.$$

Upon multiplying by $\mathfrak{a}$, a short computation shows

$$\alpha\mathcal{o} + \beta\mathcal{o} = \mathfrak{a}.$$

Therefore $\mathfrak{a}$ is generated by at most two elements. □

For our last result, let $\mathcal{o}$ be a Dedekind domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then it turns out that $\mathcal{O}$ is also a Dedekind domain.

Proposition 3.3.16. *Let $\mathcal{o}$ be a Dedekind domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then $\mathcal{O}$ is a Dedekind domain with field of fractions L.*

Proof. By Proposition 3.1.4, $\mathcal{O}$ is an integrally closed domain with field of fractions L. To show $\mathcal{O}$ is noetherian, let the degree of L/K be n and $\lambda_1, \ldots, \lambda_n$ be a basis for L/K contained in $\mathcal{O}$. Then Lemma 3.2.7 implies $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module. As $\mathcal{o}$ is noetherian, every integral ideal of $\mathcal{O}$ is also a finitely generated $\mathcal{o}$-module and therefore also a finitely generated $\mathcal{O}$-module. Whence $\mathcal{O}$ is noetherian. It remains to show that every prime $\mathfrak{P}$ of $\mathcal{O}$ is maximal. This is equivalent to showing $\mathcal{O}/\mathfrak{P}$ is a field. To this end, consider the homomorphism

$$\phi : \mathcal{o} \to \mathcal{O}/\mathfrak{P} \qquad \alpha \mapsto \alpha + \mathfrak{P},$$

induced by natural projection. Then $\ker \phi = \mathfrak{P} \cap \mathcal{o}$ and we claim $\mathfrak{P} \cap \mathcal{o}$ is a prime of $\mathcal{o}$. It is clearly an ideal of $\mathcal{o}$ and is prime because $\mathfrak{P}$ is. To see that it is not the zero ideal, let $\lambda \in \mathfrak{P}$ be nonzero. As λ is integral over $\mathcal{o}$, we have

$$\lambda^n + \alpha_{n-1}\lambda^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha_i \in \mathcal{o}$. Taking n minimal ensures $\alpha_0 \neq 0$ and isolating α_0 shows $\alpha_0 \in \mathfrak{P}$ whence $\alpha_0 \in \mathfrak{P} \cap \mathcal{o}$. Therefore $\mathfrak{P} \cap \mathcal{o} = \mathfrak{p}$ for some prime $\mathfrak{p}$. This means $\ker \phi = \mathfrak{p}$. By the first isomorphism theorem, ϕ induces an embedding $\mathbb{F}_{\mathfrak{p}} \to \mathcal{O}/\mathfrak{P}$. Therefore $\mathcal{O}/\mathfrak{P}$ is a finite dimensional $\mathbb{F}_{\mathfrak{p}}$-vector space as $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module. By primality of $\mathfrak{P}$, $\mathcal{O}/\mathfrak{P}$ is also an integral domain and thus must be a field. $\square$

Accordingly, we say $\mathcal{O}/\mathcal{o}$ is a **Dedekind extension** of a finite separable extension L/K if $\mathcal{O}$ and $\mathcal{o}$ are Dedekind domains whose field of fractions are L and K respectively and $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L. By the preceding result, it is enough to show that $\mathcal{o}$ is a Dedekind domain and $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L. These Dedekind domains are related via the identity

$$\mathcal{O} \cap K = \mathcal{o}. \tag{3.4}$$

From Lemma 3.2.7 we see that $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module of rank at most n. In view of this fact, if $\mathfrak{f}$ is a fractional ideal of $\mathcal{o}$ then $\mathfrak{f}\mathcal{O}$ is a fractional ideal of $\mathcal{O}$. In particular, $\mathfrak{a}\mathcal{O}$ is an integral ideal of $\mathcal{O}$ for any integral ideal $\mathfrak{a}$ of $\mathcal{o}$. Recall that if $\mathfrak{F}$ is a fraction ideal of $\mathcal{O}$ then we may write $\mathfrak{F} = \frac{1}{\delta}\mathfrak{A}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{A}$. Then Proposition 3.1.4 permits us take $\delta \in \mathcal{o}$. Also, it need not be true that $\mathcal{O}/\mathcal{o}$ admits an integral basis as $\mathcal{O}$ is not guaranteed to be a free $\mathcal{o}$-module of rank n. However, an integral basis will exist if $\mathcal{o}$ has finitely many primes for then it is a principal ideal domain by Proposition 3.3.14 and we may appeal to Theorem 3.2.8.

In fact, the previous result motivates the more general discussion of primes in an extension of integral domains $\mathcal{O}/\mathcal{o}$ where $\mathcal{O}$ is integral over $\mathcal{o}$. We start by introducing the notion of primes above other primes. If $\mathfrak{P}$ and $\mathfrak{p}$ are primes of $\mathcal{O}$ and $\mathcal{o}$ respectively, we say that $\mathfrak{P}$ is **above** $\mathfrak{p}$, or equivalently, $\mathfrak{p}$ is **below** $\mathfrak{P}$ if

$$\mathfrak{P} \cap \mathcal{o} = \mathfrak{p}.$$

In fact, every prime of $\mathcal{O}$ is above exactly one prime of $\mathcal{o}$. To see this, it suffices to show $\mathfrak{P} \cap \mathcal{o}$ is a prime of $\mathcal{o}$. Clearly $\mathfrak{P} \cap \mathcal{o}$ is an ideal of $\mathcal{o}$ and it is prime because $\mathfrak{P}$ is. To see that it is nonzero, let $\lambda \in \mathfrak{P}$ be nonzero. As λ is integral over $\mathcal{o}$, we have

$$\lambda^n + \alpha_{n-1}\lambda^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha_i \in \mathcal{o}$. Taking n minimal ensures $\alpha_0 \neq 0$. Isolating α_0 shows $\alpha_0 \in \mathfrak{P}$ whence $\alpha_0 \in \mathfrak{P} \cap \mathcal{o}$. Therefore $\mathfrak{P} \cap \mathcal{o}$ is a prime of $\mathcal{o}$.

We can say much more when $\mathcal{O}/\mathcal{o}$ is a Dedekind extension. For if $\mathfrak{P}$ is above $\mathfrak{p}$ then $\mathfrak{P} \mid \mathfrak{p}\mathcal{O}$ because to divide is to contain. This implies that only finitely many primes $\mathfrak{P}$ can lie above a prime $\mathfrak{p}$ and they are exactly the prime factors of $\mathfrak{p}\mathcal{O}$. Moreover, every prime $\mathfrak{p}$ satisfies

$$\mathfrak{p}\mathcal{O} \neq \mathcal{O}.$$

Assume by contradiction that $\mathfrak{p}\mathcal{O} = \mathcal{O}$. By prime factorization, let $\alpha \in \mathfrak{p} - \mathfrak{p}^2$ so that $\alpha\mathcal{o} = \mathfrak{a}\mathfrak{p}$ for some integral ideal $\mathfrak{a}$ relatively prime to $\mathfrak{p}$. Then $\mathfrak{a} + \mathfrak{p} = \mathcal{o}$ and so there exist $\beta \in \mathfrak{a}$ and $\gamma \in \mathfrak{p}$ both nonzero and satisfying $\beta + \gamma = 1$. As $\beta\mathfrak{p} \subseteq \alpha\mathcal{o}$, it follows that $\beta\mathcal{O} \subseteq \alpha\mathcal{O}$ which implies $\beta = \alpha\delta$ for some nonzero $\delta \in \mathcal{O}$. In view of Equation (3.4), $\delta \in \mathcal{o}$. Whence $\alpha\delta + \gamma = 1$ and thus $1 \in \mathfrak{p}$ contradicting that $\mathfrak{p}$ is proper. Therefore $\mathfrak{p}\mathcal{O}$ is always a integral ideal of $\mathcal{O}$. In fact, we have

$$\mathfrak{p}\mathcal{O} \cap \mathcal{o} = \mathfrak{p}.$$

The reverse inclusion is obvious. For the forward inclusion, as $\mathfrak{p}\mathcal{O} \neq \mathcal{O}$ there exists a prime factor $\mathfrak{P}$ of $\mathfrak{p}\mathcal{O}$ so that $\mathfrak{p}\mathcal{O} \subseteq \mathfrak{P}$. But then $\mathfrak{P}$ is above $\mathfrak{p}$ so that $\mathfrak{P} \cap \mathcal{o} = \mathfrak{p}$ and the forward inclusion follows. Whence the primes above $\mathfrak{p}$ are exactly the prime factors $\mathfrak{P}$ of $\mathfrak{p}\mathcal{O}$. In fact, a more general identity holds. Let $\mathfrak{f}$ be a fractional ideal of $\mathcal{o}$. Write $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{o}$ and integral ideal $\mathfrak{a}$. Expressing $\mathfrak{a}$ as its prime factorization and repeatedly applying the previous identity shows that

$$\mathfrak{f}\mathcal{O} \cap \mathcal{o} = \mathfrak{f}.$$

This is called **contraction** of the fractional ideal $\mathfrak{f}\mathcal{O}$ to $\mathcal{o}$. In practice, it is useful for relating information about the fractional ideal $\mathfrak{f}\mathcal{O}$ of $\mathcal{O}$ to the fraction ideal $\mathfrak{f}$ of $\mathcal{o}$.

Number Fields

We now turn to the case of a number field K for which our developments so far can be refined. By Proposition 3.3.16, the ring of integers $\mathcal{O}_K$ is a Dedekind domain as $\mathbb{Z}$ is a principal ideal domain. Whence $\mathcal{O}_K/\mathbb{Z}$ is a Dedekind extension. In view of this fact, we simplify some terminology. An **integral ideal** of K is simply an integral ideal of $\mathcal{O}_K$, a **prime** of K is a prime of $\mathcal{O}_K$, and a **fractional ideal** of K is a fractional ideal of $\mathcal{O}_K$. In particular, every fractional ideal $\mathfrak{f}$ is of the form

$$\mathfrak{f} = \frac{1}{d}\mathfrak{a},$$

for some nonzero $d \in \mathbb{Z}$ and integral ideal $\mathfrak{a}$. We will write $I(K)$ and $P(K)$ for the group of fractional ideals of $\mathcal{O}_K$ and the subgroup of principal fractional ideals respectively. The **ideal class group** $\mathrm{Cl}(K)$ of K is the ideal class group of $\mathcal{O}_K$. In particular,

$$\mathrm{Cl}(K) = I(K)/P(K).$$

The **class number** h_K of K is the class number of $\mathcal{O}_K$. The **unit group** of K is the unit group of $\mathcal{O}_K$ and we call any element of $\mathcal{O}_K^*$ a **unit** of K (this is slightly abusive as every element of K^* is invertible). Note that the integral ideals of K admit prime factorizations. One of our core investigations will be to understand how the principal integral ideal $p\mathcal{O}_K$ factors into a product of primes of K for any prime p. We will also leverage geometric ideas to show that the class number is finite and completely describe the unit group.

Now suppose $L/K/\mathbb{Q}$ is an extension of number fields. Then $L/K/\mathbb{Q}$ is a finite separable tower. Moreover, $\mathcal{O}_L/\mathcal{O}_K/\mathbb{Z}$ is a tower of Dedekind extensions of $L/K/\mathbb{Q}$ by Propositions 3.1.5 and 3.3.16. In particular, $\mathcal{O}_L/\mathcal{O}_K$ is a Dedekind extension of L/K.

3.4 Localization

Let $\mathcal{o}$ be an integral domain with field of fractions K. A subset D of $\mathcal{o}$ is said to be **multiplicative** if it is closed under multiplication, $1 \in D$, and $0 \notin D$. The set $\mathcal{o} - \{0\}$ is always multiplicative but more interesting examples are when we consider strict subsets. In any case, we define the **localization** $\mathcal{o}D^{-1}$ of $\mathcal{o}$ at D by

$$\mathcal{o}D^{-1} = \left\{ \frac{\eta}{\delta} \in K : \eta \in \mathcal{o} \text{ and } \delta \in D \right\}.$$

Clearly $\mathcal{o}D^{-1}$ is a subring of K which is an integral domain and is obtained from $\mathcal{o}$ by making the elements of D invertible. More generally, for any fractional ideal $\mathfrak{f}$ of $\mathcal{o}$, the **localization** $\mathfrak{f}D^{-1}$ of $\mathfrak{f}$ at D is defined by

$$\mathfrak{f}D^{-1} = \left\{ \frac{\eta}{\delta} \in K : \eta \in \mathfrak{f} \text{ and } \delta \in D \right\}.$$

In particular, $\mathfrak{a}D^{-1}$ is an integral ideal of $\mathcal{o}D^{-1}$ if $\mathfrak{a}$ is an integral ideal of $\mathcal{o}$. Writing $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{o}$ and integral ideal $\mathfrak{a}$, it follows that $\mathfrak{f}D^{-1}$ is a fractional ideal of $\mathcal{o}D^{-1}$. Note that in the case of an integral ideal $\mathfrak{a}$, we will have $\mathfrak{a}D^{-1} = \mathcal{o}D^{-1}$ if $\mathfrak{a}$ contains an element of D.

In the case of primes, we have a bijective correspondence between those of $\mathcal{o}$ disjoint from D and those of $\mathcal{o}D^{-1}$.

Proposition 3.4.1. *Let $\mathcal{o}$ be an integral domain and D be a multiplicative subset of $\mathcal{o}$. Then the maps*

$$\mathfrak{q} \mapsto \mathfrak{q}D^{-1} \quad \text{and} \quad \mathfrak{Q} \mapsto \mathfrak{Q} \cap \mathcal{o}.$$

are inverse inclusion-preserving bijections between the primes $\mathfrak{q}$ of $\mathcal{o}$ disjoint from D and the primes $\mathfrak{Q}$ of $\mathcal{o}D^{-1}$.

Proof. First suppose $\mathfrak{q}$ is a prime of $\mathcal{o}$ that is disjoint from D. Then the integral ideal $\mathfrak{q}D^{-1}$ of $\mathcal{o}D^{-1}$ is prime because $\mathfrak{q}$ is. Also, the prime $\mathfrak{q}D^{-1}$ satisfies

$$\mathfrak{q} = \mathfrak{q}D^{-1} \cap \mathcal{o},$$

because $\mathfrak{q}$ is disjoint from D. Now suppose $\mathfrak{Q}$ is a prime of $\mathcal{o}D^{-1}$. Then the integral ideal $\mathfrak{Q} \cap \mathcal{o}$ of $\mathcal{o}$ is prime because $\mathfrak{Q}$ is. Also, $\mathfrak{Q} \cap \mathcal{o}$ is disjoint from D for otherwise it contains a unit which is a contradiction as prime ideals are proper. Moreover, the prime $\mathfrak{Q} \cap \mathcal{o}$ satisfies

$$\mathfrak{Q} = (\mathfrak{Q} \cap \mathcal{o})D^{-1}.$$

All of this together shows that the mappings

$$\mathfrak{q} \mapsto \mathfrak{q}D^{-1} \quad \text{and} \quad \mathfrak{Q} \mapsto \mathfrak{Q} \cap \mathcal{O}.$$

are inverse bijections between the primes $\mathfrak{q}$ of $\mathcal{O}$ disjoint from D and the primes $\mathfrak{Q}$ of $\mathcal{O}D^{-1}$. They are clearly inclusion-preserving. $\square$

It follows from this result that the primes of $\mathcal{O}D^{-1}$ are of the form $\mathfrak{p}D^{-1}$ for primes $\mathfrak{p}$ of $\mathcal{O}$ disjoint from D. Often, D is chosen so that it is the compliment of a prime $\mathfrak{p}$. Indeed, $\mathcal{O} - \mathfrak{p}$ is a multiplicative subset precisely because $\mathfrak{p}$ is prime. In fact, let X be a set of primes of $\mathcal{O}$ and consider

$$\mathcal{O} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p}.$$

Then $\mathcal{O} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p}$ is a multiplicative subset because of the identity

$$\mathcal{O} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p} = \bigcap_{\mathfrak{p} \in X} (\mathcal{O} - \mathfrak{p}).$$

In any case, we define the **localization** $\mathcal{O}_{\mathfrak{p}}$ of $\mathcal{O}$ at $\mathfrak{p}$ by

$$\mathcal{O}_{\mathfrak{p}} = \mathcal{O}(\mathcal{O} - \mathfrak{p})^{-1}.$$

Localizing at a prime $\mathfrak{p}$ should be thought of as removing all of the algebraic information about $\mathcal{O}$ that has nothing to do with $\mathfrak{p}$. More generally, if $\mathfrak{f}$ is a fractional ideal of $\mathcal{O}$ then the **localization** $\mathfrak{f}_{\mathfrak{p}}$ of $\mathfrak{f}$ at $\mathfrak{p}$ is defined to be

$$\mathfrak{f}_{\mathfrak{p}} = \mathfrak{f}(\mathcal{O} - \mathfrak{p})^{-1}.$$

In both of these constructions we have localized at a single prime. It is possible to localize at more than one prime. Let X be a set of primes in $\mathcal{O}$. We define the **localization** $\mathcal{O}(X)$ of $\mathcal{O}$ at X by

$$\mathcal{O}(X) = \mathcal{O}\left(\mathcal{O} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p}\right)^{-1}.$$

If $\mathfrak{f}$ is a fractional ideal of $\mathcal{O}$ then the **localization** $\mathfrak{f}(X)$ of $\mathfrak{f}$ at X is defined to be

$$\mathfrak{f}(X) = \mathfrak{f}\left(\mathcal{O} - \bigcup_{\mathfrak{p} \in X} \mathfrak{p}\right)^{-1}.$$

If X consists of a single prime $\mathfrak{p}$ then $\mathcal{O}(X) = \mathcal{O}_{\mathfrak{p}}$ and $\mathfrak{f}(X) = \mathfrak{f}_{\mathfrak{p}}$.

The case of localizing at a prime is deserving of special interest. An integral domain $\mathcal{O}$ is said to be a **local** if it has a unique maximal ideal. As may be expected, the resulting integral domain after localizing at a prime will be local. In particular,

$\mathcal{O}_{\mathfrak{p}}$ is local where the maximal ideal is $\mathfrak{p}_{\mathfrak{p}}$. Indeed, recall from Proposition 3.4.1 that the map

$$\mathfrak{q} \to \mathfrak{q}_{\mathfrak{p}},$$

is an inclusion-preserving bijection between the primes of $\mathcal{O}$ contained in $\mathfrak{p}$ the primes of $\mathcal{O}_{\mathfrak{p}}$. As maximal ideals are prime, $\mathfrak{p}_{\mathfrak{p}}$ is the unique maximal ideal of $\mathcal{O}_{\mathfrak{p}}$. As for consequences, we have

$$\mathcal{O}_{\mathfrak{p}}^* = \mathcal{O}_{\mathfrak{p}} - \mathfrak{p}_{\mathfrak{p}} \quad \text{and} \quad \mathcal{O}_{\mathfrak{p}}^* + \mathfrak{p}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}^*.$$

In other words, the units of $\mathcal{O}_{\mathfrak{p}}$ are precisely the elements not in $\mathfrak{p}_{\mathfrak{p}}$ and the sum of a unit of $\mathcal{O}_{\mathfrak{p}}$ and an element of $\mathfrak{p}_{\mathfrak{p}}$ is again a unit of $\mathcal{O}_{\mathfrak{p}}$.

In order for localization to be a useful tool in algebraic investigations, it must behave well with respect to what we have already developed. To this end, we will collect some useful properties about localization. Most importantly, intersections of localizations behave well with respect to fractional ideals and units and recover the original structure before localization.

Proposition 3.4.2. *Let $\mathcal{O}$ be an integral domain with field of fractions K. Then*

$$\mathcal{O} = \bigcap_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}} \quad \textit{and} \quad \mathcal{O}^* = \bigcap_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}^*.$$

In particular, for every fractional ideal $\mathfrak{f}$ of $\mathcal{O}$, we have

$$\mathfrak{f} = \bigcap_{\mathfrak{p}} \mathfrak{f}_{\mathfrak{p}}.$$

Proof. For the first identity, the forward inclusion is obvious. For the reverse inclusion, suppose $\frac{\eta}{\delta} \in \bigcap_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}$ and set

$$\mathfrak{a} = \{\alpha \in \mathcal{O} : \alpha\eta \in \delta\mathcal{O}\}.$$

Then $\mathfrak{a}$ is an integral ideal of $\mathcal{O}$ containing δ. As δ is not contained in any prime of $\mathcal{O}$, it follows that $\mathfrak{a}$ cannot be contained in any prime of $\mathcal{O}$. But every proper ideal is contained in a maximal ideal which is necessarily prime. Whence $\mathfrak{a}$ is not proper and so $\mathfrak{a} = \mathcal{O}$. Therefore $\eta \in \delta\mathcal{O}$ which implies $\frac{\alpha}{\beta} \in \mathcal{O}$ proving the reverse inclusion. As for the second identity, the forward inclusion is clear. For the reverse inclusion, suppose $\frac{\eta}{\delta} \in \bigcap_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}^*$. We have already shown $\frac{\eta}{\delta} \in \mathcal{O}$ so it suffices to show $\frac{\eta}{\delta}$ is a unit. As δ is not contained in any prime of $\mathcal{O}$, it cannot be contained in any maximal ideal as maximal ideals are prime. This means δ is a unit. Interchanging the roles of η and δ shows that η is a unit as well. Hence $\frac{\eta}{\delta}$ is a unit and the reverse inclusion follows. The second statement is immediate from the first upon multiplying by $\mathfrak{f}$. □

It also turns out that the localization of the inverse is the inverse of the localization for any fractional ideal of $\mathcal{O}$.

Proposition 3.4.3. *Let $\mathcal{o}$ be an integral domain with field of fractions K and D be a multiplicative subset of $\mathcal{o}$. Then for any fractional ideal $\mathfrak{f}$ of $\mathcal{o}$, we have*

$$(\mathfrak{f}D^{-1})^{-1} = \mathfrak{f}^{-1}D^{-1}.$$

Proof. The definition of the inverse implies $\mathfrak{f}D^{-1}(\mathfrak{f}D^{-1})^{-1} \subseteq \mathcal{o}D^{-1}$. Multiplying by $\mathfrak{f}^{-1}$ shows $(\mathfrak{f}D^{-1})^{-1} \subseteq \mathfrak{f}^{-1}D^{-1}$. This proves the forward inclusion. Similarly, we have $\mathfrak{f}\mathfrak{f}^{-1} \subseteq \mathcal{o}$ whence $\mathfrak{f}D^{-1}\mathfrak{f}^{-1}D^{-1} \subseteq \mathcal{o}D^{-1}$ which proves the reverse inclusion. □

Equivalently, this result says that inversion and localization are commutative. Therefore when inverting and localizing we do not need to specify the order in which these operations are performed.

When $\mathcal{o}$ is an integrally closed domain, localization respects integral closure.

Proposition 3.4.4. *Let $\mathcal{o}$ be an integrally closed domain with field of fractions K, L/K be a finite separable extension, and $\mathcal{O}$ be the integral closure of $\mathcal{o}$ in L. Then for any multiplicative set D of $\mathcal{o}$, $\mathcal{O}D^{-1}$ is the integral closure of $\mathcal{o}D^{-1}$ in L. In particular, $\mathcal{O}D^{-1}$ is an integrally closed domain with field of fractions L.*

Proof. To prove the first statement, we must show $\mathcal{O}D^{-1} = \overline{\mathcal{o}D^{-1}}$. For the forward inclusion, let $\frac{\eta}{\delta} \in \mathcal{O}D^{-1}$. As $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L, η is integral over $\mathcal{o}$ so that

$$\eta^n + \alpha_{n-1}\eta^{n-1} + \cdots + \alpha_0 = 0,$$

for some positive integer n and $\alpha_i \in \mathcal{o}$. Diving by δ^n, we obtain

$$\left(\frac{\eta}{\delta}\right)^n + \frac{\alpha_{n-1}}{\delta}\left(\frac{\eta}{\delta}\right)^{n-1} + \cdots + \frac{\alpha_0}{\delta^n} = 0.$$

Thus $\frac{\eta}{\delta}$ is the root of a monic polynomial with coefficients in $\mathcal{o}D^{-1}$ whence $\frac{\eta}{\delta} \in \overline{\mathcal{o}D^{-1}}$. This proves the forward inclusion. For the reverse inclusion, suppose $\lambda \in \overline{\mathcal{o}D^{-1}}$. Then

$$\lambda^n + \frac{\alpha_{n-1}}{\delta_{n-1}}\lambda^{n-1} + \cdots + \frac{\alpha_0}{\delta_0} = 0,$$

for some positive integer n and $\frac{\alpha_i}{\delta_i} \in \mathcal{o}D^{-1}$. Then $\delta = \prod_i \delta_i$ is nonzero and upon multiplying by δ^n, we obtain

$$(\lambda\delta)^n + \frac{\alpha_{n-1}\delta}{\delta_{n-1}}(\lambda\delta)^{n-1} + \cdots + \frac{\alpha_0\delta^n}{\delta_0} = 0.$$

It follows that $\lambda\delta$ is the root of a monic polynomial with coefficients in $\mathcal{o}$. As $\mathcal{O}$ is the integral closure of $\mathcal{o}$ in L, we have $\lambda\delta \in \mathcal{O}$ and thus $\lambda \in \mathcal{O}D^{-1}$. This proves the reverse inclusion. For the second statement, observe that $\mathcal{O}D^{-1}$ is an integral domain as it is a subring of a field. Moreover, the field of fractions of $\mathcal{O}D^{-1}$ is L since the same is true for $\mathcal{O}$ by Proposition 3.1.4. The second statement follows. □

In view of the previous result, the localization of a Dedekind domain is again a Dedekind domain.

Proposition 3.4.5. *Let $\mathcal{o}$ be a Dedekind domain with field of fractions K and D be a multiplicative subset of $\mathcal{o}$. Then $\mathcal{o}D^{-1}$ is a Dedekind domain with field of fractions K.*

Proof. We see from Proposition 3.4.4 that $\mathcal{o}D^{-1}$ is an integrally closed domain with field of fractions K. To show it is noetherian, let $\mathfrak{A}$ be an integral ideal of $\mathcal{o}D^{-1}$ and set $\mathfrak{a} = \mathfrak{A} \cap \mathcal{o}$. Then

$$\mathfrak{A} = \mathfrak{a}D^{-1}.$$

Since $\mathcal{o}$ is noetherian, $\mathfrak{a}$ is a finitely generated $\mathcal{o}$-module and hence $\mathfrak{A}$ is a finitely generated $\mathcal{o}D^{-1}$-module. Therefore $\mathcal{o}D^{-1}$ is noetherian. It remains to show that every prime of $\mathcal{o}D^{-1}$ is maximal. By Proposition 3.4.1, every prime of $\mathcal{o}D^{-1}$ is of the form $\mathfrak{p}D^{-1}$ for some prime $\mathfrak{p}$ of $\mathcal{o}$. But then $\mathfrak{p}D^{-1}$ is maximal because $\mathfrak{p}$ is and the bijections in Proposition 3.4.1 are inclusion-preserving. □

An immediate consequence of this result is that the localizations $\mathcal{o}_\mathfrak{p}$ and $\mathcal{o}(X)$ are Dedekind domains if $\mathcal{o}$ is. In fact, from Propositions 3.3.16, 3.4.4 and 3.4.5 we see that $\mathcal{O}D^{-1}/\mathcal{o}D^{-1}$ is a Dedekind extension of a finite separable extension L/K if $\mathcal{O}/\mathcal{o}$ is. In this case we setup some additional notation. If $\mathfrak{p}$ is a prime of $\mathcal{o}$, we define the **localization** $\mathcal{O}_\mathfrak{p}$ of $\mathcal{O}$ at $\mathfrak{p}$ by

$$\mathcal{O}_\mathfrak{p} = \mathcal{O}(\mathcal{o} - \mathfrak{p})^{-1}.$$

More generally, if $\mathfrak{F}$ is a fractional ideal of $\mathcal{O}$ then the **localization** $\mathfrak{F}_\mathfrak{p}$ of $\mathfrak{F}$ at $\mathfrak{p}$ is defined by

$$\mathfrak{F}_\mathfrak{p} = \mathfrak{F}(\mathcal{o} - \mathfrak{p})^{-1}.$$

We can also consider the case of more than a single prime. Let X be a set of primes of $\mathcal{o}$. We define the **localization** $\mathcal{O}(X)$ of $\mathcal{O}$ at X by

$$\mathcal{O}(X) = \mathcal{O}\left(\mathcal{o} - \bigcup_{\mathfrak{p}\in X} \mathfrak{p}\right)^{-1}.$$

More generally, if $\mathfrak{F}$ is a fractional ideal of $\mathcal{O}$ then the **localization** $\mathfrak{F}(X)$ of $\mathfrak{F}$ at X is defined by

$$\mathfrak{F}(X) = \mathfrak{F}\left(\mathcal{o} - \bigcup_{\mathfrak{p}\in X} \mathfrak{p}\right)^{-1}.$$

If X consists of a single prime $\mathfrak{p}$ then $\mathcal{O}(X) = \mathcal{O}_\mathfrak{p}$ and $\mathfrak{F}(X) = \mathfrak{F}_\mathfrak{p}$. Moreover, $\mathcal{O}_\mathfrak{p}/\mathcal{o}_\mathfrak{p}$ and $\mathcal{O}(X)/\mathcal{o}(X)$ are Dedekind extensions.

We will now consider a more restrictive setting. An integral domain $\mathcal{o}$ is said to be a **discrete valuation ring** if it is a principal ideal domain with a unique maximal ideal. Equivalently, $\mathcal{o}$ is a local principal ideal domain or a Dedekind domain with exactly one prime. Let $\mathfrak{p}$ be the unique maximal ideal of $\mathcal{o}$. In particular, $\mathfrak{p}$ is prime and of the form $\mathfrak{p} = \pi\mathcal{o}$ for some prime $\pi \in \mathcal{o}$. We call π a **uniformizer** of $\mathcal{o}$ and it is uniquely defined up to multiplication by a unit. On the one hand, $\mathcal{o}$ is local so every element not in $\mathfrak{p}$ is a unit. On the other hand, $\mathcal{o}$ is a principal ideal domain so that π is the only prime of $\mathcal{o}$. Since $\mathcal{o}$ is necessarily a unique factorization domain, these facts together imply that every $\alpha \in \mathcal{o}$ is of the form

$$\alpha = \varepsilon\pi^n,$$

for some nonnegative integer n and unit ε. Moreover, α generates the integral ideal $\mathfrak{p}^n$ and every integral ideal is of this form. If K is the field of fractions of $\mathcal{o}$, it follows that every nonzero $\kappa \in K$ can be uniquely expressed as

$$\kappa = \varepsilon\pi^n,$$

for some integer n and unit ε. The most important data of a discrete valuation ring is its valuation. The **valuation** v of $\mathcal{o}$ is the function defined by

$$v : K \to \mathbb{Z} \cup \{\infty\} \qquad \kappa \mapsto v(\kappa) = \begin{cases} n & \text{if } \kappa = \varepsilon\pi^n, \\ \infty & \text{if } \kappa = 0. \end{cases}$$

We call $v(\kappa)$ the **valuation** of κ with respect to $\mathcal{o}$. Note that $v(\kappa) = 0$ if and only if κ is a unit in $\mathcal{o}$. If $\kappa \in \mathcal{o}$ then the $v(\kappa)$ is characterized by the equation

$$\kappa\mathcal{O} = \mathfrak{p}^{v(\kappa)},$$

since $\kappa = \varepsilon\pi^{v(\kappa)}$. Moreover, if $\kappa = \varepsilon\pi^n$ and $\eta = \delta\pi^m$, we have

$$\kappa\eta = \varepsilon\delta\pi^{n+m} \quad \text{and} \quad \kappa + \eta = (\varepsilon\pi^{n-\min(n,m)} + \delta\pi^{m-\min(n,m)})\pi^{\min(n,m)},$$

where $\varepsilon\pi^{n-\min(n,m)} + \delta\pi^{m-\min(n,m)}$ is a unit. These identities imply that v satisfies the properties

$$v(\kappa\eta) = v(\kappa) + v(\eta) \quad \text{and} \quad v(\kappa + \eta) \geq \min(v(\kappa), v(\eta)).$$

The first of which shows that

$$v : K^* \to \mathbb{Z} \qquad \kappa \mapsto v(\kappa),$$

is a surjective homomorphism.

A Dedekind extension $\mathcal{O}/\mathcal{o}$ is said to be **local** if $\mathcal{o}$ is a discrete valuation ring. As $\mathcal{o}$ is a principal ideal domain, $\mathcal{O}/\mathcal{o}$ admits an integral basis by Theorem 3.2.8 but we can say more. If $\mathfrak{p}$ is the unique prime of $\mathcal{o}$, then $\mathcal{O}$ has finitely many primes $\mathfrak{P}$ which are necessarily the prime factors of $\mathfrak{p}\mathcal{O}$ and are therefore all the primes above $\mathfrak{p}$. By Proposition 3.3.14, we find that $\mathcal{O}$ is also a principal ideal domain. Whence $\mathcal{O}/\mathcal{o}$ is an extension of principal ideal domains.

If $\mathcal{o}$ is a Dedekind domain then it is local if and only if it is a discrete valuation ring. Indeed, any Dedekind domain with finitely many primes is a principal ideal domain by Proposition 3.3.14. This will allow us to establish a connection between Dedekind domains and discrete valuation rings. In particular, a noetherian domain is a Dedekind domain if and only if its localization at any prime is a discrete valuation ring.

Theorem 3.4.6. *Let $\mathcal{o}$ be a noetherian domain. Then $\mathcal{o}$ is a Dedekind domain if and only if $\mathcal{o}_\mathfrak{p}$ is a discrete valuation ring for all primes $\mathfrak{p}$ of $\mathcal{o}$.*

Proof. For the forward implication, suppose $\mathcal{o}$ is a Dedekind domain and let $\mathfrak{p}$ be a prime. Then $\mathcal{o}_\mathfrak{p}$ is a local Dedekind domain and hence a discrete valuation ring as we have seen. This proves the forward implication. For the reverse implication, suppose $\mathcal{o}$ is a noetherian domain and $\mathcal{o}_\mathfrak{p}$ is a discrete valuation ring. As $\mathcal{o}_\mathfrak{p}$ is a principal ideal domain, $\mathcal{o}_\mathfrak{p}$ is an integrally closed domain by Lemma 3.1.3. Then Proposition 3.4.2 implies $\mathcal{o}$ is an integrally closed domain too. As $\mathcal{o}$ is noetherian by assumption, it remains to show that every prime $\mathfrak{q}$ of $\mathcal{o}$ is maximal. As $\mathfrak{q}$ is proper, it is contained in some maximal ideal $\mathfrak{p}$ which is necessarily prime. The inclusion-preserving bijections in Proposition 3.4.1 show that $\mathfrak{q}_\mathfrak{p}$ and $\mathfrak{p}_\mathfrak{p}$ are primes of $\mathcal{o}_\mathfrak{p}$. As $\mathcal{o}_\mathfrak{p}$ is a discrete valuation ring it is a Dedekind domain with exactly one prime. Whence $\mathfrak{q}_\mathfrak{p} = \mathfrak{p}_\mathfrak{p}$ and the aforementioned inclusion-preserving bijections imply that $\mathfrak{q} = \mathfrak{p}$. Hence $\mathfrak{q}$ is maximal. This proves the reverse implication. $\square$

In view of this result, $\mathcal{O}_\mathfrak{p}/\mathcal{o}_\mathfrak{p}$ is a local Dedekind extension. Observe that $\mathfrak{P}$ is not above $\mathfrak{p}$ if and only if $\mathfrak{P}$ contains an element of $\mathcal{o} - \mathfrak{p}$. In view of Proposition 3.4.1, it follows the finitely many primes of $\mathcal{O}_\mathfrak{p}$ are those above the unique prime of $\mathcal{o}_\mathfrak{p}$ and they correspond to the primes of $\mathcal{O}$ above $\mathfrak{p}$. Moreover, suppose $\mathfrak{A}$ is an integral ideal of $\mathcal{O}$. By prime factorization, we may write

$$\mathfrak{A} = \prod_{\mathfrak{P}} \mathfrak{P}^{e_\mathfrak{P}},$$

where $e_\mathfrak{P}$ are nonnegative integers all but finitely many of which are zero. Then we see that

$$\mathfrak{A} = \prod_{\mathfrak{P} | \mathfrak{p}\mathcal{O}} \mathfrak{P}^{e_\mathfrak{P}}.$$

In other words, localizing an integral ideal of $\mathcal{O}$ at $\mathfrak{p}$ removes all of the prime factors that are not above $\mathfrak{p}$ and localizes the primes above $\mathfrak{p}$.

Suppose $\mathcal{o}$ is a Dedekind domain with field of fractions K and let $v_\mathfrak{p}$ denote the valuation of $\mathcal{o}_\mathfrak{p}$ in view of Theorem 3.4.6. We call $v_\mathfrak{p}$ the **valuation** associated to the prime $\mathfrak{p}$. These valuations are intimately connected to prime factorizations. Indeed, prime factorization implies that for any $\kappa \in K^*$, we have

$$\kappa\mathcal{O} = \prod_{\mathfrak{q}} \mathfrak{q}^{e_\mathfrak{q}},$$

for some integers $e_\mathfrak{q}$ all but finitely many of which are zero. We claim that $v_\mathfrak{p}(\kappa) = e_\mathfrak{p}$. To see this, first observe that if $\mathfrak{p}$ and $\mathfrak{q}$ are distinct primes then $\mathfrak{q}_\mathfrak{p} = \mathcal{o}_\mathfrak{p}$. Indeed, we have

$$\kappa\mathcal{O}_\mathfrak{p} = \mathfrak{p}_\mathfrak{p}^{e_\mathfrak{p}},$$

from which $v_\mathfrak{p}(\kappa) = e_\mathfrak{p}$. In particular, $v_\mathfrak{p}(\kappa) = 0$ for all but finitely many primes $\mathfrak{p}$. In view of these properties, $v_\mathfrak{p}$ is also called an **exponential valuation**.

Continue to let $\mathcal{O}$ be a Dedekind domain with field of fractions K and let X be a set of all but finitely many primes of $\mathcal{O}$. Then $\mathcal{O}(X)$ is a Dedekind domain. By Proposition 3.4.1, the primes $\mathfrak{p}_X$ of $\mathcal{O}(X)$ are of the form $\mathfrak{p}_X = \mathfrak{p}(X)$ for $\mathfrak{p} \in X$. Moreover, $\mathcal{O}$ and $\mathcal{O}(X)$ have the same localizations at $\mathfrak{p}$ and $\mathfrak{p}_X$ respectively because the only elements which are not inverted are those of $\mathfrak{p}$. In particular, this means

$$\mathcal{O}_{\mathfrak{p}} = \mathcal{O}(X)_{\mathfrak{p}_X} \quad \text{and} \quad \mathfrak{f}_{\mathfrak{p}} = \mathfrak{f}(X)_{\mathfrak{p}_X}. \tag{3.5}$$

The relationship between $\mathcal{O}$ and $\mathcal{O}(X)$ can be expressed via a long exact sequence.

Proposition 3.4.7. *Let $\mathcal{O}$ be a Dedekind domain with field of fractions K and let X be a set of all but finitely many primes of $\mathcal{O}$. Then there is a long exact sequence*

$$1 \longrightarrow \mathcal{O}^* \longrightarrow \mathcal{O}(X)^* \longrightarrow \bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_{\mathfrak{p}}^* \longrightarrow \mathrm{Cl}(\mathcal{O}) \longrightarrow \mathrm{Cl}(\mathcal{O}(X)) \longrightarrow 1,$$

induced by natural inclusions and projections and where the fourth map and fifth maps are induced by $(\kappa_{\mathfrak{p}})_{\mathfrak{p}\notin X} \mapsto \prod_{\mathfrak{p}\notin X} \mathfrak{p}^{v_{\mathfrak{p}}(\kappa_{\mathfrak{p}})}$ and $\mathfrak{a} \mapsto \mathfrak{a}(X)$ respectively. Moreover, $K^*/\mathcal{O}_{\mathfrak{p}}^* \cong \mathbb{Z}$.

Proof. As the second map is injective and the fifth map is surjective, the sequence is exact at $\mathcal{O}^*$ and $\mathrm{Cl}(\mathcal{O}(X))$. So we must show exactness at $\mathcal{O}(X)^*$, $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_{\mathfrak{p}}^*$, and $\mathrm{Cl}(\mathcal{O})$. For exactness at $\mathcal{O}(X)^*$, $\alpha \in \mathcal{O}(X)^*$ represents the zero coset in under the third map $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_{\mathfrak{p}}^*$ if and only if $\alpha \in \mathcal{O}_{\mathfrak{p}}^*$ for $\mathfrak{p} \notin X$. But Equation (3.5) implies $\alpha \in \mathcal{O}_{\mathfrak{p}}^*$ for $\mathfrak{p} \in X$. Whence α represents the zero coset if and only if $\alpha \in \mathcal{O}^*$ by Proposition 3.4.2 proving exactness at $\mathcal{O}(X)^*$. For exactness at $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_{\mathfrak{p}}^*$, a representative $(\kappa_{\mathfrak{p}})_{\mathfrak{p}\notin X}$ of a coset in $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_{\mathfrak{p}}^*$ represents the principal class of $\mathrm{Cl}(\mathcal{O})$ under the fourth map if and only if there is a $\kappa \in K^*$ such that

$$\prod_{\mathfrak{p}\notin X} \mathfrak{p}^{v_{\mathfrak{p}}(\kappa_{\mathfrak{p}})} = \prod_{\mathfrak{p}} \mathfrak{p}^{v_{\mathfrak{p}}(\kappa)}.$$

By prime factorization, $v_{\mathfrak{p}}(\kappa) = v_{\mathfrak{p}}(\kappa_{\mathfrak{p}})$. In particular, $v_{\mathfrak{p}}(\kappa) = 0$ for all $\mathfrak{p} \in X$ and $v_{\mathfrak{p}}(\kappa) = v_{\mathfrak{p}}(\kappa_{\mathfrak{p}})$ for $\mathfrak{p} \notin X$. As $v_{\mathfrak{p}}(\kappa) = 0$ for $\mathfrak{p} \in X$, we have $\kappa \in \mathcal{O}_{\mathfrak{p}}^*$ for these primes as well. Because the primes of $\mathcal{O}(X)$ are $\mathfrak{p}_X$ for $\mathfrak{p} \in X$, Equation (3.5) and Proposition 3.4.2 together imply $\kappa \in \mathcal{O}(X)^*$. Exactness at $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_{\mathfrak{p}}^*$ follows. For exactness at $\mathrm{Cl}(\mathcal{O})$, an integral ideal $\mathfrak{a}$ representing a class of $\mathrm{Cl}(\mathcal{O})$ represents the principal class under the fifth map if and only if there is a $\kappa \in K^*$ such that

$$\mathfrak{a}(X) = \kappa\mathcal{O}(X).$$

In view of Equation (3.5) and Proposition 3.4.2 again, taking the intersection with the localizations at $\mathfrak{p}_X$ for $\mathfrak{p} \notin X$ shows

$$\mathfrak{a} = \kappa\mathcal{O}.$$

Therefore $(v_{\mathfrak{p}}(\kappa))_{\mathfrak{p}\notin X}$ is a representative of a coset in $\bigoplus_{\mathfrak{p}\notin X} K^*/\mathcal{O}_{\mathfrak{p}}^*$ whose image under the fourth map is $\mathfrak{a}$ proving exactness at $\mathrm{Cl}(\mathcal{O})$.

The last statement follows from the first isomorphism theorem since the valuation $v_{\mathfrak{p}}$ restricted to K^* is a surjective homomorphism and the kernel is exactly $\mathcal{O}_{\mathfrak{p}}^*$. This completes the proof. $\square$

Number Fields

We now turn to the setting of a number field K. Let S denote a finite set of primes of $\mathcal{O}_K$ and let X denote the set of all primes that do not belong to S. The **ring of S-integers** $\mathcal{O}_{K,S}$ of K is defined by

$$\mathcal{O}_{K,S} = \mathcal{O}_K(X).$$

We call any $\alpha \in \mathcal{O}_{K,S}$ an **algebraic S-integer**. We will write $I_S(K)$ and $P_S(K)$ for the group of fractional ideals of $\mathcal{O}_{K,S}$ and the subgroup of principal fractional ideals respectively. The **S-class group** $\mathrm{Cl}_S(K)$ of K is the ideal class group of $\mathcal{O}_{K,S}$. In particular,

$$\mathrm{Cl}_S(K) = I_S(K)/P_S(K).$$

The **S-class number** $h_{K,S}$ of K is the class number of $\mathcal{O}_{K,S}$. The **S-unit group** of K is the unit group $\mathcal{O}_{K,S}^*$ of $\mathcal{O}_{K,S}$ and we call any element of $\mathcal{O}_{K,S}^*$ an **S-unit** of K.

3.5 Orders

Recall that Dedekind domains are one-dimensional noetherian domains that are also integrally closed domains. An order will relax the condition of being integrally closed, and surprisingly, we will still be able to deduce similar information as we did for Dedekind domains. More precisely, for any integral domain $\mathcal{o}$ with field of fractions K, we define its **conductor** $\mathfrak{q}_{\mathcal{o}}$ to be

$$\mathfrak{q}_{\mathcal{o}} = \{\alpha \in \mathcal{o} : \alpha\overline{\mathcal{o}} \subseteq \mathcal{o}\}.$$

Then $\mathfrak{q}_{\mathcal{o}}$ is an ideal of $\mathcal{o}$ and it is the largest ideal which is also an ideal of $\overline{\mathcal{o}}$. Clearly $\mathcal{o}$ is an integrally closed domain if and only if the conductor is $\mathcal{o}$. In the case $\mathcal{o}$ is a noetherian domain, the conductor is an integral ideal of $\mathcal{o}$ and hence also $\overline{\mathcal{o}}$, which is to say it is nonzero, precisely when $\overline{\mathcal{o}}$ is a finitely generated $\mathcal{o}$-module.

Proposition 3.5.1. *Let $\mathcal{o}$ be a noetherian domain with field of fractions K. Then the conductor $\mathfrak{q}_{\mathcal{o}}$ is an integral ideal of $\mathcal{o}$ if and only if $\overline{\mathcal{o}}$ is a finitely generated $\mathcal{o}$-module.*

Proof. For the forward implication, suppose $\mathfrak{q}_{\mathcal{o}}$ is an integral ideal of $\mathcal{o}$. Then there is some nonzero $\alpha \in \mathfrak{q}_{\mathcal{o}}$ with $\alpha\overline{\mathcal{o}} \subseteq \mathcal{o}$. Whence $\alpha\overline{\mathcal{o}}$ is an ideal of $\mathcal{o}$ and hence a finitely generated $\mathcal{o}$-module as $\mathcal{o}$ is noetherian. But then $\overline{\mathcal{o}}$ is a finitely generated $\mathcal{o}$-module. For the reverse implication, let $\frac{\eta_1}{\delta_1}, \ldots, \frac{\eta_r}{\delta_r}$ be generators of $\overline{\mathcal{o}}$ as an $\mathcal{o}$-module. Then $\delta = \prod_i \delta_i$ is nonzero with $\delta\overline{\mathcal{o}} \subset \mathcal{o}$. Whence $\delta \in \mathfrak{q}_{\mathcal{o}}$. □

Accordingly, an **order** $\mathcal{o}$ is a one-dimensional noetherian domain whose conductor is an integral ideal of $\mathcal{o}$. Whence the conductor is also an integral ideal of $\overline{\mathcal{o}}$. In view of the previous result, the conductor being an integral ideal of $\mathcal{o}$ is equivalent to the normalization $\overline{\mathcal{o}}$ being a finitely generated $\mathcal{o}$-module. This means $\mathcal{o}$ is a Dedekind domain if and only if it is an integrally closed domain which happens if and only if the conductor is $\mathcal{o}$ itself. In fact, the normalization of an order is always a Dedekind domain.

Proposition 3.5.2. *Let $\mathcal{O}$ be an order with field of fractions K. Then $\overline{\mathcal{O}}$ is a Dedekind domain with field of fractions K.*

Proof. $\overline{\mathcal{O}}$ is an integrally closed domain with field of fractions K by definition. To see $\overline{\mathcal{O}}$ is noetherian, recall that $\overline{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-module as $\mathcal{O}$ is an order. Since $\mathcal{O}$ is noetherian, every integral ideal of $\overline{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-module and therefore also a finitely generated $\overline{\mathcal{O}}$-module. Whence $\overline{\mathcal{O}}$ is noetherian. It remains to show that every prime $\overline{\mathfrak{p}}$ of $\overline{\mathcal{O}}$ is maximal. This is equivalent to showing $\overline{\mathcal{O}}/\overline{\mathfrak{p}}$ is a field. So consider the homomorphism

$$\phi : \mathcal{O} \mapsto \overline{\mathcal{O}}/\overline{\mathfrak{p}} \qquad \alpha \mapsto \alpha + \overline{\mathfrak{p}},$$

induced by natural projection. Recall that $\overline{\mathfrak{p}}$ is above some prime $\mathfrak{p}$ of $\mathcal{O}$. Whence $\ker \phi = \mathfrak{p}$. By the first isomorphism theorem, ϕ induces an embedding $\mathbb{F}_{\mathfrak{p}} \to \overline{\mathcal{O}}/\overline{\mathfrak{p}}$. Therefore $\overline{\mathcal{O}}/\overline{\mathfrak{p}}$ is a finitely dimensional $\mathbb{F}_{\mathfrak{p}}$-vector space as $\overline{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-module. By primality of $\overline{\mathfrak{p}}$, $\overline{\mathcal{O}}/\overline{\mathfrak{p}}$ is also an integral domain and therefore must be a field. □

Perhaps not surprisingly, there is a generalization of the ideal class group to orders. As $\mathcal{O}$ is not a Dedekind domain, fractional ideals need not be invertible. The natural restriction is to consider all fractional ideals which are invertible. Let $I(\mathcal{O})$ denote the set of invertible fractional ideals of $\mathcal{O}$. Then it is easy to see that this is an abelian group.

Theorem 3.5.3. *Let $\mathcal{O}$ be an order with field of fractions K. Then $I(\mathcal{O})$ is an abelian group with identity $\mathcal{O}$.*

Proof. Associativity and commutativity of $I(\mathcal{O})$ are obvious. The identity is $\mathcal{O}$ because every invertible fractional ideal is a finitely generated $\mathcal{O}$-submodule of K. Lastly, every fractional ideal is invertible by definition of $I(\mathcal{O})$. □

Let $P(\mathcal{O})$ denote the subgroup of principal fractional ideals of $I(\mathcal{O})$. In fact, $P(\mathcal{O})$ contains every principal fraction ideal of $\mathcal{O}$ because they are all invertible. As $I(\mathcal{O})$ is abelian by the previous result, $P(\mathcal{O})$ is normal. The **Picard group** $\mathrm{Pic}(\mathcal{O})$ of $\mathcal{O}$ is defined to be the quotient

$$\mathrm{Pic}(\mathcal{O}) = I(\mathcal{O})/P(\mathcal{O}).$$

A class of $\mathrm{Pic}(\mathcal{O})$ is called an **Picard class** of $\mathcal{O}$. As every fractional ideal $\mathfrak{f}$ can be expressed as $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{a}$, we have $\delta\mathfrak{f} = \mathfrak{a}$ and it follows that every ideal class can be represented by an integral ideal. The **class number** $h_{\mathcal{O}}$ of $\mathcal{O}$ is defined by

$$h_{\mathcal{O}} = |\,\mathrm{Pic}(\mathcal{O})|.$$

The class number is a measure of the degree of failure for every invertible ideal to be principal. Of course, in the case $\mathcal{O}$ is a Dedekind domain, the Picard group is precisely the ideal class group and the corresponding class numbers are equal. This is to say that

$$\mathrm{Pic}(\mathcal{O}) = \mathrm{Cl}(\mathcal{O}).$$

The **unit group** of $\mathcal{O}$ is defined to be $\mathcal{O}^*$. That is, the unit group is the group of units in $\mathcal{O}$. The Picard and unit groups of $\mathcal{O}$ fit into a short exact sequence analogous to Proposition 3.3.8.

Proposition 3.5.4. *Let $\mathcal{O}$ be an order with field of fractions K. Then there is a short exact sequence*

$$1 \longrightarrow K^*/\mathcal{O}^* \longrightarrow I(\mathcal{O}) \longrightarrow \mathrm{Pic}(\mathcal{O}) \longrightarrow 1,$$

induced by natural inclusions and projections and where the second map is induced by $\kappa \mapsto \kappa\mathcal{O}$.

Proof. As the second map is injective and the third map is surjective, the sequence is exact at $K^*/\mathcal{O}^*$ and $\mathrm{Pic}(\mathcal{O})$. So it remains to prove exactness at $I(\mathcal{O})$. This follows because the principal fractional ideals represent the identity class of $\mathrm{Pic}(\mathcal{O})$ and these are generated by a complete set of representatives for $K^*/\mathcal{O}^*$. $\square$

It is possible to describe the Picard group in terms of principal fractional ideals for the localizations $\mathcal{O}_{\mathfrak{p}}$ at primes $\mathfrak{p}$. To prove this, we need some results specific to orders. Our first property is that localization preserves orders.

Proposition 3.5.5. *Let $\mathcal{O}$ be an order. Then for any multiplicative subset D of $\mathcal{O}$, $\mathcal{O}D^{-1}$ is an order.*

Proof. By Proposition 3.4.1, every prime of $\mathcal{O}D^{-1}$ is of the form $\mathfrak{p}D^{-1}$ for some prime $\mathfrak{p}$ of $\mathcal{O}$. But then $\mathfrak{p}D^{-1}$ is maximal because $\mathfrak{p}$ is and the bijections in Proposition 3.4.1 are inclusion-preserving. Therefore $\mathcal{O}D^{-1}$ is one-dimensional. To show it is noetherian, let $\mathfrak{A}$ be an integral ideal of $\mathcal{O}D^{-1}$ and set $\mathfrak{a} = \mathfrak{A} \cap \mathcal{O}$. Then

$$\mathfrak{A} = \mathfrak{a}D^{-1}.$$

As $\mathcal{O}$ is noetherian, $\mathfrak{a}$ is a finitely generated $\mathcal{O}$-module and hence $\mathfrak{A}$ is a finitely generated $\mathcal{O}D^{-1}$-module. This proves $\mathcal{O}D^{-1}$ is noetherian. It remains to show that the conductor of $\mathcal{O}D^{-1}$ is an integral ideal of $\mathcal{O}D^{-1}$. By Proposition 3.4.4, $\overline{\mathcal{O}}D^{-1}$ is the normalization of $\mathcal{O}D^{-1}$. So in view of Proposition 3.5.1, it suffices to show that $\overline{\mathcal{O}}D^{-1}$ is a finitely generated $\mathcal{O}D^{-1}$-submodule of K. But this is clear since $\overline{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-submodule of K by Proposition 3.5.1 again. $\square$

In particular, this result shows that $\mathcal{O}_{\mathfrak{p}}$ is an order. Our second property is a generalization of the Chinese remainder theorem for orders.

Proposition 3.5.6. *Let $\mathcal{O}$ be an order. Then for any integral ideal $\mathfrak{a}$ of $\mathcal{O}$, $\mathfrak{a}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$ for all but finitely many primes $\mathfrak{p}$ those of which satisfy $\mathfrak{a} \subseteq \mathfrak{p}$. Moreover, there is an isomorphism*

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_{\mathfrak{a} \subseteq \mathfrak{p}} \mathcal{O}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}},$$

induced by natural projection which may equivalently be expressed as

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}}.$$

Proof. Let $\widetilde{\mathfrak{a}}_{\mathfrak{p}} = \mathfrak{a}_{\mathfrak{p}} \cap \mathcal{O}$. Then $\widetilde{\mathfrak{a}}_{\mathfrak{p}}$ is an integral ideal of $\mathcal{O}$. As $\mathcal{O}$ is noetherian, Lemma 3.3.2 implies

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k \subseteq \mathfrak{a},$$

for some primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$. Since $\mathcal{O}$ is one-dimensional, every prime is maximal whence every maximal ideal containing $\mathfrak{a}$ is one of the $\mathfrak{p}_i$. So one the one hand, $\mathfrak{a} \subsetneq \mathfrak{p}$ for all but finitely many primes $\mathfrak{p}$. Whence $\mathfrak{a}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}}$ and $\widetilde{\mathfrak{a}}_{\mathfrak{p}} = \mathcal{O}$ for these primes. On the other hand, if $\mathfrak{a} \subseteq \mathfrak{p}$ then the inclusion-preserving bijections in Proposition 3.4.1 show that $\mathfrak{p}$ is the only prime containing $\widetilde{\mathfrak{a}}_{\mathfrak{p}}$. As $\mathcal{O}$ is one-dimensional, this means that if $\mathfrak{p}$ and $\mathfrak{q}$ are distinct primes then

$$\widetilde{\mathfrak{a}}_{\mathfrak{p}} + \widetilde{\mathfrak{a}}_{\mathfrak{q}} = \mathcal{O},$$

which is to say $\widetilde{\mathfrak{a}}_{\mathfrak{p}}$ and $\widetilde{\mathfrak{a}}_{\mathfrak{q}}$ are comaximal. Together, these facts show that the $\widetilde{\mathfrak{a}}_{\mathfrak{p}}$ are pairwise comaximal. Now we claim

$$\mathfrak{a} = \bigcap_{\mathfrak{p}} \widetilde{\mathfrak{a}}_{\mathfrak{p}}.$$

The forward inclusion is obvious. For the reverse inclusion, suppose $\frac{\eta}{\delta} \in \bigcap_{\mathfrak{p}} \widetilde{\mathfrak{a}}_{\mathfrak{p}}$ and set

$$\mathfrak{b} = \{\beta \in \mathcal{O} : \beta\eta \in \delta\mathfrak{a}\}.$$

Then $\mathfrak{b}$ is an integral ideal of $\mathcal{O}$ containing δ. As δ is not contained in any prime of $\mathcal{O}$, it follows that $\mathfrak{b}$ cannot be contained in any prime of $\mathcal{O}$. But every proper ideal is contained in a maximal ideal which is necessarily prime. Whence $\mathfrak{b}$ is not proper and so $\mathfrak{b} = \mathcal{O}$. Therefore $\eta \in \delta\mathfrak{a}$ which implies $\frac{\eta}{\delta} \in \mathfrak{a}$ proving the reverse inclusion. Altogether, the Chinese remainder theorem gives an isomorphism

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_{\mathfrak{a} \subseteq \mathfrak{p}} \mathcal{O}/\widetilde{\mathfrak{a}}_{\mathfrak{p}},$$

induced by natural projection. As $\widetilde{\mathfrak{a}}_{\mathfrak{p}} = \mathcal{O}$ unless $\mathfrak{a} \subseteq \mathfrak{p}$, this isomorphism can be expressed as

$$\mathcal{O}/\mathfrak{a} \cong \bigoplus_{\mathfrak{p}} \mathcal{O}/\widetilde{\mathfrak{a}}_{\mathfrak{p}}.$$

It remains to demonstrate that natural projection induces an isomorphism $\mathcal{O}/\widetilde{\mathfrak{a}}_{\mathfrak{p}} \cong \mathcal{O}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}}$. To see this, consider the homomorphism

$$\phi : \mathcal{O} \to \mathcal{O}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}} \qquad \alpha \mapsto \alpha + \mathfrak{a}_{\mathfrak{p}},$$

induced by natural projection. By the first isomorphism theorem, it suffices to show ϕ is surjective and $\ker \phi = \widetilde{\mathfrak{a}}_{\mathfrak{p}}$. As $\mathcal{O}_{\mathfrak{p}}$ is local with unique maximal ideal $\mathfrak{p}_{\mathfrak{p}}$, $\mathcal{O}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}}$ has a unique maximal ideal $\mathfrak{p}_{\mathfrak{p}}/\mathfrak{a}_{\mathfrak{p}}$ by the fourth isomorphism theorem. Whence

$$\mathcal{O} + \mathfrak{a}_{\mathfrak{p}} = \mathcal{O}_{\mathfrak{p}},$$

proving surjectivity. Moreover, $\ker \phi = \widetilde{\mathfrak{a}}_{\mathfrak{p}}$ from the definition of $\widetilde{\mathfrak{a}}_{\mathfrak{p}}$. $\square$

Our third property is that a fractional ideal of $\mathcal{O}$ is invertible if and only if its localization at a prime $\mathfrak{p}$ is principal in $\mathcal{O}_\mathfrak{p}$.

Proposition 3.5.7. *Let $\mathcal{O}$ be an order with field of fractions K. Then a fractional ideal $\mathfrak{f}$ of $\mathcal{O}$ is invertible if and only if the fractional ideal $\mathfrak{f}_\mathfrak{p}$ of $\mathcal{O}_\mathfrak{p}$ is principal for every prime $\mathfrak{p}$ of $\mathcal{O}$.*

Proof. Write $\mathfrak{f} = \frac{1}{\delta}\mathfrak{a}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{a}$. Then it suffices to show the claim for the integral ideal $\mathfrak{a}$. Set $\mathfrak{b} = \mathfrak{a}^{-1}$. For the forward implication, suppose $\mathfrak{a}$ is invertible so that $\mathfrak{a}\mathfrak{b} = \mathcal{O}$. Then we may find $\alpha_1, \ldots, \alpha_n \in \mathfrak{a}$ and $\beta_1, \ldots, \beta_n \in \mathfrak{b}$ such that

$$\sum_i \alpha_i\beta_i = 1.$$

Moreover, not all of the $\alpha_i\beta_i$ can lie in the unique maximal ideal $\mathfrak{p}_\mathfrak{p}$ of $\mathcal{O}_\mathfrak{p}$ for otherwise this ideal is not proper. Without loss of generality, we may assume $\alpha_1\beta_1 \notin \mathfrak{p}_\mathfrak{p}$ and so it is a unit in $\mathcal{O}_\mathfrak{p}$. As $\mathfrak{a}_\mathfrak{p}\mathfrak{b} = \mathcal{O}_\mathfrak{p}$ by the invertibility of $\mathfrak{a}$, it follows that $\mathfrak{a}_\mathfrak{p} = \alpha_1\mathcal{O}_\mathfrak{p}$. This shows $\mathfrak{a}_\mathfrak{p}$ is principal. For the reverse implication, suppose $\mathfrak{a}_\mathfrak{p}$ is principal for every prime $\mathfrak{p}$ of $\mathcal{O}$. Whence $\mathfrak{a}_\mathfrak{p} = \alpha_\mathfrak{p}\mathcal{O}_\mathfrak{p}$ for some nonzero $\alpha_\mathfrak{p} \in K$. As K is the field of fractions of $\mathcal{O}_\mathfrak{p}$, we may multiply by a nonzero element of $\mathcal{O}_\mathfrak{p}$, if necessary, to ensure that $\alpha_\mathfrak{p} \in \mathfrak{a}$. Now if $\mathfrak{a}$ were not invertible then there would exist a prime $\mathfrak{p}$ such that

$$\mathfrak{a}\mathfrak{a}^{-1} \subseteq \mathfrak{p},$$

since $\mathcal{O}$ is one-dimensional. Let $\alpha_1, \ldots, \alpha_r$ be generators of $\mathfrak{a}$ as an $\mathcal{O}$-module. As $\alpha_i \in \mathfrak{a}_\mathfrak{p}$, we may write $\alpha_i = \alpha_\mathfrak{p}\frac{\eta_i}{\delta_i}$ with $\frac{\eta_i}{\delta_i} \in \mathcal{O}_\mathfrak{p}$. Then $\delta_i\alpha_i \in \alpha_\mathfrak{p}\mathcal{O}$ and so $\delta_i\alpha_i\alpha_\mathfrak{p}^{-1} \in \mathcal{O}$. Then $\delta = \prod_i \delta_i$ is nonzero with $\delta\alpha_\mathfrak{p}^{-1}\mathfrak{a} \subseteq \mathcal{O}$. Thus $\delta\alpha_\mathfrak{p}^{-1} \in \mathfrak{a}^{-1}$ from which we find $\delta \in \mathfrak{a}\mathfrak{a}^{-1}$ whence $\delta \in \mathfrak{p}$. This is a contradiction and so $\mathfrak{a}$ is invertible. $\square$

We can now show how the Picard group is related to principal fractional ideals for the localizations $\mathcal{O}_\mathfrak{p}$ of an order $\mathcal{O}$ at primes $\mathfrak{p}$.

Proposition 3.5.8. *Let $\mathcal{O}$ be an order with field of fractions K. Then there is an isomorphism*

$$I(\mathcal{O}) \cong \bigoplus_\mathfrak{p} P(\mathcal{O}_\mathfrak{p}),$$

induced by localization. In particular, upon identifying $P(\mathcal{O})$ with its image under this isomorphism, we have

$$\mathrm{Pic}(\mathcal{O}) \cong \left(\bigoplus_\mathfrak{p} P(\mathcal{O}_\mathfrak{p})\right)/P(\mathcal{O}).$$

Proof. It follows from Proposition 3.5.7 that $\mathfrak{f}_\mathfrak{p}$ is a principal fractional ideal of $\mathcal{O}_\mathfrak{p}$ for every fractional ideal $\mathfrak{f}$ of $I(\mathcal{O})$ and every prime $\mathfrak{p}$ of $\mathcal{O}$. Therefore we may consider the homomorphism

$$\phi : I(\mathcal{O}) \to \bigoplus_\mathfrak{p} P(\mathcal{O}_\mathfrak{p}) \qquad \mathfrak{f} \mapsto (\mathfrak{f}_\mathfrak{p})_\mathfrak{p},$$

induced by localization. To prove the first statement, we must show that ϕ is an isomorphism. For surjectivity, let $(\kappa_\mathfrak{p}\mathcal{O}_\mathfrak{p})_\mathfrak{p} \in \bigoplus_\mathfrak{p} P(\mathcal{O}_\mathfrak{p})$. Then $\kappa_\mathfrak{p} \in K$ and all but finitely satisfy $\kappa_\mathfrak{p}\mathcal{O}_\mathfrak{p} = \mathcal{O}_\mathfrak{p}$. We claim that

$$\mathfrak{f} = \bigcap_\mathfrak{p} \kappa_\mathfrak{p}\mathcal{O}_\mathfrak{p},$$

is an invertible fractional ideal of $\mathcal{O}$ whose image under ϕ is $(\kappa_\mathfrak{p}\mathcal{O}_\mathfrak{p})_\mathfrak{p}$. As $\kappa_\mathfrak{p}\mathcal{O}_\mathfrak{p} = \mathcal{O}_\mathfrak{p}$ for all but finitely many primes $\mathfrak{p}$, there exists a nonzero $\delta \in \mathcal{O}$ for which $\delta\kappa_\mathfrak{p} \in \mathcal{O}_\mathfrak{p}$ for all primes $\mathfrak{p}$. It follows from Proposition 3.4.2 that $\delta\mathfrak{f} \subseteq \mathcal{O}$. Whence $\mathfrak{f}$ is a fractional ideal of $\mathcal{O}$. To show it is invertible and that the image under ϕ is $(\kappa_\mathfrak{p}\mathcal{O}_\mathfrak{p})_\mathfrak{p}$, it suffices to prove

$$\mathfrak{f}_\mathfrak{p} = \kappa_\mathfrak{p}\mathcal{O}_\mathfrak{p}.$$

Indeed, Proposition 3.5.7 will imply that $\mathfrak{f}$ is invertible and the definition of ϕ shows that $\mathfrak{f}$ will then map to $(\kappa_\mathfrak{p}\mathcal{O}_\mathfrak{p})_\mathfrak{p}$ under ϕ. The forward inclusion is clear by the definition of $\mathfrak{f}$. As $\delta\mathfrak{f}$ is an integral ideal, Proposition 3.5.6 implies that there exists an $\alpha \in \mathcal{O}$ satisfying

$$\alpha \equiv \delta\kappa_\mathfrak{p} \pmod{\delta\mathfrak{f}_\mathfrak{p}} \quad \text{and} \quad \alpha \equiv 0 \pmod{\delta\mathfrak{f}_\mathfrak{q}},$$

for primes $\mathfrak{q}$ with $\mathfrak{q} \neq \mathfrak{p}$. Whence $\frac{\alpha}{\delta} \in \mathfrak{f}_\mathfrak{q}$ for all primes $\mathfrak{q}$ and Proposition 3.4.2 implies $\frac{\alpha}{\delta} \in \mathfrak{f}$. But then $\kappa_\mathfrak{p} \in \mathfrak{f}_\mathfrak{p}$ and the reverse inclusion holds. Altogether, this means ϕ is surjective. To see that it is injective, suppose $\mathfrak{f} \in I(\mathcal{O})$ is such that $\mathfrak{f}_\mathfrak{p} = \mathcal{O}_\mathfrak{p}$ for all primes $\mathfrak{p}$. Then Proposition 3.4.2 implies $\mathfrak{f} = \mathcal{O}$ which means ϕ is injective. This establishes the first isomorphism. The second isomorphism is immediate by the definition of the Picard group. □

With this formulation of the Picard group, it is possible to obtain a long exact sequence relating the Picard and ideal class groups of $\mathcal{O}$ and its normalization $\overline{\mathcal{O}}$.

Proposition 3.5.9. *Let $\mathcal{O}$ be an order with field of fractions K. Then there is a long exact sequence*

$$1 \longrightarrow \mathcal{O}^* \longrightarrow \overline{\mathcal{O}}^* \longrightarrow \bigoplus_\mathfrak{p} \overline{\mathcal{O}}_\mathfrak{p}^*/\mathcal{O}_\mathfrak{p}^* \longrightarrow \mathrm{Pic}(\mathcal{O}) \longrightarrow \mathrm{Cl}(\overline{\mathcal{O}}) \longrightarrow 1,$$

induced by natural inclusions and projections, localization, as well as the maps $\kappa \mapsto \kappa\mathcal{O}$ and $\kappa \mapsto \kappa\overline{\mathcal{O}}$.

Proof. Observe that $\overline{\mathcal{O}}$ is a Dedekind domain by Proposition 3.5.2. The exact sequences in Propositions 3.3.8 and 3.5.4 combine to give a commutative diagram of short exact sequences given by

$$\begin{tikzcd}
1 \arrow[r] & K^*/\mathcal{O}^* \arrow[r] \arrow[d] & I(\mathcal{O}) \arrow[r] \arrow[d] & \mathrm{Pic}(\mathcal{O}) \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & K^*/\overline{\mathcal{O}}^* \arrow[r] & I(\overline{\mathcal{O}}) \arrow[r] & \mathrm{Cl}(\overline{\mathcal{O}}) \arrow[r] & 1.
\end{tikzcd}$$

As the first two vertical maps are surjective, exactness and commutativity together imply that the third one is surjective as well. In view of Proposition 3.5.8, we have isomorphisms

$$I(\mathcal{O}) \cong \bigoplus_{\mathfrak{p}} P(\mathcal{O}_{\mathfrak{p}}) \quad \text{and} \quad I(\overline{\mathcal{O}}) \cong \bigoplus_{\overline{\mathfrak{p}}} P(\overline{\mathcal{O}}_{\overline{\mathfrak{p}}}).$$

By Propositions 3.4.5 and 3.5.2, $\overline{\mathcal{O}}_{\mathfrak{p}}$ is a Dedekind domain. But as $\mathcal{O}_{\mathfrak{p}}$ is local and an order by Proposition 3.5.5, $\mathfrak{p}_{\mathfrak{p}}$ is the unique prime which is also the unique maximal ideal of $\mathcal{O}_{\mathfrak{p}}$. It follows that every prime $\overline{\mathfrak{p}}_{\mathfrak{p}}$ of $\overline{\mathcal{O}}_{\mathfrak{p}}$ is above $\mathfrak{p}_{\mathfrak{p}}$ which happens if and only if $\mathfrak{p} \subseteq \overline{\mathfrak{p}}$ by Proposition 3.4.1. In particular, there can only be finitely many primes whence $\overline{\mathcal{O}}_{\mathfrak{p}}$ is a principal ideal domain by Proposition 3.3.14. As $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}}$ is a discrete valuation ring by Theorem 3.4.6, we obtain an isomorphism

$$P(\overline{\mathcal{O}}_{\mathfrak{p}}) \cong \bigoplus_{\mathfrak{p} \subseteq \overline{\mathfrak{p}}} P(\overline{\mathcal{O}}_{\overline{\mathfrak{p}}}),$$

induced by localization. Since we have isomorphisms $K^*/\mathcal{O}_{\mathfrak{p}}^* \cong P(\mathcal{O}_{\mathfrak{p}})$ and $K^*/\overline{\mathcal{O}}_{\mathfrak{p}}^* \cong P(\overline{\mathcal{O}}_{\overline{\mathfrak{p}}})$ induced by the maps $\kappa \mapsto \kappa\mathcal{O}$ and $\kappa \mapsto \kappa\overline{\mathcal{O}}$ respectively, our previous work gives isomorphisms

$$I(\mathcal{O}) \cong \bigoplus_{\mathfrak{p}} K^*/\mathcal{O}_{\mathfrak{p}}^* \quad \text{and} \quad I(\overline{\mathcal{O}}) \cong \bigoplus_{\mathfrak{p}} K^*/\overline{\mathcal{O}}_{\mathfrak{p}}^*,$$

and our exact commutative diagram becomes

$$\begin{tikzcd}
1 \arrow[r] & K^*/\mathcal{O}^* \arrow[r] \arrow[d] & \bigoplus_{\mathfrak{p}} K^*/\mathcal{O}_{\mathfrak{p}}^* \arrow[r] \arrow[d] & \mathrm{Pic}(\mathcal{O}) \arrow[r] \arrow[d] & 1 \\
1 \arrow[r] & K^*/\overline{\mathcal{O}}^* \arrow[r] & \bigoplus_{\mathfrak{p}} K^*/\overline{\mathcal{O}}_{\mathfrak{p}}^* \arrow[r] & \mathrm{Cl}(\overline{\mathcal{O}}) \arrow[r] & 1.
\end{tikzcd}$$

As the vertical maps are surjective, the snake lemma gives a short exact sequence

$$1 \longrightarrow \overline{\mathcal{O}}^*/\mathcal{O}^* \longrightarrow \bigoplus_{\mathfrak{p}} \overline{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^* \longrightarrow \ker(\mathrm{Pic}(\mathcal{O}) \to \mathrm{Cl}(\overline{\mathcal{O}})) \longrightarrow 1.$$

Splicing this short exact sequence with the short exact sequences

$$1 \longrightarrow \mathcal{O}^* \longrightarrow \overline{\mathcal{O}}^* \longrightarrow \overline{\mathcal{O}}^*/\mathcal{O}^* \longrightarrow 1,$$

and

$$1 \longrightarrow \ker(\mathrm{Pic}(\mathcal{O}) \to \mathrm{Cl}(\overline{\mathcal{O}})) \longrightarrow \mathrm{Pic}(\mathcal{O}) \longrightarrow \mathrm{Cl}(\overline{\mathcal{O}}) \longrightarrow 1,$$

gives the desired long exact sequence. □

Recall that an order $\mathcal{O}$ is a Dedekind domain if and only if the conductor $\mathfrak{q}_{\mathcal{O}}$ is $\mathcal{O}$ itself. This suggests that the conductor should control the extent to which an order fails to be integrally closed. We now turn to a more detailed investigation of what this means. Recall from Proposition 3.5.5 that the localization $\mathcal{O}_{\mathfrak{p}}$ is an order for any prime $\mathfrak{p}$ of $\mathcal{O}$ and is also local. In view of Theorem 3.4.6, $\mathcal{O}$ is a Dedekind domain if and only if the localizations $\mathcal{O}_\mathfrak{p}$ are integrally closed for all primes $\mathfrak{p}$ of $\mathcal{O}$. Accordingly, $\mathfrak{p}$ is said to be **regular** if $\mathcal{O}_\mathfrak{p}$ is an integrally closed domain and is **irregular** otherwise. So if $\mathfrak{p}$ is regular then $\mathcal{O}_\mathfrak{p}$ is a Dedekind domain with exactly one prime and therefore a discrete valuation ring. This means $\mathfrak{p}$ is regular if and only if $\mathcal{O}_\mathfrak{p}$ is a discrete valuation ring. It turns out that the regular primes are exactly those which are do not contain the conductor.

Proposition 3.5.10. *Let $\mathcal{O}$ be an order. Then a prime $\mathfrak{p}$ of $\mathcal{O}$ is regular if and only if $\mathfrak{q}_\mathcal{O} \subsetneq \mathfrak{p}$. In the case $\mathfrak{p}$ is regular, there a unique prime $\overline{\mathfrak{p}}$ of $\overline{\mathcal{O}}$ above $\mathfrak{p}$ which is given by $\overline{\mathfrak{p}} = \mathfrak{p}_\mathfrak{p} \cap \overline{\mathcal{O}}$, we have $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}} = \mathcal{O}_\mathfrak{p}$, and $\mathfrak{p}$ is invertible.*

Proof. For the forward implication, if $\mathfrak{p}$ is regular then $\mathcal{O}_\mathfrak{p}$ is an integrally closed domain. As $\overline{\mathcal{O}}$ is integral over $\mathcal{O}$ and hence also $\mathcal{O}_\mathfrak{p}$, it follows that $\overline{\mathcal{O}} \subseteq \mathcal{O}_\mathfrak{p}$. But $\overline{\mathcal{O}}$ is a finitely generated $\mathcal{O}$-module by Proposition 3.5.1. So we may find generators $\frac{\eta_1}{\delta_1}, \ldots, \frac{\eta_r}{\delta_r}$ for $\overline{\mathcal{O}}$ as an $\mathcal{O}$-module with $\delta_i \notin \mathfrak{p}$. Then $\delta = \prod_i \delta_i$ is nonzero with $\delta\overline{\mathcal{O}} \subseteq \mathcal{O}$ meaning $\delta \in \mathfrak{q}_\mathcal{O}$. But primality of $\mathfrak{p}$ forces $\delta \notin \mathfrak{p}$. Whence $\mathfrak{q}_\mathcal{O} \subsetneq \mathfrak{p}$. For the reverse implication, suppose $\mathfrak{q}_\mathcal{O} \subsetneq \mathfrak{p}$. Then we may find a nonzero $\alpha \in \mathfrak{q}_\mathcal{O} - \mathfrak{p}$ with $\alpha\overline{\mathcal{O}} \subseteq \mathcal{O}$. It follows that $\overline{\mathcal{O}} \subset \mathcal{O}_\mathfrak{p}$. As $\mathfrak{p}_\mathfrak{p}$ is the unique maximal ideal of $\mathcal{O}_\mathfrak{p}$, it is necessarily a prime. In view of the previous inclusion, $\overline{\mathfrak{p}}$ is then a prime of $\overline{\mathcal{O}}$ containing $\mathfrak{p}$ and is therefore above $\mathfrak{p}$ as $\mathcal{O}$ is one-dimensional. Since $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}}$ is an integrally closed domain by Proposition 3.4.4, it is enough to show $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}} = \mathcal{O}_\mathfrak{p}$. The reverse inclusion holds because $\overline{\mathfrak{p}}$ is above $\mathfrak{p}$. For the forward inclusion, Suppose $\frac{\eta}{\delta} \in \overline{\mathcal{O}}_{\overline{\mathfrak{p}}}$. Then $\alpha\eta \in \mathcal{O}$ and $\alpha\delta \notin \mathfrak{p}$ by primality of $\mathfrak{p}$. Whence $\frac{\eta}{\delta} \in \mathcal{O}_\mathfrak{p}$ proving the forward inclusion. This completes the proof of the first statement.

As for the second statement, we have already verified $\overline{\mathfrak{p}}$ is a prime of $\overline{\mathcal{O}}$ above $\mathfrak{p}$ and that $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}} = \mathcal{O}_\mathfrak{p}$. To show it is the unique prime of $\overline{\mathcal{O}}$ above $\mathfrak{p}$, let $\overline{\mathfrak{q}}$ be another prime of $\overline{\mathcal{O}}$ above $\mathfrak{p}$. In view of the identity $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}} = \mathcal{O}_\mathfrak{p}$, it follows that $\overline{\mathcal{O}}_{\overline{\mathfrak{p}}} \subseteq \overline{\mathcal{O}}_{\overline{\mathfrak{q}}}$. The inclusion-preserving bijections in Proposition 3.4.1 show

$$\overline{\mathfrak{p}} = \overline{\mathfrak{p}}_{\overline{\mathfrak{p}}} \cap \overline{\mathcal{O}} \quad \text{and} \quad \overline{\mathfrak{q}} = \overline{\mathfrak{q}}_{\overline{\mathfrak{q}}} \cap \overline{\mathcal{O}}.$$

These identities together with the previous inequality imply $\overline{\mathfrak{p}} \subseteq \overline{\mathfrak{q}}$ and so $\overline{\mathfrak{p}} = \overline{\mathfrak{q}}$ as primes of $\overline{\mathcal{O}}$ are maximal. This proves uniqueness. It remains to show $\mathfrak{p}$ is invertible. By Proposition 3.5.7, we must show that $\mathfrak{p}_\mathfrak{q}$ is principal for every prime $\mathfrak{q}$ of $\mathcal{O}$. If $\mathfrak{q} = \mathfrak{p}$, regularity of $\mathfrak{p}$ means $\mathcal{O}_\mathfrak{p}$ is a discrete valuation ring and hence a principal ideal domain. Thus $\mathfrak{p}_\mathfrak{p}$ is principal. If $\mathfrak{q} \neq \mathfrak{p}$ then $\mathfrak{p}_\mathfrak{q} = \mathcal{O}_\mathfrak{q}$ which is also principal. This completes the proof of the second statement. □

As $\mathcal{O}$ is noetherian, Lemma 3.3.2 implies

$$\mathfrak{p}_1 \cdots \mathfrak{p}_k \subseteq \mathfrak{q}_\mathcal{O},$$

for some primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_k$. Since $\mathcal{O}$ is one-dimensional, every maximal ideal containing $\mathfrak{q}_\mathcal{O}$ is a prime and thus one of the $\mathfrak{p}_i$. In view of the previous result, this means that only finitely many primes of $\mathcal{O}$ are irregular and they are exactly the primes which are not invertible. This is a concrete description of the failure to which $\mathcal{O}$ is not integrally closed in terms of the conductor $\mathfrak{q}_{\mathcal{O}}$.

Using the concept of regular primes, it is also possible to give a more refined description of the direct sum term in Proposition 3.5.9.

Proposition 3.5.11. *Let $\mathcal{O}$ be an order with field of fractions K. Then there is an isomorphism*

$$(\overline{\mathcal{O}}/\mathfrak{q}_{\mathcal{O}})^*/(\mathcal{O}/\mathfrak{q}_{\mathcal{O}})^* \cong \bigoplus_{\mathfrak{p}} \overline{\mathcal{O}}_{\mathfrak{p}}^*/\mathcal{O}_{\mathfrak{p}}^*,$$

induced by natural projection.

Proof. As the conductor $\mathfrak{q}_{\mathcal{O}}$ is both an integral ideal of $\mathcal{O}$ and its normalization $\overline{\mathcal{O}}$, Proposition 3.5.6 gives isomorphisms

$$\mathcal{O}/\mathfrak{q}_{\mathcal{O}} \cong \bigoplus_{\mathfrak{p}} \mathcal{O}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}} \quad \text{and} \quad \overline{\mathcal{O}}/\mathfrak{q}_{\mathcal{O}} \cong \bigoplus_{\overline{\mathfrak{p}}} \overline{\mathcal{O}}_{\overline{\mathfrak{p}}}/(\mathfrak{q}_{\mathcal{O}})_{\overline{\mathfrak{p}}},$$

induced by natural projection. By Propositions 3.4.5 and 3.5.2, $\overline{\mathcal{O}}_{\mathfrak{p}}$ is a Dedekind domain. But as $\mathcal{O}_{\mathfrak{p}}$ is local and an order by Proposition 3.5.5, $\mathfrak{p}_{\mathfrak{p}}$ is the unique prime which is also the unique maximal ideal of $\mathcal{O}_{\mathfrak{p}}$. It follows that every prime $\overline{\mathfrak{p}}_{\mathfrak{p}}$ of $\overline{\mathcal{O}}_{\mathfrak{p}}$ is above $\mathfrak{p}_{\mathfrak{p}}$ which happens if and only if $\mathfrak{p} \subseteq \overline{\mathfrak{p}}$ by Proposition 3.4.1. By Proposition 3.5.6 again, we have an isomorphism

$$\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}} \cong \bigoplus_{\mathfrak{p} \subseteq \overline{\mathfrak{p}}} \overline{\mathcal{O}}_{\overline{\mathfrak{p}}}/(\mathfrak{q}_{\mathcal{O}})_{\overline{\mathfrak{p}}},$$

induced by natural projection. Whence our previous isomorphisms together imply

$$(\mathcal{O}/\mathfrak{q}_{\mathcal{O}})^* \cong \bigoplus_{\mathfrak{p}} (\mathcal{O}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^* \quad \text{and} \quad (\overline{\mathcal{O}}/\mathfrak{q}_{\mathcal{O}})^* \cong \bigoplus_{\mathfrak{p}} (\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*,$$

and we obtain an isomorphism

$$(\overline{\mathcal{O}}/\mathfrak{q}_{\mathcal{O}})^*/(\mathcal{O}/\mathfrak{q}_{\mathcal{O}})^* \cong \bigoplus_{\mathfrak{p}} (\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*/(\mathcal{O}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*.$$

Now suppose $\mathfrak{p}$ is irregular and consider the homomorphism

$$\phi : \overline{\mathcal{O}}_{\mathfrak{p}}^* \to (\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*/(\mathcal{O}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^* \qquad \alpha \mapsto (\alpha + (\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})(\mathcal{O}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^*,$$

induced by natural projection. Note that ϕ is surjective if the homomorphism

$$\pi : \overline{\mathcal{O}}_{\mathfrak{p}}^* \to (\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})^* \qquad \alpha \mapsto \alpha + (\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}},$$

given by natural projection is. To see that this map is surjective, recall that an element of a ring is invertible if and only if it is not contained in any maximal ideal. By the fourth isomorphism theorem, the preimages of the maximal ideals of $(\overline{\mathcal{O}}_{\mathfrak{p}}/(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}})$ under natural projection are maximal ideals of $\overline{\mathcal{O}}_{\mathfrak{p}}$ containing $(\mathfrak{q}_{\mathcal{O}})_{\mathfrak{p}}$. These preimages constitute all of the maximal ideals of $\overline{\mathcal{o}}_\mathfrak{p}$ since every maximal ideal of $\overline{\mathcal{o}}_\mathfrak{p}$ is a prime which must be above $\mathfrak{p}_\mathfrak{p}$ and irregularity implies $\mathfrak{q}_\mathcal{O} \subseteq \mathfrak{p}$ by Proposition 3.5.10. Therefore π, and hence ϕ as well, is surjective. Clearly $\mathcal{o}_\mathfrak{p}^* \subseteq \ker\phi$. But $\ker\phi$ is also a subgroup of $\overline{\mathcal{o}}_\mathfrak{p}^*$ which is contained in $\mathcal{o}_\mathfrak{p}$. Whence $\ker\phi = \mathcal{o}_\mathfrak{p}^*$. The first isomorphism theorem then implies

$$\overline{\mathcal{o}}_\mathfrak{p}^*/\mathcal{o}_\mathfrak{p}^* \cong (\overline{\mathcal{o}}_\mathfrak{p}/(\mathfrak{q}_\mathcal{O})_\mathfrak{p})^*/(\mathcal{o}_\mathfrak{p}/(\mathfrak{q}_\mathcal{O})_\mathfrak{p})^*,$$

provided $\mathfrak{p}$ is irregular. If $\mathfrak{p}$ is regular, then $\mathcal{o}_\mathfrak{p}$ is an integrally closed domain and Proposition 3.5.10 implies $(\mathfrak{q}_\mathcal{O})_\mathfrak{p} = \mathcal{o}_\mathfrak{p}$. Therefore the isomorphism holds as well because all quotients are trivial. This completes the proof. □

Number Fields

We now turn to the setting of a number field K of degree n. An **order** $\mathcal{o}$ of K is an order satisfying

$$\mathbb{Z} \subseteq \mathcal{o} \subseteq \mathcal{O}_K,$$

and whose normalization is $\mathcal{O}_K$. Accordingly, $\mathcal{O}_K$ is called the **maximal order** of K. Clearly it is the largest order of K. As the normalization of $\mathcal{o}$ is O_K, the field of fractions of $\mathcal{o}$ is necessarily K. In fact, $\mathcal{o}$ contains a basis for $K/\mathbb{Q}$. To see this, let $\kappa_1, \ldots, \kappa_n$ be a basis for $K/\mathbb{Q}$ and write

$$k_i = \frac{\alpha_i}{\delta_i},$$

for $\alpha_i, \delta_i \in \mathcal{o}$ with δ_i nonzero. Then $\delta = \prod_i \delta_i$ is nonzero and we find that $\delta\kappa_1, \ldots, \delta\kappa_n$ is a basis for $K/\mathbb{Q}$ contained in $\mathcal{o}$.

3.6 Prime Splitting

Suppose $\mathcal{O}/\mathcal{o}$ is a Dedekind extension of a finite separable extension L/K. Let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above a prime $\mathfrak{p}$ of $\mathcal{o}$. The **ramification index** $e_\mathfrak{p}(\mathfrak{P})$ of $\mathfrak{P}$ relative to $\mathfrak{p}$ is defined to be the power of $\mathfrak{P}$ appearing in the prime factorization of $\mathfrak{p}\mathcal{O}$. If $\mathfrak{p}\mathcal{O}$ has prime factors $\mathfrak{P}_1, \ldots, \mathfrak{P}_r$ then the prime factorization of $\mathfrak{p}\mathcal{O}$ is

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_\mathfrak{p}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_\mathfrak{p}(\mathfrak{P}_r)}.$$

Something can also be said about the residue class fields $\mathbb{F}_\mathfrak{P}$ and $\mathbb{F}_\mathfrak{p}$. Consider the homomorphism

$$\phi : \mathcal{o} \to \mathbb{F}_\mathfrak{P} \qquad \alpha \mapsto \alpha + \mathfrak{P},$$

induced by natural projection. We have $\ker\phi = \mathfrak{p}$ as $\mathfrak{P}$ is above $\mathfrak{p}$. The first isomorphism theorem implies that ϕ induces an embedding $\mathbb{F}_\mathfrak{p} \to \mathbb{F}_\mathfrak{P}$. This means $\mathbb{F}_\mathfrak{P}$ is an $\mathbb{F}_\mathfrak{p}$-vector space of dimension at most n as $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module of rank at most n. In other words, $\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}$ is a field extension of degree at most n. Accordingly, we call $\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}$ the **residue class extension** of $\mathfrak{P}$ relative to $\mathfrak{p}$. We define the **inertia degree** $f_\mathfrak{p}(\mathfrak{P})$ of $\mathfrak{P}$ relative to $\mathfrak{p}$ by

$$f_\mathfrak{p}(\mathfrak{P}) = [\mathbb{F}_\mathfrak{P} : \mathbb{F}_\mathfrak{p}].$$

That is, $f_{\mathfrak{p}}(\mathfrak{P})$ is the degree of $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ which is the dimension of the residue class field $\mathbb{F}_{\mathfrak{P}}$ as an $\mathbb{F}_{\mathfrak{p}}$-vector space. In fact, something slightly more general is true. Let $\mathfrak{A}$ and $\mathfrak{B}$ be an integral ideals of $\mathcal{O}$ with $\mathfrak{p}\mathcal{O} \subseteq \mathfrak{B} \subseteq \mathfrak{A}$. Consider the homomorphism

$$\phi : \mathfrak{A} \to \mathfrak{A}/\mathfrak{B} \qquad \alpha \mapsto \alpha + \mathfrak{B},$$

induced by natural projection. As $\mathfrak{A}$ and $\mathfrak{B}$ contain $\mathfrak{p}$, we have that $\ker \phi$ contains $\mathfrak{p}$. So ϕ induces a homomorphism $\mathbb{F}_{\mathfrak{p}} \to \mathfrak{A}/\mathfrak{B}$. Whence $\mathfrak{A}/\mathfrak{B}$ is a finite dimensional $\mathbb{F}_{\mathfrak{p}}$-vector space of dimension at most n because $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module of rank at most n. We obtain the residue class extension when $\mathfrak{A} = \mathcal{O}$ and $\mathfrak{B} = \mathfrak{P}$.

Suppose $\widetilde{\mathcal{O}}/\mathcal{O}/\mathcal{o}$ is a tower of Dedekind extensions for a finite separable tower $M/L/K$. Let $\widetilde{\mathfrak{P}}$, $\mathfrak{P}$, and $\mathfrak{p}$ be primes of $\widetilde{\mathcal{O}}$, $\mathcal{O}$, and $\mathcal{o}$ respectively with $\widetilde{\mathfrak{P}}$ above $\mathfrak{P}$ and $\mathfrak{P}$ above $\mathfrak{p}$. We illustrate this relationship via the diagram

$$\begin{array}{c} \widetilde{\mathfrak{P}} \subset \widetilde{\mathcal{O}} \subseteq M \\ \Big| \\ \mathfrak{P} \subset \mathcal{O} \subseteq L \\ \Big| \\ \mathfrak{p} \subset \mathcal{o} \subseteq K. \end{array}$$

Then $\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \mid\mid \mathfrak{p}\mathcal{O}$ and $\widetilde{\mathfrak{P}}^{e_{\mathfrak{P}}(\widetilde{\mathfrak{P}})} \mid\mid \mathfrak{P}\widetilde{\mathcal{O}}$. We also have the residue extensions $\mathbb{F}_{\widetilde{\mathfrak{P}}}/\mathbb{F}_{\mathfrak{P}}$ and $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$. It follows that

$$e_{\mathfrak{p}}(\widetilde{\mathfrak{P}}) = e_{\mathfrak{P}}(\widetilde{\mathfrak{P}})e_{\mathfrak{p}}(\mathfrak{P}) \quad \text{and} \quad f_{\mathfrak{p}}(\widetilde{\mathfrak{P}}) = f_{\mathfrak{P}}(\widetilde{\mathfrak{P}})f_{\mathfrak{p}}(\mathfrak{P}). \tag{3.6}$$

In other words, the ramification indices and inertia degrees are multiplicative with respect to towers.

The ramification indices and inertia degrees satisfy a simple relationship to the degree of L/K that is fundamental to the study of Dedekind extensions. We first prove a useful lemma.

Lemma 3.6.1. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a finite separable extension L/K and suppose $\mathfrak{P}$ a prime of $\mathcal{O}$ above a prime $\mathfrak{p}$ of $\mathcal{o}$. Then there is an isomorphism*

$$\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \cong \bigoplus_i \mathfrak{P}^i/\mathfrak{P}^{i+1},$$

induced by natural projection of a basis adapted to the filtration

$$\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \supset \mathfrak{P}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \supset \cdots \supset \mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})-1}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \supset 0,$$

of $\mathbb{F}_{\mathfrak{P}}$-vector spaces. In particular, $\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})}$ is an $\mathbb{F}_{\mathfrak{p}}$-vector space of dimension $e_{\mathfrak{p}}(\mathfrak{P})f_{\mathfrak{p}}(\mathfrak{P})$.

Proof. By the third isomorphism theorem, quotients of successive layers of the filtration are isomorphic to $\mathfrak{P}^i/\mathfrak{P}^{i+1}$. Whence we obtain a short exact sequence

$$0 \longrightarrow \mathfrak{P}^{i+1}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \longrightarrow \mathfrak{P}^{i}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \longrightarrow \mathfrak{P}^{i}/\mathfrak{P}^{i+1} \longrightarrow 0,$$

induced by natural inclusions and projections. As this is a short exact sequence of $\mathbb{F}_{\mathfrak{P}}$-vector spaces it splits. Choosing sections for all of these short exact sequences is equivalent to choosing a basis for $\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})}$ adapted to the filtration and we obtain an isomorphism

$$\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \cong \bigoplus_{i} \mathfrak{P}^{i}/\mathfrak{P}^{i+1},$$

induced by natural projection of the adapted basis. This proves the first statement. The quotients $\mathfrak{P}^{i}/\mathfrak{P}^{i+1}$ are all isomorphic to $\mathbb{F}_{\mathfrak{P}}$ by Lemma 3.3.13. As $\mathbb{F}_{\mathfrak{P}}$ is an $\mathbb{F}_{\mathfrak{p}}$-vector spaces of dimension $f_{\mathfrak{p}}(\mathfrak{P})$, it follows that $\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})}$ is an $\mathbb{F}_{\mathfrak{p}}$-vector space of dimension $e_{\mathfrak{p}}(\mathfrak{P})f_{\mathfrak{p}}(\mathfrak{P})$. This proves the second statement. □

We can now establish the relationship between the ramification indices, inertia degrees, and degree of L/K. The idea is that $\mathcal{O}/\mathfrak{p}\mathcal{O}$ is of maximum dimension as an $\mathbb{F}_{\mathfrak{p}}$-vector space, namely n, and its dimension can be computed as a sum of products of the ramification indices and inertia degrees. This relationship is known as the **fundamental equality**.

Theorem (Fundamental equality). *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K. Suppose $\mathfrak{p}$ is a prime of $\mathcal{o}$ and $\mathfrak{p}\mathcal{O}$ has prime factorization*

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_{\mathfrak{p}}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_{\mathfrak{p}}(\mathfrak{P}_r)}.$$

Then

$$n = \sum_{i} e_{\mathfrak{p}}(\mathfrak{P}_i) f_{\mathfrak{p}}(\mathfrak{P}_i).$$

Proof. Since distinct primes are relatively prime, the Chinese remainder theorem gives an isomorphism

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \bigoplus_{i} \mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)}.$$

As $\mathcal{O}/\mathfrak{p}\mathcal{O}$ and $\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)}$ are $\mathbb{F}_{\mathfrak{p}}$-vector spaces, it suffices to show $\mathcal{O}/\mathfrak{p}\mathcal{O}$ is of dimension n and $\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)}$ is of dimension $e_{\mathfrak{p}}(\mathfrak{P}_i)f_{\mathfrak{p}}(\mathfrak{P}_i)$. For $\mathcal{O}/\mathfrak{p}\mathcal{O}$, we already know it is an $\mathbb{F}_{\mathfrak{p}}$-vector space of dimension at most n as $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module of rank at most n. Therefore we must show that the dimension is exactly n. Let $\overline{\lambda_1}, \ldots, \overline{\lambda_m}$ be a basis for $\mathcal{O}/\mathfrak{p}\mathcal{O}$ as an $\mathbb{F}_{\mathfrak{p}}$-vector space and let $\lambda_1, \ldots, \lambda_m$ be any lift of this basis to $\mathcal{O}$. As $m \le n$, it suffices to show $\lambda_1, \ldots, \lambda_m$ spans L/K. Let M be the $\mathcal{o}$-module generated by $\lambda_1, \ldots, \lambda_m$ and set $N = \mathcal{O}/M$. Then $\mathcal{O} = M + \mathfrak{p}\mathcal{O}$ since $\lambda_1, \ldots, \lambda_m$ is a lift of a basis for $(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_{\mathfrak{p}}$. Whence $N = \mathfrak{p}N$ and therefore N is a finitely generated $\mathcal{o}$-module of rank at most n since $\mathcal{O}$ is. Let $\omega_1, \ldots, \omega_r$ be generators for N. As $N = \mathfrak{p}N$, we have

$$\omega_i = \sum_{j} \alpha_{i,j}\omega_j,$$

for some $\alpha_{i,j} \in \mathfrak{p}$. These r equations are equivalent to the identity

$$\begin{pmatrix} 1-\alpha_{1,1} & \alpha_{1,2} & \cdots & -\alpha_{1,r} \\ -\alpha_{2,1} & 1-\alpha_{2,2} & & \\ \vdots & & \ddots & \\ -\alpha_{r,1} & & & 1-\alpha_{r,r} \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \vdots \\ \omega_r \end{pmatrix} = \mathbf{0}.$$

Let $d = \det(I - (\alpha_{i,j}))$. Then $d \neq 0$ because expanding the determinant shows $d \equiv 1 \pmod{\mathfrak{p}}$ as $\alpha_{i,j} \in \mathfrak{p}$. Cramer's rule implies that d annihilates N which is to say that $d\mathcal{O} \subseteq M$. Equivalently,

$$d\mathcal{O} \subseteq \lambda_1 \mathcal{o} + \cdots + \lambda_m \mathcal{o}.$$

By Proposition 3.1.4, multiplication by K shows

$$L = \lambda_1 K + \cdots + \lambda_m K.$$

Hence $\lambda_1, \ldots, \lambda_m$ spans L/K whence $\mathcal{O}/\mathfrak{p}\mathcal{O}$ is an $\mathbb{F}_{\mathfrak{p}}$-vector space of dimension n. For $\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)}$, the dimensionality claim follows from Lemma 3.6.1. $\square$

It is useful to classify primes according to specific cases of the fundamental equality. We say $\mathfrak{p}$ is **inert** in $\mathcal{O}/\mathcal{o}$ if there is a prime $\mathfrak{P}$ above $\mathfrak{p}$ with $f_{\mathfrak{p}}(\mathfrak{P}) = n$. By the fundamental equality, this means $e_{\mathfrak{p}}(\mathfrak{P}) = 1$ and $r = 1$. Whence

$$\mathfrak{p}\mathcal{O} = \mathfrak{P},$$

which is to say $\mathfrak{p}\mathcal{O}$ is prime. More generally, $\mathfrak{p}$ is said to be **nonsplit** in $\mathcal{O}/\mathcal{o}$ if there is a prime $\mathfrak{P}$ above $\mathfrak{p}$ satisfying $e_{\mathfrak{p}}(\mathfrak{P})f_{\mathfrak{p}}(\mathfrak{P}) = n$ and is said to be **split** in $\mathcal{O}/\mathcal{o}$ otherwise. So

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \quad \text{or} \quad \mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_{\mathfrak{p}}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_{\mathfrak{p}}(\mathfrak{P}_r)},$$

with r at least 2, according to if $\mathfrak{p}$ is nonsplit or split accordingly. Moreover, we say $\mathfrak{p}$ is **totally split** in $\mathcal{O}/\mathcal{o}$ if every prime $\mathfrak{P}$ above $\mathfrak{p}$ satisfies $e_{\mathfrak{p}}(\mathfrak{P}) = f_{\mathfrak{p}}(\mathfrak{P}) = 1$ and so $r = n$ by the fundamental equality. Then

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1 \cdots \mathfrak{P}_n.$$

Observe that being inert or totally split correspond to antithetical factorizations in terms of inertia degrees. In particular, the smaller the inertia degrees are the greater the tendency for $\mathfrak{p}\mathcal{O}$ to factor into distinct primes.

Now let us introduce ramification of primes. If $\mathfrak{P}$ is above $\mathfrak{p}$, we say $\mathfrak{P}$ is **unramified** in $\mathcal{O}/\mathcal{o}$ if $e_{\mathfrak{p}}(\mathfrak{P}) = 1$ and the residue class extension $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is separable. Otherwise, we say $\mathfrak{P}$ is **ramified** in $\mathcal{O}/\mathcal{o}$. Moreover, $\mathfrak{P}$ is **totally ramified** in $\mathcal{O}/\mathcal{o}$ if in addition to being ramified we have $e_{\mathfrak{p}}(\mathfrak{P}) = n$ whence $f_{\mathfrak{p}}(\mathfrak{P}) = 1$. Similarly, we say that a prime $\mathfrak{p}$ of $\mathcal{o}$ is **unramified** in $\mathcal{O}/\mathcal{o}$ if every prime $\mathfrak{P}$ above it is unramified and is **ramified** in $\mathcal{O}/\mathcal{o}$ otherwise. Also, $\mathfrak{p}$ is said to be **totally ramified** in $\mathcal{O}/\mathcal{o}$

if a prime above it is totally ramified. In this case, the fundamental equality implies that there is only one prime $\mathfrak{P}$ above $\mathfrak{p}$ and

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}^n.$$

Note that being totally ramified or totally split correspond to antithetical factorizations in terms of ramification indices. In particular, the larger the maximum ramification index is the greater the tendency for $\mathfrak{p}\mathcal{O}$ to factor as a power of a single prime.

In any case, all of this suggests that it is desirable to understand how $\mathfrak{p}\mathcal{O}$ factors into primes of $\mathcal{O}$. Using orders, this is tractable for almost all primes $\mathfrak{p}$ if we know a sufficient amount of information about the minimal polynomial of a primitive element for L/K. Let θ be a primitive element for L/K so that $L = K(\theta)$. In view of Proposition 3.1.4 we may multiply by a nonzero element of $\mathcal{o}$, if necessary, to ensure $\theta \in \mathcal{O}$ and hence its minimal polynomial $m_\theta(x)$ over K has coefficients in $\mathcal{o}$. Then $\mathcal{o} \subseteq \mathcal{o}[\theta] \subseteq \mathcal{O}$ so that $\mathcal{o}[\theta]$ is an order. In fact, the field of fractions of $\mathcal{o}[\theta]$ is L since the field of fractions of $\mathcal{o}$ is K. As $\mathcal{o}[\theta]$ is integral over $\mathcal{o}$, it follows that the normalization of $\mathcal{o}[\theta]$ is $\mathcal{O}$. Therefore the conductor $\mathfrak{q}_{\mathcal{o}[\theta]}$ is an integral ideal of $\mathcal{o}[\theta]$ and $\mathcal{O}$. The **Dedekind-Kummer theorem** describes the prime factorization of $\mathfrak{p}\mathcal{O}$ provided it is relatively prime to $\mathfrak{q}_{\mathcal{o}[\theta]}$.

Theorem (Dedekind-Kummer theorem). *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K, θ be a primitive element for L/K contained in $\mathcal{O}$ with minimal polynomial $m_\theta(x)$ over K, and $\mathfrak{p}$ be a prime of $\mathcal{o}$ such that $\mathfrak{p}\mathcal{O}$ is relatively prime to $\mathfrak{q}_{\mathcal{o}[\theta]}$. Also let $\overline{m_\theta}(x) \in \mathbb{F}_\mathfrak{p}[x]$ be the image of $m_\theta(x)$ under natural projection and suppose*

$$\overline{m_\theta}(x) = \overline{m_1}(x)^{e_1} \cdots \overline{m_r}(x)^{e_r},$$

is its prime factorization. Let $m_i(x)$ be any lift of $\overline{m_i}(x)$ to $\mathcal{o}[x]$ and set

$$\mathfrak{P}_i = m_i(\theta)\mathcal{O} + \mathfrak{p}\mathcal{O}.$$

Then $\mathfrak{P}_i$ is a prime of $\mathcal{O}$ and

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r},$$

is the prime factorization of $\mathfrak{p}\mathcal{O}$. In particular,

$$e_\mathfrak{p}(\mathfrak{P}_i) = e_i \quad \textit{and} \quad f_\mathfrak{p}(\mathfrak{P}_i) = \deg(\overline{m_i}(x)).$$

Proof. We will being by demonstrating a chain of isomorphisms

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \mathcal{o}[\theta]/\mathfrak{p}[\theta] \cong \mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x)).$$

For the first isomorphism, consider the homomorphism

$$\phi : \mathcal{o}[\theta] \mapsto \mathcal{O}/\mathfrak{p}\mathcal{O} \qquad \alpha \mapsto \alpha + \mathfrak{p}\mathcal{O},$$

induced by natural projection. We have $\mathfrak{q}_{\mathcal{O}[\theta]} + \mathfrak{p}\mathcal{O} = \mathcal{O}$ because $\mathfrak{p}\mathcal{O}$ is relatively prime to $\mathfrak{q}_{\mathcal{O}[\theta]}$. Whence

$$\mathscr{O}[\theta] + \mathfrak{p}\mathcal{O} = \mathcal{O},$$

proving ϕ is surjective. We claim $\ker\phi = \mathfrak{p}[\theta]$. Since $\ker\phi = \mathfrak{p}\mathcal{O} \cap \mathscr{O}[\theta]$, it suffices to show

$$\mathfrak{p}\mathcal{O} \cap \mathscr{O}[\theta] = \mathfrak{p}[\theta].$$

The reverse inclusion is clear. For the forward inclusion, we claim $(\mathfrak{q}_{\mathcal{O}[\theta]} \cap \mathscr{O}) + \mathfrak{p} = \mathscr{O}$. For if not, $(\mathfrak{q}_{\mathcal{O}[\theta]} \cap \mathscr{O}) + \mathfrak{p} \subseteq \mathfrak{q}$ for some prime $\mathfrak{q}$ of $\mathscr{O}$ as primes are maximal. Letting $\mathfrak{Q}$ be a prime of $\mathcal{O}$ above $\mathfrak{q}$, we would then have $\mathfrak{q}_{\mathcal{O}[\theta]} + \mathfrak{p}\mathcal{O} \subseteq \mathfrak{Q}$ contradicting relative primality. A short computation shows

$$((\mathfrak{q}_{\mathcal{O}[\theta]} \cap \mathscr{O}) + \mathfrak{p})(\mathscr{O}[\theta] \cap \mathfrak{p}\mathcal{O}) \subseteq \mathfrak{p}[\theta],$$

from whence the reverse inclusion follows. So $\ker\phi = \mathfrak{p}[\theta]$. By first isomorphism theorem, we obtain

$$\mathscr{O}[\theta]/\mathfrak{p}[\theta] \cong \mathcal{O}/\mathfrak{p}\mathcal{O}.$$

This establishes the first isomorphism in the chain. For the second, recall from Proposition 3.1.4 that $m_\theta(x)$ has coefficients in $\mathscr{O}$ whence is an element of $\mathscr{O}[x]$. Therefore $\overline{m_\theta}(x)$ is an element of $\mathbb{F}_p[x]$ and we may consider the homomorphism

$$\pi : \mathscr{O}[x] \mapsto \mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x)) \qquad f(x) \mapsto \overline{f}(x) + (\overline{m_\theta}(x)),$$

given by the composition of natural projections. As such, π is surjective and $\ker\pi = (\mathfrak{p}, m_\theta(x))$. Whence the first isomorphism theorem gives

$$\mathscr{O}[x]/(\mathfrak{p}, m_\theta(x)) \cong \mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x)).$$

In view of the isomorphism $\mathscr{O}[\theta] \cong \mathscr{O}[x]/(m_\theta(x))$ induced by evaluation at θ and the third isomorphism theorem, the aforementioned isomorphism becomes

$$\mathscr{O}[\theta]/\mathfrak{p}[\theta] \cong \mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x)).$$

This establishes the second isomorphism in the chain.

We now determine the maximal ideals of $\mathcal{O}/\mathfrak{p}\mathcal{O}$ via this chain. As the $\overline{m}_i(x)^{e_i}$ are pairwise relatively prime, the Chinese remainder theorem gives an isomorphism

$$\mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x)) \cong \bigoplus_i \mathbb{F}_\mathfrak{p}[x]/(\overline{m_i}(x)^{e_i}),$$

induced by natural projection. Since $\overline{m_i}(x)$ is irreducible, $(\overline{m_i}(x))$ is maximal. By the fourth isomorphism theorem, $(\overline{m_i}(x))/(\overline{m_i}(x)^{e_i})$ is a maximal ideal of $\mathbb{F}_\mathfrak{p}[x]/(\overline{m_i}(x)^{e_i})$. So under the aforementioned isomorphism, the maximal ideals of $\mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x))$ are of the form $(\overline{m_i}(x))/(\overline{m_\theta}(x))$. As the chain of isomorphisms is induced by natural projections, the maximal ideals of $\mathcal{O}/\mathfrak{p}\mathcal{O}$ are of the form $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$.

It is now possible to show that the $\mathfrak{P}_i$ are the prime factors of $\mathfrak{p}\mathcal{O}$. To see this, consider the homomorphism

$$\pi : \mathcal{O} \to \mathcal{O}/\mathfrak{p}\mathcal{O} \qquad \alpha \mapsto \alpha + \mathfrak{p}\mathcal{O},$$

given by natural projection. As such, π is surjective. By definition of $\mathfrak{P}_i$, the image under π is $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$. As this ideal is maximal and hence prime, the preimage $\mathfrak{P}_i$ is prime too. Moreover, the $\mathfrak{P}_i$ are distinct since the $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$ are. In particular, the $\mathfrak{P}_i$ are also pairwise relatively prime. By construction, $\mathfrak{P}_i \mid \mathfrak{p}\mathcal{O}$ because to divide is to contain. Whence the $\mathfrak{P}_i$ are prime factors of $\mathfrak{p}\mathcal{O}$. They constitute all of the prime factors of $\mathfrak{p}\mathcal{O}$ because the image of any prime under π containing $\mathfrak{p}\mathcal{O}$ must be a maximal ideal of $\mathcal{O}/\mathfrak{p}\mathcal{O}$ by the fourth isomorphism theorem and every maximal ideal is one of the $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$. Together, all of this means that $\mathfrak{p}\mathcal{O}$ admits the prime factorization

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_\mathfrak{p}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_\mathfrak{p}(\mathfrak{P}_r)},$$

for some ramification indices $e_\mathfrak{p}(\mathfrak{P}_i)$.

It remains to compute the ramification indices and inertia degrees. We will first compute the inertia degrees and then use the fundamental equality to determine the ramification indices. From our previous work, we have an isomorphism

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x)).$$

Moreover, the primes $\mathfrak{P}_i$ of $\mathcal{O}$ correspond to the maximal ideals $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$ of $\mathcal{O}/\mathfrak{p}\mathcal{O}$ which correspond to the maximal ideals $(\overline{m_i}(x))/(\overline{m_\theta}(x))$ of $\mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x))$ under this isomorphism. By the third isomorphism theorem, we obtain the isomorphism of fields

$$\mathcal{O}/\mathfrak{P}\mathcal{O} \cong \mathbb{F}_\mathfrak{p}[x]/(\overline{m_i}(x)).$$

Both of these fields are $\mathbb{F}_\mathfrak{p}$-vector spaces where the latter clearly has degree $\deg(\overline{m_i}(x))$. Hence $f_\mathfrak{p}(\mathfrak{P}_i) = \deg(\overline{m_i}(x))$ which is the desired formula for the inertia degrees. For the ramification indices, as the chain of isomorphisms is induced by natural projections, the ideal $(\overline{m_i}(x)^{e_i})/(\overline{m_\theta}(x))$ of $\mathbb{F}_\mathfrak{p}[x]/(\overline{m_\theta}(x))$ corresponds to the ideal $m_i(\theta)^{e_i}\mathcal{O}/\mathfrak{p}\mathcal{O}$ of $\mathcal{O}/\mathfrak{p}\mathcal{O}$. Since the image of $\mathfrak{P}_i$ under π is $m_i(\theta)\mathcal{O}/\mathfrak{p}\mathcal{O}$, we have that $\mathfrak{P}_i^{e_i}$ is contained in the preimage of $m_i(\theta)^{e_i}\mathcal{O}/\mathfrak{p}\mathcal{O}$ under π. As $m_\theta(\theta) \in \mathfrak{p}\mathcal{O}$, it follows that

$$\mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r} \subseteq \mathfrak{p}\mathcal{O}.$$

Prime factorization forces $e_\mathfrak{p}(\mathfrak{P}_i) \le e_i$. But then

$$\sum_i e_\mathfrak{p}(\mathfrak{P}_i) f_p(\mathfrak{P}_i) = \sum_i e_i \deg(\overline{m_i}(x)),$$

as fundamental equality says that the first sum is sum is n while the prime factorization of $m_\theta(x)$ says that the second sum is n as well. Ths only way this is possible is if $e_\mathfrak{p}(\mathfrak{P}_i) = e_i$. This is the desired formula for the ramification indices. $\square$

As we have already noted, the irregular primes of $\mathcal{o}[\theta]$ correspond to the prime factors of $\mathfrak{q}_{\mathcal{o}[\theta]}$ of which there are finitely many as $\mathcal{O}$ is a Dedekind domain. This means that the Dedekind-Kummer theorem is valid for $\mathfrak{p}\mathcal{O}$ provided that $\mathfrak{p}$ is not below any of the prime factors of $\mathfrak{q}_{\mathcal{o}[\theta]}$. In other words, the Dedekind-Kummer theorem is valid for all but finitely many primes $\mathfrak{p}$. In fact, if $\mathfrak{q}_{\mathcal{o}[\theta]} = \mathcal{O}$ then we do not have to avoid any primes at all. This occurs when $\mathcal{O}/\mathcal{o}$ is monogenic. Indeed, Suppose $\mathcal{O} = \mathcal{o}[\theta]$

for some $\theta \in \mathcal{O}$. Then $1, \theta, \ldots, \theta^{n-1}$ is an integral basis for $\mathcal{O}/\scriptstyle\mathcal{O}$ which is necessarily a basis for L/K. This means θ is also a primitive element for L/K which implies $\mathfrak{q}_{\mathcal{O}[\theta]} = \mathcal{O}$.

Let us now discuss how ramification indices and inertia degrees are affected under localization. It turns out that they are preserved.

Proposition 3.6.2. *Let $\mathcal{O}/\scriptstyle\mathcal{O}$ be a Dedekind extension of a finite separable extension L/K and let D be a multiplicative subset of $\scriptstyle\mathcal{O}$. Then for the Dedekind extension $\mathcal{O}D^{-1}/\scriptstyle\mathcal{O}D^{-1}$, $\mathfrak{P}D^{-1}$ is above $\mathfrak{p}D^{-1}$ if and only if $\mathfrak{P}$ is above $\mathfrak{p}$ and there are isomorphisms*

$$\mathbb{F}_{\mathfrak{P}D^{-1}} \cong \mathbb{F}_{\mathfrak{P}} \quad \text{and} \quad \mathbb{F}_{\mathfrak{p}D^{-1}} \cong \mathbb{F}_{\mathfrak{p}}.$$

Lastly, we have equalities

$$f_{\mathfrak{p}D^{-1}}(\mathfrak{P}D^{-1}) = f_{\mathfrak{p}}(\mathfrak{P}) \quad \text{and} \quad e_{\mathfrak{p}D^{-1}}(\mathfrak{P}D^{-1}) = e_{\mathfrak{p}}(\mathfrak{P}).$$

Proof. In view of Proposition 3.4.1, $\mathfrak{P}$ and $\mathfrak{p}$ are necessarily disjoint from D. In fact, the inclusion-preserving bijections show that $\mathfrak{P}D^{-1}$ is above $\mathfrak{p}D^{-1}$ if and only if $\mathfrak{P}$ is above $\mathfrak{p}$. This proves the first statement. Now consider the homomorphisms

$$\Phi : \mathcal{O} \to \mathbb{F}_{\mathfrak{P}D^{-1}} \qquad \alpha \mapsto \alpha + \mathfrak{P}D^{-1},$$

and

$$\phi : \scriptstyle\mathcal{O} \to \mathbb{F}_{\mathfrak{p}D^{-1}} \qquad \alpha \mapsto \alpha + \mathfrak{p}D^{-1},$$

induced by natural projection. By the first isomorphism theorem, it suffices to show Φ and ϕ are surjective as well as $\ker\Phi = \mathfrak{P}$ and $\ker\phi = \mathfrak{p}$. We have

$$\mathcal{O} + \mathfrak{P}D^{-1} = \mathcal{O}D^{-1} \quad \text{and} \quad \scriptstyle\mathcal{O} + \mathfrak{p}D^{-1} = \scriptstyle\mathcal{O}D^{-1},$$

since $\mathfrak{P}D^{-1}$ and $\mathfrak{p}D^{-1}$ are maximal. This proves surjectivity of Φ and ϕ. Now $\ker\Phi = \mathfrak{P}D^{-1} \cap \mathcal{O}$ and $\ker\phi = \mathfrak{p}D^{-1} \cap \scriptstyle\mathcal{O}$. The inclusion-preserving bijections in Proposition 3.4.1 show that

$$\mathfrak{P}D^{-1} \cap \mathcal{O} = \mathfrak{P} \quad \text{and} \quad \mathfrak{p}D^{-1} \cap \scriptstyle\mathcal{O} = \mathfrak{p}.$$

Whence $\ker\Phi = \mathfrak{P}$ and $\ker\phi = \mathfrak{p}$ and we obtain the desired isomorphisms. In fact, these isomorphisms together imply

$$f_{\mathfrak{p}D^{-1}}(\mathfrak{P}D^{-1}) = f_{\mathfrak{p}}(\mathfrak{P}).$$

As $\mathfrak{P}D^{-1}$ is above $\mathfrak{p}D^{-1}$ if and only if $\mathfrak{P}$ is above $\mathfrak{p}$, the fundamental equality forces

$$e_{\mathfrak{p}D^{-1}}(\mathfrak{P}D^{-1}) = e_{\mathfrak{p}}(\mathfrak{P}). \qquad \square$$

In view of the isomorphisms from this result, the residue class extensions of $\mathcal{O}/\scriptstyle\mathcal{O}$ and $\mathcal{O}_{\mathfrak{p}}/\scriptstyle\mathcal{O}_{\mathfrak{p}}$ corresponding to the primes $\mathfrak{p}$ and $\mathfrak{p}_{\mathfrak{p}}$ respectively are isomorphic. Thus they are both extensions of $\mathbb{F}_{\mathfrak{p}}$-vector spaces. More generally, we have that if $\mathfrak{A}$ is an integral ideal of $\mathcal{O}$ containing $\mathfrak{p}$ then $\mathcal{O}/\mathfrak{A}$ and $\mathcal{O}_\mathfrak{p}/\mathfrak{A}_\mathfrak{p}$ are both a finite dimensional $\mathbb{F}_\mathfrak{p}$-vector spaces.

We are now ready to introduce the ideal norm of $\mathcal{O}/\mathcal{o}$. Suppose $\mathfrak{F}$ is a fractional ideal of $\mathcal{O}$ with prime factorization

$$\mathfrak{F} = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}.$$

Let $\mathfrak{p}_i$ be the prime below $\mathfrak{P}_i$. We define the **norm** $\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{F})$ of $\mathfrak{F}$ by

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{F}) = \mathfrak{p}_1^{e_1 f_{\mathfrak{p}_1}(\mathfrak{P}_1)} \cdots \mathfrak{p}_r^{e_r f_{\mathfrak{p}_r}(\mathfrak{P}_r)}.$$

Setting $\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathcal{O}) = \mathcal{o}$, prime factorization shows that we obtain a homomorphism

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}} : I(\mathcal{O}) \to I(\mathcal{o}) \qquad \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r} \mapsto \mathfrak{p}_1^{e_1 f_{\mathfrak{p}_1}(\mathfrak{P}_1)} \cdots \mathfrak{p}_r^{e_r f_{\mathfrak{p}_r}(\mathfrak{P}_r)},$$

called the **ideal norm** of $\mathcal{O}/\mathcal{o}$. This means that the ideal norm is completely multiplicative. It follows from the fundamental equality that

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{p}\mathcal{O}) = \mathfrak{p}^n,$$

and by multiplicativity of the ideal norm, any fractional ideal $\mathfrak{f}$ of $\mathcal{o}$ satisfies

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{f}\mathcal{O}) = \mathfrak{f}^n.$$

In any case, the ideal norm is aptly named because it is compatible with the field norm. To prove this, we first show that the ideal norm is compatible with localization.

Proposition 3.6.3. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and let D be a completely multiplicative subset of $\mathcal{o}$. Then*

$$\mathfrak{N}_{\mathcal{O}D^{-1}/\mathcal{o}D^{-1}}(\mathfrak{F}D^{-1}) = \mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{F})D^{-1}.$$

Proof. As the ideal norm is completely multiplicative, it suffices to prove the claim for a prime of $\mathcal{O}$. So let $\mathfrak{P}$ be such a prime and let $\mathfrak{p}$ be the prime below it. If $\mathfrak{P}$ is not disjoint from D then neither is $\mathfrak{p}$, and their localizations are $\mathcal{O}D^{-1}$ and $\mathcal{o}D^{-1}$ respectively. Whence the claim is immediate as both sides are $\mathcal{o}D^{-1}$. If $\mathfrak{P}$ is disjoint from D then the claim follows from Proposition 3.6.2. $\square$

With this result in hand, we can shows that the ideal norm is compatible with the field norm.

Proposition 3.6.4. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K. Then*

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\lambda\mathcal{O}) = \mathrm{N}_{L/K}(\lambda)\mathcal{o},$$

for any $\lambda \in L$.

Proof. In view of Propositions 3.4.2 and 3.6.3, we may assume that $\mathcal{O}/o$ is a local Dedekind extension. Moreover, as $\mathfrak{N}_{\mathcal{O}/o}$ and $\mathrm{N}_{L/K}$ are both completely multiplicative and L is the field of fractions of $\mathcal{O}$, it is enough to prove the claim when $\lambda \in \mathcal{O}$ and λ is prime. As $\mathcal{O}$ is a principal ideal domain, we have $\lambda\mathcal{O} = \mathfrak{P}$ for some prime $\mathfrak{P}$ of $\mathcal{O}$. Let $\mathfrak{p}$ be the prime of o below $\mathfrak{P}$ and let v be the valuation of o. Then

$$\mathfrak{N}_{\mathcal{O}/o}(\lambda\mathcal{O}) = \mathfrak{p}^{f_{\mathfrak{p}}(\mathfrak{P})} \quad \text{and} \quad \mathrm{N}_{L/K}(\lambda)o = \mathfrak{p}^{v(\mathrm{N}_{L/K}(\lambda))},$$

and we are reduced to proving

$$f_{\mathfrak{p}}(\mathfrak{P}) = v(\mathrm{N}_{L/K}(\lambda)).$$

As $f_{\mathfrak{p}}(\mathfrak{P})$ is the dimension of $\mathbb{F}_{\mathfrak{P}}$ as an $\mathbb{F}_{\mathfrak{p}}$-vector space, it suffices to show that the same is true of $v(\mathrm{N}_{L/K}(\lambda))$. To this end, $\mathrm{N}_{L/K}(\lambda) = \mathrm{N}_{\mathcal{O}/o}(\lambda)$ by Proposition 3.2.10. Recall that $\mathrm{N}_{\mathcal{O}/o}(\lambda)$ is the determinant of the o-linear operator T_λ on $\mathcal{O}$. As o is a principal ideal domain, putting T_λ in smith normal form shows that there is a basis of $\mathcal{O}/o$ for which the matrix of T_λ is

$$U \begin{pmatrix} \varepsilon_1\pi^{k_1} & & \\ & \ddots & \\ & & \varepsilon_n\pi^{k_n} \end{pmatrix} V,$$

for some invertible matrices $U, V \in \mathrm{GL}_n(o)$ of unit determinant, nonnegative integers k_i, units ε_i and uniformizer π for o. Setting $\varepsilon = \prod_i \varepsilon_i$ and $k = \sum_i k_i$ shows $\mathrm{N}_{\mathcal{O}/o}(\lambda) = \varepsilon\pi^k$ whence $v(\mathrm{N}_{L/K}(\lambda)) = k$. As the image of T_λ is $\mathfrak{P}$, the smith normal form induces an isomorphism

$$\mathbb{F}_{\mathfrak{P}} \cong \bigoplus_i o/\pi^{k_i}o.$$

Observe that the length of $o/\pi^{k_i}o$ as an o-module is k_i. Whence the length of $\mathbb{F}_{\mathfrak{P}}$ as an o-module is k. But then k is the dimension of $\mathbb{F}_{\mathfrak{P}}$ as an $\mathbb{F}_{\mathfrak{p}}$-vector space which is also $v(\mathrm{N}_{L/K}(\lambda))$. $\square$

In light of this result, the ideal norm preserves principal ideals. This means that the ideal norm induces a homomorphism

$$\mathfrak{N}_{\mathcal{O}/o} : \mathrm{Cl}(\mathcal{O}) \to \mathrm{Cl}(o) \qquad \mathfrak{a}P(\mathcal{O}) \mapsto \mathfrak{N}_{\mathcal{O}/o}(\mathfrak{a})P(o),$$

between ideal class groups.

Number Fields

We now turn to the setting where L/K is an extension of number fields. Recall that the underlying Dedekind extension is $\mathcal{O}_L/\mathcal{O}_K$. A prime $\mathfrak{p}$ of K is said to be **inert**, **nonsplit**, **split**, or **totally split** in L/K if it is in $\mathcal{O}_L/\mathcal{O}_K$. Primes $\mathfrak{P}$ and $\mathfrak{p}$ of L and K are said to be **unramified**, **ramified**, or **totally ramified** in L/K if they are in $\mathcal{O}_L/\mathcal{O}_K$. Lastly, the **ideal norm** $\mathfrak{N}_{L/K}$ of L/K is the ideal norm $\mathfrak{N}_{\mathcal{O}_L/\mathcal{O}_K}$.

In the case of a number field K, we can say more as $\mathcal{O}_K/\mathbb{Z}$ admits an integral basis and the reside class extensions are separable extensions of finite fields. A prime p of $\mathbb{Q}$ is said to be **inert**, **nonsplit**, **split**, or **totally split** in K if it is in $\mathcal{O}_K/\mathbb{Z}$. Primes $\mathfrak{p}$ and p of K and $\mathbb{Q}$ are said to be **unramified**, **ramified**, or **totally ramified** in K if they are in $\mathcal{O}_K/\mathbb{Z}$. In particular, if $\mathfrak{p}$ is above p then $\mathfrak{p}$ is unramified in K if and only if $e_p(\mathfrak{p}) = 1$. The **ideal norm** $\mathfrak{N}_K$ of K is the ideal norm $\mathfrak{N}_{\mathcal{O}_K/\mathbb{Z}}$. As $\mathbb{Z}$ is a principal ideal domain, every fractional ideal is principal. Therefore $\mathfrak{N}_K(\mathfrak{f})$ is generated by some $r_\mathfrak{f} \in \mathbb{Q}_+$ for every fractional ideal $\mathfrak{f}$ of $\mathcal{O}_K$. As the ideal norm is completely multiplicative, this induces a homomorphism

$$\mathrm{N}_K : I_K \to \mathbb{Q}_+ \qquad \mathfrak{f} \mapsto r_\mathfrak{f},$$

called the **norm** of K. We call $\mathrm{N}_K(\mathfrak{f})$ the **norm** of $\mathfrak{f}$. The relationship between this norm and the ideal norm is

$$\mathfrak{N}_K(\mathfrak{f}) = \mathrm{N}_K(\mathfrak{f})\mathbb{Z}.$$

In particular, if $\mathfrak{p}$ is a prime of K above p then

$$\mathfrak{N}_K(\mathfrak{p}) = p^{f_p(\mathfrak{p})}\mathbb{Z} \quad \text{and} \quad \mathrm{N}_K(\mathfrak{p}) = p^{f_p(\mathfrak{p})}.$$

For an integral ideal $\mathfrak{a}$, there is a simple relationship between the norm of $\mathfrak{a}$ and the quotient $\mathcal{O}_K/\mathfrak{a}$.

Proposition 3.6.5. *Let K be a number field of degree n. Then for any integral ideal $\mathfrak{a}$ of $\mathcal{O}_K$, we have*

$$\mathrm{N}_K(\mathfrak{a}) = |\mathcal{O}_K/\mathfrak{a}|.$$

Moreover,

$$\mathrm{N}_K(\alpha\mathcal{O}_K) = |\,\mathrm{N}_K(\alpha)|,$$

for any $\alpha \in \mathcal{O}_K$.

Proof. As K admits an integral basis, every integral ideal of K is a free $\mathbb{Z}$-module of rank n by Theorem 3.2.8. Whence any such quotient is finite. By the Chinese remainder theorem, the right-hand side of the desired formula is multiplicative. As the norm is also multiplicative, it suffices to prove the claim when $\mathfrak{a}$ is a prime power. So fix a prime power $\mathfrak{p}^e$ and let p be the prime below $\mathfrak{p}$. On the one hand, $\mathrm{N}_K(\mathfrak{p}^e) = p^{ef_p(\mathfrak{p})}$. On the other hand, consider the filtration

$$\mathcal{O}_K/\mathfrak{p}^e \supset \mathfrak{p}/\mathfrak{p}^e \supset \cdots \supset \mathfrak{p}^{e-1}/\mathfrak{p}^e \supset 0,$$

of $\mathcal{O}_K$-modules. By the third isomorphism theorem, quotients of successive layers of the filtration are isomorphic to $\mathfrak{p}^i/\mathfrak{p}^{i+1}$. Repeated application of this isomorphism and Lagrange's theorem together give

$$|\mathcal{O}_K/\mathfrak{p}^e| = \prod_i |\mathfrak{p}^i/\mathfrak{p}^{i+1}|.$$

The quotients $\mathfrak{p}^i/\mathfrak{p}^{i+1}$ are all isomorphic to $\mathbb{F}_\mathfrak{p}$ by Lemma 3.3.13. As $\mathbb{F}_\mathfrak{p}$ is an $\mathbb{F}_p$-vector space of dimension $f_p(\mathfrak{p})$, we find that $|\mathfrak{p}^i/\mathfrak{p}^{i+1}| = p^{f_p(\mathfrak{p})}$ and so $|\mathcal{O}_K/\mathfrak{p}^e| = p^{ef_p(\mathfrak{p})}$. This proves the first statement. The second statement follows from the first together with Proposition 3.6.4. $\square$

As a consequence of this result and Lagrange's theorem, any integral ideal $\mathfrak{a}$ contains its norm $\mathrm{N}_K(\mathfrak{a})$ and therefore the same holds for every fractional ideal $\mathfrak{f}$ as well.

3.7 Ramification

Much more can be said about the ramification of primes in a Dedekind extension $\mathcal{O}/\mathcal{o}$ when the associated extension L/K is Galois. Since L/K is assumed to be finite and separable, this amounts to further assuming L/K is normal. In this case, the Galois group is

$$\mathrm{Gal}(L/K) = \mathrm{Hom}_K(L, \overline{K}),$$

so that every K-embedding of L into $\overline{K}$ is an automorphism of L. By Proposition 3.2.1, the field trace and field norm of any $\lambda \in L$ are given by

$$\mathrm{Tr}_{L/K}(\lambda) = \sum_{\sigma} \sigma(\lambda) \quad \text{and} \quad \mathrm{N}_{L/K}(\lambda) = \prod_{\sigma} \sigma(\lambda),$$

where σ runs over the elements of $\mathrm{Gal}(L/K)$. As $\mathcal{o} \subseteq K$, $\mathrm{Gal}(L/K)$ fixes $\mathcal{o}$ pointwise and hence every fractional ideal of $\mathcal{o}$ as well. In fact, for any $\alpha \in \mathcal{O}$, we see that $\mathrm{Gal}(L/K)$ permutes the roots of the monic polynomial in $\mathcal{o}[x]$ which α satisfies. Whence there is an action

$$\mathrm{Gal}(L/K) \times \mathcal{O} \to \mathcal{O} \qquad (\sigma, \alpha) \mapsto \sigma(\alpha).$$

In particular, each K-automorphism σ restricts to an automorphism of $\mathcal{O}$ and therefore $\sigma(\mathfrak{P})$ is a prime of $\mathcal{O}$ if $\mathfrak{P}$ is. Moreover, if $\mathfrak{P}$ is above $\mathfrak{p}$ then so is $\sigma(\mathfrak{P})$ as it is a prime containing $\mathfrak{p}$. Whence

$$\sigma(\mathfrak{P}) \cap \mathcal{o} = \mathfrak{p}.$$

Accordingly, we say that $\sigma(\mathfrak{P})$ is **conjugate** to $\mathfrak{P}$. It turns out that the Galois group acts transitively on the primes above a given prime.

Proposition 3.7.1. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a finite Galois extension L/K and let $\mathfrak{p}$ be a prime of $\mathcal{o}$. Then* $\mathrm{Gal}(L/K)$ *acts transitively on the primes above $\mathfrak{p}$. Moreover, if $\mathfrak{p}\mathcal{O}$ has prime factorization*

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_{\mathfrak{p}}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_{\mathfrak{p}}(\mathfrak{P}_r)},$$

then

$$f_{\mathfrak{p}}(\mathfrak{P}_1) = \cdots = f_{\mathfrak{p}}(\mathfrak{P}_r) \quad \textit{and} \quad e_{\mathfrak{p}}(\mathfrak{P}_1) = \cdots = e_{\mathfrak{p}}(\mathfrak{P}_r).$$

Proof. Assume by contradiction that there exist distinct primes $\mathfrak{P}_i$ and $\mathfrak{P}_j$ above $\mathfrak{p}$ for which $\mathfrak{P}_i \neq \mathfrak{P}_j$ for any K-automorphism σ. As distinct primes are relatively prime, the Chinese remainder theorem implies the existence of an $\alpha \in \mathcal{O}$ such that

$$\alpha \equiv 1 \pmod{\sigma(\mathfrak{P}_i)} \quad \text{and} \quad \alpha \equiv 0 \pmod{\mathfrak{P}_j},$$

Now recall As $\mathrm{N}_{L/K}(\alpha) \in \mathcal{o}$ by Proposition 3.2.2. On the one hand, $\alpha \notin \sigma(\mathfrak{P}_i)$ whence $\sigma(\alpha) \notin \mathfrak{P}_i$. Then $\mathrm{N}_{L/K}(\alpha) \notin \mathfrak{P}_i$ by primality of $\mathfrak{P}_i$ and hence $\mathrm{N}_{L/K}(\alpha) \notin \mathfrak{p}$ because $\mathfrak{P}_i$ is above $\mathfrak{p}$. On the other hand, $\alpha \in \mathfrak{P}_j$ which implies $\alpha \in \sigma(\mathfrak{P}_j)$ when σ is the identity. Therefore $\mathrm{N}_{L/K}(\alpha) \in \mathfrak{P}_j$ and so $\mathrm{N}_{L/K}(\alpha) \in \mathfrak{p}$ because $\mathfrak{P}_j$ is above $\mathfrak{p}$. This gives a contradiction from which we conclude the action is transitive.

It remains to show that the ramification indices and inertia degrees of $\mathfrak{P}_i$ and $\mathfrak{P}_j$ are equal. As the action is transitive, there exists a K-automorphism σ satisfying $\sigma(\mathfrak{P}_i) = \mathfrak{P}_j$. Now σ is an automorphism of $\mathcal{O}$ fixing $\mathcal{o}$, and hence $\mathfrak{p}$, pointwise. Thus $\sigma(\mathfrak{p}\mathcal{O}) = \mathfrak{p}\mathcal{O}$ which implies $\mathfrak{P}_i^e \mid\mid \mathfrak{p}\mathcal{O}$ if and only if $\mathfrak{P}_j^e \mid\mid \mathfrak{p}\mathcal{O}$. Therefore the ramification indices of $\mathfrak{P}_i$ and $\mathfrak{P}_j$ are equal. Also, being an automorphism of $\mathcal{O}$, σ induces an isomorphism

$$\mathbb{F}_{\mathfrak{P}_i} \cong \mathbb{F}_{\mathfrak{P}_j}.$$

Therefore the inertia degrees of $\mathfrak{P}_i$ and $\mathfrak{P}_j$ are equal. $\square$

Another way to phrase this result is that the primes above $\mathfrak{p}$ are all conjugate to each other and their ramification indices and inertia degrees are all equal. We call the common ramification index e the **ramification index** of $\mathfrak{p}$ in $\mathcal{O}/\mathcal{o}$ and the common inertia degree f the **inertia degree** of $\mathfrak{p}$ in $\mathcal{O}/\mathcal{o}$. If there are r primes above $\mathfrak{p}$, the fundamental equality takes the particularly simple form

$$n = ref.$$

We define the **decomposition group** $D_{\mathfrak{p}}(\mathfrak{P})$ of $\mathfrak{P}$ over $\mathfrak{p}$ by

$$D_{\mathfrak{p}}(\mathfrak{P}) = \{\tau \in \mathrm{Gal}(L/K) : \tau(\mathfrak{P}) = \mathfrak{P}\}.$$

Equivalently, the decomposition group is the stabilizer subgroup of $\mathfrak{P}$ in the Galois group. The associated **decomposition field** $L^{D_{\mathfrak{p}}(\mathfrak{P})}$ of $\mathfrak{P}$ over $\mathfrak{p}$ is defined by

$$L^{D_{\mathfrak{p}}(\mathfrak{P})} = \{\lambda \in L : \tau(\lambda) = \lambda \text{ for } \tau \in D_{\mathfrak{p}}(\mathfrak{P})\}.$$

In other words, the decomposition field is the fixed field of L by the decomposition group and hence an intermediate field of L/K. In this case, the fundamental theorem of Galois theory says

$$\mathrm{Gal}(L/L^{D_{\mathfrak{p}}(\mathfrak{P})}) = D_{\mathfrak{p}}(\mathfrak{P}).$$

Let $\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ be the integral closure of $\mathcal{o}$ in $L^{D_{\mathfrak{p}}(\mathfrak{P})}$. Then Propositions 3.1.5 and 3.3.16 together imply that $\mathcal{O}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{o}$ is a tower of Dedekind extensions for the finite separable tower $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}/K$. Also, write $\mathfrak{P}^D$ for the prime of $L^{D_{\mathfrak{p}}(\mathfrak{P})}$ below $\mathfrak{P}$. We illustrate this relationship via the diagram

$$\begin{array}{c} \mathfrak{P} \subset \mathcal{O} \subseteq L \\ \Big| \\ \mathfrak{P}^D \subset \mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})} \subseteq L^{D_{\mathfrak{p}}(\mathfrak{P})} \\ \Big| \\ \mathfrak{p} \subset \mathcal{o} \subseteq K. \end{array}$$

The decomposition group encodes how $\mathfrak{p}\mathcal{O}$ splits into distinct prime factors. Indeed, by the orbit-stabilizer theorem the number of cosets in $\mathrm{Gal}(L/K)/D_{\mathfrak{p}}(\mathfrak{P})$ is equal to the size of the orbit of $\mathfrak{P}$ under the action of $\mathrm{Gal}(L/K)$. As the action is transitive by Proposition 3.7.1, $\sigma(\mathfrak{P})$ runs over the primes above $\mathfrak{p}$ as σ runs over a complete set of representatives for this coset space. Whence

$$\mathfrak{p}\mathcal{O} = \left(\prod_{\sigma} \sigma(\mathfrak{P})\right)^{e}.$$

Then the order of $\mathrm{Gal}(L/K)/D_{\mathfrak{p}}(\mathfrak{P})$ is exactly the number of prime factors of $\mathfrak{p}\mathcal{O}$. That is,

$$|\,\mathrm{Gal}(L/K)/D_{\mathfrak{p}}(\mathfrak{P})| = r.$$

In particular, $\mathfrak{p}$ is inert in $\mathcal{O}/\mathcal{o}$ if and only if $D_{\mathfrak{p}}(\mathfrak{P}) = \mathrm{Gal}(L/K)$ which is equivalent to $L^{D_{\mathfrak{p}}(\mathfrak{P})} = K$. Antithetically, $\mathfrak{p}$ is totally split in $\mathcal{O}/\mathcal{o}$ if and only if $D_{\mathfrak{p}}(\mathfrak{P}) = \{1\}$ which is equivalent to $L^{D_{\mathfrak{p}}(\mathfrak{P})} = L$. More generally, the previous identity and fundamental equality together imply

$$|D_{\mathfrak{p}}(\mathfrak{P})| = ef. \tag{3.7}$$

Therefore $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ has degree ef and $L^{D_{\mathfrak{p}}(\mathfrak{P})}/K$ has degree r. If $\mathfrak{p}$ inert in $\mathcal{O}/\mathcal{o}$ then $n = ef$ implying $e = 1$ and $f = n$ as we have seen. If $\mathfrak{p}$ is totally split in $\mathcal{O}/\mathcal{o}$ then $1 = ef$ so that $e = f = 1$ as we have also seen. Moreover, the decomposition group of a conjugate prime is the conjugate of decomposition group. More precisely, we have

$$D_{\mathfrak{p}}(\sigma(\mathfrak{P})) = \sigma D_{\mathfrak{p}}(\mathfrak{P})\sigma^{-1}.$$

This is simply because $\tau(\mathfrak{P}) = \mathfrak{P}$ if and only if $\sigma\tau\sigma^{-1}(\sigma(\mathfrak{P})) = \sigma(\mathfrak{P})$. Also, the fundamental theorem of Galois theory implies that $D_{\mathfrak{p}}(\mathfrak{P})$ is normal if and only if $L^{D_{\mathfrak{p}}(\mathfrak{P})}/K$ is Galois in which case

$$\mathrm{Gal}(L^{D_{\mathfrak{p}}(\mathfrak{P})}/K) \cong \mathrm{Gal}(L/K)/D_{\mathfrak{p}}(\mathfrak{P}).$$

Regardless, the decomposition field contains all of the information about the prime factors that $\mathfrak{p}\mathcal{O}$ splits into. In particular, $\mathfrak{P}^D$ is nonsplit in $\mathcal{O}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ and $\mathfrak{p}$ is totally split in $\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{o}$.

Proposition 3.7.2. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n Galois extension L/K, let $\mathfrak{p}$ be a prime of $\mathcal{o}$ with inertia degree f and ramification index e, and let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above $\mathfrak{p}$. Then $\mathfrak{P}^D$ is nonsplit in $\mathcal{O}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ and $\mathfrak{p}$ is totally split in $\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{o}$. Moreover,*

$$e_{\mathfrak{P}^D}(\mathfrak{P}) = e \quad \textit{and} \quad f_{\mathfrak{P}^D}(\mathfrak{P}) = f.$$

Proof. Recall $\mathrm{Gal}(L/L^{D_{\mathfrak{p}}(\mathfrak{P})}) = D_{\mathfrak{p}}(\mathfrak{P})$. It follows from Proposition 3.7.1 that the primes of $\mathcal{O}$ above $\mathfrak{P}^D$ are of the form $\tau(\mathfrak{P})$ for $\tau \in D_{\mathfrak{p}}(\mathfrak{P})$. These τ leave $\mathfrak{P}$ invariant which means $\mathfrak{P}^D$ is nonsplit in $\mathcal{O}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$. In view of this fact and that $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ has degree ef, the fundamental equality gives

$$ef = e_{\mathfrak{P}^D}(\mathfrak{P}) f_{\mathfrak{P}^D}(\mathfrak{P}).$$

As $\mathcal{O}/\mathcal{O}^{D_\mathfrak{p}(\mathfrak{P})}/\mathcal{o}$ is a tower of Dedekind extensions, Equation (3.6) implies

$$e = e_{\mathfrak{P}^D}(\mathfrak{P})e_\mathfrak{p}(\mathfrak{P}^D) \quad \text{and} \quad f = f_{\mathfrak{P}^D}(\mathfrak{P})f_\mathfrak{p}(\mathfrak{P}^D).$$

Combining all of these identities results in

$$1 = e_\mathfrak{p}(\mathfrak{P}^D)f_\mathfrak{p}(\mathfrak{P}^D).$$

Whence $e_\mathfrak{p}(\mathfrak{P}^D) = f_\mathfrak{p}(\mathfrak{P}^D) = 1$. As $\mathfrak{P}^D$ is above $\mathfrak{p}$, Proposition 3.7.1 implies that $\mathfrak{p}$ is totally split in $\mathcal{O}^{D_\mathfrak{p}(\mathfrak{P})}/\mathcal{o}$. This proves the first statement. The second statement follows at once from the identities above. $\square$

We can view this result as a statement that $\mathcal{O}^{D_\mathfrak{p}(\mathfrak{P})}/\mathcal{o}$ encodes all of the splitting information about $\mathfrak{p}\mathcal{O}$ while $\mathcal{O}/\mathcal{O}^{D_\mathfrak{p}(\mathfrak{P})}$ contains all of the ramification and inertia information. We will need to do more work to unpack the latter information. As $\tau \in D_\mathfrak{p}(\mathfrak{P})$ leaves $\mathcal{O}$ and $\mathfrak{P}$ invariant, it induces an automorphism $\overline{\tau}$ of the residue class field $\mathbb{F}_\mathfrak{P}$ defined by

$$\overline{\tau} : \mathbb{F}_\mathfrak{P} \to \mathbb{F}_\mathfrak{P} \qquad \alpha \mapsto \tau(\alpha) + \mathfrak{P}.$$

We then obtain a homomorphism

$$T_\mathfrak{p}^\mathfrak{P} : D_\mathfrak{p}(\mathfrak{P}) \to \mathrm{Aut}(\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}) \qquad \tau \mapsto \overline{\tau}.$$

It turns out that the residue class extensions $\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}$ are normal and that $T_\mathfrak{p}^\mathfrak{P}$ is surjective.

Proposition 3.7.3. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a finite Galois extension L/K, let $\mathfrak{p}$ be a prime of $\mathcal{o}$, and let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above $\mathfrak{p}$. Then the residue class extension $\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}$ is normal. Moreover, the homomorphism $T_\mathfrak{p}^\mathfrak{P}$ is surjective.*

Proof. As $f_\mathfrak{p}(\mathfrak{P}^D) = 1$ by Proposition 3.7.2, we have an isomorphism $\mathbb{F}_{\mathfrak{P}^D} \cong \mathbb{F}_\mathfrak{p}$. So without loss of generality, we may assume $L^{D_\mathfrak{p}(\mathfrak{P})} = K$ which is to say $D_\mathfrak{p}(\mathfrak{P}) = \mathrm{Gal}(L/K)$. For any $\alpha \in \mathcal{O}$ let $\overline{\alpha} \in \mathbb{F}_\mathfrak{P}$ be its image under natural projection. Also let $m_\alpha(x)$ and $m_{\overline{\alpha}}(x)$ be the minimal polynomials of α and $\overline{\alpha}$ over K and $\mathbb{F}_\mathfrak{p}$ respectively. As $m_\alpha(x)$ necessarily has coefficients in $\mathcal{o}$ and hence $\mathcal{O}$, let $\overline{m_\alpha}(x) \in \mathbb{F}_\mathfrak{P}[x]$ be its image under natural projection. Note that as $\overline{\alpha}$ is a root of $\overline{m_\alpha}(x)$, $m_{\overline{\alpha}}(x)$ divides $\overline{m_\alpha}(x)$ in $\mathbb{F}_\mathfrak{P}[x]$.

For the first statement, let $\alpha \in \mathcal{O}$. Then we must show that $m_{\overline{\alpha}}(x)$ splits into linear factors over $\mathbb{F}_\mathfrak{P}$. As $m_{\overline{\alpha}}(x)$ divides $\overline{m_\alpha}(x)$ in $\mathbb{F}_\mathfrak{P}[x]$, it suffices to show $\overline{m_\alpha}(x)$ splits into linear factors over $\mathbb{F}_\mathfrak{P}$. As L/K is Galois and thus normal, $m_\alpha(x)$ splits into linear factors over L. These linear factors belong to $\mathcal{O}[x]$ since their roots are the Galois conjugates of α which belong to $\mathcal{O}$. It follows that $\overline{m_\alpha}(x)$ splits into linear factors over $\mathbb{F}_\mathfrak{P}$ proving the first statement.

For the second statement, consider the maximal separable subextension of $\mathbb{F}_\mathfrak{P}/\mathbb{F}_\mathfrak{p}$. As this extension is finite, it is simple by the primitive element theorem so let $\theta \in \mathcal{O}$ be such that $\overline{\theta}$ is a primitive element. Then this subextension is $\mathbb{F}_\mathfrak{p}(\overline{\theta})/\mathbb{F}_\mathfrak{p}$. Now let

$\overline{\tau} \in \mathrm{Aut}(\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}})$. Since $\overline{\tau}$ fixes $\mathbb{F}_{\mathfrak{p}}$ pointwise, $\overline{\tau}(\overline{\theta})$ is a root of $m_{\overline{\theta}}(x)$. As $m_{\overline{\theta}}(x)$ divides $\overline{m_\theta}(x)$ in $\mathbb{F}_{\mathfrak{P}}[x]$, we find that $\overline{\tau}(\overline{\theta})$ is also a root of $\overline{m_\theta}(x)$. Therefore there is a root θ' of $m_\theta(x)$ whose image under natural projection is $\overline{\tau}(\overline{\theta})$. Because L/K is Galois with Galois group $D_{\mathfrak{p}}(\mathfrak{P})$, we can find $\tau \in D_{\mathfrak{p}}(\mathfrak{P})$ such that $\tau(\theta) = \theta'$. Then the image of $\overline{\theta}$ under $T_{\mathfrak{p}}^{\mathfrak{P}}(\tau)$ and $\overline{\tau}$ are the same whence these automorphisms are identical on $\mathbb{F}_{\mathfrak{p}}$. As $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}(\overline{\theta})$ is purely inseparable, it has trivial automorphism group. This forces $T_{\mathfrak{p}}^{\mathfrak{P}}(\tau) = \overline{\tau}$ proving surjectivity. □

As this result shows that $T_{\mathfrak{p}}^{\mathfrak{P}}$ is surjective, we define the **inertia group** $I_{\mathfrak{p}}(\mathfrak{P})$ of $\mathfrak{P}$ over $\mathfrak{p}$ by

$$I_{\mathfrak{p}}(\mathfrak{P}) = \ker T_{\mathfrak{p}}^{\mathfrak{P}}.$$

Being the kernel of a homomorphism, $I_{\mathfrak{p}}(\mathfrak{P})$ is a normal subgroup of $D_{\mathfrak{p}}(\mathfrak{P})$. The associated **inertia field** $L^{I_{\mathfrak{p}}(\mathfrak{P})}$ of $\mathfrak{P}$ over $\mathfrak{p}$ is defined to be

$$L^{I_{\mathfrak{p}}(\mathfrak{P})} = \{\lambda \in L : \tau(\lambda) = \lambda \text{ for } \tau \in I_{\mathfrak{p}}(\mathfrak{P})\}.$$

In other words, the inertia field of is the fixed field of L by the inertia group and hence an intermediate field of L/K. In particular, the fundamental theorem of Galois theory gives

$$\mathrm{Gal}(L/L^{I_{\mathfrak{p}}(\mathfrak{P})}) = I_{\mathfrak{p}}(\mathfrak{P}),$$

$L^{I_{\mathfrak{p}}(\mathfrak{P})}$ is an intermediate field of the Galois extension $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ as $I_{\mathfrak{p}}(\mathfrak{P})$ is a normal subgroup of $D_{\mathfrak{p}}(\mathfrak{P})$, and

$$\mathrm{Gal}(L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}) \cong D_{\mathfrak{p}}(\mathfrak{P})/I_{\mathfrak{p}}(\mathfrak{P}).$$

Let $\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$ be the integral closure of $\mathcal{o}$ in $L^{I_{\mathfrak{p}}(\mathfrak{P})}$. Then Propositions 3.1.5 and 3.3.16 together imply that $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ is a tower of Dedekind extensions for the finite separable tower $L/L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}$. We illustrate this relationship via the diagram

$$\begin{array}{c}
\mathfrak{P} \subset \mathcal{O} \subseteq L \\
\Big| \\
\mathfrak{P}^I \subset \mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})} \subseteq L^{I_{\mathfrak{p}}(\mathfrak{P})} \\
\Big| \\
\mathfrak{P}^D \subset \mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})} \subseteq L^{D_{\mathfrak{p}}(\mathfrak{P})}.
\end{array}$$

The inertia field is closely related to the automorphism group of the residue class extension $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$. Indeed, as $T_{\mathfrak{p}}^{\mathfrak{P}}$ is surjective by Proposition 3.7.3, the first isomorphism theorem shows

$$\mathrm{Gal}(L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}) \cong \mathrm{Aut}(\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}).$$

In fact, the decomposition and inertia groups of $\mathfrak{P}$ over $\mathfrak{p}$ fit into a short exact sequence.

Proposition 3.7.4. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a finite Galois extension L/K, let $\mathfrak{p}$ be a prime of $\mathcal{o}$, and let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above $\mathfrak{p}$. Then there is a short exact sequence*

$$1 \longrightarrow I_{\mathfrak{p}}(\mathfrak{P}) \longrightarrow D_{\mathfrak{p}}(\mathfrak{P}) \longrightarrow \mathrm{Aut}(\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}) \longrightarrow 1,$$

induced by natural inclusions and projections and where the third map is given by $\tau \mapsto T_{\mathfrak{p}}^{\mathfrak{P}}(\tau)$.

Proof. As the second map is injective and the third map is surjective by Proposition 3.7.3, the sequence is exact at $I_{\mathfrak{p}}(\mathfrak{P})$ and $\mathrm{Aut}(\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}})$. So it remains to prove exactness at $D_{\mathfrak{p}}(\mathfrak{P})$. This follows because $I_{\mathfrak{p}}(\mathfrak{P})$ is the kernel of $T_{\mathfrak{p}}^{\mathfrak{P}}$ by definition. □

We can say more when the residue class extension $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is separable. For in this case, Proposition 3.7.3 implies $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is Galois. Whence

$$\mathrm{Gal}(L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}) \cong \mathrm{Gal}(\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}).$$

Then Equation (3.7) and this isomorphism together imply

$$|\,\mathrm{Gal}(L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})})| = f \quad \text{and} \quad |I_{\mathfrak{p}}(\mathfrak{P})| = e.$$

It follows that $L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ has degree f and $L/L^{I_{\mathfrak{p}}(\mathfrak{P})}$ has degree e as $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ has degree ef. In particular, $\mathfrak{P}$ is totally ramified in $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$ and $\mathfrak{P}^D$ is inert in $\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$.

Proposition 3.7.5. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a finite Galois extension L/K, let $\mathfrak{p}$ be a prime of $\mathcal{o}$ with inertia degree f and ramification index e, and let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above $\mathfrak{p}$. Moreover, suppose the residue class extension $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is separable. Then $\mathfrak{P}$ is totally ramified in $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$ and $\mathfrak{P}^D$ is inert in $\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$.*

Proof. Recall that $\mathrm{Gal}(L/L^{I_{\mathfrak{p}}(\mathfrak{P})}) = I_{\mathfrak{p}}(\mathfrak{P})$. By Proposition 3.7.1, the primes of $\mathcal{O}$ above $\mathfrak{P}^I$ are of the form $\tau(\mathfrak{P})$ for $\tau \in I_{\mathfrak{p}}(\mathfrak{P})$. These τ leave $\mathfrak{P}$ invariant whence $\mathfrak{P}^I$ is nonsplit in $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$. In view of this fact and that $L/L^{I_{\mathfrak{p}}(\mathfrak{P})}$ has degree e, the fundamental equality implies

$$e = e_{\mathfrak{P}^I}(\mathfrak{P}) f_{\mathfrak{P}^I}(\mathfrak{P}).$$

So $\mathfrak{P}^I$ is totally ramified in $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$ if $f_{\mathfrak{P}^I}(\mathfrak{P}) = 1$. As $\mathfrak{P}^I$ is nonsplit in $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}$, we have $D_{\mathfrak{P}^I}(\mathfrak{P}) = I_{\mathfrak{p}}(\mathfrak{P})$. Whence $I_{\mathfrak{P}^I}(\mathfrak{P}) = I_{\mathfrak{p}}(\mathfrak{P})$. In short, the inertia and decomposition groups are the same. By Proposition 3.7.4, the only way this is possible is if $\mathrm{Gal}(\mathbb{F}_{\mathfrak{P}^I}/\mathbb{F}_{\mathfrak{P}}) = \{1\}$. This forces $f_{\mathfrak{P}^I}(\mathfrak{P}) = 1$ as desired. As $L/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ has degree ef and $\mathfrak{P}^D$ is nonsplit in $\mathcal{O}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ by Proposition 3.7.2, the fundamental equality implies

$$ef = e_{\mathfrak{P}^D}(\mathfrak{P}) f_{\mathfrak{P}^D}(\mathfrak{P}).$$

Since $\mathcal{O}/\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$ is a tower of Dedekind extensions, Equation (3.6) gives

$$e_{\mathfrak{P}^D}(\mathfrak{P}) = e_{\mathfrak{P}^I}(\mathfrak{P}) e_{\mathfrak{P}^D}(\mathfrak{P}^I) \quad \text{and} \quad f_{\mathfrak{P}^D}(\mathfrak{P}) = f_{\mathfrak{P}^I}(\mathfrak{P}) f_{\mathfrak{P}^D}(\mathfrak{P}^I).$$

Upon combining these identities and our previous work, $f_{\mathfrak{P}^D}(\mathfrak{P}^I) = f$ which is the degree of $L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}$. Therefore $\mathfrak{P}^D$ is inert in $\mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})}/\mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})}$. □

In the case the residue class extension $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is separable, we fit the decomposition and inertia fields into the diagram

$$\begin{array}{c} \mathfrak{P} \subset \mathcal{O} \subseteq L \\ {\scriptstyle e}\Big|{\scriptstyle 1} \\ \mathfrak{P}^I \subset \mathcal{O}^{I_{\mathfrak{p}}(\mathfrak{P})} \subseteq L^{I_{\mathfrak{p}}(\mathfrak{P})} \\ {\scriptstyle 1}\Big|{\scriptstyle f} \\ \mathfrak{P}^D \subset \mathcal{O}^{D_{\mathfrak{p}}(\mathfrak{P})} \subseteq L^{D_{\mathfrak{p}}(\mathfrak{P})} \\ {\scriptstyle 1}\Big|{\scriptstyle 1} \\ \mathfrak{p} \subset \mathcal{o} \subseteq K. \end{array}$$

The labels on left and right of the extensions represent the corresponding ramification indices and inertia degrees respectively which come from Propositions 3.7.2 and 3.7.5. So the extension $L^{D_{\mathfrak{p}}(\mathfrak{P})}/K$ contains all of the information about the splitting, the extension $L^{I_{\mathfrak{p}}(\mathfrak{P})}/L^{D_{\mathfrak{p}}(\mathfrak{P})}$ contains all of the information about the inertia degrees, and the extension $L/L^{I_{\mathfrak{p}}(\mathfrak{P})}$ contains all of the information about the ramification indices. In particular, $\mathfrak{p}$ is unramified in $\mathcal{O}/\mathcal{o}$ if and only if $L^{I_{\mathfrak{p}}(\mathfrak{P})} = L$ which is equivalent to $I_{\mathfrak{p}}(\mathfrak{P}) = \{1\}$.

Number Fields

In the case of a Galois extension of number fields L/K and a prime $\mathfrak{p}$ of K, the **ramification index** e and **inertia degree** f of $\mathfrak{p}$ in L/K is that of $\mathcal{O}_L/\mathcal{O}_K$. Recall that all of the residue class extensions are separable. This means that the above diagram applies to L/K.

3.8 Differents and Discriminants

Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K. We being by introducing the notion of lattices in this setting. We say that $\mathfrak{L}$ is an **$\mathcal{o}$-lattice** of L if it is a finitely generated $\mathcal{o}$-submodule of L. Moreover, we say that $\mathfrak{L}$ is **complete** if it spans L/K. That is, $\mathfrak{L}$ contains a basis of L/K. It is also possible to construct duals to complete $\mathcal{o}$-lattices. For this, we need a nondegenerate symmetric bilinear form. The natural choice is the trace form

$$\mathrm{Tr}_{L/K} : L \times L \to K \qquad (\lambda, \eta) \mapsto \mathrm{Tr}_{L/K}(\lambda\eta).$$

Then any basis $\lambda_1, \ldots, \lambda_n$ for L/K admits a dual basis $\lambda_1^\vee, \ldots, \lambda_n^\vee$ defined by

$$\mathrm{Tr}_{L/K}(\lambda_i\lambda_j^\vee) = \delta_{i,j},$$

If $\mathfrak{L}$ is complete, then we define the **dual** $\mathfrak{L}^\vee$ of $\mathfrak{L}$ by

$$\mathfrak{L}^\vee = \{\lambda \in L : \mathrm{Tr}_{L/K}(\lambda\mathfrak{L}) \subseteq \mathcal{o}\}.$$

We say that $\mathfrak{L}$ is **self-dual** if $\mathfrak{L}^\vee = \mathfrak{L}$. The dual $\mathfrak{L}^\vee$ is indeed a complete $\mathcal{o}$-lattice of L. To see this, let $\lambda_1, \ldots, \lambda_n$ be a basis for L/K contained in $\mathfrak{L}$. For any $\lambda \in L$ we may write

$$\lambda = \sum_j \kappa_j^\vee \lambda_j^\vee,$$

with $\kappa_j^\vee \in K$. As the dual basis is determined by $\mathrm{Tr}_{L/K}(\lambda_i \lambda_j^\vee) = \delta_{i,j}$, it follows that $\lambda \in \mathfrak{L}^\vee$ if and only if $\kappa_j^\vee \in \mathcal{o}$. Whence $\lambda_1^\vee, \ldots, \lambda_n^\vee$ is a basis for $\mathfrak{L}^\vee$ as an $\mathcal{o}$-submodule of L and thus $\mathfrak{L}^\vee$ is a complete $\mathcal{o}$-lattice of L. As the dual of the dual basis is the original basis, $(\mathfrak{L}^\vee)^\vee = \mathfrak{L}$. The prototypical complete $\mathcal{o}$-lattice of L is $\mathcal{O}$ whence it admits a dual $\mathcal{O}^\vee$.

Observe that an $\mathcal{o}$-lattice of L is not necessarily a fractional ideal of $\mathcal{O}$ as it need not be an $\mathcal{O}$-submodule of L. However, every fractional ideal $\mathfrak{F}$ of $\mathcal{O}$ is a complete $\mathcal{o}$-lattice of L. Indeed, $\mathfrak{F}$ is a finitely generated $\mathcal{o}$-submodule of L as $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module. Moreover, $\mathfrak{F}$ contains a basis of L/K. To see this, let $\lambda_1, \ldots, \lambda_n$ be a basis for L/K contained in $\mathcal{O}$ and let $\alpha \in \mathfrak{F}$ be nonzero. Then $\alpha\lambda_1, \ldots, \alpha\lambda_n$ is a basis for L/K contained in $\mathfrak{F}$. As $\mathfrak{F}$ is complete, it admits a dual $\mathfrak{F}^\vee$. The dual $\mathfrak{F}^\vee$ is also a fractional ideal of $\mathcal{O}$. To see that it is an $\mathcal{O}$-submodule of L, we have $\mathrm{Tr}_{L/K}(\mathcal{O}\mathfrak{F}) \subseteq \mathcal{o}$ by Proposition 3.2.2 upon writing $\mathfrak{F} = \frac{1}{\delta}\mathfrak{A}$ for some nonzero $\delta \in \mathcal{O}$ and integral ideal $\mathfrak{A}$. Whence $\mathfrak{F}^\vee$ is a finitely generated $\mathcal{O}$-submodule of L as $\mathcal{O}$ is a finitely generated $\mathcal{o}$-module. In fact, there is a simple relationship between the duals $\mathfrak{F}^\vee$ and $\mathcal{O}^\vee$ given by

$$\mathfrak{F}^\vee = \mathfrak{F}^{-1}\mathcal{O}^\vee.$$

For the forward inclusion, we have $\mathrm{Tr}_{L/K}(\mathfrak{F}^\vee\mathfrak{F}) \subseteq \mathcal{o}$. Whence $\mathfrak{F}^\vee\mathfrak{F} \subseteq \mathcal{O}^\vee$ which is equivalent to $\mathfrak{F}^\vee \subseteq \mathfrak{F}^{-1}\mathcal{O}^\vee$. This proves the forward inclusion. For the reverse inclusion, observe that $\mathrm{Tr}_{L/K}(\mathcal{O}^\vee) \subseteq \mathcal{o}$. Thus $\mathfrak{F}^{-1}\mathcal{O}^\vee \subseteq \mathfrak{F}^\vee$ proving the reverse inclusion.

Also, we might hope that localization respects duals of fractional ideals in order to make their study more approachable. This turns out to be true and is a very useful property.

Proposition 3.8.1. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and let D be a multiplicative subset of $\mathcal{o}$. Then for any fractional ideal $\mathfrak{F}$ of $\mathcal{O}$, we have*

$$(\mathfrak{F}D^{-1})^\vee = \mathfrak{F}^\vee D^{-1}.$$

Proof. As D is a subset of K, we have

$$\mathrm{Tr}_{L/K}(\mathfrak{F}^\vee\mathfrak{F}D^{-1}) = \mathrm{Tr}_{L/K}(\mathfrak{F}^\vee\mathfrak{F})D^{-1}.$$

It follows that $\mathrm{Tr}_{L/K}(\mathfrak{F}^\vee\mathfrak{F}D^{-1}) \subseteq \mathcal{o}D^{-1}$ if and only if $\mathrm{Tr}_{L/K}(\mathfrak{F}^\vee\mathfrak{F})D^{-1} \subseteq \mathcal{o}D^{-1}$. The forward and reverse inclusions both follows from this which proves the claim. $\square$

In a similar spirit to Proposition 3.4.3, this result says that dualization and localization are commutative. So when dualizing and localizing we do not need to specify the order in which these operations are performed.

We now turn to the construction of two specific integral ideals which encode many properties of fractional ideals, ramification of primes, and $\mathcal{o}$-lattices of L. In the study of these integral ideals, it will be revealed that it is an exceptionally rare phenomena for primes of $\mathcal{O}$ and $\mathcal{o}$ to ramify. The two integral ideals we will construct, one of $\mathcal{O}$ and the other of $\mathcal{o}$, will encode the ramification of the corresponding primes. These integral ideals are the different and discriminant respectively. We being by defining the **complement** $\mathfrak{C}_{\mathcal{O}/\mathcal{o}}$ of $\mathcal{O}/\mathcal{o}$ by

$$\mathfrak{C}_{\mathcal{O}/\mathcal{o}} = \mathcal{O}^\vee.$$

As we have seen, this is a fractional ideal of $\mathcal{O}$ and hence a complete $\mathcal{o}$-lattice of L. The **different** $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ of $\mathcal{O}/\mathcal{o}$ is defined to be

$$\mathfrak{D}_{\mathcal{O}/\mathcal{o}} = (\mathcal{O}^\vee)^{-1}.$$

In other words, the different is the inverse of the complement and so

$$\mathfrak{D}_{\mathcal{O}/\mathcal{o}} = \{\lambda \in L : \lambda\mathcal{O}^\vee \subseteq \mathcal{O}\}.$$

As $\mathcal{O} \subseteq \mathcal{O}^\vee$ by Proposition 3.2.2, inverting shows $\mathfrak{D}_{\mathcal{O}/\mathcal{o}} \subseteq \mathcal{O}$. Therefore the different $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ is an integral ideal of $\mathcal{O}$ and we have a chain of inclusions

$$\mathfrak{D}_{\mathcal{O}/\mathcal{o}} \subseteq \mathcal{O} \subseteq \mathcal{O}^\vee.$$

From this chain we see that the different is a measure of how much $\mathcal{O}$ fails to be self-dual since $\mathfrak{D}_{\mathcal{O}/\mathcal{o}} = \mathcal{O}^\vee$ if and only if $\mathcal{O}$ is self-dual. In terms of the different, the dual $\mathfrak{F}^\vee$ of a fractional ideal $\mathfrak{F}$ of $\mathcal{O}$ may be expressed as

$$\mathfrak{F}^\vee = \mathfrak{F}^{-1}\mathfrak{D}_{\mathcal{O}/\mathcal{o}}^{-1}. \tag{3.8}$$

As for other properties of the different, it respects localization which will be useful in further developments.

Proposition 3.8.2. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and let D be a multiplicative subset of $\mathcal{o}$. Then*

$$\mathfrak{D}_{\mathcal{O}D^{-1}/\mathcal{o}D^{-1}} = \mathfrak{D}_{\mathcal{O}/\mathcal{o}}D^{-1}.$$

Proof. By definition of the different, it is equivalent to show

$$((\mathcal{O}D^{-1})^\vee)^{-1} = (\mathcal{O}^\vee)^{-1}D^{-1}.$$

This identity follows from Propositions 3.4.3 and 3.8.1. $\square$

We define the **discriminant** $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ of $\mathcal{O}/\mathcal{o}$ to be the integral ideal of $\mathcal{o}$ generated by the discriminants $d_{L/K}(\lambda_1, \ldots, \lambda_n)$ as $\lambda_1, \ldots, \lambda_n$ run over all bases of L/K contained in $\mathcal{O}$. Recall that if $\mathcal{O}/\mathcal{o}$ admits an integral basis then $\mathcal{O}$ is a free $\mathcal{o}$-module of rank

n. In this case, bases of L/K contained in $\mathcal{O}$ are precisely the integral bases of $\mathcal{O}/\mathcal{o}$. Then

$$\mathfrak{d}_{\mathcal{O}/\mathcal{o}} = d_{\mathcal{O}/\mathcal{o}}\mathcal{o},$$

so that the discriminant $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ is generated by the discriminant $d_{\mathcal{O}/\mathcal{o}}$. In fact, this means

$$\mathfrak{d}_{\mathcal{O}/\mathcal{o}} = d_{\mathcal{O}/\mathcal{o}}(\alpha_1, \ldots, \alpha_n)\mathcal{o},$$

for any integral basis $\alpha_1, \ldots, \alpha_n$ for $\mathcal{O}/\mathcal{o}$. Whence $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ is principal if $\mathcal{O}/\mathcal{o}$ admits an integral basis. In view of Theorem 3.2.8 this will hold if $\mathcal{o}$ is a principal ideal domain.

As for properties of the discriminant, we have the very useful fact that it respects localization just as the different does.

Proposition 3.8.3. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and let D be a multiplicative subset of $\mathcal{o}$. Then*

$$\mathfrak{d}_{\mathcal{O}D^{-1}/\mathcal{o}D^{-1}} = \mathfrak{d}_{\mathcal{O}/\mathcal{o}}D^{-1}.$$

Proof. For the forward inclusion, suppose $\frac{\lambda_1}{\delta_1}, \ldots, \frac{\lambda_n}{\delta_n}$ is a basis of L/K contained in $\mathcal{O}D^{-1}$. Upon setting $\delta = \prod_i \delta_i$, we see that $\frac{\lambda_1\delta}{\delta_1}, \ldots, \frac{\lambda_n\delta}{\delta_n}$ is a basis of L/K contained in $\mathcal{O}$. Then $d_{L/K}\left(\frac{\lambda_1\delta}{\delta_1}, \ldots, \frac{\lambda_n\delta}{\delta_n}\right) \in \mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ whence $d_{L/K}\left(\frac{\lambda_1}{\delta_1}, \ldots, \frac{\lambda_n}{\delta_n}\right) \in \mathfrak{d}_{\mathcal{O}/\mathcal{o}}D^{-1}$. This shows the forward inclusion. For the reverse inclusion, suppose $\frac{\lambda_1}{\delta}, \ldots, \frac{\lambda_n}{\delta}$ is such that $\lambda_1, \ldots, \lambda_n$ is a basis of L/K contained in $\mathcal{O}$. Then $\frac{\lambda_1}{\delta}, \ldots, \frac{\lambda_n}{\delta}$ is a basis for L/K contained in $\mathcal{O}D^{-1}$. So $d_{L/K}\left(\frac{\alpha_1}{\delta}, \ldots, \frac{\alpha_n}{\delta}\right) \in \mathfrak{d}_{\mathcal{O}D^{-1}/\mathcal{o}D^{-1}}$ whence $d_{L/K}(\lambda_1, \ldots, \lambda_n) \in \mathfrak{d}_{\mathcal{O}/\mathcal{o}}D^{-1}$. This proves the reverse inclusion. $\square$

The different and discriminant are important integral ideals because they keep track of which primes ramify. In fact, we will show that primes of $\mathcal{O}$ and $\mathcal{o}$ ramify in $\mathcal{O}/\mathcal{o}$ if and only they divide the different $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ and discriminant $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ respectively provided the residue class extensions are separable. To this end, we will require a lemma.

Lemma 3.8.4. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and let $\mathfrak{A}$ be an integral ideal of $\mathcal{O}$. Then $\mathfrak{A} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $\mathrm{Tr}_{L/K}(\mathfrak{A}^{-1}) \subseteq \mathcal{o}$.*

Proof. Since to divide is to contain, $\mathfrak{A} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $\mathfrak{D}_{\mathcal{O}/\mathcal{o}} \subseteq \mathfrak{A}$. This equivalent to $(\mathcal{O}^\vee)^{-1} \subseteq \mathfrak{A}$ and inverting shows the further equivalence $\mathfrak{A}^{-1} \subseteq \mathcal{O}^\vee$. This last inclusion is equivalent to $\mathrm{Tr}_{L/K}(\mathfrak{A}^{-1}) \subseteq \mathcal{o}$. $\square$

With this result, we can now show that a prime of $\mathcal{O}$ is ramified in $\mathcal{O}/\mathcal{o}$ if and only if it divides the discriminant $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ provided the residue class extensions are separable. In fact, we will prove something slightly stronger.

Theorem 3.8.5. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and assume that all residue class extensions are separable. Let $\mathfrak{P}$ be a prime of $\mathcal{O}$ above a prime $\mathfrak{p}$ of $\mathcal{o}$. Then $\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})-1} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$. Moreover, $\mathfrak{P}^{e_\mathfrak{p}(\mathfrak{P})} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $e_\mathfrak{p}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$. In particular, $\mathfrak{P}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if $\mathfrak{P} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$.*

Proof. In view of Propositions 3.4.2, 3.6.2 and 3.8.3, we may assume that $\mathcal{O}/\mathcal{o}$ is a local Dedekind extension. Let e be a nonnegative integer with $e \le e_{\mathfrak{p}}(\mathfrak{P})$ and write $\mathfrak{p}\mathcal{O} = \mathfrak{P}^e\mathfrak{A}$ for some integral ideal $\mathfrak{A}$ of $\mathcal{O}$. By Lemma 3.8.4, $\mathfrak{P}^e \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $\mathrm{Tr}_{L/K}(\mathfrak{P}^e) \subseteq \mathcal{o}$. As $\mathfrak{P}^e = \mathfrak{p}^{-1}\mathfrak{A}$, it is further equivalent to show $\mathrm{Tr}_{L/K}(\mathfrak{A}) \subseteq \mathfrak{p}$. Since $\mathcal{O}/\mathcal{o}$ admits an integral basis, we have $\mathrm{Tr}_{K/L}(\mathfrak{A}) = \mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\mathfrak{A})$ from Proposition 3.2.10. Whence it is further equivalent to show $\mathrm{Tr}_{\mathcal{O}/\mathcal{o}}(\mathfrak{A}) \subseteq \mathfrak{p}$ which is to say

$$\mathrm{Tr}_{(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_{\mathfrak{p}}}(\overline{\alpha}) = 0,$$

for all $\alpha \in \mathfrak{A}$ where $\overline{\alpha} \in \mathfrak{A}/\mathfrak{p}\mathcal{O}$ is its image under natural projection.

To prove $\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})-1} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$, take $e = e_{\mathfrak{p}}(\mathfrak{P}) - 1$ so that $\mathfrak{P} \mid\mid \mathfrak{A}$. Then the prime factors of $\mathfrak{A}$ and $\mathfrak{p}\mathcal{O}$ are the same. Whence there is a positive integer k such that $\mathfrak{A}^k$ is divisible by $\mathfrak{p}\mathcal{O}$. So $\mathfrak{p}\mathcal{O} \mid \mathfrak{A}^k$ and therefore $\mathfrak{A}^k \subseteq \mathfrak{p}\mathcal{O}$ as to divide is to contain. It follows that the k-th power of the $\mathbb{F}_{\mathfrak{p}}$-linear operator $T_{\overline{\alpha}}$ on $\mathcal{O}/\mathfrak{p}\mathcal{O}$ is the zero operator. This means $T_{\overline{\alpha}}$ is nilpotent. As nilpotent operators have trace zero, we have $\mathrm{Tr}_{(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_{\mathfrak{p}}}(\overline{\alpha}) = 0$. This proves the first statement.

To prove $\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $e_{\mathfrak{p}}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$, take $e = e_{\mathfrak{p}}(\mathfrak{P})$ so that $\mathfrak{P}$ and $\mathfrak{A}$ are relatively prime. The Chinese remainder theorem gives an isomorphism

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong (\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})}) \oplus (\mathcal{O}/\mathfrak{A}),$$

induced by natural projection. So $\mathfrak{A}/\mathfrak{p}\mathcal{O} \cong \mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})}$ and it is further equivalent to show

$$\mathrm{Tr}_{(\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})})/\mathbb{F}_{\mathfrak{p}}}(\overline{\beta}) = 0,$$

for all $\beta \in \mathcal{O}$ where $\overline{\beta} \in \mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})}$ is its image under natural projection. We will prove that this occurs if and only if $e_{\mathfrak{p}}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$. To this end, consider the filtration

$$\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \supset \mathfrak{P}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \supset \cdots \supset \mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})-1}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})} \supset 0,$$

of $\mathbb{F}_{\mathfrak{p}}$-vector spaces. Upon choosing a basis for $\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})}$ adapted to the filtration, Lemma 3.6.1 gives an isomorphism

$$\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P}_i)} \cong \bigoplus_i \mathfrak{P}^i/\mathfrak{P}^{i+1},$$

induced by natural projection of the adapted basis. Now let $\overline{\beta}_i \in \mathfrak{P}^i/\mathfrak{P}^{i+1}$ be the images of β under the aforementioned isomorphism. Then

$$\mathrm{Tr}_{(\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})})/\mathbb{F}_{\mathfrak{p}}}(\overline{\beta}) = \sum_i \mathrm{Tr}_{(\mathfrak{P}^i/\mathfrak{P}^{i+1})/\mathbb{F}_{\mathfrak{p}}}(\overline{\beta}_i).$$

The quotients $\mathfrak{P}^i/\mathfrak{P}^{i+1}$ are all isomorphic to $\mathbb{F}_{\mathfrak{P}}$ by Lemma 3.3.13 which is induced by multiplication and natural projection. In particular, this isomorphism is an $\mathbb{F}_{\mathfrak{p}}$-linear operator induced by multiplication and therefore the $\mathbb{F}_{\mathfrak{p}}$-linear operators $T_{\overline{\beta}_i}$ and $T_{\overline{\beta}_0}$ on $\mathfrak{P}^i/\mathfrak{P}^{i+1}$ and $\mathcal{O}/\mathfrak{P}$ respectively are conjugate via this isomorphism. As the trace is invariant under conjugation, we have

$$\mathrm{Tr}_{(\mathfrak{P}^i/\mathfrak{P}^{i+1})/\mathbb{F}_{\mathfrak{p}}}(\overline{\beta}_i) = \mathrm{Tr}_{\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}}(\overline{\beta}_0),$$

which implies

$$\operatorname{Tr}_{(\mathcal{O}/\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})})/\mathbb{F}_{\mathfrak{p}}}(\overline{\beta}) = e_{\mathfrak{p}}(\mathfrak{P}) \operatorname{Tr}_{\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}}(\overline{\beta}_0).$$

Since the residue class extension $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is assumed to be separable, the trace form of $\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_{\mathfrak{p}}$ is nondegenerate. As both sides are elements of $\mathbb{F}_{\mathfrak{p}}$, the left-hand side is zero for all $\beta \in \mathcal{O}$ if and only if $e_{\mathfrak{p}}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$ proving the second statement.

It remains to show $\mathfrak{P}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if $\mathfrak{P} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$. As all residue class extensions are separable, $\mathfrak{P}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if $e_{\mathfrak{p}}(\mathfrak{P}) \geq 2$. But the previous two statements together imply that $\mathfrak{P} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ if and only if $e_{\mathfrak{p}}(\mathfrak{P}) \geq 2$. Whence $\mathfrak{P}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if $\mathfrak{P} \mid \mathfrak{D}_{\mathcal{O}/\mathcal{o}}$. □

In the case the reside class extensions are separable, note that this result implies $\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})-1}$ exactly divides $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ provided $e_{\mathfrak{p}}(\mathfrak{P}) \not\equiv 0 \pmod{\mathfrak{p}}$. However, when $e_{\mathfrak{p}}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$ we only know $\mathfrak{P}^{e_{\mathfrak{p}}(\mathfrak{P})}$ divides $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$. So $\mathfrak{P}$ very well might divide $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ to a higher power. The condition $e_{\mathfrak{p}}(\mathfrak{P}) \equiv 0 \pmod{\mathfrak{p}}$ is equivalent to the characteristic of $\mathbb{F}_{\mathfrak{p}}$ dividing $e_{\mathfrak{p}}(\mathfrak{P})$. Accordingly, if $\mathfrak{P}$ is a ramified prime of $\mathcal{O}$ then $\mathfrak{P}$ is said to be **tamely ramified** if the characteristic of $\mathbb{F}_{\mathfrak{p}}$ does not divide $e_{\mathfrak{p}}(\mathfrak{P})$ and **widely ramified** otherwise. In any case, upon factorizing $\mathfrak{D}_{\mathcal{O}/\mathcal{o}}$ into a product of primes we see that only finitely many primes of $\mathcal{O}$ can ramify in $\mathcal{O}/\mathcal{o}$.

Similarly to the previous result, a prime $\mathfrak{p}$ of $\mathcal{o}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if it divides the discriminant $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ provided the residue class extensions are separable.

Theorem 3.8.6. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K and assume that all residue class extensions are separable. Then a prime $\mathfrak{p}$ of $\mathcal{o}$ ramifies in $\mathcal{O}/\mathcal{o}$ if and only if $\mathfrak{p} \mid \mathfrak{d}_{\mathcal{O}/\mathcal{o}}$.*

Proof. In view of Propositions 3.4.2, 3.6.2 and 3.8.3, it suffices to assume $\mathcal{O}/\mathcal{o}$ is a local Dedekind extension. As $\mathcal{o}$ is a principal ideal domain, we have

$$\mathfrak{d}_{\mathcal{O}/\mathcal{o}} = d_{\mathcal{O}/\mathcal{o}}\mathcal{o}.$$

Therefore $\mathfrak{p} \mid \mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ if and only if $d_{\mathcal{O}/\mathcal{o}} \subset \mathfrak{p}$ which is to say

$$d_{(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_{\mathfrak{p}}} = 0.$$

Now let

$$\mathfrak{p}\mathcal{O} = \mathfrak{P}_1^{e_{\mathfrak{p}}(\mathfrak{P}_1)} \cdots \mathfrak{P}_r^{e_{\mathfrak{p}}(\mathfrak{P}_r)},$$

be the prime factorization of $\mathfrak{p}\mathcal{O}$. The Chinese remainder theorem gives an isomorphism

$$\mathcal{O}/\mathfrak{p}\mathcal{O} \cong \bigoplus_i \mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)},$$

and Proposition 3.2.4 further implies

$$d_{(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_{\mathfrak{p}}} = \prod_i d_{(\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)})/\mathbb{F}_{\mathfrak{p}}}.$$

Therefore $d_{(\mathcal{O}/\mathfrak{p}\mathcal{O})/\mathbb{F}_{\mathfrak{p}}} = 0$ if and only if $d_{(\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)})/\mathbb{F}_{\mathfrak{p}}} = 0$ for some prime $\mathfrak{P}_i$. We will prove that this occurs if and only if $e_{\mathfrak{p}}(\mathfrak{P}_i) \geq 2$ which will complete the proof because this is exactly when $\mathfrak{p}$ ramifies since all residue class extensions are separable by assumption.

For the forward implication, we will prove the contrapositive so suppose $e_{\mathfrak{p}}(\mathfrak{P}_i) = 1$. As the residue class extension $\mathbb{F}_{\mathfrak{P}_i}/\mathbb{F}_{\mathfrak{p}}$ is separable by assumption, Proposition 3.2.5 implies $d_{\mathbb{F}_{\mathfrak{P}_i}/\mathbb{F}_{\mathfrak{p}}} \neq 0$. For the reverse implication, suppose $e_{\mathfrak{p}}(\mathfrak{P}_i) \geq 2$. By prime factorization, there exists $\beta_1 \in \mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)-1} - \mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)}$ and let $\overline{\beta_1} \in \mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)}$ be its image under natural projection. Then $\overline{\beta_1} \neq 0$ and $\overline{\beta_1}^2 = 0$ by construction. Since $\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)}$ is an $\mathbb{F}_{\mathfrak{p}}$-vector space of dimension $e_{\mathfrak{p}}(\mathfrak{P}_i)f_{\mathfrak{p}}(\mathfrak{P}_i)$ by Lemma 3.6.1, there exists a basis of the form $\overline{\beta_1}, \dots, \overline{\beta_{e_{\mathfrak{p}}(\mathfrak{P}_i)f_{\mathfrak{p}}(\mathfrak{P}_i)}}$. Now $T^2_{\overline{\beta_1\beta_j}}$ is the zero operator by our choice of β_1. Whence $T_{\overline{\beta_1\beta_j}}$ is nilpotent. As nilpotent operators have trace zero, we have

$$\mathrm{Tr}_{(\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)})/\mathbb{F}_{\mathfrak{p}}}(\overline{\beta_1\beta_j}) = 0.$$

This means that the first row of $\mathrm{Tr}_{(\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)})/\mathbb{F}_{\mathfrak{p}}}(\overline{\beta_1}, \dots, \overline{\beta_{e_{\mathfrak{p}}(\mathfrak{P}_i)f_{\mathfrak{p}}(\mathfrak{P}_i)}})$ is zero. Therefore $d_{(\mathcal{O}/\mathfrak{P}_i^{e_{\mathfrak{p}}(\mathfrak{P}_i)})/\mathbb{F}_{\mathfrak{p}}} = 0$ completing the proof. □

In the case the reside class extensions are separable, factorizing $\mathfrak{d}_{\mathcal{O}/\mathcal{o}}$ into a product of primes shows that only finitely many primes of $\mathcal{o}$ can ramify in $\mathcal{O}/\mathcal{o}$.

Given that the prime factors of the different and discriminant correspond to the ramification of primes, provided the residue class extensions are separable, one may suspect that they are related via the ideal norm. This turns out to be true regardless of the separability of the reside class extensions.

Proposition 3.8.7. *Let $\mathcal{O}/\mathcal{o}$ be a Dedekind extension of a degree n separable extension L/K. Then*

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{D}_{\mathcal{O}/\mathcal{o}}) = \mathfrak{d}_{\mathcal{O}/\mathcal{o}}.$$

Proof. In view of Propositions 3.6.3, 3.8.2 and 3.8.3, it suffices to assume $\mathcal{O}/\mathcal{o}$ is a local Dedekind extension. Then $\mathcal{O}/\mathcal{o}$ admits an integral basis $\alpha_1, \dots, \alpha_n$ and

$$\mathfrak{d}_{\mathcal{O}/\mathcal{o}} = d_{L/K}(\alpha_1, \dots, \alpha_n)\mathcal{o},$$

in view of Proposition 3.2.10. Moreover, $\alpha_1, \dots, \alpha_n$ is a basis for $\mathcal{O}$ as an $\mathcal{o}$-submodule of L and so the dual basis $\alpha_1^\vee, \dots, \alpha_n^\vee$ is a basis for $\mathfrak{D}^{-1}_{\mathcal{O}/\mathcal{o}}$ as an $\mathcal{o}$-submodule of L. Since $\mathcal{O}$ is a principal ideal domain, $\mathfrak{D}^{-1}_{\mathcal{O}/\mathcal{o}} = \frac{1}{\delta}\mathcal{O}$ for some nonzero $\delta \in \mathcal{O}$. Whence $\frac{\alpha_1}{\delta}, \dots, \frac{\alpha_n}{\delta}$ is also a basis for $\mathfrak{D}^{-1}_{\mathcal{O}/\mathcal{o}}$ as an $\mathcal{o}$-submodule of L. Since these are bases for L/K, Equation (3.1) implies

$$d_{L/K}\left(\frac{\alpha_1}{\delta}, \dots, \frac{\alpha_n}{\delta}\right) = \mathrm{N}_{L/K}(\delta)^{-2} d_{L/K}(\alpha_1, \dots, \alpha_n),$$

and

$$d_{L/K}\left(\frac{\alpha_1}{\delta}, \dots, \frac{\alpha_n}{\delta}\right) = \varepsilon^2 d_{L/K}(\alpha_1^\vee, \dots, \alpha_n^\vee),$$

for some $\varepsilon \in \mathcal{o}^*$. In view of Propositions 3.2.1 and 3.2.6, we have

$$d_{L/K}(\alpha_1^\vee, \dots, \alpha_n^\vee) d_{L/K}(\alpha_1, \dots, \alpha_n) = 1.$$

Combining these identities results in

$$\varepsilon^2 \mathrm{N}_{L/K}(\delta)^2 = d_{L/K}(\alpha_1, \ldots, \alpha_n)^2.$$

But Proposition 3.6.4 implies $\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{D}_{\mathcal{O}/\mathcal{o}}) = \mathrm{N}_{L/K}(\delta)\mathcal{o}$. Therefore passing to integral ideals gives

$$\mathfrak{N}_{\mathcal{O}/\mathcal{o}}(\mathfrak{D}_{\mathcal{O}/\mathcal{o}})^2 = \mathfrak{d}_{\mathcal{O}/\mathcal{o}}^2.$$

The claim now follows from prime factorization. $\square$

Number Fields

We now turn to the setting where L/K is an extension of number fields. Recall that the underlying Dedekind extension is $\mathcal{O}_L/\mathcal{O}_K$. The **complement** $\mathfrak{C}_{L/K}$ of L/K is the complement of $\mathcal{O}_L/\mathcal{O}_K$, the **different** $\mathfrak{D}_{L/K}$ of L/K is the different of $\mathcal{O}_L/\mathcal{O}_K$, and the **discriminant** $\mathfrak{d}_{L/K}$ of L/K is the discriminant of $\mathcal{O}_L/\mathcal{O}_K$.

For a number field K, we can say more as $\mathcal{O}_K/\mathbb{Z}$ admits an integral basis and the residue class extensions are separable. Indeed, since $\mathbb{Z}$ is a principal ideal domain the discriminant $\mathfrak{d}_K$ is related to the discriminant Δ_K by

$$\mathfrak{d}_K = \Delta_K \mathcal{O}_K.$$

As all of the residue class extensions are separable, it follows from Theorems 3.8.5 and 3.8.6 that a prime of K ramifies in K if and only if it divides $\mathfrak{D}_K$ and a prime of $\mathbb{Q}$ ramifies in K if and only if it divides $|\Delta_K|$. In particular, finitely many primes of K and $\mathbb{Q}$ ramify in K. Also, Proposition 3.8.7 immediately implies

$$\mathrm{N}_K(\mathfrak{D}_K) = |\Delta_K|.$$

Then for any fractional ideal $\mathfrak{f}$ of $\mathcal{O}_K$, we find that

$$\mathrm{N}_K(\mathfrak{f}^\vee) = \frac{\mathrm{N}_K(\mathfrak{f}^{-1})}{|\Delta_K|},$$

from Equation (3.8).

Lastly, let $a_K(m)$ denote the number of integral ideals of $\mathcal{O}_K$ of norm m. As the norm is completely multiplicative, prime factorization implies that $a_K(m)$ is multiplicative. In fact, we can estimate the growth of $a_K(m)$.

Proposition 3.8.8. *Let K be a number field of degree n. Then $a_K(m) \le \sigma_0(m)^n$.*

Proof. Let $\mathfrak{a}$ be an integral ideal of norm m. First suppose $m = p^k$ is a prime power. By the fundamental equality, there are at most n primes $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ above p. Let $f_p(\mathfrak{p}_1), \ldots, f_p(\mathfrak{p}_n)$ be the corresponding inertia degrees. Then

$$\mathrm{N}_K(\mathfrak{a}) = p^{e_1 f_p(\mathfrak{p}_1)} \cdots p^{e_n f_p(\mathfrak{p}_n)},$$

for integers e_i with $0 \le e_i \le k$. The number of such choices is equivalent to the number of solutions

$$e_1 f_p(\mathfrak{p}_1) + \cdots + e_n f_p(\mathfrak{p}_n) = k,$$

which is at most $\sigma_0(p^k)^n$. This proves the claim in the case m is a prime power. The claim follows by multiplicativity of $a_K(m)$ and the divisor function. $\square$

As a consequence of this result, we have the slightly weaker estimate $a_K(n) \ll_\varepsilon n^\varepsilon$.

Chapter 4

Geometry of Number Fields

Throughout, all rings will be understood to be commutative and with unity. We will also make use of the theory of integral lattices in Appendix A.1 especially Minkowski's lattice point theorem and the beta function as developed in Appendix C.2.

4.1 Minkowski Space

Let K be number field of degree n. If $\sigma \in \mathrm{Hom}_{\mathbb{Q}}(K, \overline{\mathbb{Q}})$ then either σ is real or complex and in the latter case it is has a paired $\mathbb{Q}$-embedding $\overline{\sigma}$ which is the complex conjugate of σ. Accordingly, let r_1 and $2r_2$ be the number of real and complex $\mathbb{Q}$-embeddings of K into $\overline{\mathbb{Q}}$ respectively. We call the pair (r_1, r_2) the **signature** of K and it satisfies the relation

$$n = r_1 + 2r_2.$$

Setting

$$K_{\mathbb{C}} = \mathbb{C}^n,$$

makes $K_{\mathbb{C}}$ into a component-wise $\mathbb{C}$-algebra. Moreover, $K_{\mathbb{C}}$ is also a complex Hilbert space with respect to the standard complex inner product which we denote by $\langle \cdot, \cdot \rangle_{K_{\mathbb{C}}}$. We also write $d\lambda_{K_{\mathbb{C}}}$ for the induced Lebesgue measure on $K_{\mathbb{C}}$ by this inner product. Also let $\mathrm{Tr}_{K_{\mathbb{C}}}$ and $\mathrm{N}_{K_{\mathbb{C}}}$ denote the trace and norm of $K_{\mathbb{C}}/\mathbb{C}$. Because $K_{\mathbb{C}}$ is a component-wise $\mathbb{C}$-algebra, the trace and the norm are given by

$$\mathrm{Tr}_{K_{\mathbb{C}}}(\mathbf{z}) = \sum_{\sigma} z_{\sigma} \quad \text{and} \quad \mathrm{N}_{K_{\mathbb{C}}}(\mathbf{z}) = \prod_{\sigma} z_{\sigma},$$

for all $\mathbf{z} \in K_{\mathbb{R}}$. We define the **canonical embedding** j of K to be the $\mathbb{Q}$-embedding

$$j : K \to K_{\mathbb{C}} \qquad \kappa \mapsto (\sigma(\kappa))_{\sigma}.$$

In other words, the canonical embedding is a component-wise $\mathbb{Q}$-embedding of K into $K_{\mathbb{C}}$ via the σ. In view of Proposition 3.2.1, we have

$$\mathrm{Tr}_{K_{\mathbb{C}}}(j(\kappa)) = \mathrm{Tr}_K(\kappa) \quad \text{and} \quad \mathrm{N}_{K_{\mathbb{C}}}(j(\kappa)) = \mathrm{N}_K(\kappa),$$

which is to say that the compositions of the trace and norm of $K_{\mathbb{C}}/\mathbb{C}$ with the canonical embedding produces the field trace and field norm. Our primary aim is to construct an $\mathbb{R}$-subalgebra of $K_{\mathbb{C}}$ which corrects for the fact that the complex $\mathbb{Q}$-embeddings come in pairs. To this end, consider the complex conjugation map

$$F : \mathbb{C} \to \mathbb{C} \qquad z \mapsto \overline{z}.$$

This induces an automorphism of $K_{\mathbb{C}}$ given by

$$F : K_{\mathbb{C}} \to K_{\mathbb{C}} \qquad \mathbf{z} \mapsto (\overline{z_{\overline{\sigma}}})_{\sigma},$$

which is an involution. A short computation shows

$$\langle F(\mathbf{z}), F(\mathbf{w}) \rangle_{K_{\mathbb{C}}} = F(\langle \mathbf{z}, \mathbf{w} \rangle_{K_{\mathbb{C}}}),$$

for all $\mathbf{z}, \mathbf{w} \in K_{\mathbb{C}}$. In other words, inner product $\langle \cdot, \cdot \rangle_{K_{\mathbb{C}}}$ is F-equivariant. We define the **Minkowski space** $K_{\mathbb{R}}$ of K by

$$K_{\mathbb{R}} = \{\mathbf{z} \in K_{\mathbb{C}} : F(\mathbf{z}) = \mathbf{z}\}.$$

In other words, $K_{\mathbb{R}}$ consists of all of the F-invariant elements of $K_{\mathbb{C}}$. Note that $K_{\mathbb{R}}$ is an $\mathbb{R}$-subalgebra of $K_{\mathbb{C}}$. We denote the restriction of the inner product $\langle \cdot, \cdot \rangle_{K_{\mathbb{C}}}$ to $K_{\mathbb{R}}$ by $\langle \cdot, \cdot \rangle_{K_{\mathbb{R}}}$. Moreover, the inner product $\langle \cdot, \cdot \rangle_{K_{\mathbb{R}}}$ turns $K_{\mathbb{R}}$ into a real Hilbert space. Indeed, the conjugate symmetry and F-equivariance of the inner product together give

$$\overline{\langle \mathbf{z}, \mathbf{w} \rangle} = \langle \mathbf{z}, \mathbf{w} \rangle,$$

for any $\mathbf{z}, \mathbf{w} \in K_{\mathbb{R}}$. Accordingly, we call $\langle \cdot, \cdot \rangle_{K_{\mathbb{R}}}$ the **Minkowski inner product** of K. We denote the restriction of the Lebesgue measure $d\lambda_{\mathbb{C}}$ to $K_{\mathbb{R}}$ by $d\lambda_{K_{\mathbb{R}}}$ which is also the Lebesgue measure associated to $\langle \cdot, \cdot \rangle_{K_{\mathbb{R}}}$. We call $d\lambda_{K_{\mathbb{R}}}$ the **Minkowski measure**. Lastly, we denote the restrictions of $\mathrm{Tr}_{K_{\mathbb{C}}}$ and $\mathrm{N}_{K_{\mathbb{C}}}$ to $K_{\mathbb{R}}$ by $\mathrm{Tr}_{K_{\mathbb{R}}}$ and $\mathrm{N}_{K_{\mathbb{R}}}$ respectively which are the norm and trace of $K_{\mathbb{R}}/\mathbb{R}$. We call $\mathrm{Tr}_{K_{\mathbb{R}}}$ and $\mathrm{N}_{K_{\mathbb{R}}}$ the **Minkowski trace** and **Minkowski norm** of K respectively. As the complex $\mathbb{Q}$-embeddings of K into $\overline{\mathbb{Q}}$ come in conjugate pairs, $j(K) \subset K_{\mathbb{R}}$ so that the canonical embedding is a $\mathbb{Q}$-embedding of K into Minkowski space. Whence

$$\mathrm{Tr}_{K_{\mathbb{R}}}(j(\kappa)) = \mathrm{Tr}_K(\kappa) \quad \text{and} \quad \mathrm{N}_{K_{\mathbb{R}}}(j(\kappa)) = \mathrm{N}_K(\kappa).$$

So the compositions of the Minkowski trace and Minkowski norm with the canonical embedding produces the field trace and field norm.

To give a more explicit description of Minkowski space, let $N_{\sigma} = 1, 2$ according to if σ is real or complex respectively. Also let ρ and τ run over the real $\mathbb{Q}$-embeddings and a complete set of representatives for the pairs of complex $\mathbb{Q}$-embeddings of $\mathrm{Hom}_{\mathbb{Q}}(K, \overline{\mathbb{Q}})$ respectively. Note that $N_{\rho} = 1$ and $N_{\tau} = 2$. Then

$$K_{\mathbb{R}} = \{\mathbf{z} \in K_{\mathbb{C}} : \mathrm{Im}(z_{\rho}) = 0 \text{ and } z_{\overline{\tau}} = \overline{z_{\tau}} \text{ for all } \rho \text{ and } \tau\}.$$

In fact, this gives an explicit isomorphism between Minkowski space and $\mathbb{R}^n$.

Proposition 4.1.1. *Let K be a number field of degree n and signature (r_1, r_2). Then the map*

$$z_\sigma \mapsto x_\sigma = \begin{cases} z_\sigma & \text{if } \sigma = \rho, \\ \mathrm{Re}(z_\sigma) & \text{if } \sigma = \tau, \\ \mathrm{Im}(z_\sigma) & \text{if } \sigma = \overline{\tau}, \end{cases}$$

induces a component-wise isomorphism between $K_\mathbb{R}$ and $\mathbb{R}^n$ making $K_\mathbb{R}$ a real vector space of dimension n. Moreover, the inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{R}^n$ induced by the Minkowski inner product is given by

$$\langle \mathbf{x}, \mathbf{x}' \rangle = \sum_\sigma N_\sigma x_\sigma x'_\sigma,$$

for any $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$.

Proof. This map has inverse

$$x_\sigma \mapsto z_\sigma = \begin{cases} x_\sigma & \text{if } \sigma = \rho, \\ x_\sigma + i x_{\overline{\sigma}} & \text{if } \sigma = \tau, \\ x_{\overline{\sigma}} + i x_\sigma & \text{if } \sigma = \overline{\tau}. \end{cases}$$

As both maps are $\mathbb{R}$-linear, it follows that the original map induces a component-wise isomorphism between $K_\mathbb{R}$ and $\mathbb{R}^n$ proving the first statement. For the second statement, let $\mathbf{z}, \mathbf{z}' \in K_\mathbb{R}$ and let $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$ be the corresponding elements in $\mathbb{R}^n$ under this isomorphism respectively. If $\sigma = \rho$ then

$$x_\rho = z_\rho \quad \text{and} \quad x'_\rho = z'_\rho,$$

and thus

$$x_\rho x'_\rho = z_\rho \overline{z'_\rho}.$$

If $\sigma = \tau$ then

$$x_\tau = \mathrm{Re}(z_\tau), \quad x_{\overline{\tau}} = \mathrm{Im}(z_\tau), \quad x'_\tau = \mathrm{Re}(z'_\tau), \quad \text{and} \quad x'_{\overline{\tau}} = \mathrm{Im}(z'_\tau).$$

Whence a short computation shows

$$2(x_\tau x'_\tau + x_{\overline{\tau}} x'_{\overline{\tau}}) = z_\tau \overline{z'_\tau} + z_{\overline{\tau}} \overline{z'_{\overline{\tau}}}.$$

The claim about the inner product follows. □

In view of this result, we define the **Minkowski embedding** σ_K of K by

$$\sigma_K : K \to \mathbb{R}^n \qquad \kappa \mapsto ((\rho(\kappa))_\rho, (\mathrm{Re}(\tau(\kappa)), \mathrm{Im}(\tau(\kappa)))_\tau).$$

Note that the Minkowski embedding is a $\mathbb{Q}$-embedding of K into $\mathbb{R}^n$ as it is the composition of the canonical embedding and the isomorphism established in the previous result. Also, the Minkowski embedding is independent of the representatives τ because they occur in conjugate pairs.

We have now constructed Minkowski space and determined its dimension as a real vector space. Our goal now is to pass from $\mathbb{Z}$-lattices of K to integral lattices of Minkowski space. Being an inner product space, this will permit us to measure fractional ideals of K geometrically via their images inside Minkowski space. As j is a $\mathbb{Q}$-embedding and any fractional ideal $\mathfrak{f}$ of K is a complete $\mathbb{Z}$-lattice of K, $j(\mathfrak{f})$ is a complete integral lattice of $K_{\mathbb{R}}$. The covolume $V_{j(\mathfrak{f})}$ of $j(\mathfrak{f})$ is easily determined.

Proposition 4.1.2. *Let K be a number field with signature (r_1, r_2) and let $\mathfrak{f}$ be a fractional ideal of K. Then $V_{j(\mathfrak{f})}$ is the absolute value of the determinant of any embedding matrix of a basis for $K/\mathbb{Q}$ contained in $\mathfrak{f}$. In particular,*

$$V_{j(\mathfrak{f})} = \mathrm{N}_K(\mathfrak{f})\sqrt{|\Delta_K|},$$

and

$$V_{j(\mathcal{O}_K)} = \sqrt{|\Delta_K|}.$$

Proof. Upon writing $\mathfrak{f} = \frac{1}{d}\mathfrak{a}$ for some nonzero $d \in \mathbb{Z}$ and integral ideal $\mathfrak{a}$ and Proposition 3.6.5, it suffices to prove the statements for the integral ideal $\mathfrak{a}$. Let $\kappa_1, \ldots, \kappa_n$ be a basis for $K/\mathbb{Q}$ contained in $\mathfrak{a}$. Then the associated generator matrix for $j(\mathfrak{a})$ is the embedding matrix $M(\kappa_1, \ldots, \kappa_n)$. Hence

$$V_{j(\mathfrak{a})} = |\det(M(\kappa_1, \ldots, \kappa_n))|,$$

proving the first statement. To prove the first identity of the second statement, in view of Equation (3.3) it suffices to show

$$|\det(M(\kappa_1, \ldots, \kappa_n))| = \mathrm{N}_K(\mathfrak{a})|\det(M(\alpha_1, \ldots, \alpha_n))|,$$

for any integral basis $\alpha_1, \ldots, \alpha_n$ for $K/\mathbb{Q}$. Note that $\kappa_1, \ldots, \kappa_n$ is necessarily a basis for $\mathfrak{a}/\mathbb{Z}$ by Theorem 3.2.8 and write

$$\kappa_i = \sum_j a_{i,j}\alpha_j,$$

with $a_{i,j} \in \mathbb{Z}$. Then $(a_{i,j})$ is the base change matrix from $\alpha_1, \ldots, \alpha_n$ to $\kappa_1, \ldots, \kappa_n$. Whence the image of $\mathcal{O}_K$ under this matrix is $\mathfrak{a}$. As $\mathbb{Z}$ is a principal ideal domain, the smith normal form of this matrix implies that

$$\mathrm{N}_K(\mathfrak{a}) = |\det((a_{i,j}))|.$$

This fact together with Equation (3.1) and Proposition 3.2.6 gives

$$|\det(M(\kappa_1, \ldots, \kappa_n))| = \mathrm{N}_K(\mathfrak{a})|\det(M(\alpha_1, \ldots, \alpha_n))|,$$

as claimed. The second identity follows from the first upon taking $\mathfrak{f} = \mathcal{O}_K$. □

As σ_K is also a $\mathbb{Q}$-embedding, $\sigma_K(\mathfrak{f})$ is a complete integral lattice of $\mathbb{R}^n$. As a corollary of the previous result, we can determine the covolume of $V_{\sigma_K(\mathfrak{f})}$ of $\sigma_K(\mathfrak{f})$ as well.

Corollary 4.1.3. *Let K be a number field with signature (r_1, r_2) and let $\mathfrak{f}$ be a fractional ideal of K. Then*

$$V_{\sigma_K(\mathfrak{f})} = \mathrm{N}_K(\mathfrak{f})\frac{\sqrt{|\Delta_K|}}{2^{r_2}},$$

and

$$V_{\sigma_K(\mathcal{O}_K)} = \frac{\sqrt{|\Delta_K|}}{2^{r_2}}.$$

Proof. Let $\kappa_1, \dots, \kappa_n$ be a basis for $K/\mathbb{Q}$ contained in $\mathfrak{f}$. Also let $\rho_1, \dots, \rho_{r_1}$ and $\tau_1, \dots, \tau_{r_2}$ run over the real $\mathbb{Q}$-embeddings and a complete set of representatives for the pairs of complex $\mathbb{Q}$-embeddings of $\mathrm{Hom}_{\mathbb{Q}}(K, \overline{\mathbb{Q}})$ respectively. Then the associated generator matrix for $\sigma_K(\mathfrak{f})$ is

$$\begin{pmatrix} \rho_1(\kappa_1) & \cdots & \rho_{r_1}(\kappa_1) & \mathrm{Re}(\tau_1(\kappa_1)) & \mathrm{Im}(\tau_1(\kappa_1)) & \cdots & \mathrm{Re}(\tau_{r_2}(\kappa_1)) & \mathrm{Im}(\tau_{r_2}(\kappa_1)) \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ \rho_1(\kappa_n) & \cdots & \rho_{r_1}(\kappa_n) & \mathrm{Re}(\tau_1(\kappa_n)) & \mathrm{Im}(\tau_1(\kappa_n)) & \cdots & \mathrm{Re}(\tau_{r_2}(\kappa_n)) & \mathrm{Im}(\tau_{r_2}(\kappa_n)) \end{pmatrix}^t.$$

By Proposition 4.1.2 we are done if the absolute value of the determinant of this matrix is $\frac{1}{2^{r_2}}$ times the absolute value of the determinate of the embedding matrix for $\kappa_1, \dots, \kappa_n$. To show this, first add an i multiple of the imaginary columns to their corresponding real columns and then apply the identity $\mathrm{Im}(z) = \frac{z-\overline{z}}{2i}$ to the imaginary columns. We obtain the matrix

$$\begin{pmatrix} \rho_1(\kappa_1) & \cdots & \rho_{r_1}(\kappa_1) & \tau_1(\kappa_1) & \frac{\tau_1(\kappa_1)-\overline{\tau_1}(\kappa_1)}{2i} & \cdots & \tau_{r_2}(\kappa_1) & \frac{\tau_{r_2}(\kappa_1)-\overline{\tau_{r_2}}(\kappa_1)}{2i} \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ \rho_1(\kappa_n) & \cdots & \rho_{r_1}(\kappa_n) & \tau_1(\kappa_n) & \frac{\tau_1(\kappa_n)-\overline{\tau_1}(\kappa_n)}{2i} & \cdots & \tau_{r_2}(\kappa_n) & \frac{\tau_{r_2}(\kappa_n)-\overline{\tau_{r_2}}(\kappa_n)}{2i} \end{pmatrix}^t,$$

which has the same determinant as the generator matrix. Now multiply the imaginary columns by $-2i$ and then adding the preceding columns to annihilate the negative terms. This results in the matrix

$$\begin{pmatrix} \rho_1(\kappa_1) & \cdots & \rho_{r_1}(\kappa_1) & \tau_1(\kappa_1) & \overline{\tau_1}(\kappa_1) & \cdots & \tau_{r_2}(\kappa_1) & \overline{\tau_{r_2}}(\kappa_1) \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots \\ \rho_1(\kappa_n) & \cdots & \rho_{r_1}(\kappa_n) & \tau_1(\kappa_n) & \overline{\tau_1}(\kappa_n) & \cdots & \tau_{r_2}(\kappa_n) & \overline{\tau_{r_2}}(\kappa_n) \end{pmatrix}^t,$$

whose determinant is $(-2i)^{-r_2}$ that of the generator matrix. But this latter matrix is the embedding $M(\kappa_1, \dots, \kappa_n)$. Whence

$$V_{\sigma_K(\mathfrak{f})} = \frac{|\det(M(\kappa_1, \dots, \kappa_n))|}{2^{r_2}}.$$

□

4.2 Finiteness of the Class Number

Recall that the class number of a Dedekind domain is a measure of how much the Dedekind domain fails to be a principal ideal domain and even need not be finite from Remark 3.3.7. However, we will show that the class number h_K for a number field K is always finite and thus $\mathcal{O}_K$ only has finite failure from being a principal ideal domain. In doing so, we will also show that every ideal class is represented by an integral ideal of bounded norm.

Theorem 4.2.1. *Let K be a number field of degree n and signature (r_1, r_2). Also, let $X \subseteq \mathbb{R}^n$ be a compact convex symmetric set and set $M = \max_{\mathbf{x}\in X}\left(\prod_i |x_i|\right)$. Then every ideal class of $\mathrm{Cl}(K)$ is represented by an integral ideal $\mathfrak{a}$ satisfying*

$$\mathrm{N}_K(\mathfrak{a}) \leq \frac{2^{r_1+r_2}M}{\mathrm{Vol}(X)}\sqrt{|\Delta_K|}.$$

Moreover, the ideal class group $\mathrm{Cl}(K)$ is finite which is to say that the class number h_K is.

Proof. Let $\mathfrak{f}$ be a fractional ideal, n be a positive integer, and set

$$\lambda^n = 2^n\frac{V_{\sigma_K(\mathfrak{f}^{-1})}}{\mathrm{Vol}(X)}.$$

Then by construction, we have

$$\mathrm{Vol}(\lambda X) = 2^n V_{\sigma_K(\mathfrak{f}^{-1})}.$$

By Minkowski's lattice point theorem, there exists a nonzero element of $\sigma_K(\mathfrak{f}^{-1}) \cap \lambda X$. Whence there is a nonzero $\alpha \in \mathfrak{f}^{-1}$ such that $\sigma_K(\alpha) \in \lambda X$. Moreover, $\alpha\mathfrak{f} \subseteq \mathcal{O}_K$ so that $\mathfrak{a} = \alpha\mathfrak{f}$ is an integral ideal in the same class as $\mathfrak{f}$. Using Propositions 3.2.1 and 3.6.5, a short computation shows

$$\mathrm{N}_K(\mathfrak{a}) \leq \lambda^n M\, \mathrm{N}_K(\mathfrak{f}).$$

In view of our choice of λ^n and Corollary 4.1.3, this inequality becomes

$$\mathrm{N}_K(\mathfrak{a}) \leq \frac{2^{r_1+r_2}M}{\mathrm{Vol}(X)}\sqrt{|\Delta_K|},$$

which proves the first statement.

We now prove that the class group is finite. By the first statement, we can find a complete set of representatives for $\mathrm{Cl}(K)$ consisting of integral ideals of bounded norm. Since the norm is multiplicative, the prime factors of these representatives have bounded norm as well. Moreover, the norm of a prime $\mathfrak{p}$ is $p^{f_p(\mathfrak{p})}$ where p is the prime below $\mathfrak{p}$. Whence there are only finitely many possible primes p. As there are finitely many primes $\mathfrak{p}$ above any prime p, it follows that these representatives have finitely many prime factors. Altogether this means that there are finitely many representatives. Hence $\mathrm{Cl}(K)$ is finite and so the class number h_K is too. $\square$

Observe that the bound in the previous result is not explicit. To obtain an explicit bound, we need to make a choice for the compact convex symmetric set X. In order for such a bound to be sharp, we need to ensure that the volume of X is large while the constant M is small. The following lemma dictates our choice of X and computes its volume:

Lemma 4.2.2. *Suppose n is a positive integer and write $n = r_1 + 2r_2$ for some nonnegative integers r_1 and r_2. Let $X \subset \mathbb{R}^n$ be the compact convex symmetric set given by*

$$X = \left\{ (\mathbf{x}_1, \mathbf{x}_2) \in \mathbb{R}^n : \sum_i |x_{i,1}| + 2 \sum_j \sqrt{x_{2j-1,2}^2 + x_{2j,2}^2} \leq n \right\}.$$

Then

$$\mathrm{Vol}(X) = \frac{n^n}{n!} 2^{r_1} \left(\frac{\pi}{2} \right)^{r_2}.$$

Proof. Making the change of variables $x_{2j-1,2} \mapsto r_j \sin(\theta_j)$ and $x_{2j,2} \mapsto r_j \cos(\theta_j)$ gives

$$\mathrm{Vol}(X) = \int_{X'} r_1 \cdots r_{r_2} \, d\mathbf{x}_1 \, d\mathbf{r} \, d\boldsymbol{\theta},$$

because the Jacobian determinant is $r_1 \cdots r_{r_2}$, and where

$$X' = \left\{ (\mathbf{x}_1, \mathbf{r}, \boldsymbol{\theta}) \in [-n, n]^{r_1} \oplus [0, n]^{r_2} \oplus [0, 2\pi]^{r_2} : \sum_i |x_i| + 2 \sum_j r_j \leq n \right\}.$$

Making the change of variables $r_j \mapsto \frac{r_j}{2}$ and using the facts that the integrand is symmetric in the x_i and independent of θ_j gives

$$\mathrm{Vol}(X) = 2^{r_1} \left(\frac{\pi}{2} \right)^{r_2} \int_{X''} r_1 \cdots r_{r_2} \, d\mathbf{x}_1 \, d\mathbf{r},$$

where

$$X'' = \left\{ (\mathbf{x}_1, \mathbf{r}) \in [0, n]^{r_1 + r_2} : \sum_i x_i + 2 \sum_j r_j \leq n \right\}.$$

To compute the remaining integral, for nonnegative integers ℓ and k and positive t, let

$$X''_{\ell,k}(t) = \left\{ (\mathbf{x}_1, \mathbf{r}) \in [0, t]^{\ell + k} : \sum_i x_i + 2 \sum_j r_j \leq t \right\},$$

and define

$$I_{\ell,k}(t) = \int_{X''_{\ell,k}(t)} r_1 \cdots r_\ell \, d\mathbf{x}_1 \, d\mathbf{r}.$$

In this notation, we have

$$\mathrm{Vol}(X) = 2^{r_1} \left(\frac{\pi}{2} \right)^{r_2} I_{r_1, r_2}(n).$$

To compute the remaining integral, the change of variables $x_i \mapsto tx_i$ and $r_j \mapsto tr_j$ gives

$$I_{\ell,k}(t) = t^{\ell+2k} I_{\ell,k}(1).$$

Whence

$$I_{\ell,k}(1) = \int_0^1 (1 - x_\ell)^{\ell-1+2k} I_{\ell-1,k}(1)\, dx_\ell = \frac{1}{\ell+2k} I_{\ell-1,k}(1).$$

Upon repeating this procedure $\ell - 1$ times results in

$$I_{\ell,k}(1) = \frac{1}{(\ell+2k)\cdots(2k+1)} I_{0,k}(1).$$

Similarly, we have

$$I_{0,k}(1) = \int_0^1 r_k(1 - r_k)^{2k-2} I_{0,k-1}(1)\, dr_k = \frac{1}{2k} I_{0,k-1}(1),$$

because the latter integral is the beta function $B(1, 2k-1)$. Upon repeating this procedure $k-1$ times results in

$$I_{0,k}(1) = \frac{1}{k!}.$$

Altogether, this means

$$I_{\ell,k}(t) = \frac{t^{\ell+2k}}{(\ell+2k)!},$$

and therefore

$$\mathrm{Vol}(X) = \frac{n^n}{n!} 2^{r_1} \left(\frac{\pi}{2}\right)^{r_2}.$$

□

Observe that the compact convex symmetric set X consists of those elements in $\mathbb{R}^n$ whose norm corresponding to the inner product induced by the Minkowski inner product from Proposition 4.1.1 at most n. We can now obtain an explicit bound known as the **Minkowski bound**.

Theorem (Minkowski bound). *Let K be a number field of degree n and signature (r_1, r_2). Then every ideal class of $\mathrm{Cl}(K)$ is represented by an integral ideal $\mathfrak{a}$ satisfying*

$$\mathrm{N}_K(\mathfrak{a}) \le \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|\Delta_K|}.$$

Proof. Theorem 4.2.1 and Lemma 4.2.2 together show that every ideal class of $\mathrm{Cl}(K)$ is represented by an integral ideal $\mathfrak{a}$ satisfying

$$\mathrm{N}_K(\mathfrak{a}) \le M \left(\frac{4}{\pi}\right)^{r_2} \frac{n!}{n^n} \sqrt{|\Delta_K|},$$

where $M = \max_{\mathbf{x}\in X} \left(\prod_i |x|\right)$. But for all $\mathbf{x} \in X$, the arithmetic-geometric mean inequality implies

$$\left(\prod_i |x_i|\right)^{\frac{1}{n}} \le 1,$$

Whence $M \le 1$ completing the proof. □

As a corollary we can obtain a lower bound for the discriminant of a number field and show that every number field K, other than $\mathbb{Q}$, has at least one ramified prime.

Corollary 4.2.3. *Let K be a number field of degree n. Then*

$$|\Delta_K| \geq \left(\frac{\pi}{4}\right)^{\frac{n}{2}} \frac{n^n}{n!}.$$

In particular, there is at least one ramified prime of K provided the degree of K is at least 2.

Proof. The inequality follows from the Minkowski bound since the norm of every integral ideal is at least 1 and r_2 is at most n. This proves the first statement. For the second statement, suppose $n \geq 2$. In the case $n = 2$, the lower bound is larger than 1 so that $|\Delta_K|$ is at least 2. Whence $|\Delta_K|$ has a prime divisor and so some prime above this divisor is a ramified prime of K. Now suppose $n \geq 3$. As $n^n \geq n!$, $\left(\frac{\pi}{4}\right)^{\frac{n}{2}} \frac{n^n}{n!}$ is an increasing function in n. Therefore $|\Delta_K| \geq 2$ and, as before, there is at least one ramified prime of K . □

Generally speaking, the class number of K is one of the most difficult pieces of arithmetic data about K to compute. For example, it is still unknown if there are infinitely many number fields of class number 1. That is to say, it is unknown if there are infinitely many number fields such that their ring of integers are principal ideal domains.

Let S be a finite set of primes of K. The S-class number $h_{K,S}$ is also finite. In this setting, the long exact sequence in Proposition 3.4.7 takes the form

$$1 \longrightarrow \mathcal{O}_K^* \longrightarrow \mathcal{O}_{K,S}^* \longrightarrow \mathbb{Z}^{|S|} \longrightarrow \mathrm{Cl}(K) \longrightarrow \mathrm{Cl}_S(K) \longrightarrow 1,$$

implying

$$\mathrm{Cl}_S(K) \cong \mathrm{Cl}(K)/\langle \mathfrak{p}P(K) : \mathfrak{p} \in S \rangle.$$

Whence

$$h_{K,S} = \frac{h_K}{|\langle \mathfrak{p}P(K) : \mathfrak{p} \in S \rangle|},$$

which shows that the S-class number of K is finite and generally smaller than the class number of K. In fact, finiteness implies that it is possible to choose S large enough (consisting of all the prime factors for a complete set of integral ideal representatives) such that $h_{K,S} = 1$. This means we can always find a finite set of primes S such that the ring of S-integers of K is a principal ideal domain.

Let $\mathcal{O}$ be an order of K. Then the class number $h_{\mathcal{O}}$ is finite as well. In this setting and using Proposition 3.5.11, the long exact sequence in Proposition 3.5.9 takes the form

$$\begin{array}{ccccc}
1 \longrightarrow \mathcal{O}^* \longrightarrow & \mathcal{O}_K^* & & \\
& \downarrow & & \\
& (\mathcal{O}_K/\mathfrak{q}_{\mathcal{O}})^*/(\mathcal{O}/\mathfrak{q}_{\mathcal{O}})^* & & \\
& \downarrow & & \\
& \mathrm{Pic}(\mathcal{O}) & \longrightarrow \mathrm{Cl}(K) \longrightarrow 1. &
\end{array}$$

By Proposition 3.6.5, the quotient $\mathcal{O}_K/\mathfrak{q}_{\mathcal{O}}$ is finite whence $(\mathcal{O}_K/\mathfrak{q}_{\mathcal{O}})^*/(\mathcal{O}/\mathfrak{q}_{\mathcal{O}})^*$ is as well. Exactness then implies $\mathcal{O}_K^*/\mathcal{O}^*$ is also finite and that

$$h_{\mathcal{O}} = \frac{h_K}{|\mathcal{O}_K^*/\mathcal{O}^*|}\frac{|(\mathcal{O}_K/\mathfrak{q}_{\mathcal{O}})^*|}{|(\mathcal{O}/\mathfrak{q}_{\mathcal{O}})^*|},$$

so $h_{\mathcal{O}}$ is finite too. This identity also shows that the class number of an order of K is generally larger than the class number of K.

4.3 Dirichlet's Unit Theorem

Let K be a number field of degree n and signature (r_1, r_2). We define the **rank** r_K of K by

$$r_K = r_1 + r_2 - 1.$$

The rank of K is intimately related to the structure of the unit group $\mathcal{O}_K^*$. Observe that we can express the unit group as

$$\mathcal{O}_K^* = \{\alpha \in \mathcal{O}_K : \mathrm{N}_K(\alpha) = \pm 1\}.$$

Our main goal will be to describe the group structure of $\mathcal{O}_K^*$ completely. Let $\mu(K)$ be the subgroup of K^* consisting of all the roots of unity in K. As any m-th root of unity is a root of $x^m - 1$, it follows that roots of unity are algebraic integers. Whence $\mu(K)$ is a subgroup of $\mathcal{O}_K^*$. Clearly $\{\pm 1\} \subseteq \mu(K)$. In fact, $\mu(K)$ always is finite. To see this, let $\omega \in \mu(K)$ and recall that any root of unity is a primitive root of unity of some, potentially smaller, order. So we may assume ω is a primitive m-th root of unity. The minimal polynomial of ω over $\mathbb{Q}$ is the m-th cyclotomic polynomial $\Phi_m(x)$ which has degree $\varphi(m)$. Whence $\varphi(m) \le n$. Since $\varphi(m) \to \infty$ as $m \to \infty$, there can only be finitely many such ω. This means $\mu(K)$ is finite. Accordingly, set

$$w_K = |\mu(K)|.$$

Our primary aim will be to show that $\mathcal{O}_K^*$ is a direct product of $\mu(K)$ and a free Z-module of rank r_K. Determining that the rank of the free Z-module is exactly r_K will be the most difficult part of the argument. To accomplish this, we will require a map on K^* that transitions between the Minkowski trace and Minkowski norm. To define this map, let ρ and τ run over the real $\mathbb{Q}$-embeddings and a complete set of representatives for the pairs of complex $\mathbb{Q}$-embeddings of $\mathrm{Hom}_{\mathbb{Q}}(K, \overline{\mathbb{Q}})$ respectively. Consider the homomorphism

$$\ell : K_{\mathbb{R}}^* \to \mathbb{R}^{r_K+1} \qquad (z_\sigma)_\sigma \mapsto ((\log|z_\rho|)_\rho, (2\log|z_\tau|)_\tau).$$

This map is independent of the choice of representatives τ by definition of Minkowski space. Then the Minkowski trace and Minkowski norm and related via the identity

$$\mathrm{Tr}_{K_{\mathbb{R}}}(\ell(\mathbf{z})) = \log|\mathrm{N}_{K_{\mathbb{R}}}(\mathbf{z})|,$$

for all $\mathbf{z} \in K_{\mathbb{R}}$. We define the **logarithmic embedding** $\log_K$ of K to be the homomorphism given by

$$\log_K : K^* \to \mathbb{R}^{r_K+1} \qquad \kappa \mapsto ((\log|\rho(\kappa)|)_\rho, (2\log|\rho(\tau)|)_\tau).$$

Then $\log_K$ is just the restriction of j to K^* composed with ℓ. As such, it is also independent of the representatives τ. We distinguish the subsets of $\mathbb{R}^{r_K+1}$ given by

$$S = \{\mathbf{x} \in \mathbb{R}_+^{r_K+1} : \mathrm{N}_{\mathbb{R}^{r_K+1}/\mathbb{R}}(\mathbf{x}) = 1\} \quad \text{and} \quad H = \{\mathbf{x} \in \mathbb{R}^{r_K+1} : \mathrm{Tr}_{\mathbb{R}^{r_K+1}/\mathbb{R}}(\mathbf{x}) = 0\}.$$

We call S and H the **norm-one hypersurface** and **trace-zero hyperplane** of $\mathbb{R}^{r_K+1}$ respectively. Note that H is a subspace of $\mathbb{R}^{r_K+1}$ of dimension r_K. We will also make use of the subset

$$U = \{\mathbf{z} \in K_{\mathbb{R}}^* : \mathrm{N}_{K_{\mathbb{R}}}(\mathbf{z}) = \pm 1\}.$$

Observe j maps $\mathcal{O}_K^*$ into U and ℓ maps U into H. Let λ denote the restriction of $\log_K$ to $\mathcal{O}_K^*$ and set

$$\Lambda_K = \log_K(\mathcal{O}_K^*),$$

so that Λ_K is the image of λ. In particular, $\Lambda_K \subset H$. We call Λ_K the **unit lattice** of K. It is not immediately obvious that Λ_K is an integral lattice, but we will show this and more. We first show that the unit group and unit lattice of K fit into a short exact sequence.

Proposition 4.3.1. *Let K be a number field. Then there is a short exact sequence*

$$1 \longrightarrow \mu(K) \longrightarrow \mathcal{O}_K^* \longrightarrow \Lambda_K \longrightarrow 1,$$

induced by natural inclusions and where the third map is given by λ.

Proof. As the second map is injective and the third map is surjective, the sequence is exact at $\mu(K)$ and Λ_K. So it remains to prove exactness at $\mathcal{O}_K^*$. For this we must show $\ker\lambda = \mu(K)$. The reverse inclusion holds since the elements of $\mathrm{Hom}_{\mathbb{Q}}(K, \overline{\mathbb{Q}})$ preserve roots of unity. For the forward inclusion, the elements of $j(\ker\lambda)$ are such that each component is absolutely bounded. So one the one hand, this set belongs to a compact subset of $K_{\mathbb{R}}$. On the other hand, this set is also a subset of the integral lattice $j(\mathcal{O}_K)$ which is a discrete subset of $K_{\mathbb{R}}$. Together, this means $j(\ker\lambda)$ is a subset of a compact and discrete set which is necessarily finite. As j is a $\mathbb{Q}$-embedding, it follows that the subgroup $\ker\lambda$ of $\mathcal{O}_K^*$ contains finitely many elements and hence only roots of unity since these are the only elements of $\mathbb{C}$ with finite order. This proves the forward inclusion. $\square$

Our aim now is to show that the unit lattice Λ_K is a a free Z-module of rank r_K. For this, we will require a lemma.

Lemma 4.3.2. *Let K be a number field. There are finitely many elements in $\mathcal{O}_K$ of a given field norm up to multiplication by units.*

Proof. The claim is true for algebraic integers of norm ± 1 as these are exactly the units of K. Now suppose $m \geq 2$. By Proposition 3.6.5, $\mathcal{O}_K/m\mathcal{O}_K$ is finite with $\mathrm{N}_K(m)$ many elements. Therefore, it suffices to show that in each coset there is at most finitely many element of norm n up to multiplication by units. In fact, we will show that there is at most one such element. So let α and β be two representatives in the same coset which are of field norm n. Writing $\alpha = \beta + n\gamma$ for some $\gamma \in \mathcal{O}_K$, we may further write

$$\frac{\alpha}{\beta} = 1 + \frac{\mathrm{N}_K(\beta)}{\beta}\gamma.$$

As any integral ideal contains its norm, Proposition 3.6.5 implies $\frac{\mathrm{N}_K(\beta)}{\beta} \in \mathcal{O}_K$. Whence $\frac{\alpha}{\beta} \in \mathcal{O}_K$ and interchanging the roles of α and β shows that $\frac{\beta}{\alpha} \in \mathcal{O}_K$ too. But then $\frac{\alpha}{\beta}$ is a unit in $\mathcal{O}_K$ and thus α and β differ up to multiplication by a unit. $\square$

With this lemma, we can show Λ_K is a complete integral lattice of H and compute its rank as a free Z-module.

Theorem 4.3.3. *Let K be a number field of degree n and signature (r_1, r_2). Then Λ_K is a complete integral lattice of H. In particular, Λ_K is a free Z-module of rank r_K.*

Proof. We will begin by showing that Λ_K is a lattice of H. Since λ is a homomorphism, Λ_K is a subgroup of H. Therefore Λ_K is a lattice if and only if it is discrete. To prove this, we will show that for any positive constant c, the compact subset

$$X = \{\mathbf{x} \in \mathbb{R}^{r_K+1} : |x_\rho| \leq c \text{ and } |x_\tau| \leq 2c \text{ for all } \rho \text{ and } \tau\},$$

contains only finitely many elements of Λ_K. The preimage of X under ℓ is the compact subset

$$\ell^{-1}(X) = \{\mathbf{z} \in K_{\mathbb{R}} : e^{-c} \leq |z_\sigma| \leq e^c \text{ for all } \sigma\},$$

and hence it contains finitely many elements of $j(\mathcal{O}_K^*)$ which is a subset of the complete integral lattice $j(\mathcal{O}_K)$ of $K_{\mathbb{R}}$. Whence $\ell^{-1}(X) \cap j(\mathcal{O}_K^*)$ is finite. The image of this intersection under ℓ is exactly $X \cap \Lambda_K$. Whence $X \cap \Lambda_K$ is finite too. Therefore Λ_K is discrete and thus a lattice of H.

We now show that Λ_K is a complete integral lattice of H and since H is a real vector space of dimension r_K, this will prove the claim about the rank of Λ_K as well. It is enough to show that there is a bounded subset M of H whose translates by Λ_K cover H. Since ℓ is surjective it suffices to construct a bounded subset T of U such that

$$U = \bigcup_{\varepsilon} j(\varepsilon)T.$$

Indeed, if such a T exists then any $\mathbf{z} \in T$ satisfies $|\mathrm{N}_{K_{\mathbb{R}}}(\mathbf{z})| = 1$ which forces every component of $\mathbf{z}$ is absolutely bounded above and away from zero because T is bounded. Setting $M = \ell(T)$, it follows that M is also bounded and

$$H = \bigcup_{\lambda} (M + \lambda).$$

To construct T, fix constants $c_\sigma > 0$ satisfying $c_\sigma = c_{\overline{\sigma}}$ and

$$\prod_\sigma c_\sigma > \left(\frac{2}{\pi}\right)^{r_2} \sqrt{|\Delta_K|}.$$

Also set $C = \prod_\sigma c_\sigma$. Now consider the bounded subset

$$Z = \{\mathbf{z} \in K_\mathbb{R} : |z_\sigma| < c_\sigma \text{ for all } \sigma\}.$$

By construction, the absolute value of the norm of any element of Z is at most C. For any $\mathbf{w} \in K_\mathbb{R}^*$, we can form the bounded subset

$$\mathbf{w}Z = \{\mathbf{z} \in K_\mathbb{R} : |z_\sigma| < |w_\sigma| c_\sigma \text{ for all } \sigma\}.$$

Letting $\mathbf{u} \in U$, the absolute value of the norm of any element of $\mathbf{u}Z$ is at most C. By Proposition 4.1.1, the volume of $\mathbf{u}Z$ is then 2^{r_2} times the volume of

$$X = \{\mathbf{x} \in \mathbb{R}^n : x_\rho^2 < c_\rho^2 \text{ and } x_\tau^2 + x_{\overline{\tau}}^2 < c_\tau^2 \text{ for all } \rho \text{ and } \tau\},$$

which is $C2^{r_1}\pi^{r_2}$ because X is the product of r_1 many intervals each of length $2c_\rho$ and r_2 many disks each of of radius c_τ. Thus $\mathrm{Vol}(\mathbf{u}Z) = C2^{r_K+1}\pi^{r_2}$. By Proposition 4.1.2, $V_{j(\mathcal{O}_K)} = \sqrt{|\Delta_K|}$ and our choice of C gives

$$\mathrm{Vol}(\mathbf{u}Z) > 2^n V_{j(\mathcal{O}_K)}.$$

By Minkowski's lattice point theorem, there exists a nonzero element of $j(\mathcal{O}_K) \cap \mathbf{u}Z$. Now from Lemma 4.3.2, let $\alpha_1, \ldots, \alpha_m$ be a complete set of representatives of $\mathcal{O}_K$ whose field norm is at most C up to multiplication by a unit. Set

$$T = U \cap \left(\bigcup_i j(\alpha_i)^{-1} Z\right).$$

Then T is a bounded subset of U since the $j(\alpha_i)^{-1}Z$ are. We now claim that

$$U = \bigcup_\varepsilon j(\varepsilon)T.$$

The reverse inclusion holds because each $j(\varepsilon)T$ is a subset of U. For the forward inclusion, let $\mathbf{u} \in U$. Then $\mathbf{u}^{-1} \in U$ whence there exists a nonzero element of $j(\mathcal{O}_K) \cap \mathbf{u}^{-1}Z$. This means $j(\alpha) = \mathbf{u}^{-1}\mathbf{z}$ for some $\alpha \in \mathcal{O}_K$ and $\mathbf{z} \in Z$ with

$$|\,\mathrm{N}_K(\alpha)| < C.$$

But then there exists an α_i and $\varepsilon \in \mathcal{O}_K^*$ such that $\alpha_i = \alpha\varepsilon$. Writing $\mathbf{u} = j(\alpha)^{-1}\mathbf{z}$, a short computation shows

$$\mathbf{u} = j(\varepsilon)j(\alpha_i)^{-1}\mathbf{z},$$

As $\mathbf{u}, j(\varepsilon) \in U$, we see that $j(\alpha_i)^{-1}\mathbf{z} \in U$ and thus $j(\alpha_i)^{-1}\mathbf{z} \in T$. But then $\mathbf{u} \in j(\varepsilon)T$ proving the forward inclusion. $\square$

In view of this result, there exist units $\varepsilon_1, \ldots, \varepsilon_{r_K}$ of K such that $\lambda(\varepsilon_1), \ldots, \lambda(\varepsilon_{r_K})$ is a basis for the unit lattice Λ_K. Indeed, such elements are obtained by taking the preimage under j of any basis for $\Lambda_K/\mathbb{Z}$. Accordingly, we say that $\varepsilon_1, \ldots, \varepsilon_{r_K}$ is a **system of fundamental units** for K and we call any such element a **fundamental unit** for K. **Dirichlet's unit theorem** determines the group structure of the units of K and says that $\mathcal{O}_K^*$ is a direct product of the root of unity in K and products of powers of fundamental units.

Theorem (Dirichlet's unit theorem). *Let K be a number field of degree n and signature (r_1, r_2). Then*

$$\mathcal{O}_K^* \cong \mu(K) \times \mathbb{Z}^{r_K}.$$

In particular, if $\varepsilon_1, \ldots, \varepsilon_{r_K}$ is a system of fundamental units for K then any unit ε of K is of the form

$$\varepsilon = \omega \varepsilon_1^{\nu_1} \cdots \varepsilon_{r_K}^{\nu_{r_K}},$$

for some $\omega \in \mu(K)$ and integers ν_i.

Proof. As Λ_K is a free $\mathbb{Z}$-module by Theorem 4.3.3, the short exact sequence in Proposition 4.3.1 splits. Whence

$$\mathcal{O}_K^* \cong \mu(K) \times \mathbb{Z}^{r_K},$$

because the rank of Λ_K is r_K. This proves the first statement. For the second statement, let $\varepsilon_1, \ldots, \varepsilon_{r_K}$ be a system of fundamental units for K and let E be the subgroup of $\mathcal{O}_K^*$ generated by them. Then Proposition 4.3.1 and the first isomorphism theorem together show that λ induces an isomorphism $E \cong \mathbb{Z}^{r_K}$. Then

$$\mathcal{O}_K^* \cong \mu(K) \times E,$$

from which the second statement follows. □

Recall from Theorem 4.3.3 that Λ_K is a complete integral lattice of H. The determination of its covolume V_{Λ_K} it important as it measures how dense the units of K are. To this end, let $\varepsilon_1, \ldots, \varepsilon_{r_K}$ be a system of fundamental units for K so that $\lambda(\varepsilon_1), \ldots, \lambda(\varepsilon_{r_K})$ is a basis for $\Lambda_K/\mathbb{Z}$. Setting

$$\lambda_0 = \frac{\mathbf{1}}{\sqrt{r_K + 1}},$$

we see that λ_0 is a unit vector in $\mathbb{R}^{r_K+1}$ orthogonal to H. Whence λ_0 is orthogonal to Λ and it follows that $\lambda_0, \lambda(\varepsilon_1), \ldots, \lambda(\varepsilon_{r_K})$ is a basis for the complete integral lattice $\Lambda'_K = \mathbb{Z}\lambda_0 + \Lambda_K$ of $\mathbb{R}^{r_K+1}$. Since λ_0 is a unit vector, the volume of a fundamental domain for Λ'_K in $\mathbb{R}^{r_K+1}$ is equal to the volume of a fundamental domain for Λ_K in H. This means that the corresponding covolumes are equal which is to say $V_{\Lambda_K} = V_{\Lambda'_K}$. So it suffices to compute $V_{\Lambda'_K}$. The associated generator matrix for Λ'_K is given by

$$\begin{pmatrix} \frac{1}{\sqrt{r_K+1}} & \lambda(\varepsilon_1)_1 & \cdots & \lambda(\varepsilon_{r_K})_1 \\ \vdots & \vdots & & \vdots \\ \frac{1}{\sqrt{r_K+1}} & \lambda(\varepsilon_1)_{r_K+1} & \cdots & \lambda(\varepsilon_{r_K})_{r_K+1} \end{pmatrix}.$$

Adding all of the rows to an arbitrary fixed row results in

$$\begin{pmatrix} \frac{1}{\sqrt{r_K+1}} & \lambda(\varepsilon_1)_1 & \cdots & \lambda(\varepsilon_{r_K})_1 \\ \vdots & \vdots & & \vdots \\ \sqrt{r_K+1} & 0 & & 0 \\ \vdots & \vdots & & \vdots \\ \frac{1}{\sqrt{r_K+1}} & \lambda(\varepsilon_1)_{r_K+1} & \cdots & \lambda(\varepsilon_{r_K})_{r_K+1} \end{pmatrix},$$

which has the same determinant as that of the generator matrix. Cofactor expanding along the row with all zeros except the first entry shows

$$V_{\Lambda_K} = \sqrt{r_K+1}R_K,$$

where R_K is the absolute value of the determinant of any rank r_K minor of

$$\begin{pmatrix} \lambda(\varepsilon_1)_1 & \cdots & \lambda(\varepsilon_{r_K})_1 \\ \vdots & & \vdots \\ \lambda(\varepsilon_1)_{r_K+1} & \cdots & \lambda(\varepsilon_{r_K})_{r_K+1} \end{pmatrix}.$$

Indeed, we may choose any rank r_K minor since the cofactor expansion was along an arbitrary row. We call R_K the **regulator** of K. Since the covolume V_{Λ_K} of Λ_K is independent of the choice of basis for $\Lambda_K/\mathbb{Z}$, the regulator R_K is independent of the choice of a system of fundamental units for K. The regulator is a rough measure of the density of the units for K in the sense that smaller the regulator is the more dense the units are.

Let S be a finite set of primes of K. Then it is a simple matter to determine the structure of the S-unit group of $\mathcal{O}_{K,S}$. As the ideal class group is finite, the long exact sequence Proposition 3.4.7 in this setting contracts to the short exact sequence

$$1 \longrightarrow \mathcal{O}_K^* \longrightarrow \mathcal{O}_{K,S}^* \longrightarrow \mathbb{Z}^{|S|} \longrightarrow 1,$$

which splits because $\mathbb{Z}^{|S|}$ is a free $\mathbb{Z}$-module. This fact and Dedekind's unit theorem together imply

$$\mathcal{O}_{K,S}^* \cong \mu(K) \times \mathbb{Z}^{r_K+|S|}.$$

So the rank of the S-unit group of K is generally larger than the rank of the unit group of K.

Let $\mathcal{O}$ be an order of K. Then we can also describe the structure of the unit group of $\mathcal{O}$. Indeed, we have already shown $\mathcal{O}_K^*/\mathcal{O}^*$ is finite. By the structure theorem for finitely generated modules over principal ideal domains, $\mathcal{O}^*$ must have the same rank as $\mathcal{O}_K^*$. Then Dirichlet's unit theorem implies

$$\mathcal{O}^* \cong \mu(\mathcal{O}) \times \mathbb{Z}^{r_K},$$

for some subgroup $\mu(\mathcal{O})$ of $\mu(K)$. Therefore the unit group of $\mathcal{O}$ is generally smaller than the unit group of K.

Part III

Analytic Number Theory

Chapter 5

Dirichlet Series

Throughout, $s = \sigma + it$ and $s_0 = \sigma_0 + it_0$ will stand for complex variables with σ, σ_0, t, and t_0 real. We will also use the Mellin transform and the corresponding inversion theorem developed in Appendix C.1.

5.1 Convergence Properties

A **Dirichlet series** $D(s)$ is a sum of the form

$$D(s) = \sum_{n \geq 1} \frac{a(n)}{n^s},$$

with $a(n) \in \mathbb{C}$. Our first aim is to understand where Dirichlet series converge and where they converge absolutely. It does not take much for $D(s)$ to converge uniformly in a sector.

Theorem 5.1.1. *Suppose $D(s)$ is a Dirichlet series that converges at $s_0 = \sigma_0 + it_0$. Then for any $H > 0$, $D(s)$ converges uniformly in the sector*

$$\{s \in \mathbb{C} : \sigma \geq \sigma_0 \text{ and } |t - t_0| \leq H(\sigma - \sigma_0)\}.$$

In particular, $D(s)$ converges in the half-plane $\sigma > \sigma_0$.

Proof. Write $R(u) = \sum_{n \geq u} \frac{a(n)}{n^{s_0}}$ for the tail of $D(s_0)$ so that

$$\frac{a(n)}{n^{s_0}} = (R(n) - R(n+1)).$$

Then for positive integers N and M with $M \leq N$, summation by parts implies

$$\sum_{M \leq n \leq N} \frac{a(n)}{n^s} = R(M)M^{s_0-s} - R(N+1)(N+1)^{s_0-s} - \sum_{M \leq n \leq N} R(n+1)(n^{s_0-s} - (n+1)^{s_0-s}).$$

We will express the sum on the right-hand side as an integral. To do this, observe that

$$n^{s_0-s} - (n+1)^{s_0-s} = -(s_0 - s) \int_n^{n+1} u^{s_0-s-1}\, du.$$

As $R(u)$ is constant on the interval $(n, n+1]$ a short computation shows

$$\sum_{M \leq n \leq N} R(n+1)(n^{s_0-s} - (n+1)^{s_0-s}) = -(s_0 - s) \int_M^{N+1} R(u) u^{s_0-s-1} \, du,$$

Whence

$$\sum_{M \leq n \leq N} \frac{a(n)}{n^s} = R(M)M^{s_0-s} - R(N+1)(N+1)^{s_0-s} + (s_0 - s) \int_M^{N+1} R(u) u^{s_0-s-1} \, du.$$

As $D(s)$ converges at $s = s_0$, we can choose M sufficiently large such that $|R(u)| < \varepsilon$ for $u \geq M$. It follows that $|R(u)u^{s_0-s}| < \varepsilon$ for such u provided s in the desired sector. For such s, we have

$$|s - s_0| \leq (\sigma - \sigma_0) + |t - t_0| \leq (H+1)(\sigma - \sigma_0).$$

These estimates together imply

$$\sum_{M \leq n \leq N} \frac{a(n)}{n^s} = O\left(2\varepsilon + \varepsilon(H+1)(\sigma - \sigma_0) \int_M^{N+1} u^{\sigma_0 - \sigma - 1} \, du\right).$$

Since the integral is $O\left(\frac{1}{\sigma - \sigma_0}\right)$, the sum is $o(1)$ for s in the desired sector. The first statement follows by uniform Cauchy's criterion. Taking the limit as $H \to \infty$ proves the second statement. $\square$

We will want to keep track of where Dirichlet series converge absolutely. Let σ_c be the infimum of all σ for which $D(s)$ converges. We call σ_c the **abscissa of convergence** of $D(s)$. Similarly, let σ_a be the infimum of all σ for which $D(s)$ converges absolutely. We call σ_a the **abscissa of absolute convergence** of $D(s)$. As the summands of $D(s)$ are holomorphic, convergence is locally absolutely uniform for $\sigma > \sigma_a$ (actually uniform in sectors) and so $D(s)$ is holomorphic in this half-plane.

The abscissas σ_c and σ_a act as the boundaries of convergence and absolute convergence respectively. Anything can happen on the lines $\sigma = \sigma_c$ and $\sigma = \sigma_a$, but to the right of them we have convergence and absolute convergence of $D(s)$ respectively. It turns out that σ_a is never far from σ_c provided σ_c is finite.

Theorem 5.1.2. *If $D(s)$ is a Dirichlet series with finite abscissa of convergence σ_c then*

$$\sigma_c \leq \sigma_a \leq \sigma_c + 1.$$

Proof. The first inequality is clear since absolute convergence implies convergence. For the second inequality, the terms $a(n)n^{-(\sigma_c+\varepsilon)}$ tend to zero as $n \to \infty$ because $D(s)$ converges at $s = \sigma_c + \varepsilon$. Therefore $a(n) \ll_\varepsilon n^{\sigma_c+\varepsilon}$ and so $D(s)$ is absolutely convergent at $s = \sigma_c + 1 + 2\varepsilon$. This means $\sigma_a \leq \sigma_c + 1 + 2\varepsilon$. Taking the limit as $\varepsilon \to 0$ gives the second inequality. $\square$

We now turn to the question of uniqueness of Dirichlet series. In particular, we would like to show that Dirichlet series are uniquely determined by their coefficient as this would allow us compare coefficient analogous to that of Taylor series. This is indeed possible.

Proposition 5.1.3. *Suppose $D(s)$ is a Dirichlet series with finite abscissa of convergence σ_c such that*

$$D(s) = \sum_{n \geq 1} \frac{a(n)}{n^s} \quad \textit{and} \quad D(s) = \sum_{n \geq 1} \frac{b(n)}{n^s}.$$

Then $a(n) = b(n)$.

Proof. Set $c(n) = a(n) - b(n)$ so that it suffices to prove $c(n) = 0$. Take $\sigma > \sigma_a$ so that $D(s)$ converges uniformly. Letting $\sigma \to \infty$, we have

$$\lim_{\sigma \to \infty} D(\sigma) = a(1) \quad \text{and} \quad \lim_{\sigma \to \infty} D(\sigma) = b(1),$$

upon interchanging the limit and the sum by uniform convergence. Whence $c(1) = 0$. Assume by induction that $c(n) = 0$ for $n \leq N$ and consider the Dirichlet series

$$\sum_{n \geq 1} \frac{c(n)}{n^s}.$$

Its abscissa of absolute convergence is σ_a. As

$$\sum_{n \leq N} \frac{c(n)}{n^s} = 0,$$

by assumption, it follows that

$$c(N+1) = -\sum_{n > N} c(n) \left(\frac{N}{n} \right)^{\sigma}.$$

Take $\sigma > \sigma_a$ so that sum on the right-hand side converges uniformly. Letting $\sigma \to \infty$, we have

$$\lim_{\sigma \to \infty} \left(-\sum_{n > N} c(n) \left(\frac{N}{n} \right)^{\sigma} \right) = 0,$$

upon interchanging the limit and sum by uniform convergence. Hence $c(N+1) = 0$ which completes the proof. $\square$

We will now introduce several resulting concerning the growth and average growth of the coefficients of a Dirichlet series. For legibility, it will be useful to introduce some notation. If $D(s)$ is a Dirichlet series with coefficients $a(n)$ then for $X > 0$, we set

$$A(X) = \sum_{n \leq X} a(n) \quad \text{and} \quad |A|(X) = \sum_{n \leq X} |a(n)|.$$

These are the partial sums of the coefficients $a(n)$ and their absolute values up to X respectively. Many of our results will be in terms of these sums. Our first result shows that the coefficients of a Dirichlet series grown at most polynomially provided the Dirichlet series converges absolutely at some point.

Proposition 5.1.4. *Suppose $D(s)$ is a Dirichlet series with coefficients $a(n)$ that converges absolutely at $s = \alpha$ for some real α. Then*

$$a(n) \ll_\varepsilon n^{\alpha+\varepsilon}.$$

In particular, if $D(s)$ has finite abscissa of absolute convergence σ_a then

$$a(n) \ll_\varepsilon n^{\sigma_a+\varepsilon}.$$

Proof. Necessarily $\sigma_a \leq \alpha$ and so $D(s)$ converges absolutely in the half-plane $\sigma > \alpha$. Write

$$|a(n)| \leq n^{\alpha+\varepsilon} \sum_{n \geq 1} \frac{|a(n)|}{n^{\alpha+\varepsilon}}.$$

The sum is $O_\varepsilon(1)$ because $D(s)$ is absolutely convergent for $\sigma > \alpha$. The first estimate follows at once. The second is immediate from the first and the definition of the abscissa of absolute convergence. $\square$

It is natural to be curious about the converse, namely, is it possible to determine an upper bound on the abscissa of absolute convergence if we know a polynomial bound for the coefficients. This is indeed possible.

Proposition 5.1.5. *Suppose $D(s)$ is a Dirichlet series whose coefficients satisfy the estimate $a(n) \ll_\alpha n^\alpha$ for some real α. Then the abscissa of absolute convergence satisfies $\sigma_a \leq \alpha + 1$.*

Proof. Let $\sigma > \alpha + 1$. It suffices to show convergence of the series

$$\sum_{n \geq 1} \frac{|a(n)|}{n^\sigma}.$$

As

$$\sum_{n \geq 1} \frac{|a(n)|}{n^\sigma} \ll_\alpha \sum_{n \geq 1} \frac{1}{n^{\sigma-\alpha}},$$

and the latter series converges, the proof is complete. $\square$

Let us now turn to the same questions but using averages of the coefficients. Our first result is analogous to Proposition 5.1.4.

Proposition 5.1.6. *Suppose $D(s)$ is a Dirichlet series with coefficients $a(n)$ that converges absolutely at $s = \alpha$ for some real α. Then*

$$A(X) \ll X^{\alpha+\varepsilon} \quad \textit{and} \quad |A|(X) \ll X^{\alpha+\varepsilon}.$$

In particular,

$$A(X) \ll_\varepsilon X^{\sigma_a+\varepsilon} \quad \textit{and} \quad |A|(X) \ll_\varepsilon X^{\sigma_a+\varepsilon}.$$

Proof. Necessarily $\sigma_a \leq \alpha$ and so $D(s)$ converges absolutely in the half-plane $\sigma > \alpha$. Write

$$\sum_{n \leq X} |a(n)| \leq X^{\alpha+\varepsilon} \sum_{n \geq 1} \frac{|a(n)|}{n^{\alpha+\varepsilon}}.$$

The sum is $O_\varepsilon(1)$ because $D(s)$ is absolutely convergent for $\sigma > \alpha$. As $A(x) \ll |A|(x)$, the first statement follows at once. The second is immediate from the first and the definition of the abscissa of absolute convergence. □

It turns out that if $A(X)$ is bounded then σ_c is negative.

Proposition 5.1.7. *Suppose $D(s)$ is a Dirichlet series and that $A(X) \ll 1$. Then $\sigma_c \leq 0$.*

Proof. Let $\sigma > 0$. Abel summation gives

$$\sum_{n \leq X} \frac{a(n)}{n^s} = \frac{A(X)}{X^s} + \int_1^X A(u)u^{-(s-1)}\,du.$$

As $A(X) \ll 1$, take the limit as $X \to \infty$ to obtain

$$D(s) = s\int_1^\infty A(u)u^{-(s+1)}\,du.$$

This expresses $D(s)$ as an integral. Direct evaluation shows that the integral is finite. Therefore $D(s)$ converges for $\sigma > 0$ which means $\sigma_c \leq 0$. □

Unfortunately, it is not often the case that $A(X)$ is bounded as this is quite a strong condition for most Dirichlet series. Fortunately, we can still obtain nice result analogous to that of Proposition 5.1.5 if we assume $|A|(X)$ grows at most polynomially.

Proposition 5.1.8. *Suppose $D(s)$ is a Dirichlet series such that $|A|(X) \ll_\alpha X^\alpha$ for some $\alpha \geq 0$. Then the abscissa of absolute convergence satisfies $\sigma_a \leq \alpha$.*

Proof. Let $\sigma > \alpha$. It suffices to show convergence of the series

$$\sum_{n \geq 1} \frac{|a(n)|}{n^\sigma}.$$

Abel summation gives

$$\sum_{n \leq X} \frac{|a(n)|}{n^\sigma} = \frac{|A|(X)}{X^\sigma} + \sigma\int_1^X |A|(u)u^{-(\sigma+1)}\,du.$$

In view of the bound $|A|(X) \ll_\alpha X^\alpha$, take the limit as $X \to \infty$ to obtain

$$\sum_{n \geq 1} \frac{|a(n)|}{n^\sigma} = \sigma\int_1^\infty |A|(u)u^{-(\sigma+1)}\,du.$$

Direct evaluation shows that the integral is bounded. Therefore our series is bounded and hence must converge. □

Unfortunately, Proposition 5.1.8 does not immediately imply a sharper upper bound for the abscissa of absolute convergence than that of Proposition 5.1.5. This is because if $a(n) \ll_\alpha n^\alpha$ then the trivial bounds for $A(X)$ and $|A(X)|$ are

$$A(X) \ll_\alpha X^{\alpha+1} \quad \text{and} \quad |A|(X) \ll_\alpha X^{\alpha+1}.$$

In particular, using the second bound in Proposition 5.1.8 will give $\sigma_a \leq \alpha + 1$. So if we want to obtain sharper upper bounds for the abscissa of absolute convergence then we must improve the polynomial bound for the coefficients directly. Unfortunately, this is often a daunting task in practice especially if the Dirichlet series is connected to a deep algebraic or arithmetic object. However, if we assume that the coefficients are nonnegative then **Landau's theorem** provides a way of locating the abscissa of absolute convergence exactly:

Theorem (Landau's theorem). *Suppose $D(s)$ is a Dirichlet series with nonnegative coefficients $a(n)$ and finite abscissa of absolute convergence σ_a. Then σ_a is a singularity of $D(s)$.*

Proof. Replacing $a(n)$ by $a(n)n^{-\sigma_a}$, if necessary, we may assume $\sigma_a = 0$. Now suppose to the contrary that $D(s)$ was holomorphic at $s = 0$. Then $D(s)$ is holomorphic in the domain

$$\mathcal{D} = \{s \in \mathbb{C} : \sigma_a > 0\} \cup \{s \in \mathbb{C} : |s| < \delta\},$$

for some $\delta > 0$. Let $P(s)$ be the power series expansion of $D(s)$ at $s = 1$. Then

$$P(s) = \sum_{k \geq 0} \frac{c_k}{k!}(s-1)^k,$$

where

$$c_k = \sum_{n \geq 1} \frac{a(n)(-\log(n))^k}{n},$$

upon differentiating $D(s)$ termwise. The radius of convergence of $P(s)$ is the distance from $s = 1$ to the nearest singularity of $P(s)$. Since $P(s)$ is holomorphic on $\mathcal{D}$, the closest possible singularities are at $s = \pm i\delta$. Therefore, the radius of convergence is at least $\sqrt{1+\delta^2}$. Write $\sqrt{1+\delta^2} = 1 + \delta'$ for some $\delta' > 0$. Then for $|s-1| < 1+\delta'$, $P(s)$ is holomorphic and can be expressed as

$$P(s) = \sum_{k \geq 0} \frac{(s-1)^k}{k!} \sum_{n \geq 1} \frac{a(n)(-\log(n))^k}{n}.$$

Now choose s in this region to be real with with $-\delta' < s < 1$. Then this double sum is a sum of positive terms by assumption of the $a(n)$ being nonnegative. As $P(s)$ is necessarily absolutely convergent, interchanging the sums and a short computation shows

$$P(s) = D(s).$$

As $-\delta' < s < 1$ and $D(s)$ has nonnegative coefficients, it converges absolutely for some $s < 0$. This contradicts $\sigma_a = 0$. □

There is a very important distinction in Landau's theorem that abscissa of absolute convergence is a singularity and not just a point where the Dirichlet series does not converge. Indeed, this means that if $D(s)$ could be analytically continued to a region containing $\sigma = \sigma_a$ then the continuation must have a pole at $s = \sigma_a$. In particular, the abscissa of absolute convergence would then be a pole with the largest possible real part. This signals that the singularity is an inherent property of the analytic continuation and not one of the representation of the function as a Dirichlet series.

5.2 Euler Products

Generally speaking, if the coefficients $a(n)$ are chosen at random, $D(s)$ is not guaranteed to good properties outside of absolute convergence in some half-plane (provided it converges at a point). However, many Dirichlet series of interest will have coefficients that are multiplicative. These Dirichlet series admits product expressions.

Proposition 5.2.1. *Suppose $D(s)$ is a Dirichlet series with finite abscissa of absolute convergence σ_a and whose coefficients $a(n)$ are multiplicative. Then*

$$D(s) = \prod_p \left(\sum_{k \geq 0} \frac{a(p^k)}{p^{ks}} \right),$$

for $\sigma > \sigma_a$. If the prime power coefficients satisfy

$$a(p^k) = \sum_{j_1 \leq \cdots \leq j_k} \alpha_{j_1}(p) \cdots \alpha_{j_k}(p),$$

for some $\alpha_1(p), \ldots, \alpha_d(p) \in \mathbb{C}$ and positive integer d, then the product takes the form

$$D(s) = \prod_p (1 - \alpha_1(p)p^{-s})^{-1} \cdots (1 - \alpha_d(p)p^{-s})^{-1}.$$

Proof. Let $\sigma > \sigma_a$ and consider the series

$$\sum_{k \geq 0} \frac{a(p^k)}{p^{ks}}.$$

This series converges absolutely because $D(s)$ does. Now let N be a positive integer. By absolute convergence and the fundamental theorem of arithmetic,

$$\prod_{p \leq N} \left(\sum_{k \geq 0} \frac{a(p^k)}{p^{ks}} \right) = \sum_{n \leq N} \frac{a(n)}{n^s} + {\sum_{n > N}}^{*} \frac{a(n)}{n^s},$$

where the $*$ indicates that we are summing over only those additional terms $\frac{a(n)}{n^s}$ that appear in the expanded product. As $N \to \infty$, the first sum on the right-hand side tends to $D(s)$ and the second sum tends to zero because it is part of the tail of $D(s)$.

This proves that the product converges and is equal to $D(s)$. If the prime power coefficients are of the form

$$a(p^k) = \sum_{j_1 \leq \cdots \leq j_k} \alpha_{j_1}(p) \cdots \alpha_{j_k}(p),$$

then

$$\sum_{k \geq 0} \frac{a(p^k)}{p^{ks}} = (1 - \alpha_1(p)p^{-s})^{-1} \cdots (1 - \alpha_d(p)p^{-s})^{-1},$$

from which the product formula follows. $\square$

The representation

$$D(s) = \prod_p \left(\sum_{k \geq 0} \frac{a(p^k)}{p^{ks}} \right),$$

the called the **Euler product** of $D(s)$. By Proposition 5.2.1, multiplicativity of the coefficients is enough to ensure that the Euler product exists and is equal to the Dirichlet series in the half-plane of absolute convergence. If the Euler product takes the form

$$D(s) = \prod_p (1 - \alpha_1(p)p^{-s})^{-1} \cdots (1 - \alpha_d(p)p^{-s})^{-1},$$

then it is said to be of **degree** d. The special case of complete multiplicative coefficients corresponds to degree 1 Euler products as

$$D(s) = \prod_p (1 - a(p)p^{-s})^{-1}.$$

In any case, the existence of $\alpha_1(p), \ldots, \alpha_d(p) \in \mathbb{C}$ with

$$a(p^k) = \sum_{j_1 \leq \cdots \leq j_k} \alpha_{j_1}(p) \cdots \alpha_{j_k}(p),$$

guarantees that the Dirichlet series has an Euler product of degree d.

Remark 5.2.2. Replacing $D(s)$ with its absolute series in Proposition 5.2.1 shows that

$$\sum_{n \geq 1} \frac{|a(n)|}{n^\sigma} = \prod_p \left(\sum_{k \geq 0} \frac{|a(p^k)|}{p^{k\sigma}} \right)$$

for $\sigma > \sigma_a$. Under the stronger assumption of an Euler product of degree d this identity becomes

$$\sum_{n \geq 1} \frac{|a(n)|}{n^\sigma} = \prod_p (1 - |\alpha_1(p)|p^{-\sigma})^{-1} \cdots (1 - |\alpha_d(p)|p^{-\sigma})^{-1}.$$

Either of these equalities is stronger than mere absolute convergence of the infinite product since each factor is replaced with its absolute series not just the absolute value of the series. In particular, this implies that the Euler product converges locally absolutely uniformly in the same region that the Dirichlet series does.

If $D(s)$ admits a Euler product, we write $D^{(N)}(s)$ to denote the Dirichlet series with the factors $p \mid N$ in the Euler product removed. This means

$$D^{(N)}(s) = D(s) \prod_{p \mid N} \left(\sum_{k \geq 0} \frac{a(p^k)}{p^{ks}} \right).$$

Dually, we let $D_{(N)}(s)$ denote the Dirichlet series consisting only of the factors $p \mid N$ in the Euler product. This means

$$D_{(N)}(s) = \prod_{p \mid N} \left(\sum_{k \geq 0} \frac{a(p^k)}{p^{ks}} \right).$$

With this notation, we have the relationship

$$D(s) = D^{(N)}(s) D_{(N)}(s).$$

5.3 Dirichlet Convolution

We now turn to discussing how Dirichlet series behave with respect to products. Let $D_f(s)$ and $D_g(s)$ be the Dirichlet series defined by

$$D_f(s) = \sum_{n \geq 1} \frac{f(n)}{n^s} \quad \text{and} \quad D_g(s) = \sum_{n \geq 1} \frac{g(n)}{n^s},$$

for some arithmetic functions f and g. Let $f * g$ be the Dirichlet convolution of f and g. A short computation shows

$$D_f(s) D_g(s) = D_{f*g}(s).$$

In other words, $D_f(s)D_g(s)$ is again a Dirichlet series whose coefficients are given by the Dirichlet convolution of f and g. This relation also shows that $D_{f*g}(s)$ converges absolutely wherever both $D_f(s)$ and $D_g(s)$ do. Since Dirichlet convolution preserves multiplicativity, $D_{f*g}(s)$ will have multiplicative coefficients if both $D_f(s)$ and $D_g(s)$ do. Moreover, from the Möbius inversion formula we immediately find that

$$D_g(s) = D_{f*\mathbf{1}}(s),$$

if and only if

$$D_f(s) = D_{g*\mu}(s).$$

These identities can be used to compute the Dirichlet series for many types of arithmetic functions.

5.4 Perron Formulas

With Mellin inversion, it is possible to relate the sums of coefficients of a Dirichlet series to an integral of its associated Dirichlet series. Such formulas are desirable because they allow for the examination of these sums by methods in complex analysis. First, we setup some general notation. Let $D(s)$ be a Dirichlet series with coefficients $a(n)$. For $X > 0$, we set

$$A^*(X) = \sum_{n \le X}{}^{*} a(n),$$

where the $*$ indicates that the last term is multiplied by $\frac{1}{2}$ if X is an integer. This slight modification of $A(X)$ accounts for the fact that Mellin inversion returns the average of the left-hand and right-hand limits at a jump discontinuity. We would like to relate $A^*(X)$ to the inverse Mellin transform of $D(s)$. We will prove several variants of this basic idea. The first being (classical) **Perron's formula** which is a consequence of Abel summation and Mellin inversion applied to Dirichlet series.

Theorem (Perron's formula, classical). *Let $D(s)$ be a Dirichlet series with coefficient $a(n)$ and finite and nonnegative abscissa of absolute convergence σ_a. Then for any $c > \sigma_a$, we have*

$$A^*(X) = \frac{1}{2\pi i} \int_{(c)} D(s) X^s \, \frac{ds}{s}.$$

Proof. Let $\sigma > \sigma_a$. By Proposition 5.1.6, $A(X) \ll_\varepsilon X^{\sigma_a + \varepsilon}$. Taking ε sufficiently small, we find that $A(X)X^{-s} \to 0$ as $X \to \infty$. Abel summation then gives

$$D(s) = s \int_1^\infty A(u) u^{-(s+1)} \, du. \tag{5.1}$$

As $A(u) = 0$ in the interval $[0, 1)$, we can write the previous identity in the form

$$\frac{D(s)}{s} = \int_0^\infty A(u) u^{-(s+1)} \, du.$$

Mellin inversion immediately gives the result. □

In the proof of classical Perron's formula, we obtained a useful integral representation for a Dirichlet series. We collect this in the following corollary:

Corollary 5.4.1. *Let $D(s)$ be a Dirichlet series with finite and nonnegative abscissa of absolute convergence σ_a. Then for $\sigma > \sigma_a$, we have*

$$D(s) = s \int_1^\infty A(u) u^{-(s+1)} \, du.$$

Proof. The identity is Equation (5.1). □

Classical Perron's formula is not always useful in applications because often it is necessary to estimate the integral of the Dirichlet series. As the integral need not be absolutely convergent, this would require nontrivial estimates for the Dirichlet series in vertical strips. Fortunately, there are two ways to correct for this defect each of which leads to a different variant of Perron's formula. The first is to truncate the integral while the latter is to introduce a smoothing function. In the former case, a careful estimation of the error term is often necessary while the latter case requires estimates for the smoothing function. Both variants can be equally useful and the choice of which to use is often dependent upon what methods are available in the current setting.

Let us first focus on the truncated variant. To state it, we will need to setup some notation and prove a lemma. For $c > 0$ and $T > 0$, consider the integrals

$$\delta(y) = \frac{1}{2\pi i}\int_{(c)} y^s \frac{ds}{s} \quad \text{and} \quad I(y,T) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} y^s \frac{ds}{s},$$

defined for $y > 0$. Note that $I(y,T)$ is just $d(y)$ truncated outside height T. The lemma we require gives an explicit evaluation of $\delta(y)$ and gives an approximation for the error between $\delta(y)$ and its truncation $I(y,T)$. The proof is quite laborious but standard.

Lemma 5.4.2. *We have*

$$\delta(y) = \begin{cases} 0 & \text{if } y < 1, \\ \frac{1}{2} & \text{if } y = 1, \\ 1 & \text{if } y > 1. \end{cases}$$

Moreover,

$$I(y,T) - \delta(y) = \begin{cases} O\left(y^c \min\left(1, \frac{1}{T\log(y)}\right)\right) & \text{if } y \neq 1, \\ O\left(\frac{c}{T}\right) & \text{if } y = 1. \end{cases}$$

Proof. Since $I(y,T) \to \delta(y)$ as $T \to \infty$, it suffices to estimate $I(y,T)$ and then take the limit as $T \to \infty$. First consider the case $y = 1$. A short computation shows

$$I(1,T) = \frac{1}{\pi}\tan^{-1}\left(\frac{T}{c}\right).$$

Truncate the Laurent series of the inverse tangent after the first term to write $\tan^{-1}(t) = \frac{\pi}{2} + O\left(\frac{1}{t}\right)$. Then

$$I(1,T) = \frac{1}{2} + O\left(\frac{c}{T}\right),$$

and we see that $\delta(y) = \frac{1}{2}$. This proves everything when $y = 1$. Now suppose $y < 1$ and let $d > c$. Let $\eta = \eta_1 + \cdots + \eta_4$ be the rectangular contour in Figure 5.1 where the horizontal lines are along $t = \pm T$ and the vertical lines are along $\sigma = c$ and $\sigma = d$. Consider

$$\frac{1}{2\pi i}\int_{\eta} y^s \frac{ds}{s}.$$

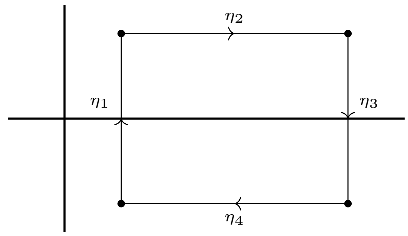

Figure 5.1: A rectangular contour.

We will evaluate this integral in the limit as $d \to \infty$. The contour does not enclose the only pole of the integrand which is at $s = 0$. So on the one hand, the residue theorem gives

$$\frac{1}{2\pi i}\int_{\eta} y^s \, \frac{ds}{s} = 0.$$

On the other hand, the integral is a sum along all four contours. Observe that

$$\frac{1}{2\pi i}\int_{\eta_1} y^s \, \frac{ds}{s} = I(y,T).$$

For the integrals over η_2 and η_4, the parameterizations $s \mapsto \sigma \pm iT$ show

$$\frac{1}{2\pi i}\int_{\eta_2} y^s \, \frac{ds}{s} + \frac{1}{2\pi i}\int_{\eta_4} y^s \, \frac{ds}{s} = O\left(\frac{y^c}{\log(y)T}\right),$$

as $y < 1$. For the integral over η_3, the parameterization $s \mapsto d + it$ shows

$$\frac{1}{2\pi i}\int_{\eta_3} y^s \, \frac{ds}{s} = O\left(y^d \log(T)\right).$$

Whence

$$I(y,T) = O\left(\frac{y^c}{\log(y)T}\right),$$

upon taking the limit as $d \to \infty$ since $y < 1$. It follows that $\delta(y) = 0$. We now obtain another estimate for $I(y,T)$ this time using a modified contour. Let $\eta = \eta_1 + \eta_2$ be the semicircular contour in Figure 5.2 where the vertical line is along $\sigma = c$ with endpoints at $s = c \pm iT$.

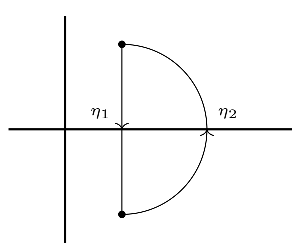

Figure 5.2: A semicircular contour.

As before, the residue theorem gives

$$\frac{1}{2\pi i}\int_{\eta} y^s \, \frac{ds}{s} = 0.$$

However, now

$$\frac{1}{2\pi i}\int_{\eta_1} y^s \, \frac{ds}{s} = -I(y,T).$$

The parameterization $s \mapsto \sqrt{(c^2+T^2)}e^{i\theta}$ shows that

$$\frac{1}{2\pi i}\int_{\eta_2} y^s \, \frac{ds}{s} = O(y^c).$$

Whence

$$I(y,T) = O(y^c).$$

Combining these two estimates for $I(y,T)$ proves everything when $y < 1$. Now suppose $y > 1$ and let $d < 0$. We argue as in the case $y < 1$ except we use the the rectangular contour in Figure 5.3.

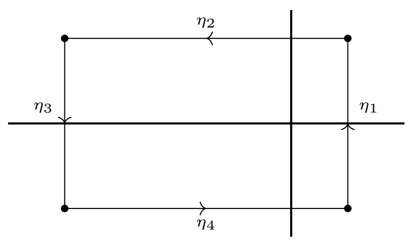

Figure 5.3: A rectangular contour.

This time the the contour encloses the simple pole of the integrand at $s = 0$ whose residue is 1. Arguing analogously, we find

$$I(y,T) = 1 + O\left(\frac{y^c}{\log(y)T}\right),$$

and so $\delta(y) = 1$.

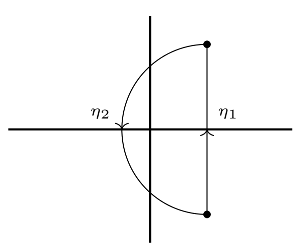

Figure 5.4: A semicircular contour.

We now modify the contour by using the semicircular one in Figure 5.4. Again, the contour encloses the simple pole of the integrand at $s = 0$ whose residue is 1. Arguing analogously, we obtain

$$I(y,T) = 1 + O(y^c).$$

Combining these two estimates for $I(y,T)$ proves everything when $y < 1$ and completes the proof. $\square$

This result will provide the estimate we need to prove (truncated) **Perron's formula**:

Theorem (Perron's formula, truncated). *Let $D(s)$ be a Dirichlet series with coefficient $a(n)$ and finite and nonnegative abscissa of absolute convergence σ_a. Then for any $c > \sigma_a$ and $T > 0$, we have*

$$\begin{aligned} A^*(X) &= \frac{1}{2\pi i}\int_{c-iT}^{c+iT} D(s)X^s\,\frac{ds}{s} \\ &\quad + O\left(X^c \sum_{\substack{n\geq 1\\ n\neq X}} \frac{|a(n)|}{n^c}\min\left(1, \frac{1}{T\left|\log\left(\frac{X}{n}\right)\right|}\right) + \delta_X |a(X)|\frac{c}{T}\right), \end{aligned}$$

where $\delta_X = 1, 0$ according to if X is an integer or not.

Proof. By Lemma 5.4.2, we may write

$$A^*(X) = \sum_{n\geq 1} a(n)\delta\left(\frac{X}{n}\right),$$

and

$$\delta(y) = I(y,T) - \begin{cases} O\left(y^c \min\left(1, \frac{1}{T\log(y)}\right)\right) & \text{if } y \neq 1, \\ O\left(\frac{c}{T}\right) & \text{if } y = 1. \end{cases}$$

Substituting the second identity into the first and combining the O-estimates gives

$$\begin{aligned} A^*(X) &= \sum_{n\geq 1} a(n)\frac{1}{2\pi i}\int_{c-iT}^{c+iT} \frac{X^s}{n^s}\,\frac{ds}{s} \\ &\quad + O\left(X^c \sum_{\substack{n\geq 1\\ n\neq X}} \frac{|a(n)|}{n^c}\min\left(1, \frac{1}{T\left|\log\left(\frac{X}{n}\right)\right|}\right) + \delta_X |a(X)|\frac{c}{T}\right). \end{aligned}$$

As $D(s)$ converges absolutely, we may interchange the sum and the integral. The statement follows. $\square$

Since the integral in truncated Perron's formula is over a finite vertical line, the integral is automatically absolutely convergent. There is also a mor crude variant of truncated Perron's formula that follows as a corollary:

Corollary 5.4.3. *Let $D(s)$ be a Dirichlet series with coefficient $a(n)$ and finite and nonnegative abscissa of absolute convergence σ_a. Then for any $c > \sigma_a$ and $0 < T < X^c$, we have*

$$A^*(X) = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} D(s)X^s\,\frac{ds}{s} + O\left(\frac{X^c}{T}\right).$$

Proof. For $0 < T < X^c$, we have

$$\min\left(1, \frac{1}{T\log\left(\frac{X}{n}\right)}\right) \ll \frac{X^c}{T}.$$

Also, Proposition 5.1.4 guarantees $a(X) \ll X^c$. These estimates and that $D(s)$ is absolutely convergent at $s = c$ together imply that error term in truncated Perron's formula is $O\left(\frac{X^c}{T}\right)$. The claim follows. $\square$

Having discussed truncated Perron's formula, we turn to the variant where we instead introduce a smoothing function. We call $\psi(y)$ a **smooth weight** if it is a nonnegative bump function whose support is bounded away from zero. For any $X > 0$, we set

$$A_\psi(X) = \sum_{n\geq 1} a(n)\psi\left(\frac{n}{X}\right).$$

We distinguish between two cases. The first is when we choose $\psi(y)$ to assign weight 1 or 0 to the coefficients and we obtain sums such as

$$\sum_{\frac{X}{2}\leq n<X} a(n) \quad \text{or} \quad \sum_{X\leq n<X+Y} a(n).$$

Sums of this type are called **unweighted**. As an example of an unweighted sum, suppose $\psi(y)$ is a smooth weight that is identically 1 on $\left[\frac{1}{2}, 1\right]$ and supported in $\left[\frac{1}{2} - \frac{1}{X}, 1 + \frac{1}{X}\right]$. Then

$$A_\psi(X) = \sum_{\frac{X}{2}\leq n\leq X} a(n).$$

In the second case, we want $\psi(y)$ to dampen the coefficients with a weight other than 1 or 0. Sums of this type are called **weighted**. In either case, the Mellin transform $\Psi(s)$ of $\psi(y)$ is given by

$$\Psi(s) = \int_0^\infty \psi(y)y^s\,\frac{dy}{y}.$$

As $\psi(y)$ has compact support bounded away from zero, the integral defining $\Psi(s)$ is a locally absolutely uniformly convergent. In particular, $\Psi(s)$ is entire and Mellin inversion implies that $\psi(y)$ is the Mellin inverse of $\Psi(s)$. As for nice properties, $\Psi(s)$ exhibit rapid decay.

Proposition 5.4.4. *Suppose $\psi(y)$ is smooth weight and let $\Psi(s)$ be its Mellin transform. Then for bounded σ, we have*

$$\Psi(s) \ll (|s|+1)^{-N},$$

for any positive integer N.

Proof. Consider
$$\Psi(s) = \int_0^\infty \psi(y)y^s \,\frac{dy}{y}.$$
Since $\psi(y)$ is compactly supported, integration by parts yields
$$\Psi(s) = \frac{1}{s}\int_0^\infty \psi'(y)y^{s+1} \,\frac{dy}{y}.$$
Repeated integration by parts gives
$$\Psi(s) = \frac{1}{s(s+1)\cdots(s+N-1)}\int_0^\infty \psi^{(N)}(y)y^{s+N} \,\frac{dy}{y}.$$
Therefore
$$\Psi(s) \ll (|s|+1)^{-N}\int_0^\infty \psi^{(N)}(y)y^{\sigma+N} \,\frac{dy}{y}.$$
Now $\psi^{(N)}(y)$ is compactly supported in the same region as $\psi(y)$. In particular, $\psi^{(N)}(y)$ has compact support away from zero. Therefore
$$\int_0^\infty \psi^{(N)}(y)y^{\sigma+N} \,\frac{dy}{y} \ll 1,$$
and the estimate follows. □

The following result is (smoothed) **Perron's formula**:

Theorem (Perron's formula, smoothed). *Let $D(s)$ be a Dirichlet series with coefficient $a(n)$ and finite and nonnegative abscissa of absolute convergence σ_a. Let $\psi(y)$ be a smooth weight and let $\Psi(s)$ be its Mellin transform. Then for any $c > \sigma_a$, we have*
$$A_\psi(X) = \frac{1}{2\pi i}\int_{(c)} D(s)\Psi(s)X^s \,ds.$$
In particular,
$$\sum_{n\geq 1} a(n)\psi(n) = \frac{1}{2\pi i}\int_{(c)} D(s)\Psi(s) \,ds.$$

Proof. By Mellin inversion, we may write
$$A_\psi(X) = \sum_{n\geq 1}\frac{a(n)}{2\pi i}\int_{(c)} \Psi(s)\left(\frac{n}{X}\right)^{-s} ds.$$
We may interchange the sum and integral by the absolute convergence of $D(s)$ and Proposition 5.4.4. This gives
$$A_\psi(X) = \frac{1}{2\pi i}\int_{(c)} D(s)\Psi(s)X^s \,ds,$$
which is the first statement. For the second, take $X = 1$. □

The integral in smoothed Perron's formula is absolutely convergent (this permitted the interchange of the sum and integral). In practice, this means that the integral can be estimated directly and we do not need to truncate. The compensation for this is that we have introduced a weighting factor to the sum of coefficients.

Chapter 6

Analytic L-functions

Throughout, $s = \sigma + it$ and $u = \tau + ir$ will stand for complex variables with σ, τ, t, and r real. We also write

$$(u)_k = u(u+1)\cdots(u+(k-1)),$$

for the Pochhammer symbol. We will make use of the gamma function developed in Appendix C.2 especially Stirling's formula as well as the Mellin transform and the corresponding inversion theorem developed in Appendix C.1. Also, all sums and products over zeros of a function will be counted with multiplicity and ordered symmetrically with respect to the size of the ordinate if they are not absolutely convergent.

6.1 Analytic Data

An **analytic L-function** $L(s)$ is a Dirichlet series

$$L(s) = \sum_{n \geq 1} \frac{a_L(n)}{n^s},$$

with coefficients $a_L(n) \in \mathbb{C}$ such that $a_L(1) = 1$ and satisfying the following properties:

Analyticity: There exists a nonnegative integer m_L such that the Dirichlet series of $L(s)$ is absolutely convergent for $\sigma > 1$ and admits meromorphic continuation to $\mathbb{C}$ with at most a pole at $s = 1$ of order m_L. Moreover, $(s-1)^{m_L} L(s)$ is order 1.

Functional Equation: There is a positive integer q_L, called the **conductor** of $L(s)$, a positive integer d_L called the **degree** of $L(s)$, complex numbers $(\mu_j)_{j \leq d_L}$ called the **gamma parameters** of $L(s)$ that are either real or occur in conjugate pairs and satisfy $\mathrm{Re}(\mu_j) \geq 0$, and a complex number ε_L called the **root number** of $L(s)$ with $|\varepsilon_L| = 1$, all of which determine functions

$$\gamma(s, L) = \pi^{-\frac{d_L s}{2}} \prod_j \Gamma\left(\frac{s+\mu_j}{2}\right) \quad \text{and} \quad \Lambda(s, L) = q_L^{\frac{s}{2}} \gamma(s, L) L(s),$$

called the **gamma factor** of $L(s)$ and the **completion** of $L(s)$ respectively, where the latter satisfies

$$\Lambda(s,L) = \varepsilon_L \overline{\Lambda(1-\overline{s},L)}.$$

This is called the **functional equation** for $L(s)$. The tuple $(\varepsilon_L, q_L, \mu_1, \ldots, \mu_{d_L})$ is called the **functional equation data** of $L(s)$.

Euler product: For every prime p, there exist complex numbers $(\alpha_j(p))_{j \le d_L}$ called the **Euler parameters** of $L(s)$ at p satisfying $|\alpha_j(p)| \le p^{\theta_L}$ for some $0 \le \theta_L < 1$ and are such that $L(s)$ admits the Euler product

$$L(s) = \prod_p (1-\alpha_1(p)p^{-s})^{-1} \cdots (1-\alpha_{d_L}(p)p^{-s})^{-1},$$

for $\sigma > 1$. The polynomial L_p defined by

$$L_p(T) = (1-\alpha_1(p)T)\cdots(1-\alpha_{d_L}(p)T),$$

is called the **Euler factor** of $L(s)$ at p. If $p \nmid q_L$ then L_p is of degree d_L while if $p \mid q_L$ then L_p is of degree less than d_L. In the former case we say p is **unramified** while in the latter we say p is **ramified**.

An analytic L-function is said to be **Selberg class** if it satisfies the following additional property:

Ramanujan bound: We may take $\theta_L = 0$ so that $|\alpha_j(p)| \le 1$.

Some comments on the definition of analytic L-functions are in order.

(i) As the Dirichlet series of $L(s)$ converges absolutely for $\sigma > 1$ its converges locally absolutely uniformly in this half-plane and therefore defines a holomorphic function there. Moreover, the Euler product necessarily converges locally absolutely uniformly in the same half-plane and hence defines a holomorphic function there as well which agrees with the Dirichlet series.

(ii) The bound $\mathrm{Re}(\mu_j) \ge 0$ ensures that the gamma factor is holomorphic in the half-plane $\sigma > 0$. As this factor is then guaranteed to be finite and nonzero at $s = 1$, the completion possesses a pole at $s = 1$ of order r. By the functional equation, the completion also has a pole at $s = 0$ of the same order.

(iii) If we merely assume $(s-1)^{m_L}L(s)$ is finite order then the functional equation forces the order to be 1. This can be deduced by considering the entire function

$$s^{m_L}(s-1)^{m_L}\Lambda(s,L),$$

and applying the Phragmén-Lindelöf convexity principle in vertical strips.

(iv) The condition that the root number satisfies $|\varepsilon_L| = 1$ isn't strictly necessary. For applying the functional equation twice shows $|\varepsilon_L|^2 = 1$.

(v) As the gamma function is conjugation-equivariant and the gamma parameters are real or occur in conjugate pairs, the gamma factor is also conjugation-equivariant. This means that we can write the functional equation in the form

$$q_L^{\frac{s}{2}}\gamma(s,L)L(s) = \varepsilon_L q_L^{\frac{1-s}{2}}\gamma(1-s,L)\overline{L(1-\overline{s})}.$$

(vi) The Euler product implies that the coefficients $a_L(n)$ are multiplicative and are determined on prime powers by

$$a_L(p^k) = \sum_{j_1\le\cdots\le j_k} \alpha_{j_1}(p)\cdots\alpha_{j_k}(p).$$

In other words, $a_L(p^k)$ is the complete symmetric polynomial of degree k in the Euler parameters at p. If the Ramanujan bound holds, then it follows that $a_L(n) \ll \sigma_{d_L}(n)$. This implies the slightly weaker, but more practical, estimate $a_L(n) \ll_\varepsilon n^\varepsilon$.

We will also define the **dual** $\overline{L}(s)$ of $L(s)$ by

$$\overline{L}(s) = \overline{L(\overline{s})}.$$

Analytic L-functions are closed under duals. Indeed, $\overline{L}(s)$ is an analytic L-function with coefficients $a_{\overline{L}}(n) = \overline{a_L(n)}$, a pole at $s = 1$ of order $m_{\overline{L}} = m_L$, conductor $q_{\overline{L}} = q_L$, degree $d_{\overline{L}} = d_L$, gamma parameters $(\overline{\mu_j})_{j\le d_L}$, root number $\varepsilon_{\overline{L}} = \overline{\varepsilon_L}$, gamma factor $\gamma(s,\overline{L}) = \gamma(s,L)$, completion $\Lambda(s,\overline{L}) = \overline{\Lambda(\overline{s},L)}$, and Euler parameters $(\overline{\alpha_j(p)})_{j\le d_L}$ at p. Moreover, the dual of $L(s)$ is Selberg class if $L(s)$ is. We also say $L(s)$ is **self-dual** if it is equal to its dual. This is equivalent to the coefficients $a_L(n)$ being real. In any case, we can express the functional equation as

$$\Lambda(s,L) = \varepsilon_L\Lambda(1-s,\overline{L}),$$

so the functional equation relates an analytic L-function with its dual. This formula will also be useful when taking derivatives of the functional equation.

It is clear from the definition that analytic L-functions are closed under multiplication and that the degree is additive. Moreover, this closure respects Selberg class L-functions. This makes the set of analytic L-functions into a graded commutative semigroup where the grading is induced by degree. It follows that there exist irreducible elements with respect to the grading and we say that an analytic L-function is **primitive** if it such an irreducible. Clearly any analytic L-function of degree 1 is primitive. Moreover, every analytic L-function factors into a product of primitive L-functions. Such a factorization is only conjectured to be unique for Selberg class L-functions. A sufficient condition would be **Selberg's orthogonality conjecture**.

Conjecture (Selberg's orthogonality conjecture). *For any two primitive Selberg class L-functions $L_1(s)$ and $L_2(s)$, we have*

$$\sum_{p\le x} \frac{a_{L_1}(p)\overline{a_{L_2}(p)}}{p} = \delta_{L_1,L_2}\log\log(x) + O(1).$$

Indeed, we have the following result:

Proposition 6.1.1. *Assume Selberg's orthogonality conjecture. Then every Selberg class L-function factors uniquely into a product of primitive Selberg class L-functions.*

Proof. If the factorization into primitive unity L-functions were not unique, then we would have distinct factorizations satisfying

$$L_1(s)\cdots L_n(s) = M_1(s)\cdots M_m(s),$$

for some primitive Selberg class L-functions L_i and M_j. By uniqueness of coefficients of Dirichlet series, compare the p-th coefficient to see that

$$\sum_i a_{L_i}(p) = \sum_j a_{M_j}(p).$$

Now consider

$$\sum_{p<x}\frac{1}{p}\left(\sum_i a_{L_i}(p)\right)\left(\sum_j \overline{a_{M_j}(p)}\right) = \sum_{i,j}\sum_{p<x}\frac{a_{L_i}(p)\overline{a_{M_j}(p)}}{p}.$$

Selberg's orthogonality conjecture implies

$$O(1) = c\log\log(x) + O(1),$$

for some positive integer c since the factorizations are distinct. This is impossible and thus the factorization is unique. $\square$

To an analytic L-function $L(s)$, we associate its **analytic conductor** $\mathfrak{q}(s,L)$ defined by

$$\mathfrak{q}(s,L) = q_L\mathfrak{q}_\infty(s,L),$$

where

$$\mathfrak{q}_\infty(s,L) = \prod_j(|s+\mu_j|+3).$$

For convenience, we will also set

$$\mathfrak{q}(L) = \mathfrak{q}(0,L) \quad \text{and} \quad \mathfrak{q}_\infty(L) = \mathfrak{q}_\infty(0,L).$$

The choice of 3 in $|s+\mu_j|+3$ is a matter of convenience as could use any positive constant. In particular, it is useful when taking logarithms as $\log(|s+\mu_j|+3) \geq 1$. Estimates for the analytic conductor follow from those of the gamma function. Recall from Stirling's formula that

$$\Gamma(s) \ll_\varepsilon (|t|+3)^{\sigma-\frac{1}{2}}e^{-\frac{\pi}{2}|t|} \quad \text{and} \quad \frac{1}{\Gamma(s)} \ll (|t|+3)^{\frac{1}{2}-\sigma}e^{\frac{\pi}{2}|t|}, \tag{6.1}$$

for bounded σ provided that in the former estimate s is at least distance ε away from the poles of the gamma function. Hence

$$\frac{\Gamma(1-s)}{\Gamma(s)} \ll_\varepsilon (|t|+3)^{1-2\sigma},$$

for bounded σ provided s is at least distance ε away from the poles of $\Gamma(1-s)$. Then the estimates

$$\frac{\gamma(1-s,L)}{\gamma(s,L)} \ll_\varepsilon \mathfrak{q}_\infty(s,L)^{\frac{1}{2}-\sigma} \quad \text{and} \quad q_L^{\frac{1}{2}-s}\frac{\gamma(1-s,L)}{\gamma(s,L)} \ll_\varepsilon \mathfrak{q}(s,L)^{\frac{1}{2}-\sigma}, \tag{6.2}$$

hold for bounded σ provided s is at least distance ε away from the poles of $\gamma(1-s,L)$. An associated estimate can be obtain for the logarithmic derivative of the analytic conductor. Recall from the logarithm of Stirling's formula that

$$\frac{\Gamma'}{\Gamma}(s) \ll \log(|s|+3),$$

provided σ is bounded and s is at least distance ε away from the poles of the gamma function. Then the estimates

$$\frac{\gamma'}{\gamma}(s,L) \ll_\varepsilon \log \mathfrak{q}_\infty(s,L) \quad \text{and} \quad \log(q_L) + \frac{\gamma'}{\gamma}(s,L) \ll_\varepsilon \log \mathfrak{q}(s,L), \tag{6.3}$$

hold for bounded σ provided s is at least distance ε away from the poles of the gamma factor.

The **critical strip** of an analytic L-function is the vertical strip left invariant by the transformation $s \mapsto 1-s$. This region can also be described as

$$\left\{ s \in \mathbb{C} : \left|\sigma - \frac{1}{2}\right| \leq \frac{1}{2} \right\}.$$

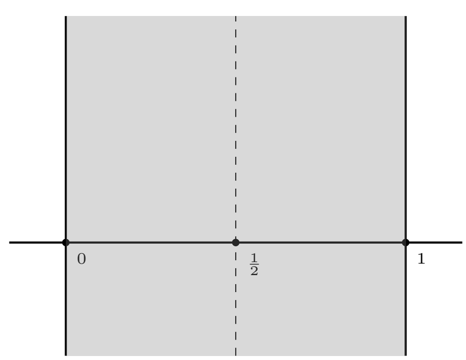

Figure 6.1: The critical strip.

The **critical line** is the vertical line left invariant by the transformation $s \mapsto 1-s$ which is also given by $\sigma = \frac{1}{2}$. The critical line bisects the critical strip vertically. The **central point** is the fixed point of the transformation $s \mapsto 1-s$, in other words, the point $s = \frac{1}{2}$. Clearly the central point is also the center of the critical line. The critical strip, critical line, and central point are all displayed in Figure 6.1.

In the half-plane $\sigma > 1$ we may study the analytic properties of $L(s)$ via its Dirichlet series. Using the functional equation, we may write

$$L(s) = \varepsilon_L q_L^{\frac{1}{2}-s} \frac{\gamma(1-s, L)}{\gamma(s, L)} \overline{L(1-\overline{s})}.$$

This permits the study of $L(s)$ in the half-plane $\sigma < 0$ by using the Dirichlet series of $\overline{L(1-\overline{s})}$ and the the functional equation data. The interior of the critical strip is exactly the region where we cannot use either of these methods to study the analytic properties of $L(s)$. Of course, anything may be possible on the boundary lines $\sigma = 0$ and $\sigma = 1$ (for example Landau's theorem).

6.2 The Approximate Functional Equation

Despite not being able to study an analytic L-function $L(s)$ in the critical strip by means of Dirichlet series, there is formula which acts as a compromise between the functional equation and the Dirichlet series. This formula is known as the approximate functional equation. The usefulness comes from the fact that the approximate functional equation is valid inside of the critical strip and therefore can be used to obtain analytic properties about $L(s)$ in that region. We will first derive a preliminary result showing $L(s)$ has at most polynomial growth.

Proposition 6.2.1. *Let $L(s)$ be an analytic L-function with σ bounded and σ at least distance ε away from the possible pole at $s = 1$. Then there is a positive constant A such that*

$$L(s) \ll_\varepsilon (|t| + 3)^A.$$

Proof. Observe that $(s-1)^{m_L} L(s) \ll_\varepsilon (|t|+3)^{m_L}$ on the vertical line $\sigma = \max(1 + \varepsilon, \sigma_2)$. Now suppose $\sigma = \min(-\varepsilon, \sigma_1)$. On this vertical line, the functional equation and Equation (6.2) together show

$$L(s) \ll_\varepsilon \mathfrak{q}(s, L)^{\frac{1}{2}-\sigma} \overline{L(1-\overline{s})}.$$

Hence there exists a positive constant A'' with

$$(s-1)^{m_L} L(s) \ll_\varepsilon (|t|+3)^{A''},$$

on the vertical line $\sigma = \min(-\varepsilon, \sigma_1)$. As $(s-1)^{m_L} L(s)$ is entire and of order 1, we can apply the Phragmén-Lindelöf convexity principle to $(s-1)^{m_L} L(s)$ the vertical strip $\min(-\varepsilon, \sigma_1) \leq \sigma \leq \max(1+\varepsilon, \sigma_2)$. Whence there is a positive constant A' such that

$$(s-1)^{m_L} L(s) \ll_\varepsilon (|t|+3)^{A'},$$

provided σ is bounded. Assuming s is at least distance ε away from the possible pole at $s = 1$, diving by $(s-1)^{m_L}$ completes the proof. ∎

We are almost ready to prove the approximate function equation for an analytic L-function $L(s)$. The formula itself consists of two sums representing the Dirichlet series at s and a dualized Dirichlet series at $1-s$ as well as a potential residue term. The dual sum comes equip with a term containing the data of the functional equation. Both sums will also be dampened by a smooth cutoff function. In the statement of the approximate function equation, we will make use of a test function $\Phi(u)$. We require $\Phi(u)$ be an even holomorphic function bounded in the vertical strip $|\tau| < a+1$ for some $a > 1$ and such that $\Phi(0) = 1$. For s in the critical strip, we let $V_s(y, L)$ be the inverse Mellin transform

$$V_s(y, L) = \frac{1}{2\pi i}\int_{(a)} \frac{\gamma(s+u, L)}{\gamma(s, L)}\Phi(u)y^{-u}\frac{du}{u},$$

defined for $y > 0$. Stirling's formula implies

$$\frac{\Gamma(s+u)}{\Gamma(s)} \ll_\varepsilon \frac{(|t+r|+3)^{\sigma+\tau-\frac{1}{2}}}{(|t|+3)^{\sigma-\frac{1}{2}}}e^{-\frac{\pi}{2}(|t+r|-|t|)},$$

for s in the critical strip and bounded τ provided $s+u$ is at least distance ε away from the poles of the gamma function. Whence

$$\frac{\gamma(s+u, L)}{\gamma(s, L)} \ll_\varepsilon \frac{\mathfrak{q}_\infty(s+u, L)^{\frac{\sigma+\tau}{2}-\frac{1}{4}}}{\mathfrak{q}_\infty(s, L)^{\frac{\sigma}{2}-\frac{1}{4}}}e^{-d_L\frac{\pi}{2}(|t+r|-|t|)}, \tag{6.4}$$

for s in the critical strip and bounded τ provided $s+u$ is at least distance ε away from the poles of the gamma factor. Since $\Phi(u)$ is bounded in the vertical strip $|\tau| < a+1$, the integrand exhibits exponential decay. Therefore the integral is locally absolutely uniformly convergent and hence $V_s(y, L)$ is smooth. The function $V_s(y, L)$ is the smooth cutoff function mentioned previously. We will also let $\varepsilon(s, L)$ be given by

$$\varepsilon(s, L) = \varepsilon_L q_L^{\frac{1}{2}-s}\frac{\gamma(1-s, L)}{\gamma(s, L)}.$$

This term appears as a factor in the dual sum accounting for the functional equation data. Lastly, we set

$$R(s, X, L) = \operatorname*{Res}_{u=1-s}\frac{\Lambda(s+u, L)\Phi(u)X^u}{u} + \operatorname*{Res}_{u=-s}\frac{\Lambda(s+u, L)\Phi(u)X^u}{u}.$$

This is the potential residue term that appears in the approximate functional equation. Observe that it vanishes provided $s \neq 0, 1$. We now prove the **approximate function equation** for $L(s)$.

Theorem (Approximate functional equation). *Suppose $L(s)$ is an analytic L-function, $\Phi(u)$ is an even holomorphic function bounded in the vertical strip $|\tau| < a+1$ for some $a > 1$ and such that $\Phi(0) = 1$, and let $X > 0$. Then for s in the critical strip, we have*

$$L(s) = \sum_{n\geq 1}\frac{a_L(n)}{n^s}V_s\left(\frac{n}{\sqrt{q_L}X}, L\right) + \varepsilon(s, L)\sum_{n\geq 1}\frac{\overline{a_L(n)}}{n^{1-s}}V_{1-s}\left(\frac{nX}{\sqrt{q_L}}, L\right) + \frac{R(s, X, L)}{q_L^{\frac{s}{2}}\gamma(s, L)}.$$

Proof. Consider the integral

$$\frac{1}{2\pi i}\int_{(a)} \Lambda(s+u,L)\Phi(u)X^u\,\frac{du}{u}.$$

Stirling's formula shows that $\gamma(s+u,L)$ exhibits exponential decay while $L(s+u)$ has at most polynomial growth by Proposition 6.2.1. Since $\Phi(u)$ is bounded in the vertical strip $|\tau|<a+1$, it follows that the integrand exhibits exponential decay. Therefore the integral is locally absolutely uniformly convergent. We will evaluate the integral in two ways. On the one hand, we can expand $L(s+u)$ inside the integrand as a Dirichlet series and by absolute boundedness of the integral we may interchange the sum and integral. A short computation shows that this is

$$q_L^{\frac{s}{2}}\gamma(s,L)\sum_{n\geq 1}\frac{a_L(n)}{n^s}V_s\left(\frac{n}{\sqrt{q_L}X},L\right).$$

On the other hand, we can shift the line of integration to $(-a)$. In doing so we pass by a simple pole at $u=0$ and possible poles at $u=1-s$ and $u=-s$ giving

$$\frac{1}{2\pi i}\int_{(-a)} \Lambda(s+u,L)\Phi(u)X^u\,\frac{du}{u}+\Lambda(s,L)+R(s,X,L).$$

Apply the functional equation and perform the change of variables $u\mapsto -u$ to rewrite this as

$$-\varepsilon_L\frac{1}{2\pi i}\int_{(a)} \Lambda(1-(s+u),\overline{L})\Phi(u)X^{-u}\,\frac{du}{u}+\Lambda(s,L)+R(s,X,L).$$

Analogous to the above, we can now expand $\overline{L}(1-(s+u))$ inside the integrand as a Dirichlet series and by absolute boundedness of the integral we may interchange the sum and integral. A short computation shows that our previous expression becomes

$$-\varepsilon_L q_L^{\frac{1-s}{2}}\gamma(1-s,L)\sum_{n\geq 1}\frac{\overline{a_L(n)}}{n^{1-s}}V_s\left(\frac{nX}{\sqrt{q_L}},L\right)+\Lambda(s,L)+R(s,X,L).$$

Equating these two evaluations and isolating the completion gives

$$\begin{aligned}\Lambda(s,L)&=q_L^{\frac{s}{2}}\gamma(s,L)\sum_{n\geq 1}\frac{a_L(n)}{n^s}V_s\left(\frac{n}{\sqrt{q_L}X},L\right)\\&\quad+\varepsilon_L q_L^{\frac{1-s}{2}}\gamma(1-s,L)\sum_{n\geq 1}\frac{\overline{a_L(n)}}{n^{1-s}}V_{1-s}\left(\frac{nX}{\sqrt{q_L}},L\right)+R(s,X,L).\end{aligned}$$

Diving by $q_L^{\frac{s}{2}}\gamma(s,L)$ completes the proof. □

Let us now show how $V_s(y,L)$ has the effect of dampening the two dual sums appearing on the right-hand side of the approximate functional equation. In practice, it is common to choose $\Phi(u)$ such that it has exponential decay and we can make the vertical strip on which it is bounded arbitrarily wide. For example, let

$$\Phi(u) = \cos^{-4d_L M}\left(\frac{\pi u}{4M}\right),$$

for some positive integer M. Clearly $\Phi(u)$ is an even holomorphic function in the vertical strip $|\tau| < (2M-1)+1$ and satisfies $\Phi(0) = 1$. In view of the identity $\cos(u) = \frac{e^{iu}+e^{-iu}}{2}$, we find that

$$\cos^{-4d_L M}\left(\frac{\pi u}{4M}\right) \ll_\varepsilon e^{-d_L \pi |r|}, \tag{6.5}$$

for $|\tau| < (2M-1)+1$ provided u is at least distance ε away from the poles of $\Phi(u)$. Therefore $\Phi(u)$ admits exponential decay. For this choice of $\Phi(u)$, $V_s(y,L)$ and its derivatives will possess rapid decay.

Proposition 6.2.2. *Let $L(s)$ be an analytic L-function, set $\Phi(u) = \cos^{-4d_L M}\left(\frac{\pi u}{4M}\right)$ for some positive integer M, and let $V_s(y,L)$ be the inverse Mellin transform defined by*

$$V_s(y,L) = \frac{1}{2\pi i}\int_{(2M-1)} \frac{\gamma(s+u,L)}{\gamma(s,L)}\Phi(u) y^{-u}\frac{du}{u}.$$

Then for s in the critical strip, any nonnegative integer k, and positive integer N with $N < 2M$, $V_s(y,L)$ satisfies the estimates

$$(-y)^k V_s^{(k)}(y,L) = \begin{cases} \delta_{k,0} + O_\varepsilon\left(\left(\frac{y}{\sqrt{\mathfrak{q}_\infty(s,L)}}\right)^N\right) & \text{if } y \ll \sqrt{\mathfrak{q}_\infty(s,L)}, \\ O_\varepsilon\left(\left(\frac{y}{\sqrt{\mathfrak{q}_\infty(s,L)}}\right)^{-N}\right) & \text{if } y \gg \sqrt{\mathfrak{q}_\infty(s,L)}. \end{cases}$$

In particular,

$$(-y)^k V_s^{(k)}(y,L) \ll_\varepsilon \left(1 + \frac{y}{\sqrt{\mathfrak{q}_\infty(s,L)}}\right)^{-N}.$$

Proof. As we have already seen, the integrand defining $V_s(y,L)$ admits exponential decay. This permits us to differentiate under the integral sign and shift the line of integration. The former shows

$$(-y)^k V_s^{(k)}(y,L) = \frac{1}{2\pi i}\int_{(2M-1)} \frac{\gamma(s+u,L)}{\gamma(s,L)}\Phi(u)(u)_k y^{-u}\frac{du}{u}.$$

Shifting to $(-N)$, we pass by a simple pole at $u=0$ of residue 1 if and only if $k=0$. This gives

$$(-y)^k V_s^{(k)}(y,L) = \delta_{k,0} + \frac{1}{2\pi i}\int_{(-N)} \frac{\gamma(s+u,L)}{\gamma(s,L)}\Phi(u)(u)_k y^{-u}\frac{du}{u}.$$

This integrand also exhibits exponential decay since the Pochhammer symbol grows polynomially. Therefore the integral is dominated by the contribution when $u \ll 1$ and the Equations (6.4) and (6.5) together show

$$(-y)^k V_s^{(k)}(y, L) = \delta_{k,0} + O_\varepsilon\left(\left(\frac{y}{\sqrt{\mathfrak{q}_\infty(s, L)}}\right)^N\right).$$

If we instead shift to (N), we do not pass by any poles and obtain

$$(-y)^k V_s^{(k)}(y, L) = \frac{1}{2\pi i}\int_{(N)} \frac{\gamma(s+u, L)}{\gamma(s, L)}\Phi(u)(u)_k y^{-u}\frac{du}{u}.$$

An analogous argument to estimate the remaining integral shows

$$(-y)^k V_s^{(k)}(y, L) = O_\varepsilon\left(\left(\frac{y}{\sqrt{\mathfrak{q}_\infty(s, L)}}\right)^{-N}\right).$$

From these O-estimates we obtain nontrivial bounds in the ranges $y \ll \sqrt{\mathfrak{q}_\infty(s, L)}$ and $y \gg \sqrt{\mathfrak{q}_\infty(s, L)}$ respectively. Combining both of these estimates produces the bound

$$(-y)^k V_s^{(k)}(y, L) \ll_\varepsilon \left(1 + \frac{y}{\sqrt{\mathfrak{q}_\infty(s, L)}}\right)^{-N}.$$

□

With our choice of $\Phi(u)$, this result shows that $V_s(y, L)$ is essentially 1 for $y \ll \sqrt{\mathfrak{q}_\infty(s, L)}$ and has rapid decay thereafter as we can take M (and hence N) to be arbitrarily large.

In a similar spirit to the approximate functional equation, a useful summation formula can be derived from the functional equation. Let $\psi(y)$ be a smooth weight where $\Psi(s)$ is its Mellin transform. Then we will let $\psi(y, L)$ be the inverse Mellin transform

$$\psi(y, L) = \frac{1}{2\pi i}\int_{(a)} q_L^s \frac{\gamma(s, L)}{\gamma(1-s, L)} y^{-s}\Psi(1-s)\, ds,$$

defined for $y > 0$ where $a > 1$. By our choice of $\psi(y)$, its inverse Mellin transform $\Psi(s)$ has rapid decay via Proposition 5.4.4. Since $L(s)$ has at most polynomial growth by Proposition 6.2.1, the integrand has rapid decay as well. Therefore the integral is locally absolutely uniformly convergent and hence $\psi(y, L)$ is smooth for $y > 0$. We will also set

$$R_L = \operatorname*{Res}_{s=1} L(s).$$

Our result is the following:

Theorem 6.2.3. *Let $L(s)$ be an analytic L-function and let $\psi(y)$ be a smooth weight where $\Psi(s)$ is its Mellin transform. Then*

$$\sum_{n \ge 1} a_L(n)\psi(n) = \frac{\varepsilon_L}{\sqrt{q_L}}\sum_{n\ge 1}\overline{a_L(n)}\psi(n, L) + R_L\Psi(1).$$

Proof. Smoothed Perron's formula implies

$$\sum_{n\geq 1} a_L(n)\psi(n) = \frac{1}{2\pi i}\int_{(a)} L(s)\Psi(s)\,ds.$$

By our choice of $\psi(y)$, its inverse Mellin transform $\Psi(s)$ has rapid decay via Proposition 5.4.4. Since $L(s)$ has at most polynomial growth by Proposition 6.2.1, the integrand has rapid decay as well. Therefore the integral is locally absolutely uniformly convergent which permits us to shift the line of integration. Shifting to $(1-a)$, we pass by a potential pole at $s = 1$ and obtain

$$\sum_{n\geq 1} a_L(n)\psi(n) = \frac{1}{2\pi i}\int_{(1-a)} L(s)\Psi(s)\,ds + R_L\Psi(1).$$

Apply the functional equation to rewrite this equality in the form

$$\sum_{n\geq 1} a_L(n)\psi(n) = \frac{1}{2\pi i}\int_{(1-a)} \varepsilon_L q_L^{\frac{1}{2}-s}\frac{\gamma(1-s,L)}{\gamma(s,L)}\overline{L(1-\overline{s})}\Psi(s)\,ds + R_L\Psi(1).$$

Performing the change of variables $s \mapsto 1-s$ in this latter integral gives

$$\sum_{n\geq 1} a_L(n)\psi(n) = \frac{1}{2\pi i}\int_{(a)} \varepsilon_L q_L^{s-\frac{1}{2}}\frac{\gamma(s,L)}{\gamma(1-s,L)}\overline{L(\overline{s})}\Psi(1-s)\,ds + R_L\Psi(1).$$

The proof is complete upon expanding $\overline{L(\overline{s})}$ as a Dirichlet series, interchanging the sum and integral by absolute boundedness of the integral, and factoring out $\frac{\varepsilon_L}{\sqrt{q_L}}$. $\square$

6.3 The Riemann Hypothesis and Nontrivial Zeros

The zeros of an analytic L-functions $L(s)$ has interesting behavior. From the Euler product we immediately see that $L(s)$ has no zeros in the half-plane $\sigma > 1$. We can use the functional equation to determine the zeros for $\sigma < 0$. Indeed, write the functional equation in the form

$$L(s) = \varepsilon_L q_L^{\frac{1}{2}-s}\frac{\gamma(1-s,L)}{\gamma(s,L)}\overline{L(1-\overline{s})}.$$

So for $\sigma < 0$, we see that $\overline{L(1-\overline{s})}$ is nonzero. Moreover, $\gamma(1-s,L)$ is as well. Together this means that for $\sigma < 0$ the poles of $\gamma(s,L)$ are zeros of $L(s)$. Such a zero is called a **trivial zero** of $L(s)$. From the definition of the gamma factor, they are all simple and of the form $s = -(\mu_j + 2n)$ for some gamma parameter μ_j and nonnegative integer n.

Any other zero of $L(s)$ is called a **nontrivial zero** and it necessarily lies inside of the critical strip. Let ρ be a nontrivial zero of $L(s)$. Then the functional equation implies that $1-\overline{\rho}$ is also a nontrivial zero of $L(s)$. This means that nontrivial zeros occur in pairs

$$\rho \quad \text{and} \quad 1-\overline{\rho}.$$

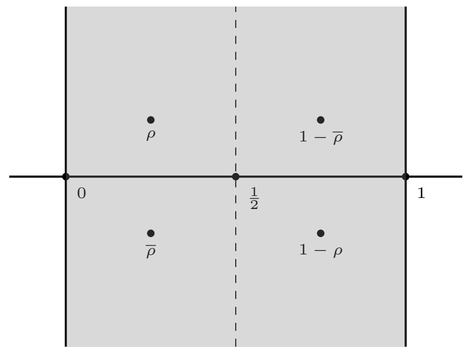

Figure 6.2: Symmetric nontrivial zeros.

It is possible to say more when $L(s)$ takes real values for $s > 1$. For in this case, the Schwarz reflection principle implies $L(\overline{s}) = \overline{L(s)}$ and that $L(s)$ takes real values on the entire real line save for the possible pole at $s = 1$. It follows from the functional equation that $\overline{\rho}$ and $1-\rho$ are also nontrivial zeros. Therefore the nontrivial zeros of $L(s)$ come in sets of four

$$\rho, \quad \overline{\rho}, \quad 1-\rho, \quad \text{and} \quad 1-\overline{\rho},$$

as displayed in Figure 6.2.

The **Riemann hypothesis** for $L(s)$ says that this symmetry should be as simple as possible.

Conjecture (Riemann hypothesis). *All of the nontrivial zeros of the analytic L-function $L(s)$ lie on the vertical line $\sigma = \frac{1}{2}$.*

While stated as a conjecture, we not expect the Riemann hypothesis to hold for just any analytic L-function. We do however expect it to hold for Selberg class L-functions. In particular, the **Selberg class Riemann hypothesis** says that this symmetry should hold for any Selberg class L-function:

Conjecture (Selberg class Riemann hypothesis). *All of nontrivial zeros of any Selberg class L-function $L(s)$ lie on the vertical line $\sigma = \frac{1}{2}$.*

So far, the Riemann hypothesis remains completely out of reach for any analytic L-function and thus the Selberg class Riemann hypothesis does as well.

6.4 The Lindelöf Hypothesis and Moments

Instead of asking about the zeros of an analytic L-function $L(s)$ on the critical line we can ask about its growth there. Let us begin this investigation by deriving an upper bound for $L(s)$ on the critical line using the Phragmén-Lindelöf convexity principle. This amounts to a more refined version of the argument used in Proposition 6.2.1 for the critical strip.

Theorem 6.4.1. *Let $L(s)$ be an analytic L-function. Then for $-\varepsilon \le \sigma \le 1+\varepsilon$, we have*

$$L(s) \ll_\varepsilon \mathfrak{q}(s,L)^{\frac{1-\sigma}{2}+\varepsilon},$$

provided s is at least distance ε away from the possible pole at $s=1$.

Proof. Since $(s-1)^{m_L}L(s)$ is entire and of order 1, we will apply the Phragmén-Lindelöf convexity principle just outside the edges of the critical strip. Just past the right edge along the vertical line $\sigma = 1+\varepsilon$, we have

$$(s-1)^{m_L}L(s) \ll_\varepsilon (s-1)^{m_L}.$$

Just past the left edge along the vertical line $\sigma = -\varepsilon$, the functional equation and Equation (6.2) together imply the bound

$$(s-1)^{m_L}L(s) \ll_\varepsilon (s-1)^{m_L}\mathfrak{q}(s,L)^{\frac{1}{2}+\varepsilon}.$$

By the Phragmén-Lindelöf convexity principle, we obtain

$$(s-1)^{m_L}L(s) \ll_\varepsilon (s-1)^{m_L}\mathfrak{q}(s,L)^{\frac{1-\sigma}{2}+\varepsilon},$$

provided $-\varepsilon \le \sigma \le 1+\varepsilon$. Assuming s is at least distance ε away from the possible pole at $s=1$, dividing by $(s-1)^{m_L}$ completes the proof. □

At the critical line this theorem produces the bound

$$L\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\frac{1}{4}+\varepsilon}.$$

This is known as the **convexity bound** for $L(s)$. The **Lindelöf hypothesis** for $L(s)$ says that the exponent can be reduced to ε:

Conjecture (Lindelöf hypothesis). *The analytic L-function $L(s)$ satisfies*

$$L\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\varepsilon}.$$

Unlike the Riemann hypothesis, the Lindelöf hypothesis is expected to hold for any analytic L-function. In any case, the **Selberg class Lindelöf hypothesis** says that the exponent can be reduced to ε for any Selberg class L-function:

Conjecture (Selberg class Lindelöf hypothesis). *Any Selberg class L-function $L(s)$ satisfies*

$$L\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\varepsilon}.$$

Like the Riemann hypothesis, we have been unable to prove the Lindelöf hypothesis for any analytic L-function. However, in practice the Lindelöf hypothesis is more tractable. Generally speaking, any improvement upon the exponent in the convexity bound in any aspect of the analytic conductor is called a **subconvexity estimate** (or a **convexity breaking bound**). In the uniform case, we would like to prove bounds of the form

$$L\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\delta+\varepsilon},$$

for some $0 \le \delta \le \frac{1}{4}$. The convexity bound says that we may take $\delta = \frac{1}{4}$ while the Lindelöf hypothesis for $L(s)$ implies that we may take $\delta = 0$. Any uniform subconvexity estimate would give a δ strictly less than $\frac{1}{4}$. Subconvexity estimates are viewed with great interest. This is primarily because of their connection to the Lindelöf hypothesis, but also because any improvement upon the convexity bound in any aspect of the analytic conductor often has drastic consequences in applications.

Remark 6.4.2. Some subconvexity bounds are deserving of names. The cases $\delta = \frac{3}{16}$ and $\delta = \frac{1}{6}$ are referred to as the **Burgess bound** and **Weyl bound** respectively.

With a little more work, we can obtain a a similar bound for the derivatives of analytic L-functions. First observe that Theorem 6.4.1 gives the estimate

$$L(s) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\frac{1}{4}+\varepsilon},$$

in the vertical strip $\left|\sigma - \frac{1}{2}\right| \le \frac{\varepsilon}{2}$. This is just slightly stronger than the convexity bound. By Cauchy's integral formula, we may write

$$L^{(k)}\left(\frac{1}{2}+it\right) = \frac{k!}{2\pi i}\int_\eta \frac{L(s)}{(s-\frac{1}{2}-it)^{k+1}}\,ds,$$

where η is the circle about $\frac{1}{2}+it$ of radius $\frac{\varepsilon}{2}$. The parameterization $u \mapsto \frac{1}{2}+it+\frac{\varepsilon}{2}e^{i\theta}$ for η shows that

$$L^{(k)}\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^{\frac{1}{4}+\varepsilon}.$$

This is known as the **convexity bound** for $L^{(k)}(s)$.

While both the Riemann and Lindelöf hypotheses remain out of reach for any analytic L-function $L(s)$, the latter is more tractable because the convexity bound gives a weak estimate. Unfortunately, breaking the convexity bound for any analytic L-function is a very daunting task requiring extremely modern techniques. This approach of breaking convexity to study the Lindelöf hypothesis is quite direct. However, there is an indirect approach by considering averages of the analytic L-function in question. This is the study of moments of L-functions. For any analytic L-function $L(s)$ and positive integer k, we define the k-th **moment** of $L(s)$, namely $M_k(T, L)$, by

$$M_k(T, L) = \int_0^T \left|L\left(\frac{1}{2}+it\right)\right|^k dt,$$

for any positive T. It is useful to think of $M_k(T,L)$ as a k-th power average of $L\left(\frac{1}{2}+it\right)$. One of the fundamental reasons why the study of moments is useful is that sufficient estimates for $M_{2k}(T,L)$ in T is equivalent to the Lindelöf hypothesis.

Proposition 6.4.3. *Let $L(s)$ be an analytic L-function. The truth of the Lindelöf hypothesis for $L(s)$ is equivalent to the estimate*

$$M_{2k}(T,L) \ll_\varepsilon T^{1+\varepsilon},$$

for all positive integers k.

Proof. First suppose the Lindelöf hypothesis for $L(s)$ holds. Then we obtain the estimate

$$M_{2k}(T,L) \ll_\varepsilon T^{1+\varepsilon},$$

at once for all k. Now suppose that $M_{2k}(T,L) \ll_\varepsilon T^{1+\varepsilon}$ for all k. If the Lindelöf hypothesis for $L(s)$ is false then there is some $\delta > 0$ and positive unbounded sequence $(t_n)_{n\ge 1}$ such that

$$\left|L\left(\frac{1}{2}+it_n\right)\right| > C\mathfrak{q}\left(\frac{1}{2}+it_n, L\right)^{\delta},$$

for any positive constant C. In particular, we may write

$$\left|L\left(\frac{1}{2}+it_n\right)\right| > Ct_n^{d_L\delta}.$$

The convexity bound for $L(s)$ implies the weaker estimate $\left|L'\left(\frac{1}{2}+it\right)\right| \le C't^{d_L}$ for some positive constant C' provided t is bounded away from zero. Now write

$$\left|L\left(\frac{1}{2}+it\right) - L\left(\frac{1}{2}+it_n\right)\right| = \left|\int_{t_n}^{t} L'\left(\frac{1}{2}+ir\right)\,dr\right|.$$

Taking $C = 2C'$, a short computation using these results shows that

$$\left|L\left(\frac{1}{2}+it\right) - L\left(\frac{1}{2}+it_n\right)\right| \le \frac{C}{2}t^{d_L\delta},$$

provided t is bounded away from zero and satisfies $|t-t_n| \le t_n^{d_L(\delta-1)}$. For such t, it follows that

$$\left|L\left(\frac{1}{2}+it\right)\right| > \frac{C}{2}t_n^{d_L\delta}.$$

Now take $T = \frac{4}{3}t_n$ so that the interval $(t_n - t_n^{d_L(\delta-1)}, t_n + t_n^{d_L(\delta-1)})$ is contained in $\left(\frac{T}{2}, T\right)$. Then

$$M_{2k}(T,L) > 2\left(\frac{C}{2}\right)^{2k}\left(\frac{3}{4}\right)^{(2k+1)d_L\delta-d_L}T^{(2k+1)d_L\delta-d_L}.$$

This is a contradiction for sufficiently large k which completes the proof. ☐

The usefulness of this result is that the difficulty of proving the Lindelöf hypothesis for $L(s)$ has been transferred to proving sufficient asymptotics for the moments $M_{2k}(T,L)$ each of which should, heuristically speaking, be an easier problem to resolve on its own.

6.5 Estimates on the Critical Line

We now turn to the question of how to obtain estimates on the critical line. There are many methods one can apply often depending upon the particular analytic L-function of interest. Here we will provide a method that works in vast generality and reduces the estimation of $L\left(\frac{1}{2}+it\right)$ to that of estimating smoothed finite sums of coefficients. To achieve this we need to establish the existence of certain bump functions. We say that a function $\omega(y)$ is a **smooth dyadic weight** if it is a nonnegative bump function supported in $[1-\delta, 2+\delta]$ and such that

$$\sum_{k\in\mathbb{Z}} \omega\left(\frac{y}{2^k}\right) = 1,$$

for positive y. This last condition means $\left(\omega\left(\frac{y}{2^k}\right)\right)_{k\in\mathbb{Z}}$ is a partition of unity on $(0,\infty)$. To see that dyadic weights exist, let $\psi(y)$ be a smooth weight that is identically 1 on $[1,2]$ and supported in $[1-\delta, 2+\delta]$. Write

$$\sigma(y) = \sum_{k\in\mathbb{Z}} \psi\left(\frac{y}{2^k}\right).$$

Then $\sigma(y)$ is finite as finitely many of its summands are nonzero since any positive y satisfies $2^k \le y < 2^{k+1}$ for some unique integer k. This forces $\sigma(y)$ to be smooth and bounded. It is also bounded away from zero as $\sigma(y) \ge 1$ because $\psi(y)$ is identically 1 on $[1,2]$. Then we may take

$$\omega(y) = \frac{\psi(y)}{\sigma(y)}.$$

This shows that smooth dyadic weights exist. For any $t \in \mathbb{R}$ and $X > 0$, set

$$A_\omega(t, X, L) = \sum_{n\ge1} a_L(n) n^{-it} \omega\left(\frac{n}{X}\right).$$

For convenience, we will also let

$$A_\omega(X, L) = A_\omega(0, X, L).$$

Our result estimates $L(s)$ at $s = \frac{1}{2}+it$ in terms of $A_\omega(t, X, L)$.

Theorem 6.5.1. *Let $L(s)$ be an analytic L-function and let $\omega(y)$ be a smooth dyadic weight. Then*

$$L\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it, L\right)^\varepsilon \sup_{X \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}+it, L\right)}} \left(\frac{|A_\omega(t, X, L)|}{\sqrt{X}}\right).$$

Proof. Take $s = \frac{1}{2}+it$, $X = 1$, and $\Phi(u) = \cos^{-4d_L M}\left(\frac{\pi u}{4M}\right)$ for some large positive integer M in the approximate functional equation. This permits us to write

$$\begin{aligned} L\left(\frac{1}{2}+it\right) &= \sum_{n\ge1} \frac{a_L(n)n^{-it}}{\sqrt{n}} V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}}, L\right) \\ &\quad + \varepsilon\left(\frac{1}{2}+it, L\right) \sum_{n\ge1} \frac{\overline{a_L(n)}n^{it}}{\sqrt{n}} V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}}, L\right) + \frac{R\left(\frac{1}{2}+it, 1, L\right)}{q_L^{\frac{1}{4}+i\frac{t}{2}} \gamma\left(\frac{1}{2}+it, L\right)}. \end{aligned}$$

Equation (6.2) implies $\varepsilon\left(\frac{1}{2}+it,L\right) \ll 1$ while Equations (6.1) and (6.5) together imply that the residue term exhibits exponential decay. This implies that right-hand side of the approximate functional equation is

$$O\left(\sum_{n\geq 1}\frac{a_L(n)n^{-it}}{\sqrt{n}}V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}},L\right)\right)+O\left(\sum_{n\geq 1}\frac{\overline{a_L(n)}n^{it}}{\sqrt{n}}V_{\frac{1}{2}-it}\left(\frac{n}{\sqrt{q_L}},L\right)\right)+O(e^{-|t|}).$$

We now turn to estimating the remaining sums which will be accomplished by applying the smooth dyadic weight. Consider the first sum. By our choice of $\Phi(u)$, the estimates in Proposition 6.2.2 are valid which implies that the summands exhibit rapid decay for $n \gg \sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}$. Whence the sum is say

$$\sum_{n\ll\sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}}\frac{a_L(n)n^{-it}}{\sqrt{n}}V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}},L\right)+O_\varepsilon\left(\mathfrak{q}\left(\frac{1}{2}+it,L\right)^{\varepsilon}\right)$$

Now write

$$V_s(y,L)=\sum_{k\in\mathbb{Z}}\omega\left(\frac{y}{2^k}\right)V_s(y,L).$$

Apply this identity to the sum and interchange the resulting two sums by local finiteness to obtain

$$\sum_{k\in\mathbb{Z}}\sum_{n\ll\sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}}\frac{a_L(n)n^{-it}}{\sqrt{n}}\omega\left(\frac{n}{2^k\sqrt{q_L}}\right)V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}},L\right).$$

By compact support of the smooth dyadic weight, for each k the inner sum over n is supported on the dyadic block $n \asymp X_k$ where $X_k = 2^k\sqrt{q_L}$. As $n \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}$, the summands which contribute must satisfy $k \ll \log\mathfrak{q}_\infty\left(\frac{1}{2}+it,L\right)$. Therefore our double sum can be expressed as

$$\sum_{k\ll\log\mathfrak{q}_\infty\left(\frac{1}{2}+it,L\right)}\sum_{n\asymp X_k}\frac{a_L(n)n^{-it}}{\sqrt{n}}\omega\left(\frac{n}{X_k}\right)V_{\frac{1}{2}+it}\left(\frac{n}{\sqrt{q_L}},L\right).$$

Let $V(y)=\frac{1}{\sqrt{y}}V_{\frac{1}{2}+it}\left(\frac{y}{\sqrt{q_L}},L\right)$. By Abel summation, the inner sum is O of

$$\sup_{n\asymp X_k}|A_\omega(X_k,t,L)|\,|V(n)|+\int_{u\asymp X_k}|A_\omega(u,t,L)||V'(u)|\,du.$$

In view of Proposition 6.2.2, we have the estimates

$$V(y)=O\left(\frac{1}{\sqrt{y}}\right)\quad\text{and}\quad V'(y)=O\left(\frac{1}{y^{\frac{3}{2}}}\right),$$

for $y \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}$. Whence our previous expression is O of

$$\sup_{X\asymp X_k}\left(\frac{|A_\omega(t,X,L)|}{\sqrt{X}}\right).$$

As there are $O_\varepsilon\left(\mathfrak{q}\left(\frac{1}{2}+it,L\right)^\varepsilon\right)$ many such k, the bound above implies that our sum is O_ε of

$$\mathfrak{q}\left(\frac{1}{2}+it,L\right)^\varepsilon \sup_{X \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}} \left(\frac{|A_\omega(t,X,L)|}{\sqrt{X}}\right).$$

The estimate for the second sum is treated analogously. Thus

$$L\left(\frac{1}{2}+it\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it,L\right)^\varepsilon \sup_{X \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}} \left(\frac{|A_\omega(t,X,L)|}{\sqrt{X}}\right),$$

upon absorbing smaller order O-estimates. $\square$

If $A_\omega(t,X,L) \ll_\varepsilon X^{\frac{1}{2}+\delta+\varepsilon}$ for some $\delta \geq 0$ then the supremum in Theorem 6.5.2 will automatically pick out the largest possible bound when X is the square root of the analytic conductor. This means that the savings of $\sqrt{X}$ in the denominator is essentially as large as possible. For example, the convexity bound follows upon showing that we may take $\delta = \frac{1}{2}$. Since the Dirichlet series of $L(s)$ is absolutely convergent for $\sigma < 1$, we have a bound of shape

$$\sum_{n \leq X} |a_L(n)| \ll_\varepsilon X^{1+\varepsilon}.$$

Then $A_\omega(t,X,L) \ll_\varepsilon X^{1+\varepsilon}$ as well and we obtain

$$\sup_{X \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}+it,L\right)}} \frac{|A_\omega(t,X,L)|}{\sqrt{X}} \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}+it,L\right)^{\frac{1}{4}+\varepsilon}, \tag{6.6}$$

which gives the convexity bound. As we believe $L(s)$ satisfies the Lindelöf hypothesis, we should expect lots of cancellation in the sum $A_\omega(t,X,L)$. In fact, the expected cancellation should be $O\left(\mathfrak{q}\left(\frac{1}{2}+it,L\right)^{\frac{1}{4}}\right)$ in light of the aforementioned bound. This means we expect $A_\omega(t,X,L)$ to exhibit square-root cancellation in the sense

$$A_\omega(t,X,L) \ll_\varepsilon X^{\frac{1}{2}+\varepsilon},$$

which is to say we may take $\delta = 0$. As the coefficients $a_L(n)$ may very well be real, the expected cancellation must come from the terms $n^{-it} = e^{-it\log(n)}$ which act on the coefficients $a_L(n)$ by rotation. In effect, we should think of the terms $a_L(n)n^{-it}$ as modeling identically distributed random variables. In order to exploit this cancellation via analytic methods, one can make use of smoothed Perron's formula. Letting $\Omega(s)$ be the Mellin transform of $\omega(y)$, smoothed Perron's formula gives

$$A_\omega(t,X,L) = \frac{1}{2\pi i}\int_{(c)} L\left(u+it\right)\Omega(u)X^u\,du,$$

for any $c > 1$. Then estimates for $A_\omega(t,X,L)$ can be obtained from those of this inverse Mellin transform.

The case when $t = 0$ deserves special interest as we are estimating the value of an analytic L-function at the central point. The value of $L(s)$ at the central point is called the **central value** of $L(s)$. Many important properties about an analytic L-function can be connected to its central value. Any argument used to estimate the central value of an analytic L-function is called a **central value estimate**. Taking $t = 0$ in Theorem 6.5.1 gives a way to obtain central value estimates.

Theorem 6.5.2. *Let $L(s)$ be an analytic L-function and let $\omega(y)$ be a smooth dyadic weight. Then*

$$L\left(\frac{1}{2}\right) \ll_\varepsilon \mathfrak{q}\left(\frac{1}{2}, L\right)^{\varepsilon} \max_{X \ll \sqrt{\mathfrak{q}\left(\frac{1}{2}, L\right)}} \left(\frac{|A_\omega(X, L)|}{\sqrt{X}}\right).$$

Proof. Take $t = 0$ in Theorem 6.5.1. □

6.6 Logarithmic Derivatives

Recall that $L(s)$ nonzero in the half-plane $\sigma > 1$. As this region is simply connected, there exists a unique holomorphic logarithm $\log L(s)$ there. By absolute convergence of the Euler product, we may write

$$\log L(s) = -\sum_p \sum_j \log(1 - \alpha_j(p)p^{-s}).$$

Moreover, the bound $|\alpha_j(p)| \le p^{\theta_L}$ ensures that $|\alpha_j(p)p^{-s}| < 1$ in this half-plane. Therefore the Taylor series of the logarithm is valid in this region and we may further write

$$\log L(s) = \sum_p \sum_j \sum_{k \ge 1} \frac{\alpha_j(p)^k}{kp^{ks}}.$$

While this triple sum converges for $\sigma > 1$, the bound $|\alpha_j(p)p^{-s}| < p^{\theta_L - \sigma}$ only guarantees absolutely convergence for $\sigma > 1 + \theta_L$. It is in this half-plane that we may differentiate termwise. A short computation shows

$$\frac{L'}{L}(s) = -\sum_{n \ge 1} \frac{\Lambda_L(n)}{n^s},$$

where

$$\Lambda_L(n) = \begin{cases} \sum_j \alpha_j(p)^k \log(p) & \text{if } n = p^k, \\ 0 & \text{otherwise.} \end{cases}$$

We call $\Lambda_L(n)$ the **von Mangoldt coefficients** of $L(s)$. Also observe that the dual satisfies $\Lambda_{\overline{L}}(n) = \overline{\Lambda_L(n)}$.

There is an incredibly useful formula for the logarithmic derivative of $L(s)$ which is often the starting point for deeper analytic investigations. To deduce it, we require a more complete understanding of the completion $\Lambda(s, L)$. First observe that the zeros ρ of $\Lambda(s, L)$ are contained within the critical strip. Indeed, by our earlier discussion of zeros, $\Lambda(s,L)$ is nonzero for $\sigma > 1$ and the functional equation implies that $\Lambda(s,L)$ is nonzero for $\sigma < 0$ too. This means that the zeros of $\Lambda(s,L)$ are exactly nontrivial zeros of $L(s)$. Now let us setup some notation. We define $\xi(s,L)$ by

$$\xi(s,L) = (s(1-s))^{m_L}\Lambda(s,L).$$

Observe that the dual satisfies $\xi(s,\overline{L}) = \overline{\xi(\overline{s},L)}$. In any case, the functional equation implies

$$\xi(s,L) = \varepsilon_L\overline{\xi(1-\overline{s},L)}, \tag{6.7}$$

which we refer to as the **functional equation** for $\xi(s,L)$. Now $\xi(s,L)$ is just $\Lambda(s,L)$ with the potential poles at $s=0$ and $s=1$ removed. This means $\xi(s,L)$ is entire. In fact, it admits a Hadamard factorization. To see this, first observe that $\xi(s,L)$ is of order 1 by Stirling's formula and that $(s-1)^{m_L}L(s)$ is of order 1. Then the Hadamard factorization theorem implies

$$\xi(s,L) = e^{A_L+B_Ls}\prod_{\rho\neq 0,1}\left(1-\frac{s}{\rho}\right)e^{\frac{s}{\rho}},$$

for some constants A_L and B_L and that the sum

$$\sum_{\rho\neq 0,1}\frac{1}{|\rho|^{1+\varepsilon}}, \tag{6.8}$$

is convergent. Our next result will utilize the Hadamard factorization of $\xi(s,L)$ to produce a formula relating the logarithmic derivative of $L(s)$ to its zeros.

Proposition 6.6.1. *For any analytic L-function $L(s)$, we have*

$$-\frac{L'}{L}(s) = \frac{m_L}{s} + \frac{m_L}{s-1} + \frac{1}{2}\log(q_L) + \frac{\gamma'}{\gamma}(s,L) - B_L - \sum_{\rho\neq 0,1}\left(\frac{1}{s-\rho}+\frac{1}{\rho}\right).$$

Proof. We will compute the logarithmic derivative of $\xi(s,L)$ in two ways. On the one hand, taking the logarithmic derivative of the definition of $\xi(s,L)$ yields

$$\frac{\xi'}{\xi}(s,L) = \frac{m_L}{s} + \frac{m_L}{s-1} + \frac{1}{2}\log(q_L) + \frac{\gamma'}{\gamma}(s,L) + \frac{L'}{L}(s).$$

On the other hand, taking the logarithmic derivative of the Hadamard factorization gives

$$\frac{\xi'}{\xi}(s,L) = B_L + \sum_{\rho\neq 0,1}\left(\frac{1}{s-\rho}+\frac{1}{\rho}\right).$$

Equating these expressions to obtain

$$B_L + \sum_{\rho\neq 0,1}\left(\frac{1}{s-\rho}+\frac{1}{\rho}\right) = \frac{m_L}{s} + \frac{m_L}{s-1} + \frac{1}{2}\log(q_L) + \frac{\gamma'}{\gamma}(s,L) + \frac{L'}{L}(s).$$

This is equivalent to the desired formula. $\square$

Explicit evaluation of the constants A_L and B_L can be challenging and heavily depend upon the particular L-function under investigation. However, useful formulas are not too difficult to obtain. For example, B_L is related to a sum over nontrivial zeros. To see this, using the Hadamard factorization of $\xi(s, L)$ to take the logarithmic derivative of both sides of the functional equation gives

$$B_L + \sum_{\rho \neq 0,1} \left(\frac{1}{s-\rho} + \frac{1}{\rho} \right) = -\overline{B_L} + \sum_{\rho \neq 0,1} \left(\frac{1}{s-(1-\overline{\rho})} - \frac{1}{\overline{\rho}} \right).$$

As the nontrivial zeros occur in pairs ρ and $1-\overline{\rho}$, isolating B_L shows that

$$\mathrm{Re}(B_L) = -\sum_{\rho \neq 0,1} \mathrm{Re}\left(\frac{1}{\rho} \right). \tag{6.9}$$

6.7 Zero Density Estimates

The distribution of the zeros of an analytic L-function is central to how the L-function encodes arithmetic information about its coefficients. Here we introduce a method of counting zeros of L-functions which gives a simple understanding of their density. We first require an immensely useful lemma.

Lemma 6.7.1. *Let $L(s)$ be an analytic L-function. Then the following properties hold:*

(i) For any nonnegative T, we have

$$\sum_{\rho \neq 0,1} \frac{1}{1+(T-\gamma)^2} = O(\log \mathfrak{q}(iT, L)).$$

In particular, the number of nontrivial zeros $\rho = \beta + i\gamma$ with $\rho \neq 0,1$ and such that $|T-\gamma| \leq 1$ is $O(\log \mathfrak{q}(iT, L))$.

(ii) Suppose $s = \sigma + iT$ for bounded σ and nonnegative T. Then

$$-\frac{L'}{L}(s) = \frac{m_L}{s} + \frac{m_L}{s-1} - \sum_{\substack{\rho \neq 0,1 \\ |T-\gamma| \leq 1}} \frac{1}{s-\rho} + O_\varepsilon(\log \mathfrak{q}(s, L)),$$

provided s is at least distance ε away from the nontrivial zeros of $L(s)$ and the poles of the gamma factor and $s \neq 0,1$.

Proof. We will prove the properties separately.

(i) Let $s = \sigma + iT$ for some $\sigma > 1 + \theta_L$ so that the logarithmic derivative of $L(s)$ is absolutely bounded. Taking the real part of the formula in Proposition 6.6.1 and using Equations (6.3) and (6.9) together gives

$$\sum_{\rho \neq 0,1} \mathrm{Re}\left(\frac{1}{s-\rho} \right) = O(\log \mathfrak{q}(iT, L)),$$

upon absorbing all other terms into the O-estimate. For our choice of s, we have

$$\mathrm{Re}\left(\frac{1}{s-\rho}\right) \gg \frac{1}{1+(T-\gamma)^2},$$

which implies the first statement. The second statement follows from the first as all of the terms in the sum are positive and at least $\frac{1}{2}$.

(ii) Let $s = \sigma + iT$ with σ bounded and nonnegative T. Also suppose s is at least distance ε away from the nontrivial zeros of $L(s)$ and the poles of the gamma factor and $s \neq 0, 1$. Invoking Proposition 6.6.1 and using Equation (6.3) gives

$$-\frac{L'}{L}(s) = \frac{m_L}{s} + \frac{m_L}{s-1} - \sum_{\rho \neq 0,1}\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right) + O_\varepsilon(\log \mathfrak{q}(s, L)).$$

A short computation shows

$$\frac{1}{s-\rho} + \frac{1}{\rho} \ll \frac{1}{|\gamma|^2} + \frac{1}{|T-\gamma|^2},$$

and it follows that

$$\sum_{\substack{\rho \neq 0,1 \\ |T-\gamma|>1}}\left(\frac{1}{s-\rho} + \frac{1}{\rho}\right) \ll \sum_{\substack{\rho \neq 0,1 \\ |T-\gamma|>1}} \frac{1}{|\gamma|^2} + \sum_{\substack{\rho \neq 0,1 \\ |T-\gamma|>1}} \frac{1}{|T-\gamma|^2}.$$

The first sum on the right-hand side is $O(1)$ by Equation (6.8). As we are restricting to nontrivial zeros such that $|T-\gamma| > 1$ in the second sum, it is $O(\log \mathfrak{q}(s, L))$ by (i). Also, (i) implies

$$\sum_{\substack{\rho \neq 0,1 \\ |T-\gamma|\leq 1}} \frac{1}{\rho} = O_\varepsilon(\log \mathfrak{q}(s, L)).$$

Altogether, this shows

$$-\frac{L'}{L}(s) = \frac{m_L}{s} + \frac{m_L}{s-1} - \sum_{\substack{\rho \neq 0,1 \\ |T-\gamma|\leq 1}} \frac{1}{s-\rho} + O_\varepsilon(\log \mathfrak{q}(s, L)),$$

as claimed. □

With this lemma we can deduce a result which estimates the number of nontrivial zeros of $L(s)$ in a box. For any nonnegative T, define

$$N(T, L) = |\{\rho = \beta + i\gamma \in \mathbb{C} : L(\rho) = 0 \text{ with } 0 \leq \beta \leq 1 \text{ and } |\gamma| \leq T\}|.$$

In other words, $N(T, L)$ is the number of nontrivial zeros of $L(s)$ whose ordinate belongs to $[-T, T]$. We will use the argument principle to deduce an asymptotic formula for $N(T, L)$.

Theorem 6.7.2. *For any analytic L-function $L(s)$ and $T \geq 1$, we have*

$$N(T,L) = \frac{T}{\pi}\log\left(\frac{q_L T^{d_L}}{(2\pi e)^{d_L}}\right) + O(\log \mathfrak{q}(iT,L)).$$

In particular,

$$\frac{N(T,L)}{T} \sim \frac{1}{\pi}\log\left(\frac{q_L T^{d_L}}{(2\pi e)^{d_L}}\right).$$

Proof. Let $T \geq 1$ and set

$$N'(T,L) = |\{\rho = \beta + i\gamma \in \mathbb{C} : L(\rho) = 0 \text{ with } 0 \leq \beta \leq 1 \text{ and } 0 < \gamma \leq T\}|.$$

As the nontrivial zeros occur in pairs ρ and $1-\overline{\rho}$, it follows that

$$N(T,L) = 2N'(T,L) + O(1),$$

where $O(1)$ accounts for the possible real nontrivial zeros of which there are finitely many. We will estimate $N'(T,L)$ and in doing so we may assume $L(s)$ does not have a nontrivial zero in the horizontal strip $T-\delta \leq t \leq T+\delta$ for some $\delta > 0$ by varying T by a sufficiently small constant, if necessary, and observing that $N'(T,L)$ is modified by a quantity of size $O(\log \mathfrak{q}(iT,L))$ by Lemma 6.7.1 (i). Taking δ smaller, if necessary, we may additionally assume $\Lambda(s,L)$ has no nontrivial zeros in the horizontal strip $-\delta \leq t < 0$.

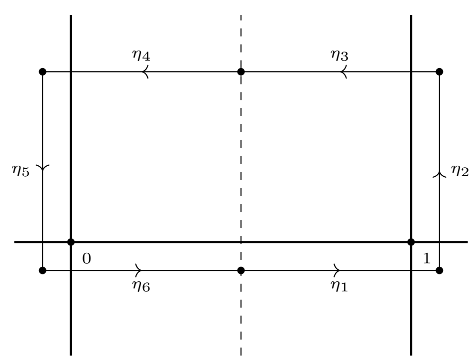

Figure 6.3: A zero counting contour.

Let $\eta = \sum_i \eta_i$ be the rectangular contour Figure 6.3 where the horizontal lines are along $t = T$ and $t = -\delta$ and the vertical lines are along $\sigma = -(\theta_L + \delta)$ and $\sigma = 1 + \theta_L + \delta$. Then by our previous comments and the argument principle, we may write

$$N'(T,L) = \frac{1}{2\pi i}\int_\eta \frac{\xi'}{\xi}(s,L)\,ds + O(\log \mathfrak{q}(iT,L)),$$

and where

$$\frac{1}{2\pi i}\int_\eta \frac{\xi'}{\xi}(s,L)\,ds = \frac{1}{2\pi}\Delta_\eta \arg \xi(s,L).$$

For convenience, set $\eta_R = \eta_1 + \eta_2 + \eta_3$ and $\eta_L = \eta_4 + \eta_5 + \eta_6$. Observe that η_R is taken to $-\eta_L$ under the change of variables $s \mapsto 1 - \overline{s}$. Using this fact and the functional equation, a short computation shows

$$\Delta_{\eta_R} \arg \xi(s, L) = \Delta_{\eta_L} \arg \xi(s, L).$$

In other words, the change in argument along η_R is equal to the change in argument along η_L. Whence

$$\frac{1}{2\pi i} \int_{\eta} \frac{\xi'}{\xi}(s, L)\, ds = \frac{1}{\pi} \Delta_{\eta_R} \arg \xi(s, L).$$

We will now estimate the integral by estimating the change in argument along η_R of each factor in

$$\xi(s, L) = (s(1-s))^{m_L} q_L^{\frac{s}{2}} \pi^{-\frac{d_L s}{2}} \prod_j \Gamma\left(\frac{s + \mu_j}{2}\right) L(s).$$

For the contribution from $(s(1-s))^{m_L}$, first recall that $\arg(s) = \tan^{-1}\left(\frac{t}{\sigma}\right)$ provided $\sigma > 0$. In this case, we have the estimate $\arg(s) = \frac{\pi}{2} + O\left(\frac{1}{t}\right)$ by truncating the Laurent series of the inverse tangent after the first term. A short computation shows

$$\Delta_{\eta_R} \arg((s(1-s))^{m_L}) = O\left(\frac{1}{T}\right).$$

For the remaining factors, we will use $\arg(s) = \operatorname{Im}(\log(s))$. The contributions from $q_L^{\frac{s}{2}}$ and $\pi^{-\frac{d_L s}{2}}$ are easily estimated to be

$$\Delta_{\eta_R} \arg\left(q_L^{\frac{s}{2}}\right) = \frac{T}{2} \log(q_L) + O(1),$$

and

$$\Delta_{\eta_R} \arg\left(\pi^{-\frac{d_L s}{2}}\right) = \frac{T}{2} \log\left(\frac{1}{\pi^{d_L}}\right) + O(1),$$

respectively. For the product of gamma functions, first observe that the logarithm of Stirling's formula gives

$$\log \Gamma(s) = \frac{1}{2}\log(2\pi) + \left(s - \frac{1}{2}\right)\log(s) - s + O\left(\frac{1}{s}\right),$$

provided $t \geq 1$ say. A short computation shows

$$\Delta_{\eta_R} \arg \Gamma\left(\frac{s + \mu_j}{2}\right) = \frac{T}{2} \log\left(\frac{T}{2e}\right) + O(1),$$

and it follows that the contribution from the product of gamma functions is

$$\Delta_{\eta_R} \arg\left(\prod_j \Gamma\left(\frac{s + \mu_j}{2}\right)\right) = \frac{T}{2} \log\left(\frac{T^{d_L}}{(2e)^{d_L}}\right) + O(1).$$

For $L(s)$, being by observing that

$$\Delta_{\eta_R} \arg(L(s)) = \operatorname{Im}\left(\int_{\eta_R} \frac{L'}{L}(s)\,ds\right).$$

As the logarithmic derivative of $L(s)$ is absolutely bounded for $\sigma = 1 + \theta_L + \delta$, the integral is $O(1)$ on η_2. On η_1 and η_3, use Lemma 6.7.1 (ii) to write

$$\operatorname{Im}\left(\int_{\eta_1 \cup \eta_3} \frac{L'}{L}(s)\,ds\right) = \sum_{\substack{\rho \neq 0,1 \\ |T-\gamma| \le 1}} \operatorname{Im}\left(\int_{\eta_1 \cup \eta_3} \frac{1}{s-\rho}\,ds\right) + O(\log \mathfrak{q}(iT, L)),$$

where we have absorbed the terms corresponding to 0 and 1 into the O-estimate. Each summand is $\Delta_{\eta_1 \cup \eta_3} \arg(s - \rho) = O(1)$. As there are $O(\log \mathfrak{q}(iT, L))$ many terms by Lemma 6.7.1 (i), it follows that

$$\Delta_{\eta_R} \arg(L(s)) = O(\log \mathfrak{q}(iT, L)).$$

Combining our change in argument estimates gives

$$N'(T, L) = \frac{T}{2\pi} \log\left(\frac{q_L T^{d_L}}{(2\pi e)^{d_L}}\right) + O(\log \mathfrak{q}(iT, L)),$$

from which the asymptotic formula follows. The asymptotic equivalence is immediate since it is a strictly weaker asymptotic. $\square$

It is worth noting that the main term in the asymptotic formula of the previous result comes from the change in argument of $q_L^{\frac{s}{2}} \gamma(s, L)$ along η_3 (and η_4 by the functional equation). In particular, the contribution from $L(s)$ is only to the error term. This is a good example of how information of an analytic L-function is intrinsically connected to its gamma factor.

Also observe that the asymptotic equivalence can be interpreted as saying that for large T the density of $N(T, L)$ is approximately $\frac{1}{\pi} \log\left(\frac{q_L T^{d_L}}{(2\pi e)^{d_L}}\right)$. Since this grows as $T \to \infty$, we see that the nontrivial zeros tend to accumulate farther up the critical strip with logarithmic growth. To account for this accumulation, if $\rho = \beta + i\gamma$ is a nontrivial zero of $L(s)$ then we call $\rho_{\text{unf}} = \beta + i\omega$ the **unfolded nontrivial zero** corresponding to ρ where we set

$$\omega = \frac{\gamma}{\pi} \log\left(\frac{q_L |\gamma|^{d_L}}{(2\pi e)^{d_L}}\right).$$

For any nonnegative W, define

$$N_{\text{unf}}(W, L) = |\{\rho_{\text{unf}} = \beta + i\omega \in \mathbb{C} : L(\rho) = 0 \text{ with } 0 \le \beta \le 1 \text{ and } |\omega| \le W\}|.$$

In other words, $N_{\text{unf}}(T, L)$ is the number of unfolded nontrivial zeros of $L(s)$ whose ordinate belongs to $[-W, W]$. The point of considering unfolded nontrivial zeros is that they are uniformly dense along the critical strip.

Proposition 6.7.3. *For any analytic L-function $L(s)$ and $W \geq \frac{1}{\pi}\log\left(\frac{q_L}{(2\pi e)^{d_L}}\right)$, we have*

$$\frac{N_{unf}(W, L)}{W} \sim 1.$$

Proof. Consider the function $f(t)$ defined by

$$f(t) = \frac{t}{\pi}\log\left(\frac{q_L|t|^{d_L}}{(2\pi e)^{d_L}}\right),$$

for real t. Since $f(t)$ is a strictly increasing continuous function, it has an inverse $g(w)$. It follows that $|\omega| \leq W$ if and only if $|\rho| \leq g(W)$ whence

$$N_{\text{unf}}(W, L) = N(g(W), L).$$

But by Theorem 6.7.2 and that $g(w)$ is the inverse of $f(t)$, we have $N(g(W), L) \sim W$. Then

$$N_{\text{unf}}(W, L) \sim W,$$

which is equivalent to the claim. □

We interpret this result as saying that the unfolded nontrivial zeros are evenly spaced opposed to Theorem 6.7.2 which say that they tend to accumulate up the critical strip.

6.8 Zero-free Regions

Although the Riemann hypothesis for any analytic L-function remains out of reach, some progress has been made to understand regions inside of the critical strip for which there are no nontrivial zeros except for possibly one real exception. Such a region is said to be a **zero-free region**. We will derive a standard zero-free region for any analytic L-function under a mild assumption about the real part of its logarithmic derivative. First we require a lemma which bounds the possible amount of real zeros.

Lemma 6.8.1. *Let $L(s)$ be an analytic L-function such that the real part of the logarithmic derivative of $L(s)$ is nonpositive for $\sigma > 1$. Then $L(1) \neq 0$ whence $m_L \geq 0$. Moreover, there exists a positive constant c such that $L(s)$ has at most m_L real zeros in the region*

$$\sigma \geq 1 - \frac{c}{(m_L + 1)\log \mathfrak{q}(L)}.$$

Proof. Observe that the real parts of the terms corresponding to the nontrivial zeros in Lemma 6.7.1 (ii) are positive provided $\sigma > 1$. Letting $s = \sigma$ and taking the real part of this formula, we obtain the inequality

$$\sum_{\beta} \text{Re}\left(\frac{1}{\sigma - \beta}\right) \leq \frac{m_L}{\sigma - 1} + \text{Re}\left(\frac{L'}{L}(\sigma)\right) + O(\log \mathfrak{q}(L)),$$

upon discarding all of the terns corresponding to nontrivial zeros except those which are real and absorbing the term corresponding to 0 into the O-estimate. Also note that the left-hand side is positive. As the real part of the logarithmic derivative of $L(s)$ is nonpositive for $\sigma > 1$ by assumption, we further have

$$\sum_{\beta} \mathrm{Re}\left(\frac{1}{\sigma - \beta}\right) \le \frac{m_L}{\sigma - 1} + O(\log \mathfrak{q}(L)).$$

From this inequality we see that $\beta \neq 1$. For if some $\beta = 1$, we have $m_L < 0$ whence the right-hand side is negative for σ sufficiently close to 1 contradicting positivity of the left-hand side. Thus $L(1) \neq 0$ and hence $m_L \ge 0$ proving the first statement.

For the second statement, recall that $L(s)$ is nonzero for $\sigma > 1$. Letting c be a positive constant, there are finitely many, say n, real zeros β satisfying

$$\beta \ge 1 - \frac{c}{(m_L + 1)\log \mathfrak{q}(L)}.$$

Setting $\sigma = 1 + \frac{2c}{\log \mathfrak{q}(L)}$, a short computation using the previous two inequalities gives

$$\frac{n \log \mathfrak{q}(L)}{2c + \frac{c}{(m_L+1)}} \le \left(\frac{m_L}{2c} + O(1)\right) \log \mathfrak{q}(L).$$

Isolating n yields

$$n \le m_L + \frac{m_L}{2(m_L + 1)} + O(c).$$

Taking c smaller, if necessary, shows $n \le m_L$. As $L(s)$ is nonzero for $\sigma > 1$, it follows that there are at most m_L real zeros in the region

$$\sigma \ge 1 - \frac{c}{(m_L + 1)\log \mathfrak{q}(L)}.$$ □

We can now prove our zero-free region result.

Theorem 6.8.2. *Let $L(s)$ be an analytic L-function such that the real part of the logarithmic derivative of $L(s)$ is nonpositive for $\sigma > 1$. Then there exists a positive constant c such that $L(s)$ has no zeros in the region*

$$\sigma \ge 1 - \frac{c}{\log(\mathfrak{q}(L)(|t| + 3))},$$

except for at most m_L many real zeros β each with $\beta < 1$.

Proof. For real r, let $M(s)$ be the L-function defined by

$$M(s) = L^{m_L}(s)\overline{L}^{m_L}(s)L^{2m_L}(s + ir)\overline{L}^{2m_L}(s - ir).$$

Then $d_M = 6m_L d_L$ and $\mathfrak{q}(M)$ satisfies

$$\mathfrak{q}(M) < \mathfrak{q}(L)^{6m_L}(|r| + 3)^{4m_L}.$$

As the real part of the logarithmic derivative of $L(s)$ is nonpositive for $\sigma > 1$, the same holds $M(s)$ as well. Now let $\rho = \beta + i\gamma$ be a complex nontrivial zero of $L(s)$. Setting $r = \gamma$, $M(s)$ has a real nontrivial zero at $s = \beta$ of multiplicity at least $4m_L$ and a pole at $s = 1$ of order at most $2m_L$. From Lemma 6.8.1, it follows that there is a positive constant c for which $M(s)$ can have at most $2m_L$ real nontrivial zeros in the region

$$\sigma \geq 1 - \frac{c}{\log(\mathfrak{q}(L)(|\gamma| + 3))}.$$

Therefore the region

$$\sigma \geq 1 - \frac{c}{\log(\mathfrak{q}(L)(|t| + 3))},$$

does not contain any complex nontrivial zeros of $L(s)$. Now let β be a real nontrivial zero of $L(s)$. Setting $r = 0$, $M(s)$ has a real nontrivial zero at $s = \beta$ of multiplicity at least $6m_L$ and a pole at $s = 1$ of order at most $6m_L^2$. Whence Lemma 6.8.1 implies upon taking c smaller, if necessary, that there are at most m_L many real zeros in this region. If such a real zero β exists then $\beta < 1$ because $L(s)$ is nonzero for $\sigma > 1$ and must have a pole at $s = 1$. This completes the proof. $\square$

Some comments about this result are in order. First observe that the larger c can be taken the closer the curve bounding the zero-free region approaches the critical line. As we do not have a nonzero lower bound for c, this constant could be quite small and so we are only guaranteed a zero-free region very close to the line $\sigma = 1$. This means that the possible real zeros could be very close to 1. Moreover, as the nontrivial zeros of $L(s)$ occur in pairs ρ and $1 - \overline{\rho}$, the zero-free region implies a symmetric zero-free region about the critical line with at most one real zero in each half as displayed in Figure 6.4. It is possible to obtain stronger zero-free regions in certain cases but this is heavily dependent upon the particular analytic L-function under investigation. In many cases it is possible to augment the proof of Theorem 6.8.2 to make the constant c effective and such results have important applications.

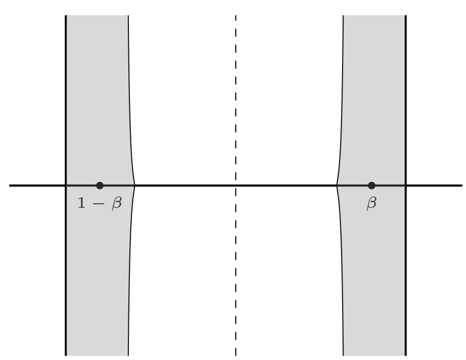

Figure 6.4: A symmetric zero-free region.

A possible real zero β in Theorem 6.8.2 is referred to as a **Siegel zero**. Such real zeros are conjectured to not exist. For example, Siegel zeros are conjectured to not exist for Selberg class L-functions in view of the Selberg class Riemann hypothesis.

Nevertheless, it is an interesting question to see how the presence of a Siegel zero can affect the behavior of the associated analytic L-function.

6.9 The Explicit Formula

There is a formula, somewhat analogous to the approximate functional equation, that relates a weighted sum of the von Mangoldt coefficients of any analytic L-function $L(s)$ to a sum over its nontrivial zeros. This is called the **explicit formula** for $L(s)$.

Theorem (Explict Formula). *Let $\psi(y)$ be a smooth weight where $\Psi(s)$ is its Mellin transform. Set $\phi(y) = y^{-1}\psi(y^{-1})$ so that its Mellin transform satisfies $\Phi(s) = \Psi(1-s)$. Then for any analytic L-function $L(s)$, we have*

$$\sum_{n\geq 1}\left(\Lambda_L(n)\psi(n) + \overline{\Lambda_L(n)}\phi(n)\right) = \psi(1)\log(q_L) + m_L\Psi(1) - \sum_{\rho}\Psi(\rho) + \frac{1}{2\pi i}\int_{(\frac{1}{2})}\left(\frac{\gamma'}{\gamma}(s,L) + \frac{\gamma'}{\gamma}(1-s,L)\right)\Psi(s)\,ds.$$

Proof. By smoothed Perron's formula implies

$$\sum_{n\geq 1}\Lambda_L(n)\psi(n) = \frac{1}{2\pi i}\int_{(a)} -\frac{L'}{L}(s)\Phi(s)\,ds,$$

for any $a > 1+\theta_L$. Let $\varepsilon > 0$ be such that there are no trivial zeros in the vertical strip $-2\varepsilon \leq \sigma < 0$. As the logarithmic derivative is absolutely bounded for $\sigma > 1+\theta_L$, Proposition 6.2.1 implies that the integrand has rapid decay and therefore the integral is locally absolutely uniformly convergent. This permits us to shift the line of integration. Shifting to $(-\varepsilon)$, we pass by simple poles at $s=1$ and $s=\rho$ for every nontrivial zero ρ obtaining

$$\sum_{n\geq 1}\Lambda_L(n)\psi(n) = m_L\Psi(1) - \sum_{\rho}\Psi(\rho) + \frac{1}{2\pi i}\int_{(-\varepsilon)} -\frac{L'}{L}(s)\Psi(s)\,ds.$$

Now take the logarithmic derivative of the functional equation to obtain

$$-\frac{L'}{L}(s) = \log(q_L) + \frac{\gamma'}{\gamma}(s,L) + \frac{\gamma'}{\gamma}(1-s,L) + \frac{\overline{L}'}{\overline{L}}(1-s).$$

Thus the remaining integral above may be expressed as

$$\frac{1}{2\pi i}\int_{(-\varepsilon)}\left(\log(q_L) + \frac{\gamma'}{\gamma}(s,L) + \frac{\gamma'}{\gamma}(1-s,L) + \frac{\overline{L}'}{\overline{L}}(1-s)\right)\Psi(s)\,ds.$$

The integral over the first term on the right-hand side is $\psi(1)\log(q_L)$ by Mellin inversion. As for the last term, smoothed Perron's formula and that $\Phi(s) = \Psi(1-s)$ together give

$$\frac{1}{2\pi i}\int_{(-\varepsilon)} -\frac{\overline{L}'}{\overline{L}}(1-s)\Psi(s)\,ds = \sum_{n\geq 1}\overline{\Lambda_L(n)}\phi(n).$$

Whence

$$\sum_{n\geq 1}\left(\Lambda_L(n)\psi(n)+\overline{\Lambda_L(n)}\phi(n)\right)=\psi(1)\log(q_L)+m_L\Psi(1)-\sum_{\rho}\Psi(\rho)$$
$$+\frac{1}{2\pi i}\int_{(-\varepsilon)}\left(\frac{\gamma'}{\gamma}(s,L)+\frac{\gamma'}{\gamma}(1-s,L)\right)\Psi(s)\,ds.$$

The remaining integrand has rapid decay by Proposition 5.4.4 and Equation (6.3) and so the integral is locally absolutely uniformly convergent. Therefore we may shift the line of integration. Shifting to $\left(\frac{1}{2}\right)$, we pass over no poles. This completes the proof. $\square$

Chapter 7

The Riemann Zeta Function

Throughout, $s = \sigma + it$ will stand for complex variables with σ and t real. We will also make use of the gamma function developed in Appendix C.2. Also, all sums and products over zeros of a function will be counted with multiplicity and ordered symmetrically with respect to the size of the ordinate if they are not absolutely convergent.

7.1 Analytic Properties

The **Riemann zeta function** $\zeta(s)$ is defined by the Dirichlet series

$$\zeta(s) = \sum_{n \geq 1} \frac{1}{n^s}.$$

As the coefficients are uniformly bounded and completely multiplicative, $\zeta(s)$ is absolutely convergent for $\sigma > 1$ and admits the following degree 1 Euler product

$$\zeta(s) = \prod_p (1 - p^{-s})^{-1},$$

by Proposition 5.2.1.

Our primary aim is to show that the Riemann zeta function is a Selberg class L-function. In order to do this, we need to establish meromorphic continuation to the entire complex plane and prove a functional equation. These will be accomplished in tandem by developing and analyzing an integral representation for the Riemann zeta function that evolves from a Mellin transform. The function which we will take the Mellin transform of is closely related to the **Jacobi theta function** $\vartheta(z)$ defined by

$$\vartheta(z) = \sum_{n \in \mathbb{Z}} e^{2\pi i n^2 z},$$

for $z \in \mathbb{H}$. This function is locally absolutely uniformly convergent in this region by the Weierstrass M-test. Moreover, the sum of a geometric series implies

$$\vartheta(z) - 1 = O(e^{-2\pi y}),$$

so that $\vartheta(z) - 1$ exhibits exponential decay. The essential result we will need about the Jacobi theta function is that it satisfies a transformation property. This will follow by an application of Poisson summation.

Theorem 7.1.1. *For $z \in \mathbb{H}$,*

$$\vartheta(z) = \frac{1}{\sqrt{-2iz}}\vartheta\left(-\frac{1}{4z}\right).$$

Proof. We will apply Poisson summation to the Jacobi theta function. By the identity theorem it suffices to verify the transformation formula for $z = iy$ with y positive. So set

$$f(x) = e^{-2\pi x^2 y}.$$

Then $f(x)$ is Schwartz class. By Proposition B.3.2, we have

$$(\mathcal{F}f)(t) = \frac{e^{-\frac{\pi t^2}{2y}}}{\sqrt{2y}}.$$

Poisson summation together with this identity gives the result. □

We can now introduce the function whose Mellin transform will essentially give the Riemann zeta function. This function $\omega(z)$ is defined by

$$\omega(z) = \sum_{n \geq 1} e^{\pi i n^2 z},$$

for $z \in \mathbb{H}$. It is related to the Jacobi theta function via

$$\omega(z) = \frac{\vartheta\left(\frac{z}{2}\right) - 1}{2},$$

and hence is locally absolutely uniformly convergent in this region. In particular, we have the estimates

$$\omega(iy) = \begin{cases} O\left(y^{-\frac{1}{2}}\right) & \text{if } y \ll 1, \\ O(e^{-\pi y}) & \text{if } y \gg 1, \end{cases}$$

where the second estimate follows from the exponential decay of the Jacobi theta function and the first estimate from Theorem 7.1.1. Now consider the Mellin transform

$$\int_0^\infty \omega(iy) y^{\frac{s}{2}} \, \frac{dy}{y}.$$

By the previous estimates, this integral is locally absolutely uniformly convergent for $\sigma > 1$ and hence defines an analytic function there. This also permits the us to interchange the integral and the sum inside the integrand resulting in

$$\int_0^\infty \omega(iy) y^{\frac{s}{2}} \, \frac{dy}{y} = \sum_{n \geq 1} \int_0^\infty e^{-\pi n^2 y} y^{\frac{s}{2}} \, \frac{dy}{y}.$$

Under the change of variables $y \mapsto \frac{y}{\pi n^2}$, the integral is realized as $\Gamma\left(\frac{s}{2}\right)$ and the sum becomes $\zeta(s)$ giving

$$\sum_{n\geq 1}\int_0^\infty e^{-\pi n^2 y} y^{\frac{s}{2}}\,\frac{dy}{y} = \frac{\Gamma\left(\frac{s}{2}\right)}{\pi^{\frac{s}{2}}}\zeta(s).$$

As the reciprocal of the gamma function is entire, we obtain an integral representation

$$\zeta(s) = \frac{\pi^{\frac{s}{2}}}{\Gamma\left(\frac{s}{2}\right)}\int_0^\infty \omega(iy) y^{\frac{s}{2}}\,\frac{dy}{y}. \tag{7.1}$$

This integral representation does not establish the meromorphic continuation of the Riemann zeta function to the entire complex plane because the integral does not converge for $\sigma \leq 1$ in the region where $y \ll 1$ in view of the estimates we have just establishes. In order to bypass this obstruction, decompose the integral as

$$\int_0^\infty \omega(iy) y^{\frac{s}{2}}\,\frac{dy}{y} = \int_0^1 \omega(iy) y^{\frac{s}{2}}\,\frac{dy}{y} + \int_1^\infty \omega(iy) y^{\frac{s}{2}}\,\frac{dy}{y}. \tag{7.2}$$

We will write the first piece in the same form as the second and symmetrize the result as much as possible. To this end, performing the change of variables $y \mapsto \frac{1}{y}$ to the first piece gives

$$\int_1^\infty \omega\left(\frac{i}{y}\right) y^{-\frac{s}{2}}\,\frac{dy}{y}.$$

A short computation using Theorem 7.1.1 shows that

$$\omega\left(\frac{i}{y}\right) = \sqrt{y}\,\omega(iy) + \frac{\sqrt{y}}{2} - \frac{1}{2}.$$

The previous integral may then be expressed as

$$\int_1^\infty \left(\sqrt{y}\,\omega(iy) + \frac{\sqrt{y}}{2} - \frac{1}{2}\right) y^{-\frac{s}{2}}\,\frac{dy}{y},$$

which is further equivalent to

$$\int_1^\infty \omega(iy) y^{\frac{1-s}{2}}\,\frac{dy}{y} - \frac{1}{s(1-s)},$$

upon integrating the lat two terms. Substituting this result back into Equation (7.2) and combining with Equation (7.1) yields the integral representation

$$\zeta(s) = \frac{\pi^{\frac{s}{2}}}{\Gamma\left(\frac{s}{2}\right)}\left[-\frac{1}{s(1-s)} + \int_1^\infty \omega(iy) y^{\frac{1-s}{2}}\,\frac{dy}{y} + \int_1^\infty \omega(iy) y^{\frac{s}{2}}\,\frac{dy}{y}\right]. \tag{7.3}$$

This integral representation gives meromorphic continuation since the two symmetric integrals are locally absolutely uniformly convergent on $\mathbb{C}$ via the exponential decay of the integrands and that they are bounded near 1. The fractional term is holomorphic except for simple poles at $s = 0$ and $s = 1$. The meromorphic continuation to $\mathbb{C}$

follows with possible simple poles at $s = 0$ and $s = 1$. In fact, there is no pole at $s = 0$ because $\Gamma\left(\frac{s}{2}\right)$ has a simple pole at $s = 0$ and so its reciprocal has a simple zero. At $s = 1$, $\Gamma\left(\frac{s}{2}\right)$ is nonzero and so $\zeta(s)$ has a simple pole. Therefore $\zeta(s)$ has meromorphic continuation to all of $\mathbb{C}$ with a simple pole at $s = 1$.

A convenient consequence of this integral representation is that the functional equation is almost immediate. Indeed, applying the symmetry $s \mapsto 1 - s$ to Equation (7.3) gives the functional equation

$$\frac{\Gamma\left(\frac{s}{2}\right)}{\pi^{\frac{s}{2}}}\zeta(s) = \frac{\Gamma\left(\frac{1-s}{2}\right)}{\pi^{\frac{1-s}{2}}}\zeta(1-s).$$

We identify the gamma factor and completion are

$$\gamma(s,\zeta) = \pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right) \quad \text{and} \quad \Lambda(s,\zeta) = \pi^{-\frac{s}{2}}\Gamma\left(\frac{s}{2}\right)\zeta(s),$$

respectively. In particular, the conductor is 1 so that no primes ramify, the root number is also 1, and $\zeta(s)$ is self-dual.

It is straightforward to see that the order of $(s-1)\zeta(s)$ is 1. Indeed, as the integrals in Equation (7.3) are locally absolutely uniformly convergent, computing the order amounts to estimating the gamma factor. Since the reciprocal of the gamma function is of order 1, it follows that $(s-1)\zeta(s)$ is of order 1 as well.

We now compute the residue of $\zeta(s)$ at $s = 1$. First observe that the only term in Equation (7.3) contributing to the pole is $-\frac{\pi^{\frac{s}{2}}}{\Gamma\left(\frac{s}{2}\right)}\frac{1}{s(1-s)}$. As this pole is simple, the residue is given by the limit as $s \to 1$ after removing the polar divisor. Whence a short computation shows

$$\operatorname*{Res}_{s=1}\zeta(s) = 1.$$

We summarize all of our work into the following theorem:

Theorem 7.1.2. *$\zeta(s)$ is a Selberg class L-function with Euler product*

$$\zeta(s) = \prod_p (1-p^{-s})^{-1}.$$

Moreover, it admits meromorphic continuation to $\mathbb{C}$ with a simple pole at $s = 1$ of residue 1 and possesses the functional equation

$$\frac{\Gamma\left(\frac{s}{2}\right)}{\pi^{\frac{s}{2}}}\zeta(s) = \frac{\Gamma\left(\frac{1-s}{2}\right)}{\pi^{\frac{1-s}{2}}}\zeta(1-s).$$

7.2 Todo: [The Prime Number Theorem]

7.3 A Second Moment Estimate

We will discuss the second moment of the Riemann zeta function. Of primary aim is to prove an infamous result of Hardy and Littlewood. Throughout, we will also illustrate where main obstructions appear in the proof and how inputting more refined analytic data will lead to improved asymptotics. Throughout, it will be useful to recall that

$$\sum_{n\ll N}\frac{1}{n}=\log(N)+O(1), \tag{7.4}$$

for any $N\geq 1$, which follows from the definition of the Euler-Mascheroni constant. The aforementioned result of Hardy and Littlewood is an asymptotic equivalence for $M_2(T,\zeta)$. Actually, we prove something slightly stronger:

Theorem 7.3.1. *For $T>2$,*

$$M_2(T,\zeta)=T\log(T)+O\left(T\log^{\frac{3}{4}}(T)\right).$$

In particular,

$$M_2(T,\zeta)\sim T\log(T).$$

Proof. Take $s=\frac{1}{2}+it$, $X=\sqrt{\frac{t}{\log(t)}}$ for $t\geq 2$, and $\Phi(u)=\cos^{-4M}\left(\frac{\pi u}{4M}\right)$ for some large positive integer M in the approximate functional equation. Then we may write

$$\begin{aligned}\zeta\left(\frac{1}{2}+it\right)&=\sum_{n\geq 1}\frac{n^{-it}}{n^{\frac{1}{2}}}V_{\frac{1}{2}+it}\left(n\sqrt{\frac{\log(t)}{t}},\zeta\right)\\&\quad+\varepsilon\left(\frac{1}{2}+it,\zeta\right)\sum_{n\geq 1}\frac{n^{it}}{n^{\frac{1}{2}}}V_{\frac{1}{2}-it}\left(n\sqrt{\frac{t}{\log(t)}},\zeta\right)+\frac{R\left(\frac{1}{2}+it,\sqrt{\frac{t}{\log(t)}},\zeta\right)}{\gamma\left(\frac{1}{2}+it,\zeta\right)}.\end{aligned}$$

First observe $\varepsilon\left(\frac{1}{2}+it,\zeta\right)\ll 1$ by Equation (6.2) while Equations (6.1) and (6.5) together show that the residue term exhibits exponential decay. By our choice of $\Phi(u)$, the estimates in Proposition 6.2.2 are valid. Whence the summands are bounded for $n\ll\frac{t}{\sqrt{\log(t)}}$ and $n\ll\sqrt{\log(t)}$ respectively and then exhibit rapid decay thereafter. Altogether, this means

$$\zeta\left(\frac{1}{2}+it\right)=\sum_{n\ll\frac{t}{\sqrt{\log(t)}}}\frac{1}{n^{\frac{1}{2}+it}}+O\left(\sum_{n\ll\sqrt{\log(t)}}\frac{1}{\sqrt{n}}\right),$$

The estimate

$$\sum_{n\ll\sqrt{\log(t)}}\frac{1}{\sqrt{n}}=O\left(\log^{\frac{1}{4}}(t)\right),$$

follows upon bounding the sum by its associated integral. Whence we arrive at

$$\zeta\left(\frac{1}{2}+it\right)=\sum_{n\ll\frac{t}{\sqrt{\log(t)}}}\frac{1}{n^{\frac{1}{2}+it}}+O\left(\log^{\frac{1}{4}}(t)\right). \tag{7.5}$$

Applying this estimate to the definition of $M_2(T,\zeta)$ and recalling that $\zeta\left(\frac{1}{2}+it\right)$ is absolutely bounded for $0 \le t \le 2$, yields

$$\begin{aligned} M_2(T,\zeta) = &\int_2^T \left| \sum_{n \ll \frac{t}{\sqrt{\log(t)}}} \frac{1}{n^{\frac{1}{2}+it}} \right|^2 dt \\ &+ O\left(\int_2^T \left| \sum_{n \ll \frac{t}{\sqrt{\log(t)}}} \frac{1}{n^{\frac{1}{2}+it}} \right| \log^{\frac{1}{4}}(t)\, dt \right) + O\left(\int_2^T \sqrt{\log(t)}\, dt \right). \end{aligned} \tag{7.6}$$

We will now estimate each term in the right-hand side. For the second term, the Cauchy-Schwarz inequality gives

$$\int_2^T \left| \sum_{n \ll \frac{t}{\sqrt{\log(t)}}} \frac{1}{n^{\frac{1}{2}+it}} \right| \log^{\frac{1}{4}}(t)\, dt = \left(\int_2^T \left| \sum_{n \ll \frac{t}{\sqrt{\log(t)}}} \frac{1}{n^{\frac{1}{2}+it}} \right|^2 dt \right)^{\frac{1}{2}} \left(\int_2^T \sqrt{\log(t)}\, dt \right)^{\frac{1}{2}}.$$

Observe that the second integral on the right-hand side is the error term of Equation (7.6). To estimate it, a short application of integration by parts yields

$$\int_2^T \sqrt{\log(t)}\, dt = T\sqrt{\log(T)} + O(T).$$

These estimates applied to Equation (7.6) gives

$$\begin{aligned} M_2(T,\zeta) = &\int_2^T \left| \sum_{n \ll \frac{t}{\sqrt{\log(t)}}} \frac{1}{n^{\frac{1}{2}+it}} \right|^2 dt \\ &+ O\left(\left(\int_2^T \left| \sum_{n \ll \frac{t}{\sqrt{\log(t)}}} \frac{1}{n^{\frac{1}{2}+it}} \right|^2 dt \right)^{\frac{1}{2}} \sqrt{T} \log^{\frac{1}{4}}(T) \right) \\ &+ O\left(T\sqrt{\log(T)} \right). \end{aligned} \tag{7.7}$$

We now show that the common integral appearing in the first and second terms is asymptotically equivalent to $T\log(T)$. Since the integral converges absolutely, we may expand the sum and interchange it with the integral to obtain

$$\int_2^T \left| \sum_{n \ll \frac{t}{\sqrt{\log(t)}}} \frac{1}{n^{\frac{1}{2}+it}} \right|^2 dt = \sum_{n,m \ll \frac{T}{\sqrt{\log(T)}}} \int_{\max(2,t(n),t(m))}^T \frac{1}{n^{\frac{1}{2}+it} m^{\frac{1}{2}-it}}\, dt,$$

where $t(x) = \inf\left\{t \geq 2 : \frac{t}{\sqrt{\log(t)}} \gg x\right\}$. Separating the terms for which $n = m$ or not, the right-hand side becomes

$$\sum_{n \ll \frac{T}{\sqrt{\log(T)}}} \frac{T-n}{n} + \sum_{\substack{n,m \ll \frac{T}{\sqrt{\log(T)}} \\ n \neq m}} \frac{1}{\sqrt{nm}} \int_{\max(2,t(n),t(m))}^{T} \left(\frac{m}{n}\right)^{it} dt.$$

Upon splitting the first sum into two sums and computing the remaining integral, we find that

$$\int_2^T \left| \sum_{n \ll \frac{t}{\sqrt{\log(t)}}} \frac{1}{n^{\frac{1}{2}+it}} \right|^2 dt = T \sum_{n \ll \frac{T}{\sqrt{\log(T)}}} \frac{1}{n} + O\left(\sum_{\substack{n,m \ll \frac{T}{\sqrt{\log(T)}} \\ n \neq m}} \frac{1}{\sqrt{nm}\log\left(\frac{m}{n}\right)} \right) + O\left(\frac{T}{\sqrt{\log(T)}}\right). \tag{7.8}$$

We will now estimate the first two terms on the right-hand side. For the first term, an application of Equation (7.4) gives

$$T \sum_{n \ll \frac{T}{\sqrt{\log(T)}}} \frac{1}{n} = T \log\left(\frac{T}{\sqrt{\log(T)}}\right) + O(T). \tag{7.9}$$

For the second sum, separate the terms for which $n < \frac{m}{2}$ or not so that

$$\sum_{\substack{n,m \ll \frac{T}{\sqrt{\log(T)}} \\ n \neq m}} \frac{1}{\sqrt{nm}\log\left(\frac{m}{n}\right)} = \sum_{\substack{n,m \ll \frac{T}{\sqrt{\log(T)}} \\ n < \frac{m}{2}}} \frac{1}{\sqrt{nm}\log\left(\frac{m}{n}\right)} + \sum_{\substack{n,m \ll \frac{T}{\sqrt{\log(T)}} \\ n \geq \frac{m}{2} \\ n \neq m}} \frac{1}{\sqrt{nm}\log\left(\frac{m}{n}\right)}.$$

In the first piece on the right-hand side, we have $\log\left(\frac{m}{n}\right) \geq \log(2)$ so that $\log\left(\frac{m}{n}\right)$ is bounded from below. Hence

$$\sum_{\substack{n,m \ll \frac{T}{\sqrt{\log(T)}} \\ n < \frac{m}{2}}} \frac{1}{\sqrt{nm}\log\left(\frac{m}{n}\right)} = O\left(\left(\sum_{n \ll \frac{T}{\sqrt{\log(T)}}} \frac{1}{\sqrt{n}}\right)^2\right).$$

The estimate

$$\sum_{n \ll \frac{T}{\sqrt{\log(T)}}} \frac{1}{\sqrt{n}} = O\left(\frac{\sqrt{T}}{\log^{\frac{1}{4}}(T)}\right),$$

follows upon bounding the sum by its associated integral. Whence

$$\sum_{\substack{n,m \ll \frac{T}{\sqrt{\log(T)}} \\ n < \frac{m}{2}}} \frac{1}{\sqrt{nm}\log\left(\frac{m}{n}\right)} = O\left(\frac{T}{\sqrt{\log(T)}}\right). \tag{7.10}$$

For the second piece, write $n = m - r$ where $1 \leq r \leq \frac{m}{2}$ so that $\log\left(\frac{m}{n}\right) = -\log(1 - \frac{r}{m})$. Using Taylor series of the logarithm, it follows that $\log\left(\frac{m}{n}\right) \geq \frac{r}{m}$. Whence

$$\sum_{\substack{n,m \ll \frac{T}{\sqrt{\log(T)}} \\ n \geq \frac{m}{2} \\ n \neq m}} \frac{1}{\sqrt{nm}\log\left(\frac{m}{n}\right)} = O\left(\sum_{m \ll \frac{T}{\sqrt{\log(T)}}} \sum_{r \leq \frac{m}{2}} \frac{1}{r}\right).$$

To estimate the double sum, use Equation (7.4) and observe that

$$\sum_{m \ll \frac{T}{\sqrt{\log(T)}}} (\log(m) + O(1)) = \frac{T}{\sqrt{\log(T)}} \log\left(\frac{T}{\sqrt{\log(T)}}\right) + O\left(\frac{T}{\sqrt{\log(T)}}\right).$$

Thus

$$\sum_{\substack{n,m \ll \frac{T}{\sqrt{\log(T)}} \\ n \geq \frac{m}{2} \\ n \neq m}} \frac{1}{\sqrt{nm}\log\left(\frac{m}{n}\right)} = O\left(T\sqrt{\log(T)}\right). \tag{7.11}$$

Combining Equations (7.10) and (7.11) yields

$$\sum_{\substack{n,m \ll \frac{T}{\sqrt{\log(T)}} \\ n \neq m}} \frac{1}{\sqrt{nm}\log\left(\frac{m}{n}\right)} = O\left(T\sqrt{\log(T)}\right). \tag{7.12}$$

Then Equations (7.9) and (7.12) together with Equation (7.8) yields

$$\int_2^T \left| \sum_{n \ll \frac{t}{\sqrt{\log(t)}}} \frac{1}{n^{\frac{1}{2}+it}} \right|^2 = T\log(T) + O\left(T\sqrt{\log(T)}\right).$$

Applying this estimate to Equation (7.7), we at last obtain

$$M_2(T, \zeta) = T\log(T) + O\left(T\log^{\frac{3}{4}}(T)\right).$$

which is the desired asymptotic formula. The asymptotic equivalence follows at once as it is a strictly weaker asymptotic. $\square$

A few comments about the proof of this second moment result. The error term in the asymptotic formula only saves $\log^{\frac{1}{4}}(T)$ from the main term which is not much. In fact, much better error terms can be obtained using more refined analytic techniques. Nevertheless, as $\log(T) \ll_\varepsilon T^\varepsilon$, the asymptotic equivalence implies that the necessary bound for the Riemann zeta function is satisfied for Proposition 6.4.3 in the case of the second moment.

The main analytic data being fed into Theorem 7.3.1 is the approximate functional equation for the Riemann zeta function. The main contribution from the resulting asymptotic, namely Equation (7.5), in $M_2(T,\zeta)$ is $T\log(T)$ which comes from the diagonal contribution in

$$\int_2^T \left| \sum_{n \ll \frac{t}{\sqrt{\log(t)}}} \frac{1}{n^{\frac{1}{2}+it}} \right|^2 = \int_2^T \sum_{n,m \ll \frac{t}{\sqrt{\log(t)}}} \frac{1}{n^{\frac{1}{2}+it} m^{\frac{1}{2}-it}}\, dt,$$

occurring when $n = m$, and is given by

$$T \sum_{n \ll \frac{T}{\sqrt{\log(T)}}} \frac{1}{n}.$$

This is also the longest sum arising from Equation (7.6) of highest order (due to the factor of T). It should also be noted that the contribution for those terms when $n \neq m$ and $n \geq \frac{m}{2}$ comes very close to the main term. This sum is

$$\sum_{\substack{n,m \ll \frac{T}{\sqrt{\log(T)}} \\ n \geq \frac{m}{2} \\ n \neq m}} \frac{1}{\sqrt{nm}\log\left(\frac{m}{n}\right)},$$

and it was estimated to be $O\left(T\sqrt{\log(T)}\right)$. If the length of our sum in Equation (7.5) was increased by a factor of $\sqrt{\log(t)}$ from $\frac{t}{\sqrt{\log(t)}}$ to t, then this term would contribute $O\left(T\log(T)\right)$ which would ruin both the asymptotic formula and asymptotic equivalence. In essence, this means that the choice of the length of our sum in the proof of Theorem 7.3.1 is essentially optimal.

Appendix A

Discrete Geometry

A.1 Integral Lattices

Let F be a characteristic zero field and V be an $\mathbb{F}$-vector space of dimension n with nondegenerate symmetric inner product $\langle \cdot, \cdot \rangle$. We say that a subset Λ of V is a **integral lattice** if Λ is a free $\mathbb{Z}$-module. In particular, any integral lattice Λ is of the form

$$\Lambda = \mathbb{Z}v_1 + \cdots + \mathbb{Z}v_m,$$

for some linearly independent vectors $v_1, \ldots, v_m$ of V with $m \leq n$. We say Λ is **complete** if $n = m$. This means $v_1, \ldots, v_n$ is necessarily a basis of V. If $e_1, \ldots, e_n$ is an orthonormal basis for V, write

$$v_i = \sum_i v_{i,j} e_j,$$

with $v_{i,j} \in F$, and define the associated **generator matrix** P by

$$P = \begin{pmatrix} v_{1,1} & \cdots & v_{n,1} \\ \vdots & & \vdots \\ v_{1,n} & \cdots & v_{n,n} \end{pmatrix}.$$

Then P is the base change matrix from $e_1, \ldots, e_n$ to $v_1, \ldots, v_n$. The **covolume** V_Λ of a complete integral lattice Λ is defined to be

$$V_\Lambda = |\det(P)|.$$

As the base change matrix between any two orthonormal basses is an orthogonal matrix and hence has determinant ± 1, the covolume is independent of the choice of bases. In the case where V is a real or complex vector space, we will always take $e_1, \ldots, e_n$ to be the standard basis.

If Λ is a complete integral lattice of V then the **dual** $\Lambda^\vee$ of Λ is defined by

$$\Lambda^\vee = \{v \in V : \langle \lambda, v \rangle \in \mathbb{Z} \text{ for all } \lambda \in \Lambda\}.$$

In other words, the dual integral lattice consists of all of the vectors in V whose inner product with elements of the integral lattice are integers. Clearly the dual integral lattice is an abelian group. Less clear is that the dual integral lattice turns out to always be complete.

Proposition A.1.1. *If $v_1, \dots, v_n$ is a basis for a complete integral lattice Λ then the dual basis $v_1^\vee, \dots, v_n^\vee$ is a basis for $\Lambda^\vee$. In particular, $\Lambda^\vee$ is a complete integral lattice.*

Proof. Let $v \in V$ and write

$$v = \sum_j a_i v_j^\vee,$$

with $v_j^\vee \in \mathbb{R}$. Since the dual basis is determined by $\langle v_i, v_j^\vee \rangle = \delta_{i,j}$, it follows that $v \in \Lambda^\vee$ if and only if $v_j^\vee \in \mathbb{Z}$. This means $v_1^\vee, \dots, v_n^\vee$ is a basis for $\Lambda^\vee$ as a free $\mathbb{Z}$-module and thus is a complete integral lattice. $\square$

Since the dual of the dual basis is the original basis, $(\Lambda^\vee)^\vee = \Lambda$. We say that Λ is **self-dual** if $\Lambda^\vee = \Lambda$. For example, the integral lattice $\mathbb{Z}^n$ is self-dual because the standard basis is self-dual. It also turns out that the covolume of the dual integral lattice is the inverse of the volume of the original integral lattice.

Proposition A.1.2. *Let Λ be a complete integral lattice of V. Then*

$$V_{\Lambda^\vee} = \frac{1}{V_\Lambda}.$$

Proof. Let $e_1, \dots, e_n$ be an orthonormal basis for V, let $v_1, \dots, v_n$ be a basis for Λ, and let P be the associated generator matrix. By Proposition A.1.1, $v_1^\vee, \dots, v_n^\vee$ is a basis for $\Lambda^\vee$ and $(P^{-1})^t$ is the base change matrix from $e_1, \dots, e_n$ to $v_1^\vee, \dots, v_n^\vee$. As $\det((P^{-1})^t) = \det(P)^{-1}$ the claim follows. $\square$

We now turn to the case when V is a real inner product space of dimension n. Let $d\lambda$ be the Lebesgue measure induced by the inner product with associated volume Vol given by

$$\mathrm{Vol}(M) = \int_M d\lambda,$$

for any measurable set M. Note that Λ acts on V by automorphisms given by translation. In other words, we have a group action

$$\Lambda \times V \to V \qquad (\lambda, v) \mapsto \lambda + v.$$

Moreover, Λ acts properly discontinuously on V. To see this, let $v \in V$ and let δ_v be such that $0 < \delta_v < \min_i(v - v_i)$. Taking U_v to be the ball of radius δ_v about v, the intersection $\lambda + U_v \cap U_v$ is empty unless $\lambda = 0$. As Λ is also discrete, it follows that V/Λ is also connected Hausdorff. Whence V/Λ admits a fundamental domain

$$\mathcal{M} = \{t_1 v_1 + \cdots + t_n v_n \in V : 0 \leq t_i \leq 1 \text{ for all } i\}.$$

Moreover, any translation of $\mathcal{M}$ by an element of Λ is also a fundamental domain. As we might expect, the covolume of Λ is equal to the volume of $\mathcal{M}$.

Proposition A.1.3. *Let Λ be a complete integral lattice of a finite dimensional real inner product space V and let $\mathcal{M}$ be a fundamental domain for Λ. Then*

$$V_\Lambda = \mathrm{Vol}(\mathcal{M}).$$

Proof. Let $e_1, \ldots, e_n$ be an orthonormal basis of V, and $v_1, \ldots, v_n$ be a basis for Λ. Also let P be the associated generator matrix. The change of variables

$$x_1 e_1 + \cdots + x_n e_n \mapsto x_1 v_1 + \cdots + x_n v_n,$$

has Jacobian determinant V_Λ. Whence

$$\mathrm{Vol}(\mathcal{M}) = \int_{\mathcal{M}} d\lambda = V_\Lambda \int_{[0,1]^n} d\mathbf{x} = V_\Lambda.$$

□

This result shows that the covolume of a complete integral lattice of a real inner product space is a measure of the density of the integral lattice. The smaller the covolume the smaller the fundamental domain and the more dense the integral lattice is.

It turns out that for a real inner product space, being an integral lattice is equivalent to being a discrete subgroup.

Proposition A.1.4. *Let Λ be a subset of a finite dimensional real inner product space V. Then Λ is an integral lattice if and only if it is a discrete subgroup.*

Proof. It is clear that if Λ is an integral lattice then it is a discrete subgroup. So suppose Λ is a discrete subgroup. Then Λ is closed. Let V' be the subspace spanned by Λ and let its dimension be m. Choosing a basis $v_1', \ldots, v_m'$ of V', set

$$\Lambda' = \mathbb{Z}v_1' + \cdots + \mathbb{Z}v_m'.$$

Then Λ' is a complete integral lattice of V' and $\Lambda' \subseteq \Lambda \subset V$. We claim Λ/Λ' is finite. To see this, let λ be a representative of a coset in Λ/Λ' and let $\mathcal{M}'$ be a fundamental domain for Λ' in V'. As $\mathcal{M}'$ is a fundamental domain, there exists unique $w' \in \mathcal{M}'$ and $\lambda' \in \Lambda'$ such that $\lambda = w' + \lambda'$. But then $w' = \lambda - \lambda' \in \mathcal{M}' \cap \Lambda$ and since $\mathcal{M}' \cap \Lambda$ is closed, discrete, and compact, it must be finite. Hence there are finitely many w' and thus finitely many cosets in Λ/Λ'. Letting $|\Lambda/\Lambda'| = q$, we have $q\Lambda \subseteq \Lambda'$ and therefore

$$\Lambda \subseteq \Lambda' = \mathbb{Z}\frac{1}{q}v_1' + \cdots + \mathbb{Z}\frac{1}{q}v_m'.$$

In particular, Λ is a subgroup of a free $\mathbb{Z}$-module and therefore is a free $\mathbb{Z}$-module itself. This means Λ is an integral lattice. □

As for complete integral lattices in V, they are equivalent to the existence of a bounded set whose translates cover V.

Proposition A.1.5. *Let Λ be an integral lattice of a finite dimensional real inner product space V. Then Λ complete if and only if there exists a bounded subset M of V whose translates by Λ cover V.*

Proof. First suppose Λ is complete. Then we may take $M = \mathcal{M}$ to be a fundamental domain of Λ which is bounded and whose translates by Λ cover V. Now suppose Λ is an integral lattice and there exists a bounded subset M of V whose translates by Λ cover V. Let W be the subspace of V spanned by Λ. Then W is closed. Moreover, Λ is complete if and only if $V = W$ and this is what we will show. So let $v \in V$. Since the translates of M by Λ cover V, for every positive integer n we may write

$$nv = w_n + \lambda_n,$$

with $v_n \in M$ and $\lambda_n \in \Lambda$. As M is bounded, $\lim_{n\to\infty} \frac{1}{n} v_n = 0$. Moreover, $\frac{1}{n}\lambda_n \in W$ and since W is closed we must have that $\lim_{n\to\infty} \frac{1}{n}\lambda_n$ exists and belongs to W. These two limits together imply

$$v = \lim_{n\to\infty} \frac{1}{n}\lambda_n,$$

and is an element of W. Thus $V = W$ which means Λ is complete. □

The most important result we will require about integral lattices is **Minkowski's lattice point theorem** which states that, under some mild conditions, a set of sufficiently large volume in V contains a nonzero point of a complete integral lattice.

Theorem (Minkowski's lattice point theorem). *Suppose Λ is a complete integral lattice of a real inner product space V of dimension n. Let $X \subset V$ is a compact convex symmetric set. If*

$$\mathrm{Vol}(X) \geq 2^n V_\Lambda,$$

then there exists a nonzero $\lambda \in \Lambda \cap X$.

Proof. We will prove the claim depending on if the inequality is strict or not. First suppose $\mathrm{Vol}(X) > 2^n V_\Lambda$. Consider the linear map

$$\phi : \frac{1}{2}X \to V/\Lambda \qquad \frac{1}{2}x \mapsto \frac{1}{2}x + \Lambda.$$

If ϕ were injective then we would have

$$\mathrm{Vol}\left(\frac{1}{2}X\right) = \frac{1}{2^n}\mathrm{Vol}(X) \leq V_\Lambda,$$

so that $\mathrm{Vol}(X) \leq 2^n V_\Lambda$. This is a contradiction, so ϕ cannot be injective. Hence there exists distinct $x_1, x_2 \in \frac{1}{2}X$ such that $\phi(x_1) = \phi(x_2)$. Thus $2x_1, 2x_2 \in X$. In particular, since X is symmetric we must have $-2x_2 \in X$. Convexity of X implies

$$\left(1 - \frac{1}{2}\right)2x_1 + \frac{1}{2}(-2x_2) = x_1 - x_2,$$

is an element of X. But $x_1 - x_2 \in \Lambda$ because $\phi(x_1) = \phi(x_2)$ and ϕ is linear. Then $\lambda = x_1 - x_2$ is nonzero with $\lambda \in \Lambda \cap X$. Now suppose $\mathrm{Vol}(X) = 2^n V_\Lambda$. Then

$$\mathrm{Vol}((1+\varepsilon)X) = (1+\varepsilon)^n 2^n V_\Lambda > 2^n V_\Lambda.$$

What we have just proved shows that there exists a nonzero $\lambda_\varepsilon \in \Lambda \cap (1+\varepsilon)X$. In particular, if $\varepsilon \leq 1$ then $\lambda_\varepsilon \in 2X \cap \Lambda$. The set $2X \cap \Lambda$ is compact and discrete and therefore finite. Taking $\varepsilon = \frac{1}{n}$, the sequence $(\lambda_{\frac{1}{n}})_{n\geq 1}$ belongs to the finite set $2X \cap \Lambda$ and so must converge to a point λ. Since Λ is discrete and the $\lambda_{\frac{1}{n}}$ are nonzero so too is λ. As λ is an element of

$$\bigcap_{n\geq 1} \left(1 + \frac{1}{n}\right) X,$$

and X is closed, $\lambda \in X$ as well. Thus we have found a nonzero $\lambda \in \Lambda \cap X$. $\square$

Appendix B

Real Analysis

Throughout, all sums over $\mathbb{Z}$ are assumed to be ordered symmetrically if they are not absolutely convergent. Also, all piecewise functions are assumed to have finitely many discontinuities.

B.1 The Fourier Transform

Let $f(x)$ be a function on $\mathbb{R}$. The **Fourier transform** $(\mathcal{F}f)(\xi)$ of f is defined by

$$(\mathcal{F}f)(\xi) = \int_{\mathbb{R}} f(x)e^{-2\pi i\xi x}\,dx.$$

In general, this integral need not be convergent but under mild conditions it is. For if $f(x)$ is absolutely integrable on $\mathbb{R}$ then this integral is locally absolutely uniformly convergent. Whence the Fourier transform is continuous. More precisely, this condition holds if $f(x) = O_\varepsilon((|x|+3)^{-(1+\varepsilon)})$. Let us first demonstrate some properties of the Fourier transform.

Proposition B.1.1. *Let $f(x)$ and $g(x)$ be absolutely integrable on $\mathbb{R}$. Then the following properties hold:*

(i) For any real α and β, we have

$$(\mathcal{F}(\alpha f + \beta g))(\xi) = \alpha(\mathcal{F}f)(\xi) + \beta(\mathcal{F}g)(\xi).$$

(ii) Suppose $g(x) = f(x+\alpha)$ for some real α. Then

$$(\mathcal{F}g)(\xi) = e^{2\pi i\langle\alpha,\xi\rangle}(\mathcal{F}f)(\xi).$$

(iii) Suppose $g(x) = f(\alpha x)$ for some nonzero real α. Then

$$(\mathcal{F}g)(\xi) = \frac{1}{|\alpha|}(\mathcal{F}f)\left(\frac{\xi}{\alpha}\right).$$

Proof. Property (i) is immediate from linearity of the integral while property (ii) follows by applying the change of variables $x \mapsto x - \alpha$. Property (iii) is proved by performing the change of variables $x \mapsto \alpha x$. $\square$

There is also an inverse transform. Let $F(\xi)$ be a function on $\mathbb{R}$. Then the **inverse Fourier transform** $(\mathcal{F}^{-1}F)(x)$ of $F(x)$ is defined by

$$(\mathcal{F}^{-1}F)(x) = \int_{\mathbb{R}} F(\xi)e^{2\pi i\xi x}\,d\xi.$$

It is absolutely convergent because $F(\xi)$ is absolutely integrable on $\mathbb{R}$. Again, there is no reason to assume this integral need converge. However, if $F(x)$ is absolutely integrable on $\mathbb{R}$ then this integral is locally absolutely uniformly convergent. Whence the inverse Fourier transform is continuous. As for the Fourier transform, this condition holds if $F(x) = O_\varepsilon((|x|+3)^{-(1+\varepsilon)})$.

It is useful to know under what conditions the Fourier transform and the inverse Fourier transform are genuinely inverse operators. This result is known as Fourier inversion.

Theorem (Fourier inversion). *Let $f(x)$ be a piecewise smooth function on $\mathbb{R}$ satisfying $f(x) = O_\varepsilon((|x|+3)^{-(1+\varepsilon)})$. Then*

$$f(x) = (\mathcal{F}^{-1}F)(x),$$

where $f(x)$ is understood to mean the average of the left-hand and right-hand limits at jump discontinuities. Conversely, let $F(\xi)$ be a piecewise smooth function $\mathbb{R}$ satisfying $F(x) = O_\varepsilon((|x|+3)^{-(1+\varepsilon)})$. Then

$$F(\xi) = (\mathcal{F}f)(\xi),$$

where it is understood that $F(\xi)$ represents the average of the left-hand and right-hand limits at jump discontinuities

B.2 Fourier Series

Suppose $f(x)$ a function on $\mathbb{R}$. Then the t-th **Fourier coefficient** $\hat{f}(t)$ of $f(x)$ is defined by

$$\hat{f}(t) = \int_0^1 f(x)e^{-2\pi itx}\,dx.$$

The Fourier coefficients of $f(x)$ need not exist in general but they do under mild conditions, for example, if $f(x)$ is absolutely integrable on $[0,1]$. Accordingly, the **Fourier series** $S_f(x)$ of $f(x)$ is given by

$$S_f(x) = \sum_{t\in\mathbb{Z}} \hat{f}(t)e^{2\pi itx}.$$

A crucial question is where the Fourier series of $f(x)$ converges, if at all, and if so does it even converge to $f(x)$ itself. Under quite reasonable conditions the Fourier series converges uniformly to the function itself everywhere it is continuous.

Theorem B.2.1. *Suppose $f(x)$ is a 1-periodic piecewise smooth function on $\mathbb{R}$. Then the Fourier series of $f(x)$ converges everywhere. Moreover, it converges uniformly to $f(x)$ on compact sets where $f(x)$ is continuous and to the average of the left-hand and right-hand limits at jump discontinuities.*

B.3 Poisson Summation

Let $f(x)$ be a function on $\mathbb{R}$. There are two natural methods of building a function from $f(x)$ which is 1-periodic. The first is to average $f(x)$ over all translates by elements of $\mathbb{Z}$ while the second is to consider the Fourier series of $f(x)$. This gives us the two series

$$\sum_{n\in\mathbb{Z}} f(x+n) \quad \text{and} \quad \sum_{t\in\mathbb{Z}} \hat{f}(t)e^{2\pi itx}.$$

The link between the Fourier transform and Fourier series is given by **Poisson summation** which says that these two expressions are the same, provided we replace the Fourier coefficients of $f(x)$ with its Fourier transform, under some mild assumptions.

Theorem B.3.1. *Let $f(x)$ be a piecewise smooth function on $\mathbb{R}$ such that $f^{(k)}(x) = O_\varepsilon((|x|+3)^{-(1+\varepsilon)})$ for all nonnegative integers k. Then*

$$\sum_{n\in\mathbb{Z}} f(x+n) = \sum_{t\in\mathbb{Z}} (\mathcal{F}f)(t)e^{2\pi itx},$$

where $f(x+n)$ is understood to mean the average of the left-hand and right-hand limits at jump discontinuities. In particular,

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{t\in\mathbb{Z}} (\mathcal{F}f)(t).$$

Proof. As $f(x) = O_\varepsilon((|x|+3)^{-(1+\varepsilon)})$, we see that $f(x)$ is absolutely integrable on $\mathbb{R}$. Now let

$$F(x) = \sum_{n\in\mathbb{Z}} f(x+n),$$

and observe that the same asymptotic implies $F(x)$ is locally absolutely uniformly convergent on $\mathbb{R}$ via the Weierstrass M-test. In fact, the since $f^{(k)}(x) = O_\varepsilon((|x|+3)^{-(1+\varepsilon)})$ for all nonnegative integers k, repeated application of the Weierstrass M-test and uniform limit theorem implies that $F(x)$ is piecewise smooth. As $F(x)$ is also clearly 1-periodic, it admits a Fourier series converging everywhere by Theorem B.2.1. In particular, the Fourier series converges uniformly to $F(x)$ on compact sets where $F(x)$ is continuous and to the average of the left-hand and right-hand limits at jump discontinuities. In the latter case, we may interchange the limit and the sum by uniform convergence to see that this is simply the sum of the average of the left-hand and right-hand limits of $f(x+n)$. Since $F(x)$ is also absolutely integrable on $[0,1]$, we compute its t-th Fourier coefficient given by

$$\hat{F}(t) = \int_0^1 F(x)e^{-2\pi itx}\,dx.$$

We may interchange of sum and integral by absolute convergent to obtain

$$\hat{F}(t) = (\mathcal{F}f)(t).$$

Whence

$$\sum_{n\in\mathbb{Z}} f(x+n) = \sum_{t\in\mathbb{Z}} (\mathcal{F}f)(t)e^{2\pi itx}.$$

This proves the first statement. Setting $x = 0$ proves the second statement. $\square$

In practical settings, we need a class of functions for which Poisson summation is valid and whose Fourier transforms can be easily computed. We say that $f(x)$ is of **Schwartz class** if $f(x)$ is smooth and has rapid decay along with all of its derivatives. The prototypical Schwartz class function is $e^{-\pi x^2}$. Conveniently, this function is its own Fourier transform.

Proposition B.3.2. *Let α be positive and and set $f(x) = e^{-2\pi\alpha x^2}$. Then*

$$(\mathcal{F}f)(\xi) = \frac{e^{-\frac{\pi\xi^2}{2\alpha}}}{\sqrt{2\alpha}}.$$

In particular, $e^{-\pi x^2}$ is its own Fourier transform.

Proof. Consider

$$(\mathcal{F}f)(\xi) = \int_{\mathbb{R}} e^{-2\pi(\alpha x^2 + i\xi x)}\, dx.$$

The change of variables $x \mapsto \frac{x}{\sqrt{\alpha}}$ transforms this integral into

$$\frac{1}{\sqrt{\alpha}} \int_{\mathbb{R}} e^{-2\pi\left(x^2 + \frac{i\xi x}{\sqrt{\alpha}}\right)}\, dx.$$

Complete the square in the exponent by observing

$$x^2 + \frac{i\xi x}{\sqrt{\alpha}} = \left(x + \frac{i\xi}{2\sqrt{\alpha}}\right)^2 + \frac{\xi^2}{4\alpha},$$

so that the previous integral is equal to

$$\frac{e^{-\frac{\pi\xi^2}{2\alpha}}}{\sqrt{\alpha}} \int_{\mathbb{R}} e^{-2\pi\left(x + \frac{i\xi}{2\sqrt{\alpha}}\right)^2}\, dx.$$

By rapid decay of the integrand, the change of variables $x \mapsto \frac{x}{\sqrt{2}} - \frac{i\xi}{\sqrt{\alpha}}$ is permitted giving

$$\frac{e^{-\frac{\pi\xi^2}{2\alpha}}}{\sqrt{2\alpha}} \int_{\mathbb{R}} e^{-\pi x^2}\, dx = \frac{e^{-\frac{\pi\xi^2}{2\alpha}}}{\sqrt{2\alpha}},$$

because the remaining integral is the Gaussian integral. This proves the first statement. The second statement follows by taking $\alpha = \frac{1}{2}$. $\square$

Clearly any Schwartz class function satisfies the conditions of Poisson summation. In fact, if $f(x)$ is compactly supported and equal to a Schwartz class function on its support then the conditions of Poisson summation are also satisfied. Moreover, the previous result also suggests that it should be relatively simple to compute the required Fourier transform appearing in Poisson summation.

Appendix C

Complex Analysis

Throughout, $s = \sigma + it$ and $u = \tau + ir$ will stand for complex variables with σ, τ, t, and r real.

C.1 The Mellin Transform

Let $\psi(x)$ be a function on $(0, \infty)$. Then the **Mellin transform** $(\mathcal{M}\psi)(s)$ of $\psi(x)$ is defined by

$$(\mathcal{M}\psi)(s) = \int_0^\infty \psi(x)x^{s-1}\,dx.$$

This integral need not be convergent. However, if

$$\psi(x) = \begin{cases} O_\varepsilon(x^{-a-\varepsilon}) & \text{if } x \ll 1, \\ O_\varepsilon(x^{-b+\varepsilon}) & \text{if } x \gg 1, \end{cases}$$

for some $a < b$ then it is locally absolutely uniformly convergent for $a < \sigma < b$ upon writing

$$(\mathcal{M}\psi)(s) = \int_0^1 \psi(x)x^{s-1}\,dx + \int_1^\infty \psi(x)x^{s-1}\,dx.$$

Of course, if we may take a or b arbitrary large then the Mellin transform is locally absolutely uniformly convergent in the half-plane $\sigma < b$ or $\sigma > a$ respectively. In any case, the Mellin transform is holomorphic in the corresponding region.

There is also an inverse transform. Let $\Psi(s)$ be a holomorphic function on the vertical strip $a < \sigma < b$. Then the **inverse Mellin transform** $(\mathcal{M}^{-1}\Psi)(x)$ of $\Psi(s)$ is defined by

$$(\mathcal{M}^{-1}\Psi)(x) = \frac{1}{2\pi i}\int_{(c)} \Psi(s)x^{-s}\,ds,$$

for any $a < c < b$. This integral need not converge or even be independent of c. Yet if $\Psi(s) = O_\varepsilon((|t|+3)^{-(1+\varepsilon)})$ then the integral is locally absolutely uniformly convergent and independent of c.

It is useful to know under what conditions the Mellin transform and the inverse Mellin transform are genuinely inverse operators. This result is known as Mellin inversion.

Theorem (Mellin inversion). *Let $\Psi(s)$ be a holomorphic function on the vertical strip $a < \sigma < b$ and such that $\Psi(s) = O_\varepsilon((|t|+3)^{-(1+\varepsilon)})$. If $\psi(x)$ is the inverse Mellin transform of $\Psi(x)$, then*

$$\Psi(s) = (\mathcal{M}\psi)(s).$$

Conversely, suppose $\psi(x)$ is a piecewise smooth function on $(0, \infty)$ such that

$$\psi(x) = \begin{cases} O_\varepsilon(x^{-a-\varepsilon}) & \text{if } x \ll 1, \\ O_\varepsilon(x^{-b+\varepsilon}) & \text{if } x \gg 1, \end{cases}$$

with $a < b$. If $\Psi(s)$ is the Mellin transform of $\psi(x)$, then

$$\psi(x) = (\mathcal{M}^{-1}\Psi)(x),$$

where $\psi(x)$ is understood to mean the average of the left-hand and right-hand limits at jump discontinuities.

C.2 The Gamma Function

The **gamma function** $\Gamma(s)$ is defined by the integral

$$\Gamma(s) = \int_0^\infty e^{-x} x^{s-1}\, dx.$$

Equivalently, $\Gamma(s)$ is the Mellin transform of e^{-x}. As e^{-x} exhibits exponential decay and is bounded near zero, this defines a holomorphic function for $\sigma > 0$. Also note that $\Gamma(s)$ is real for $s > 0$. The most basic properties of $\Gamma(s)$ are collected in the following result:

Proposition C.2.1. *The gamma function satisfies*

$$\Gamma(1) = 1 \quad \text{and} \quad \Gamma(s+1) = s\Gamma(s).$$

In particular,

$$\Gamma(n) = (n-1)!,$$

for any positive integer n. Also, the gamma function is conjugation-equivariant which is to say

$$\Gamma(\overline{s}) = \overline{\Gamma(s)}.$$

Proof. The first identity follow by direct evaluation. The second is a direct computation requiring an application of integration by parts. The following statement is obtained by repeated application of the latter identity. The last statement follows from the identity theorem and that $\Gamma(s)$ is real for $\sigma > 1$. $\square$

In light of $\Gamma(n) = (n-1)!$, we can view $\Gamma(s)$ as a holomorphic extension of the factorial function. An interesting observation is the value that this extension takes at $s = \frac{1}{2}$. The change of variables $x \mapsto \pi x^2$ transforms the integral defining $\Gamma\left(\frac{1}{2}\right)$ into a Gaussian integral giving

$$\Gamma\left(\frac{1}{2}\right) = \pi \int_{-\infty}^{\infty} e^{-\pi x^2}\, dx = \sqrt{\pi}. \tag{C.1}$$

The aforementioned relation

$$\Gamma(s+1) = s\Gamma(s),$$

is referred to as the **functional equation** for the gamma function. The functional equation will allow for the meromorphic conduction of the Gamma function to all of $\mathbb{C}$.

Theorem C.2.2. *For any nonnegative integer n, $\Gamma(s)$ admits meromorphic continuation to the half-plane $\sigma > -(n+1)$ given by*

$$\Gamma(s) = \frac{\Gamma(s+n+1)}{s(s+1)\cdots(s+n)}.$$

In particular, $\Gamma(s)$ admits meromorphic continuation to $\mathbb{C}$ with simple poles at $s = -n$ of residues $\frac{(-1)^n}{n!}$.

Proof. Applying the functional equation repeatedly gives

$$\Gamma(s) = \frac{\Gamma(s+n+1)}{s(s+1)\cdots(s+n)},$$

The right-hand side defines an meromorphic function in the half-plane $\sigma > -(n+1)$. The only possible poles are $0, -1, \ldots, -n$ and they are all simple. Then

$$\operatorname*{Res}_{s=-n} \Gamma(s) = \lim_{s \to -n} \frac{\Gamma(s+n+1)}{s(s+1)\cdots(s+n)} = \frac{(-1)^n}{n!}.$$

As n was arbitrary, this completes the proof. $\square$

Important residues are

$$\operatorname*{Res}_{s=0} \Gamma(s) = 1 \quad \text{and} \quad \operatorname*{Res}_{s=1} \Gamma(s) = -1.$$

In order to deduce more analytic properties of the gamma function, we will introduce an auxiliary function that will be more amenable to study. The **beta function** $B(s,u)$ is defined by

$$B(s,u) = \int_0^1 x^{s-1}(1-x)^{u-1}\, dx.$$

This defines a holomorphic function for $\sigma > 0$ and $\tau > 0$ since the integral is locally absolutely uniformly convergent in this region. At present, it does not seem apparent that the beta function is related to the gamma function at all. This will be resolved shortly. The beta function is symmetric, namely

$$B(s,u) = B(u,s),$$

as it seen by performing the change of variables $x \mapsto 1-x$ to the integral. The following result establishes the relationship between the beta function and the gamma function:

Proposition C.2.3. *For $\sigma > 0$ and $\tau > 0$, we have*

$$\Gamma(s)\Gamma(u) = B(s,u)\Gamma(s+u).$$

Proof. By local absolute uniform convergence of the integrals, we may write

$$\Gamma(s)\Gamma(u) = \int_0^\infty \int_0^\infty e^{-(x+y)} x^{s-1} y^{u-1}\,dx\,dy.$$

Change variables $(x,y) \mapsto (x, y-x)$ to obtain

$$\int_0^\infty \int_0^y e^{-y} x^{s-1} (y-x)^{u-1}\,dx\,dy,$$

The further change of variables $x \mapsto xy$ gives

$$\int_0^\infty \int_0^1 e^{-y} (xy)^{s-1} (y-xy)^{u-1}\,dx\,dy.$$

Then

$$\int_0^\infty \int_0^1 e^{-y} (xy)^{s-1} (y-xy)^{u-1}\,dx\,dy = B(s,u)\Gamma(s+u),$$

by interchanging integrals again. Therefore

$$\Gamma(s)\Gamma(u) = B(s,u)\Gamma(s+u),$$

as claimed. □

It is not yet possible to meromorphically continue the beta function to $\mathbb{C}^2$ because we cannot divide by the gamma function as we do not yet know where it possess zeros. It turns out that the gamma function has no zeros and we will prove this shortly. The essential result is to prove Euler's reflection formula which relates the gamma function to a periodic function of which we know its zeros explicitly. To accomplish this we will make use of Proposition C.2.3 and a formula for the beta function which will allow us to compute it by means of the residue theorem. To deduce this formula, apply the change of variables $x \mapsto \frac{x}{x+1}$ to the beta function to obtain

$$B(s,u) = \int_0^\infty \frac{x^{s-1}}{(x+1)^{s+u}}\,dx. \tag{C.2}$$

This shows that beta function is the Mellin transform of $\frac{1}{(s+1)^{s+u}}$. This formula is immensely useful when $s+u=1$ because we may then evaluate $B(s,u)$ by means of the residue theorem. We will make use of this fact to prove **Euler's reflection formula**.

Theorem (Euler's reflection formula). *For any $s \in \mathbb{C} - \mathbb{Z}$, we have*

$$\Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}.$$

Proof. By the identity theorem we may assume $0 < s < 1$. Proposition C.2.3 gives

$$\Gamma(s)\Gamma(1-s) = B(s, 1-s)\Gamma(1).$$

Using Equation (C.2), we may rewrite this identity as

$$\Gamma(s)\Gamma(1-s) = \int_0^\infty \frac{x^{s-1}}{x+1}\,dx,$$

so it suffices to show that the remaining integral is $\frac{\pi}{\sin(\pi s)}$.

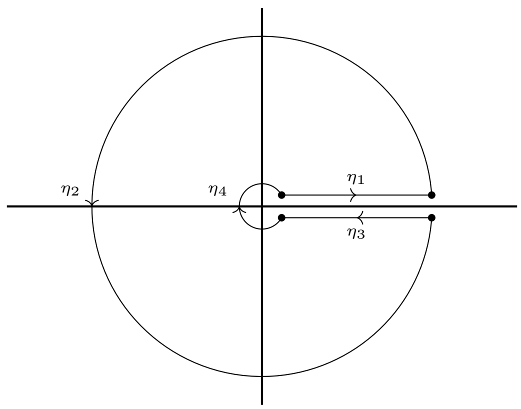

Figure C.1: A keyhole contour.

Let $\eta = \sum_i \eta_i$ be the keyhole contour Figure C.1 where the large arc has radius R and the small arc has radius ε. Consider

$$\int_\eta \frac{z^{s+1}}{z+1}\,dz.$$

We will evaluate this integral in the limit as $R \to \infty$ and then as $\varepsilon \to 0$. The contour encloses the only pole of the integrand which is simple at $z = -1$ and of residue $e^{\pi i(s-1)}$ (after taking the branch cut $\log_0$ of the logarithm along the positive real line). So, on the one hand, the residue theorem gives

$$\int_\eta \frac{z^{s-1}}{z+1}\,dz = 2\pi i e^{\pi i(s-1)},$$

where in the last equality we have used that $\log_0(-1) = i\pi$ (since this is the branch cut of the logarithm along the positive real line). On the other hand, the integral is a sum along all four contours. Clearly

$$\lim_{\varepsilon \to 0} \lim_{R \to \infty} \int_{\eta_1} \frac{z^{s+1}}{z+1}\,dz = \int_0^\infty \frac{x^{s-1}}{x+1}\,dx.$$

The argument of s along η_3 is 2π that of it along η_1. Therefore

$$\lim_{\varepsilon\to 0}\lim_{R\to\infty}\int_{\eta_3}\frac{z^{s+1}}{z+1}\,dz = e^{2\pi i(s-1)}\int_0^\infty \frac{x^{s-1}}{x+1}\,dx.$$

The parameterization $z \mapsto Re^{i\theta}$ for η_2 shows that the integral along this contour tends to zero as $R \to \infty$. Whence

$$\lim_{\varepsilon\to 0}\lim_{R\to\infty}\int_{\eta_2}\frac{z^{s+1}}{z+1}\,dz = 0.$$

Similarly, the parameterization $z \mapsto \varepsilon e^{i\theta}$ for η_4 shows that the integral along this contour tends to zero as $\varepsilon \to 0$ and is independent of R. Hence

$$\lim_{\varepsilon\to 0}\lim_{R\to\infty}\int_{\eta_4}\frac{z^{s+1}}{z+1}\,dz = 0.$$

All of this shows

$$\lim_{\varepsilon\to 0}\lim_{R\to\infty}\int_{\eta}\frac{z^{s-1}}{z+1}\,dz = (1-e^{2\pi i(s-1)})\int_0^\infty \frac{x^{s-1}}{x+1}\,dx,$$

and together with our previous residue computation gives

$$(1-e^{2\pi i(s-1)})\int_0^\infty \frac{x^{s-1}}{x+1}\,dx = 2\pi i e^{\pi i(s-1)}.$$

Using the formula $\sin(\pi s) = \frac{e^{i\pi s}+e^{-i\pi s}}{2i}$, a short computation shows

$$\int_0^\infty \frac{x^{s-1}}{x+1}\,dx = \frac{\pi}{\sin(\pi s)},$$

as claimed. $\square$

It quickly follows from Euler's reflection formula that the gamma function has no zeros. Euler's reflection formula implies

$$\Gamma(s)\Gamma(1-s)\frac{\sin(\pi s)}{\pi} = 1,$$

provided s is not an integer. Since $\sin(\pi s)$ is nonzero for such s, $\Gamma(s)$ is nonzero except possibly at integers. For any positive integer $\Gamma(s)$ is given by a factorial which is nonzero while for any nonpositive integer $\Gamma(s)$ has a simple pole. Therefore $\Gamma(s)$ has no zeros. This implies that the reciprocal of the gamma function is entire. As a nice consequence, we can obtain the meromorphic continuation of the beta function:

Theorem C.2.4. *$B(s,u)$ admits meromorphic continuation to $\mathbb{C}^2$ given by*

$$B(s,u) = \frac{\Gamma(s)\Gamma(u)}{\Gamma(s+u)}.$$

Proof. As $\Gamma(s)$ has no zeros, Proposition C.2.3 implies

$$B(s,u) = \frac{\Gamma(s)\Gamma(u)}{\Gamma(s+u)},$$

for all $\sigma > 0$ and $\tau > 0$. Then completes the proof since the gamma function admits meromorphic continuation to $\mathbb{C}$. $\square$

Now knowing that the gamma function has no zeros, we can use the beta function to prove an important multiplicativity property of the gamma function. This is known as **Legendre's duplication formula**.

Theorem (Legendre's duplication formula). *For any $s \in \mathbb{C} - \mathbb{Z}_{0-}$, we have*

$$\Gamma(s)\Gamma\left(s+\frac{1}{2}\right) = 2^{1-2s}\sqrt{\pi}\Gamma(2s).$$

Proof. By the identity theorem we may assume $s > 1$. Consider

$$B(s,s) = \int_0^1 x^{s-1}(1-x)^{s-1}\,dx.$$

The change of variables $x \mapsto \frac{x+1}{2}$ results in

$$2^{1-2s}\int_{-1}^1 (1-x^2)^{s-1}\,dx.$$

As the integrand is even, we can express it as

$$2^{2-2s}\int_0^1 (1-x^2)^{s-1}\,dx.$$

The change of variables $x \mapsto \sqrt{x}$ yields

$$2^{1-2s}\int_0^1 x^{-\frac{1}{2}}(1-x)^{s-1}\,dx = 2^{1-2s}B\left(\frac{1}{2},s\right).$$

Therefore

$$B(s,s) = 2^{1-2s}B\left(\frac{1}{2},s\right).$$

Invoking Theorem C.2.4 to replace the beta functions with gamma functions gives

$$\frac{\Gamma(s)^2}{\Gamma(2s)} = 2^{1-2s}\frac{\Gamma\left(\frac{1}{2}\right)\Gamma(s)}{\Gamma\left(s+\frac{1}{2}\right)}.$$

As $\Gamma(s)$ is nonzero, we may cancel the common $\Gamma(s)$ factor and use Equation (C.1) to obtain

$$\frac{\Gamma(s)}{\Gamma(2s)} = \frac{2^{1-2s}\sqrt{\pi}}{\Gamma\left(s+\frac{1}{2}\right)}.$$

This is equivalent to the desired formula. $\square$

As the reciprocal of the gamma function is entire it admits a Hadamard factorization. We will compute this factorization. To this end, recall the limit

$$\lim_{n\to\infty}\left(\frac{x}{n}+1\right)^n = e^x, \tag{C.3}$$

provided $x > 0$, and that $\left(\frac{x}{n}+1\right)^n$ is monotonically increasing in n. We first prove the following lemma:

Lemma C.2.5. *For any $s \in \mathbb{C} - \mathbb{Z}_{0-}$, we have*

$$\lim_{n\to\infty}\frac{n^{-s}\Gamma(s+n)}{\Gamma(n)} = 1.$$

Proof. Apply the functional equation to write

$$\frac{n^{-s}\Gamma(s+n)}{\Gamma(n)} = \left(\frac{s+n-1}{n}\right)\frac{n^{1-s}\Gamma(s-1+n)}{\Gamma(n)},$$

As the limit of the first factor on the right-hand side is 1 as $n \to \infty$, we see that the claim holds for s if and only if it holds for $s-1$. Therefore we may assume $\sigma < 0$. In other words, it suffices to show

$$\lim_{n\to\infty}\frac{n^{s}\Gamma(n-s)}{\Gamma(n)} = 1,$$

for $\sigma > 0$. Appealing to Theorem C.2.4, observe

$$\frac{n^{s}\Gamma(n-s)}{\Gamma(n)} = \frac{n^{s}B(s,n-s)}{\Gamma(s)} = \frac{1}{\Gamma(s)}n^{s}\int_0^\infty \frac{x^{s-1}}{(x+1)^n}\,dx.$$

The change of variables $x \mapsto \frac{x}{n}$ gives

$$\frac{1}{\Gamma(s)}\int_0^\infty \frac{x^{s-1}}{\left(\frac{x}{n}+1\right)^n}\,dx.$$

We now take the limit as $n \to \infty$. By Equation (C.3) and that $\left(\frac{x}{n}+1\right)^n$ is monotonically increasing in n, we may interchange the limit and integral by monotone convergence giving

$$\lim_{n\to\infty}\frac{1}{\Gamma(s)}\int_0^\infty \frac{x^{s-1}}{\left(\frac{x}{n}+1\right)^n}\,dx = \frac{1}{\Gamma(s)}\int_0^\infty e^{-x}x^{s-1}\,dx = 1.$$

This means

$$\lim_{n\to\infty}\frac{n^{s}\Gamma(n-s)}{\Gamma(n)} = 1,$$

as desired. $\square$

Obtaining the Hadamard factorization of the reciprocal of the gamma function is now a simple matter. Recall that the Euler-Mascheroni constant γ is defined as the limit

$$\gamma = \lim_{N\to\infty}\left(\sum_{n\le N}\frac{1}{n} - \log(N)\right). \tag{C.4}$$

The Hadamard factorization is the following:

Theorem C.2.6. *For any $s \in \mathbb{C} - \mathbb{Z}_{0-}$, we have*

$$\frac{1}{\Gamma(s)} = se^{\gamma s}\prod_{n\ge1}\left(1+\frac{s}{n}\right)e^{-\frac{s}{n}},$$

where γ is the Euler-Mascheroni constant.

Proof. Let k be a positive integer. Apply the functional equation repeatedly to write

$$\frac{1}{\Gamma(s)} = \frac{s(s+1)\cdots(s+k-1)}{\Gamma(s+k)},$$

In view of the identity

$$s+n = e^{\frac{s}{n}}e^{-\frac{s}{n}}n\left(1+\frac{s}{n}\right),$$

we obtain

$$\frac{1}{\Gamma(s)} = se^{\left(1+\frac{1}{2}+\cdots+\frac{1}{k}\right)s}\prod_{1\le n\le k}\left(1+\frac{s}{n}\right)e^{-\frac{s}{n}}\frac{(k-1)!}{\Gamma(s+k)}.$$

As $k^{-s} = e^{-s\log(k)}$ and $\Gamma(k) = (k-1)!$, we may further write

$$\frac{1}{\Gamma(s)} = se^{\left(1+\frac{1}{2}+\cdots+\frac{1}{k}-\log(k)\right)s}\frac{\Gamma(k)}{k^{-s}\Gamma(s+k)}\prod_{1\le n\le k}\left(1+\frac{s}{n}\right)e^{-\frac{s}{n}}.$$

Taking the limit as $k\to\infty$, Lemma C.2.5 and Equation (C.4) together imply

$$\frac{1}{\Gamma(s)} = se^{\gamma s}\prod_{n\ge1}\left(1+\frac{s}{n}\right)e^{-\frac{s}{n}},$$

as desired. $\square$

Upon inspection of the Hadamard factorization, we find that the reciprocal of the gamma function is of order 1. We now turn to analyzing the growth rate of the gamma function itself. In fact, we can obtain a well-known asymptotic formula for the gamma function known as **Stirling's formula**.

Theorem (Stirling's formula). *Let $|\arg(s)| < \pi-\varepsilon$ and $|s| > \delta$ for some $\varepsilon, \delta > 0$. Then*

$$\Gamma(s) = \sqrt{2\pi}s^{s-\frac{1}{2}}e^{-s}\left(1+O_{\varepsilon,\delta}\left(\frac{1}{s}\right)\right).$$

In particular,

$$\Gamma(s) \sim \sqrt{2\pi}s^{s-\frac{1}{2}}e^{-s}.$$

Proof. We begin by additionally assuming $\sigma > 0$. Express $\Gamma(s+1)$ in the complex form

$$\Gamma(s+1) = \int_0^\infty e^{-z} e^{s\log(z)}\,dz.$$

The change of variables $z \to s(1+z)$ gives

$$\Gamma(s+1) = s^{s+1}e^{-s}\int_{-1}^\infty e^{s(\log(1+z)-z)}\,dz,$$

upon homotoping the transformed contour to $(-1,\infty)$. We will estimate the present integral. Define

$$I(s) = \int_{-1}^\infty e^{s(\log(1+z)-z)}\,dz,$$

along with

$$I_1(s) = \int_{-1}^1 e^{s(\log(1+z)-z)}\,dz, \quad \text{and} \quad I_2(s) = \int_1^\infty e^{s(\log(1+z)-z)}\,dz.$$

Then

$$\Gamma(s+1) = s^{s+1}e^{-s}I(s), \tag{C.5}$$

with

$$I(s) = I_1(s) + I_2(s).$$

Making the change of variables $z \mapsto 2+z$ to $I_2(s)$ shows

$$I_2(s) = e^{-2s}\int_{-1}^\infty e^{s(\log(3+z)-z)}\,dz.$$

Since $\log(3+z) = \log(1+z) + O(1)$, $I_2(s)$ is related to $I(s)$ via

$$I_2(s) = e^{-2s}\int_{-1}^\infty e^{s(\log(1+z)-z)}(1+O_{\varepsilon,\delta}(s))\,dz = I(s)(e^{-2s} + O_{\varepsilon,\delta}(se^{-2s})).$$

Whence

$$I(s) = I_1(s) + I(s)(e^{-2s} + O_{\varepsilon,\delta}(se^{-2s})).$$

Now isolate $I(s)$ to see that

$$I(s) = I_1(s)(1 + O_{\varepsilon,\delta}(se^{-2s})). \tag{C.6}$$

Therefore it remains to estimate $I_1(s)$. To this end, notice

$$\frac{d}{dz}(\log(1+z) - z) = \frac{1}{1+z} - 1 \quad \text{and} \quad \frac{d^2}{dz^2}(\log(z) - z) = -\frac{1}{(1+z)^2}.$$

At $z = 0$, the first derivative is zero and the second derivative is negative. So taking the Taylor series of $\log(1+z) - z$ and truncating after the third term gives

$$\log(1+z) - z = -\frac{1}{2}z^2 + \frac{1}{3}z^3 + O(z^4).$$

Use this estimate in $I_1(s)$ to write

$$I_1(s) = \int_{-1}^{1} e^{-\frac{s}{2}z^2} e^{\frac{1}{3}sz^3 + O_{\varepsilon,\delta}(sz^4)}\, dz.$$

Considering the Taylor series of e^z and truncating after the second term, we find that

$$e^{\frac{1}{3}sz^3 + O_{\varepsilon,\delta}(sz^4)} = 1 + \frac{1}{3}sz^3 + O_{\varepsilon,\delta}(sz^4) + O_{\varepsilon,\delta}(s^2z^6).$$

Plugging this estimate into the previous integral yields

$$I_1(s) = \int_{-1}^{1} e^{-\frac{s}{2}z^2} \left(1 + \frac{1}{3}sz^3 + O_{\varepsilon,\delta}(sz^4) + O_{\varepsilon,\delta}(s^2z^6)\right)\, dz.$$

Since the first and second terms in the asymptotic are even and odd in z respectively, symmetry forces

$$I_1(s) = 2\int_{0}^{1} e^{-\frac{s}{2}z^2} \left(1 + O_{\varepsilon,\delta}(sz^4) + O_{\varepsilon,\delta}(s^2z^6)\right)\, dz.$$

Performing the change of variables $z \mapsto \sqrt{\frac{2z}{s}}$, and noting that z is bounded (so that the O-estimates are independent of z), results in

$$I_1(s) = \sqrt{2}s^{-\frac{1}{2}} \int_{0}^{\frac{|s|}{2}} e^{-z}z^{-\frac{1}{2}}\, dz \left(1 + O_{\varepsilon,\delta}\left(\frac{1}{s}\right)\right), \tag{C.7}$$

where we have homotoped the transformed contour to $\left(0, \frac{|s|}{2}\right)$. We will further estimate the remaining integral. Observe that

$$\int_{0}^{\frac{|s|}{2}} e^{-z}z^{-\frac{1}{2}}\, dz = \Gamma\left(\frac{1}{2}\right) - \int_{\frac{|s|}{2}}^{\infty} e^{-z}z^{-\frac{1}{2}}\, dz = \sqrt{\pi} - \int_{\frac{|s|}{2}}^{\infty} e^{-z}z^{-\frac{1}{2}}\, dz,$$

where we have used Equation (C.1) to evaluate the gamma function. By the exponential decay of the integrand, we have say

$$\int_{\frac{|s|}{2}}^{\infty} e^{-z}z^{-\frac{1}{2}}\, dz = O_{\varepsilon,\delta}\left(\int_{\frac{|s|}{2}}^{\infty} \frac{1}{z^3}\, dz\right) = O_{\varepsilon,\delta}\left(\frac{1}{s^2}\right),$$

so that

$$\int_{0}^{\frac{|s|}{2}} e^{-z}z^{-\frac{1}{2}}\, dz = \sqrt{\pi} + O_{\varepsilon,\delta}\left(\frac{1}{s^2}\right). \tag{C.8}$$

Substituting Equation (C.8) into Equation (C.7) gives

$$I_1(s) = \sqrt{2\pi}s^{-\frac{1}{2}} \left(1 + O_{\varepsilon,\delta}\left(\frac{1}{s}\right)\right). \tag{C.9}$$

Combining Equations (C.5), (C.6) and (C.9) yields

$$\Gamma(s+1) = \sqrt{2\pi}s^{s+\frac{1}{2}}e^{-s}\left(1 + O_{\varepsilon,\delta}\left(\frac{1}{s}\right)\right).$$

Using the functional equation gives

$$\Gamma(s) = \sqrt{2\pi}s^{s-\frac{1}{2}}e^{-s}\left(1 + O_{\varepsilon,\delta}\left(\frac{1}{s}\right)\right).$$

This proves the asymptotic formula when $\sigma > 0$. In fact, applying the functional equation repeatedly shows

$$\Gamma(s-n) = \sqrt{2\pi}\frac{s^{s-\frac{1}{2}}}{(s-1)(s-2)\dots(s-n)}e^{-s}\left(1 + O_{\varepsilon,\delta}\left(\frac{1}{s}\right)\right),$$

for any nonnegative integer n. As $\frac{1}{s-k} = \frac{1}{s} + O_{\varepsilon,\delta}\left(\frac{1}{s^2}\right)$, it follows that

$$\Gamma(s-n) = \sqrt{2\pi}s^{s-n-\frac{1}{2}}e^{-s}\left(1 + O_{\varepsilon,\delta}\left(\frac{1}{s}\right)\right).$$

Since n was arbitrary, this proves the asymptotic formula when $\sigma \le 0$ as well. The asymptotic equivalence follows immediately since it is a strictly weaker asymptotic. □

We now turn to estimates for the gamma function in vertical strips. The first of which is useful asymptotic equivalence showing that the gamma function admits exponential decay.

Corollary C.2.7. *Let $|\arg(s)| < \pi - \varepsilon$ and $|s| > \delta$ for some $\varepsilon, \delta > 0$ and suppose σ is bounded. Then*

$$\Gamma(s) \sim \sqrt{2\pi}|t|^{\sigma-\frac{1}{2}}e^{-\frac{\pi}{2}|t|}.$$

Proof. Stirling's formula implies

$$\Gamma(s) \sim \sqrt{2\pi}e^{\left(s-\frac{1}{2}\right)\log(s)}e^{-s}.$$

Writing $\log(s) = \log|s| + i\arg(s)$, this asymptotic becomes

$$\Gamma(s) \sim \sqrt{2\pi}e^{\left(s-\frac{1}{2}\right)\log|s|}e^{\left(s-\frac{1}{2}\right)i\arg(s)}e^{-s}.$$

Taking absolute values further implies

$$\Gamma(s) \sim \sqrt{2\pi}e^{\left(\sigma-\frac{1}{2}\right)\log|s|}e^{-(t\arg(s)+\sigma)}.$$

As σ is bounded, we have the asymptotics $|s| = |t| + o(1)$ and $\arg(s) = \operatorname{sgn}(t)\frac{\pi}{2} + o(1)$. These estimates along with the fact $-t\operatorname{sgn}(t) = -|t|$ together imply

$$\Gamma(s) \sim \sqrt{2\pi}e^{\left(\sigma-\frac{1}{2}\right)\log|t|}e^{-\frac{\pi}{2}|t|}.$$

This is equivalent to the desired asymptotic. □

We can use Corollary C.2.7 to obtain weaker estimates valid on the negative real line away from the poles of the gamma function.

Corollary C.2.8. *Suppose σ is bounded. Then*

$$\Gamma(s) \ll_\varepsilon (|t|+3)^{\sigma-\frac{1}{2}}e^{-\frac{\pi}{2}|t|} \quad \text{and} \quad \frac{1}{\Gamma(s)} \ll (|t|+3)^{\frac{1}{2}-\sigma}e^{\frac{\pi}{2}|t|},$$

where in the former estimate we assume s is at least distance ε away from the poles of the gamma function in this vertical strip.

Proof. Let s be in the given region and assume $|t| > \varepsilon$. Corollary C.2.7 implies the strictly weaker estimates

$$\Gamma(s) \ll_\varepsilon |t|^{\sigma-\frac{1}{2}}e^{-\frac{\pi}{2}|t|} \quad \text{and} \quad \frac{1}{\Gamma(s)} \ll_\varepsilon |t|^{\frac{1}{2}-\sigma}e^{\frac{\pi}{2}|t|}.$$

Since the gamma function is holomorphic and bounded on the compact set defined by σ being bounded and $|t| \le \varepsilon$ with s at least distance ε away from its poles, we may relax the condition $|t| \ge 1$ provided we replace $|t|$ with $|t|+3$. This proves the first estimate. The second estimate follows by an analogous argument but holds in the entire vertical strip as the reciprocal of the gamma function is entire. $\square$

The choice of 3 in $|t|+3$ in Corollary C.2.8 is a matter of convenience as we could use any large positive constant. In particular, it is useful when taking logarithms as $\log(|t|+3) \ge 1$.

Having discussed the major properties of the gamma function, we now introduce a related function. We call $\frac{\Gamma'}{\Gamma}(s)$ the **digamma function**. In other words, the digamma function is the logarithmic derivative of the gamma function. From Theorem C.2.2, it is meromorphic on $\mathbb{C}$ with simple poles at the nonpositive integers whose residues are all -1. Taking the logarithmic derivative of the functional equation shows

$$\frac{\Gamma'}{\Gamma}(s+1) = \frac{\Gamma'}{\Gamma}(s) + \frac{1}{s}.$$

In similar spirit, we may take the logarithmic derivative of the Hadamard factorization of the reciprocal of the gamma function to obtain a summation formula for the digamma function.

Corollary C.2.9. *For all $s \in \mathbb{C} - \mathbb{Z}_{0-}$, we have*

$$\frac{\Gamma'}{\Gamma}(s+1) = -\gamma + \sum_{n\ge1}\left(\frac{1}{n} - \frac{1}{s+n}\right),$$

where γ is the Euler-Mascheroni constant.

Proof. As the gamma function has no zeros, we may use the functional equation to write

$$\frac{1}{\Gamma(s+1)} = \frac{1}{s\Gamma(s)}.$$

Taking the logarithmic derivative by using the Hadamard factorization shows

$$-\frac{\Gamma'}{\Gamma}(s+1) = \gamma + \sum_{n\geq 1}\left(\frac{1}{s+n} - \frac{1}{n}\right).$$

This is equivalent to the desired formula. $\square$

As for estimates involving the digamma function, we can appeal to Stirling's formula to obtain as asymptotic formula.

Proposition C.2.10. *Let $|\arg(s)| < \pi - \varepsilon$ and $|s| > \delta$ for some $\varepsilon, \delta > 0$. Then*

$$\frac{\Gamma'}{\Gamma}(s) = \log(s) + O_{\varepsilon,\delta}\left(\frac{1}{s}\right).$$

In particular,

$$\frac{\Gamma'}{\Gamma}(s) \sim \log(s).$$

Proof. Taking the logarithm of Stirling's formula gives

$$\log\Gamma(s) = \frac{1}{2}\log(2\pi) + \left(s - \frac{1}{2}\right)\log(s) - s + O_{\varepsilon,\delta}\left(\frac{1}{s}\right),$$

Set $g(s) = \frac{1}{2}\log(2\pi) + \left(s - \frac{1}{2}\right)\log(s) - s$. Then we may write

$$\log\Gamma(s) - g(s) = O_{\varepsilon,\delta}\left(\frac{1}{s}\right).$$

Now write

$$\log\Gamma(s) = g(s) + \log\Gamma(s) - g(s),$$

and note that $g'(s) = \log(s) - \frac{1}{2s}$. Taking the derivative of this identity and using Cauchy's integral formula to express the derivative of $\log\Gamma(s) - g(s)$, we obtain

$$\frac{\Gamma'}{\Gamma}(s) = \log(s) - \frac{1}{2s} + \frac{1}{2\pi i}\int_{\eta}\frac{\log\Gamma(u) - g(u)}{(u-s)^2}\,du,$$

where η is a circle about s of sufficiently small radius r depending upon ε and δ. The parameterization $u \mapsto s + re^{i\theta}$ for η shows that

$$\frac{1}{2\pi i}\int_{\eta}\frac{\log\Gamma(u) - g(u)}{(u-s)^2}\,du = O_{\varepsilon,\delta}\left(\frac{1}{s}\right),$$

in view of our estimate for $\log\Gamma(s) - g(s)$. Whence

$$\frac{\Gamma'}{\Gamma}(s) = \log(s) + O_{\varepsilon,\delta}\left(\frac{1}{s}\right).$$

This proves the asymptotic formula. The asymptotic equivalence is immediate as it is a strictly weaker asymptotic. $\square$

Just as for the gamma function, it is possible to obtain estimates for the digamma function in vertical strips.

Corollary C.2.11. *Let* $|\arg(s)| < \pi - \varepsilon$ *and* $|s| > \delta$ *for some* $\varepsilon, \delta > 0$ *and suppose* σ *is bounded. Then*

$$\frac{\Gamma'}{\Gamma}(s) \sim \log|s|.$$

Proof. Writing $\log(s) = \log|s| + i\arg(s)$, Proposition C.2.10 gives

$$\frac{\Gamma'}{\Gamma}(s) \sim (\log|s| + i\arg(s)).$$

As σ is bounded, we have $\arg(s) = O(1)$. This asymptotic implies

$$\frac{\Gamma'}{\Gamma}(s) \sim \log|s|,$$

as desired. □

We can use Corollary C.2.11 to obtain a weaker estimate valid on the negative real line away from the poles of the gamma function.

Corollary C.2.12. *Suppose* σ *is bounded and* s *is at least distance* ε *away from the poles of the gamma function in this vertical strip. Then*

$$\frac{\Gamma'}{\Gamma}(s) \ll_\varepsilon \log(|s| + 3).$$

Proof. Let s be in the given region and assume $|t| > \varepsilon$. Corollary C.2.11 implies the strictly weaker estimate

$$\frac{\Gamma'}{\Gamma}(s) \ll_\varepsilon \log|s|.$$

Since the digamma function is holomorphic and bounded on the compact set defined by σ being bounded and $|t| \leq \varepsilon$ with s at least distance ε away from its poles, we may relax the condition $|t| \geq 1$ provided we replace $|s|$ with $|s| + 3$. The estimate follows. □

The choice of 3 in $|s| + 3$ at the end of Corollary C.2.12 is a matter of convenience as could use any large positive constant. In particular, it is useful when taking logarithms as $\log(|s| + 3) \geq 1$.

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